

Day-1

A. FRONTEND

Frontend is the **visual and interactive layer** of a website or application that users directly see and interact with through a web browser or mobile app.

It is also called the **Client Side** because it runs on the user's device (browser/mobile).

Responsibilities of Frontend

- Displays web pages and user interface (UI)
- Accepts user input (forms, buttons, search boxes)
- Validates user input before sending it to the server
- Calls Backend APIs to fetch or send data
- Displays the response received from the backend
- Provides animations and interactive effects
- Ensures responsive design for mobile, tablet, and desktop devices

Common Frontend Technologies

Technology	Purpose
HTML	Structure of the webpage
CSS	Styling and layout
JavaScript	Interactivity and dynamic behavior
Bootstrap	Responsive UI framework
React.js	JavaScript library for building UI

Angular	Frontend framework
Vue.js	Progressive JavaScript framework
TypeScript	JavaScript with static typing

Frontend Examples

- Login page
- Registration form
- Shopping cart
- Dashboard
- Profile page
- Online payment page

Real-Time Example

When you click the **Login** button:

1. Frontend collects username and password.
 2. Sends them to the backend through an API.
 3. Waiting for the response.
 4. Displays either:
 - Login Successful
 - Invalid Username/Password
-

B. BACKEND

Backend is the **server-side** part of an application.

Users cannot see it directly, but it performs all the business logic, security, data processing, and database operations.

It acts as the **brain of the application**.

Responsibilities of Backend

- Processes client requests
- Implements business logic
- Connects with databases
- Performs CRUD operations
- Authenticates and authorizes users
- Encrypts passwords and sensitive data
- Generates reports
- Sends emails and notifications
- Handles file uploads/downloads
- Returns responses to the frontend

Common Backend Technologies

Technology	Purpose
Java	Backend Programming Language
Spring Boot	Java Framework
Node.js	JavaScript Runtime
Python (Django/Flask)	Backend Development
PHP	Web Development
.NET	Microsoft Backend Framework
Go	High-performance Backend

Databases Used

- MySQL
- PostgreSQL
- Oracle
- SQL Server
- MongoDB
- Firebase

Backend Tasks Example

Suppose a user logs in:

1. Receives username/password.
2. Checks the database.
3. Verifies password.
4. Generates JWT Token or Session.
5. Sends success/failure response.

Real-Time Example

In an Online Banking System:

- Checks account balance
- Validates transaction
- Transfers money
- Updates database
- Sends SMS/Email confirmation

All these happen in the backend.

C. API (Application Programming Interface)

An **API** is a set of rules and protocols that allows two different applications or systems to communicate and exchange data without exposing their internal implementation.

It acts as a **bridge** between the Frontend and Backend or between two different software systems.

Simple Definition

API = Communication Bridge

Restaurant Analogy

You (Client) → gives order to Waiter (API) → Kitchen (Server) prepares food
Kitchen → gives food to Waiter (API) → You receive your meal

Types of APIs:

Type	Description	Example
REST API	Uses HTTP, most common	Weather app fetching data
SOAP API	XML-based, used in banking	Net banking transactions
GraphQL	Fetch only what you need	Facebook, GitHub
WebSocket API	Real-time, two-way	WhatsApp, Live scores

Real-Time Scenario:

Booking a flight on MakeMyTrip:

- MakeMyTrip doesn't own airline data
- It calls IndiGo's API, Air India's API etc.
- Each airline API returns available seats & prices
- MakeMyTrip displays all results in one place
- You never saw the API — but it did all the connecting!

Storage Areas

as part of our applications we required to store data like Customers information, or billing information, calls information...etc

to store this data we required storage areas.

there are 2 types of storage areas such as

1. Temporary Storage Areas
2. Permanent Storage Areas.

1. Temporary Storage Areas

These are the memory areas where the data will be stored temporally.

EG: all JVM memory areas(like Heap Area, Stack Area, Method area, PC Registers, Native Method stacks). Once JVM Shutdown all these memory areas will be cleared automatically.

2. Permanent Storage Areas.

also known as persistent storage areas. Here we can store data permanently

example: Files Systems, Databases, Data warehouse, Bigdata..etc

File Management System(FMS)

def: A File Management System is a System where data is stored in files on the operation systems, each application program must handle its own storage data, retrieval and updating. File Systems can be provided by the Local Operating System.

File systems are best suitable for to store very less amount of information(like KB)

example:- a library management system storing books in separate text files likes, books.txt, members.txt

Limitations of File Management Systems(FMS)

1. Data Redundancy :- same data is stored in multiple files
2. Data Inconsistency :- updates in one file may not be reflected in another.
3. poor Data Security :- No proper accessed control.

4. Difficult Data Retrieval : searching requires custom programs.
5. Integrity issues :- No Constraints(eg: rollnumber uniqueness)
6. Scalability issues : hard to manage as data grows.

To overcome the above problems of File systems , we should go for databases.

Database Management System

definition: a DBMS is a Collection of Programs that enables users to create,manage, and manipulate databases.

it acts as an interface between user and database.

Advantages

1. We can store huge amounts of information in databases.
2. Query language support is available for every database and hence we can perform database operations very easily.
3. To access data present in the database, compulsory username and password must be required. hence data is secured.
4. inside database data will be stored in the form of tables. While developing database table schemas, database admin follow various normalization techniques and we can implement various constraints like Unique key constraint, primary key constraints, ..etc which prevent data duplication. hence there is no chance of data inconsistency problems.

Limitation of Databases

- 1) Databases cannot store huge amounts of information like terabytes(1000gb)of data.
- 2) Database can provide support for only structured data(tabular data or relational data) and cannot provide support for semi structured data like(xml files)and unstructured data (like video files, audio files,images..etc)

To overcome these problems we should go for advanced storage areas like bigdata,Data warehouses. Etc.

what are the differences between File Management System(FMS) and Database Management System(DBMS)

=====

=====

1. data is stored in separate files

- data is stored in a structured database.
- 2. High data redundancy(duplicate data)
reduces data redundancy
- 3. difficult to maintain and update
easy to maintain and update
- 4. security is limited
provides strong security mechanisms.
- 5. data sharing is difficult
multiple users can access data simultaneously.
- 6. no relationship between files.
supports relationships between tables.
- 7. Backup and recovery are difficult
provides backup and recovery features
- 8. Data inconsistency may cause anomalies.
maintains data consistency and integrity.
- 9. suitable for small applications
suitable for small, medium and large applications.

Q: What is the difference between a File System and a DBMS?(IQ)

A file system stores data in separate files and may lead to data redundancy, and security issues, A DBMS stores data in structured manner using tables, reduces redundancy, provides security, supports multiple users, and ensures data consistency and integrity.

Day-2

Day-2

What is data, field, record, database?

What is Data?

raw facts or figures without context.

example: 101,Ravi,50000,hyd

What is the field?

smallest unit of data in a database(column/attribute)

example: EmpName,EmpId,EmpSalary,EmpAddress

What is Record ?

collection of related fields(row/tuple)

example :(111,ravi,60000,vjy)

What is a database ?

organized collection of related records stored together.

example : an employee database containing all employee records.

Client-server

client : the role of the client in client-server architecture is to send requests to the server. and to get a response from the server.

Server : - server is software the role of the server in client-server arch is to take requests from clients.

if the request is a valid request then it will generate some response we can handover that response to the client. If the client request is not a valid request then it will raise the errors.

protocol :- it has a set of rules and regulations. The role of the protocol in client-server arch is to send requests from client to server.

Data Models In DBMS

def : a Data Model defines how data is stored, organized, and manipulated in a database.

1. Physical Data Model

def: describes how data is physically stored in storage devices.

Audience : Database administrators.

2. Logical Data Model

defines the structure of data (entities, attributes, relationships) independent of physical storage areas.

Audience : Developers and data architects.

3. Hierarchical Data Model :

data is represented as tree Structure(parent child relationship)

example :- company-->departments-->employees

4. Network Data Model :

Data is represented as graph with records(nodes) and relationships(edges)

example : student enrolled in multiple courses, courses taught by multiple professors.

5. Relational Data Models(RDBMS)

data is stored in tables(relations) with rows(tuples) and columns(attributes)

example

student(id, name, marks, age, dept_id)

department(dept_id, dept_name)

6. Entity Relationship(E-R Model)

a conceptual model that shows entities(things), attributes(properties) and relationships

MySQL Installation

<https://dev.mysql.com/downloads/installer/>

after installation go to location

just double click ---> choose the yes option.

4 option

server

client

full

custom ---> choose custom option--> choose next button

it will show available products

such as

MySQL server

application

documentation

go to my sql server

<https://tinyurl.com/MySQL-Batch-03>

Day-3

SQL Identifiers

SQL identifiers are names used to identify database objects such as.

Database names, Table names, Column names, view names, index names, constraints, triggers, stored procedures.

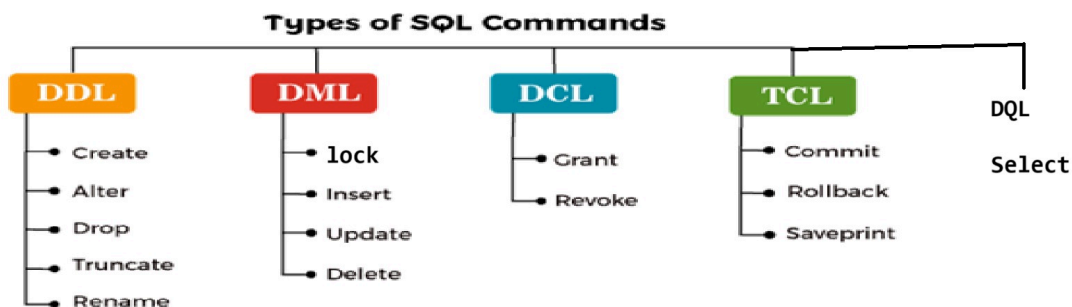
While we are declaring an identifier we have to follow some rules and regulations.

1. the only allowed characters to declare identifiers a-z or A-Z or 0-9,_, \$
2. must begin with a letter or underscore(_).
3. we can't declare identifiers with reserved keywords.
4. spaces are not considered to declare identifiers.
5. identifiers are case sensitive
6. max length to declare identifiers are 15 characters.

SQL (Structured Query Language)

Structured query language(SQL) is standardized programming language specially designed for managing and manipulating relation databases. it provides a declarative syntax that allows users to define what data they want retrieve, insert, update or delete without specifying exactly how database should execute these operations.

Sql serves as the universal interface between applications and relational database management systems like MYSQL, Postgre Sql, Oracle and SQL Server. the language consists of several components.



SQL (Structured Query Language) commands are broadly classified into five main categories based on their functionality: DDL, DML, DQL, DCL, and TCL. Each category and every command within it is explained below with its purpose and usage.

Command	Description / Definition
DDL	Data Definition Language - defines and modifies the structure of database objects (tables, schemas, indexes).

DML	Data Manipulation Language - manages data stored within database objects (insert, update, delete).
DQL	Data Query Language - used to fetch/retrieve data from the database.
DCL	Data Control Language - manages permissions and access control on database objects.
TCL	Transaction Control Language - manages transactions to maintain data integrity.

1. DDL - Data Definition Language

DDL commands are used to define, create, alter, and remove the structure of database objects such as tables, schemas, and indexes. Changes made by DDL commands are auto-committed and permanent.

Command	Description / Definition
CREATE	Creates a new database object such as a table, database, view, index, or schema. Example: CREATE TABLE Students (ID INT, Name VARCHAR(50));
ALTER	Modifies the structure of an existing database object, such as adding, deleting, or modifying columns in a table. Example: ALTER TABLE Students ADD Email VARCHAR(100);
DROP	Permanently deletes an existing database object (table, database, index, view) along with all its data and structure. This action cannot be rolled back in most databases.
TRUNCATE	Removes all rows/records from a table quickly while keeping the table structure intact for future use. Faster than DELETE as it does not log individual row deletions.
RENAME	Renames an existing database object, such as a table or column, to a new name without changing its structure or data.

COMMENT	Adds explanatory notes or comments to database objects (tables, columns) for documentation purposes; does not affect data or structure.
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2. DML - Data Manipulation Language

DML commands are used to insert, update, delete, and manage data stored inside existing database tables. Unlike DDL, DML changes can typically be rolled back if not committed.

Command	Description / Definition
INSERT	Adds new rows/records of data into a table. Example: INSERT INTO Students (ID, Name) VALUES (1, 'Ravi');
UPDATE	Modifies existing records in a table by changing the values of one or more columns based on a specified condition. Example: UPDATE Students SET Name='Kumar' WHERE ID=1;
DELETE	Removes one or more existing rows from a table based on a specified condition, while keeping the table structure. Can be rolled back if used within a transaction.
MERGE	Combines INSERT, UPDATE, and DELETE operations into a single statement by comparing source and target tables (also called an 'upsert').
CALL	Invokes or executes a stored procedure to perform a set of predefined operations.
EXPLAIN PLAN	Describes the access path or execution plan that the database will use to run a given SQL statement, used mainly for performance tuning.
LOCK TABLE	Controls concurrency by locking a table to restrict access from other transactions while it is being modified.

3. DQL - Data Query Language

DQL consists of commands used purely to query and retrieve data from one or more tables without modifying it.

Command	Description / Definition
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SELECT	Retrieves data from one or more tables or views based on specified columns and conditions. It is the most frequently used SQL command. Example: SELECT Name FROM Students WHERE ID=1;
---------------	---

4. DCL - Data Control Language

DCL commands deal with the rights, permissions, and other access controls of the database system, determining which users can access or manipulate data.

Command	Description / Definition
GRANT	Gives specific privileges or permissions (such as SELECT, INSERT, UPDATE, DELETE) to a database user or role. Example: GRANT SELECT ON Students TO User1;
REVOKE	Removes previously granted privileges or permissions from a database user or role, restricting their access to database objects.

5. TCL - Transaction Control Language

TCL commands manage transactions within a database to ensure data integrity, consistency, and to group DML statements into logical units of work.

Command	Description / Definition
COMMIT	Permanently saves all changes made during the current transaction to the database, making them visible to other users.
ROLLBACK	Undoes/reverts changes made during the current transaction back to the last committed state or a savepoint, in case of an error.
SAVEPOINT	Sets a temporary marker within a transaction to which you can later roll back, without undoing the entire transaction.
SET TRANSACTION	Defines properties for a transaction, such as its isolation level, to control how it interacts with other concurrent transactions.
RELEASE SAVEPOINT	Removes a previously created savepoint from the current transaction, freeing associated resources.

Note: Some references also group SELECT under DML instead of a separate DQL category, and command availability/syntax may vary slightly across database systems (MySQL, PostgreSQL, Oracle, SQL Server).

```
mysql> show databases;
```

```
+-----+
| Database          |
+-----+
| information_schema |
| jeevan_raksha_pharmacy |
| mysql             |
| performance_schema |
| sys               |
+-----+
5 rows in set (0.08 sec)
```

```
mysql> CREATE DATABASE PFS3;
Query OK, 1 row affected (0.06 sec)
```

```
mysql> SHOW DATABASES;
```

```
+-----+
| Database          |
+-----+
| information_schema |
| jeevan_raksha_pharmacy |
| mysql             |
| performance_schema |
| pfs3              |
| sys               |
+-----+
6 rows in set (0.00 sec)
```

```
mysql> USE PFS3;
Database changed
```

```
mysql> CREATE TABLE EMPLOYEES(
-> EMPID CHAR(5),
-> FNAME VARCHAR(100),
```

```

-> LNAME VARCHAR(30),
-> AGE INT,
-> DOJ DATE,
-> ADDRESS TINYTEXT,
-> DEPT VARCHAR(20)
-> );

```

Query OK, 0 rows affected (0.27 sec)

```
mysql> ^C
```

```
mysql> -- TO SHOW THE TABLE DESCRIPTION
```

```
mysql> DESC EMPLOYEES;
```

Field	Type	Null	Key	Default	Extra
EMPID	char(5)	YES		NULL	
FNAME	varchar(100)	YES		NULL	
LNAME	varchar(30)	YES		NULL	
AGE	int	YES		NULL	
DOJ	date	YES		NULL	
ADDRESS	tinytext	YES		NULL	
DEPT	varchar(20)	YES		NULL	

7 rows in set (0.02 sec)

```
mysql> SHOW COLUMNS FROM EMPLOYEES;
```

Field	Type	Null	Key	Default	Extra
EMPID	char(5)	YES		NULL	
FNAME	varchar(100)	YES		NULL	
LNAME	varchar(30)	YES		NULL	
AGE	int	YES		NULL	
DOJ	date	YES		NULL	
ADDRESS	tinytext	YES		NULL	
DEPT	varchar(20)	YES		NULL	

7 rows in set (0.00 sec)

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```
mysql> USE PFS3;
Database changed
mysql> CREATE TABLE EMPLOYEES(
  -> EMPID CHAR(5),
  -> FNAME VARCHAR(100),
  -> LNAME VARCHAR(60),
  -> AGE INT,
  -> DOJ DATE,
  -> ADDRESS TINYTEXT,
  -> DEPT VARCHAR(50)
  -> );
Query OK, 0 rows affected (0.30 sec)
```

```
mysql> DESC EMPLOYEES;
+-----+-----+-----+-----+-----+-----+
| Field | Type      | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| EMPID | char(5)   | YES  |     | NULL    |       |
| FNAME | varchar(100) | YES  |     | NULL    |       |
| LNAME | varchar(60) | YES  |     | NULL    |       |
| AGE   | int       | YES  |     | NULL    |       |
| DOJ   | date      | YES  |     | NULL    |       |
| ADDRESS | tinytext  | YES  |     | NULL    |       |
| DEPT  | varchar(50) | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
7 rows in set (0.04 sec)
```

```
mysql> -- ALTER
mysql> -- MODIFY THE FNAME COLUMN IN EMPLOYEES TABLE TO VARCHAR(70)
mysql> -- SYNTAX : ALTER TABLE TABLE-NAME MODIFY COLUMN-NAME
DATATYPE SIZE.
mysql> ALTER TABLE EMPLOYEES MODIFY FNAME VARCHAR(70);
Query OK, 0 rows affected (0.34 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> DESC EMPLOYEES;
+-----+-----+-----+-----+-----+-----+
| Field | Type      | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| EMPID | char(5)   | YES  |     | NULL    |       |
```

FNAME	varchar(70)	YES		NULL	
LNAME	varchar(60)	YES		NULL	
AGE	int	YES		NULL	
DOJ	date	YES		NULL	
ADDRESS	tinytext	YES		NULL	
DEPT	varchar(50)	YES		NULL	

7 rows in set (0.01 sec)

```
mysql> ALTER TABLE EMPLOYEES MODIFY LNAME VARCHAR(80);
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that
corresponds to your MySQL server version for the right syntax to use near 'MODIFY LNAME
VARCHAR(80)' at line 1
mysql> ALTER TABLE EMPLOYEES MODIFY LNAME VARCHAR(80);
Query OK, 0 rows affected (2.27 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> DESC EMPLOYEES;
```

Field	Type	Null	Key	Default	Extra
EMPID	char(5)	YES		NULL	
FNAME	varchar(70)	YES		NULL	
LNAME	varchar(80)	YES		NULL	
AGE	int	YES		NULL	
DOJ	date	YES		NULL	
ADDRESS	tinytext	YES		NULL	
DEPT	varchar(50)	YES		NULL	

7 rows in set (0.01 sec)

```
mysql> -- ADD A NEW COLUMN NAAMED WITH LOCATION OF TYPE TINYTEXT TO
THE EMPLOYEES TABLE.
mysql> -- SYNTAX : ALTER TABLE TABLE-NAME ADD COLUMN-NAME
DATATYPE(SIZE)
mysql> ALTER TABLE EMPLOYEE ADD LOCATION TINYTEXT;
ERROR 1146 (42S02): Table 'pfs3.employee' doesn't exist
mysql> ALTER TABLE EMPLOYEES ADD LOCATION TINYTEXT;
Query OK, 0 rows affected (0.34 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> DESC EMPLOYEES;
```

```
+-----+-----+-----+-----+-----+-----+
| Field | Type      | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| EMPID | char(5)    | YES  |     | NULL    |       |
| FNAME | varchar(70)| YES  |     | NULL    |       |
| LNAME | varchar(80)| YES  |     | NULL    |       |
| AGE   | int        | YES  |     | NULL    |       |
| DOJ   | date       | YES  |     | NULL    |       |
| ADDRESS | tinytext  | YES  |     | NULL    |       |
| DEPT  | varchar(50)| YES  |     | NULL    |       |
| LOCATION | tinytext | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
```

```
8 rows in set (0.04 sec)
```

```
mysql> -- TO ADD A COLUMN AFTER A PARTICULAR COLUMN
```

```
mysql> -- SYNTAX : ALTER TABLE TABLE-NAME ADD COLUMN-NAME AFTER
SPECIFIED-COLUMN NAME
```

```
mysql> ALTER TABLE EMPLOYEES ADD PFID INT AFTER ADDRESS;
```

```
Query OK, 0 rows affected (0.14 sec)
```

```
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> DESC EMPLOYEES;
```

```
+-----+-----+-----+-----+-----+-----+
| Field | Type      | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| EMPID | char(5)    | YES  |     | NULL    |       |
| FNAME | varchar(70)| YES  |     | NULL    |       |
| LNAME | varchar(80)| YES  |     | NULL    |       |
| AGE   | int        | YES  |     | NULL    |       |
| DOJ   | date       | YES  |     | NULL    |       |
| ADDRESS | tinytext  | YES  |     | NULL    |       |
| PFID  | int        | YES  |     | NULL    |       |
| DEPT  | varchar(50)| YES  |     | NULL    |       |
| LOCATION | tinytext | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
```

```
9 rows in set (0.02 sec)
```

```
mysql> -- RENAME THE FNAME COLUMN TO FIRSTNAME IN THE EMPLOYEES
TABLE;
```

```
mysql> -- SYNTAX : ALTER TABLE TABLE-NAME CHANGE
```

```
EXISTING-COLUMN-NAME NEW COLUMN-NAME DATATYPE(SIZE)
```

```
mysql> ALTER TABLE EMPLOYEES CHANGE COLUMN FNAME FIRSTNAME
VARCHAR(30);
```

```
Query OK, 0 rows affected (0.11 sec)
```

```
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> DESC EMPLOYEES;
```

Field	Type	Null	Key	Default	Extra
EMPID	char(5)	YES		NULL	
FIRSTNAME	varchar(30)	YES		NULL	
LNAME	varchar(80)	YES		NULL	
AGE	int	YES		NULL	
DOJ	date	YES		NULL	
ADDRESS	tinytext	YES		NULL	
PFID	int	YES		NULL	
DEPT	varchar(50)	YES		NULL	
LOCATION	tinytext	YES		NULL	

```
9 rows in set (0.00 sec)
```

```
mysql> ALTER TABLE EMPLOYEES CHANGE COLUMN LNAME LASTNAME
VARCHAR(30);
```

```
Query OK, 0 rows affected (0.20 sec)
```

```
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> DESC EMPLOYEES;
```

Field	Type	Null	Key	Default	Extra
EMPID	char(5)	YES		NULL	
FIRSTNAME	varchar(30)	YES		NULL	
LASTNAME	varchar(30)	YES		NULL	
AGE	int	YES		NULL	
DOJ	date	YES		NULL	
ADDRESS	tinytext	YES		NULL	

PFID	int	YES	NULL	
DEPT	varchar(50)	YES	NULL	
LOCATION	tinytext	YES	NULL	

9 rows in set (0.01 sec)

mysql> -- HOW TO DROP A COLUMN

mysql> -- SYNTAX : ALTER TABLE TABLE-NAME DROP COLUMN COLUMN-NAME

mysql> -- DROP THE COLUMN PFID IN EMPLOYEES TABLE

mysql> ALTER TABLE EMPLOYEES DROP COLUMN PFI;

ERROR 1091 (42000): Can't DROP 'PFI'; check that column/key exists

mysql> ALTER TABLE EMPLOYEES DROP COLUMN PFID;

Query OK, 0 rows affected (0.10 sec)

Records: 0 Duplicates: 0 Warnings: 0

mysql> DESC EMPLOYEES;

Field	Type	Null	Key	Default	Extra
EMPID	char(5)	YES	NULL		
FIRSTNAME	varchar(30)	YES	NULL		
LASTNAME	varchar(30)	YES	NULL		
AGE	int	YES	NULL		
DOJ	date	YES	NULL		
ADDRESS	tinytext	YES	NULL		
DEPT	varchar(50)	YES	NULL		
LOCATION	tinytext	YES	NULL		

8 rows in set (0.03 sec)

mysql> ALTER TABLE EMPLOYEES DROP COLUMN LOCATION;

Query OK, 0 rows affected (0.11 sec)

Records: 0 Duplicates: 0 Warnings: 0

mysql> DESC EMPLOYEES;

Field	Type	Null	Key	Default	Extra
EMPID	char(5)	YES	NULL		
FIRSTNAME	varchar(30)	YES	NULL		

LASTNAME	varchar(30)	YES	NULL
AGE	int	YES	NULL
DOJ	date	YES	NULL
ADDRESS	tinytext	YES	NULL
DEPT	varchar(50)	YES	NULL

7 rows in set (0.00 sec)

```
mysql> -- HOW TO CHANGE TABLE -NAME
mysql> -- SYNTAX : ALTER TABLE TABLE-NAME RENAME TO NEW-TABLE-NAME
mysql> -- CHANGE THE TABLE NAME EMPLOYEES TO CODEGNAN_EMP TABLE
mysql> ALTER TABLE EMPLOYEES RENAME TO CODEGNAN_EMP;
Query OK, 0 rows affected (0.06 sec)
```

```
mysql> DESC EMPLOYEES;
ERROR 1146 (42S02): Table 'pfs3.employees' doesn't exist
mysql> SHOW TABLES;
+-----+
| Tables_in_pfs3 |
+-----+
| codegnan_emp   |
+-----+
1 row in set (0.03 sec)
```

```
mysql> DESC CODEGNAN_EMP;
+-----+-----+-----+-----+-----+
| Field | Type   | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+
| EMPID | char(5) | YES  |     | NULL    |       |
| FIRSTNAME | varchar(30) | YES  |     | NULL    |       |
| LASTNAME | varchar(30) | YES  |     | NULL    |       |
| AGE    | int     | YES  |     | NULL    |       |
| DOJ    | date    | YES  |     | NULL    |       |
| ADDRESS | tinytext | YES  |     | NULL    |       |
| DEPT   | varchar(50) | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+
```

7 rows in set (0.00 sec)

```
mysql> ALTER TABLE CODEGNAN_EMP RENAME TO EMPLOYEES;
Query OK, 0 rows affected (0.07 sec)
```

```
mysql> SHOW TABLES;
```

```
+-----+  
| Tables_in_pfs3 |  
+-----+  
| employees      |  
+-----+  
1 row in set (0.01 sec)
```

```
mysql> NOTE
```

Part A: Create Table Assignment

Create a table named company_staff.

Add the following columns to the table:

emp_id, age, experience, gender, emp_name, address, salary, department, skills, joining_date.

Use INT datatype for emp_id.

Use TINYINT datatype for age.

Use SMALLINT datatype for experience.

Use CHAR(1) datatype for gender.

Use VARCHAR(50) datatype for emp_name.

Use TINYTEXT datatype for address.

Use DECIMAL(10,2) datatype for salary.

Use ENUM('HR','IT','Sales','Finance','Support') datatype for the department.

Use SET('Java','Python','SQL','Excel','Communication') datatype for skills.

Use DATE datatype for joining_date.

Make emp_id as PRIMARY KEY.

Display the structure of the table after creation.

Show all tables available in the database.

```
CREATE TABLE company_staff (  
  emp_id INT ,  
  age TINYINT,  
  experience SMALLINT,  
  gender CHAR(1),  
  emp_name VARCHAR(50),  
  address TINYTEXT,  
  salary DECIMAL(10,2),  
  department ENUM('HR','IT','Sales','Finance','Support'),  
  skills SET('Java','Python','SQL','Excel','Communication'),
```



```
    joining_date DATE  
);
```

```
-- 1. Add email column  
-- 2. Add phone_number column  
-- 3. Add bonus column after salary  
-- 4. Add status column after department  
-- 5. Add company_name column first -->first  
alter table table-name add column-name datatype size first;  
-- 6. Add qualification column after emp_name  
-- 7. Modify emp_name column  
-- 8. Modify salary column  
-- 9. Rename emp_name to employee_name  
-- 10. Rename joining_date to date_of_joining  
-- 11. Drop bonus column  
-- 12. Drop phone_number column  
-- 13. Rename table company_staff to staff_records  
-- 14. Rename table staff_records back to company_staff  
-- 15. Add last_updated column before joining_date
```

```
-- 1. Add email column  
ALTER TABLE company_staff  
ADD email VARCHAR(100);
```

```
-- 2. Add phone_number column  
ALTER TABLE company_staff  
ADD phone_number BIGINT;
```

```
-- 3. Add bonus column after salary  
ALTER TABLE company_staff  
ADD bonus DECIMAL(8,2) AFTER salary;
```

```
-- 4. Add status column after department  
ALTER TABLE company_staff  
ADD status VARCHAR(20) AFTER department;
```

```
-- 5. Add company_name column first  
ALTER TABLE company_staff  
ADD company_name VARCHAR(50) FIRST;
```

```
-- 6. Add qualification column after emp_name
ALTER TABLE company_staff
ADD qualification VARCHAR(50) AFTER emp_name;
```

```
-- 7. Modify emp_name column
ALTER TABLE company_staff
MODIFY emp_name VARCHAR(100);
```

```
-- 8. Modify salary column
ALTER TABLE company_staff
MODIFY salary DECIMAL(12,2);
```

```
-- 9. Rename emp_name to employee_name
ALTER TABLE company_staff
RENAME COLUMN emp_name TO employee_name;
```

```
-- 10. Rename joining_date to date_of_joining
ALTER TABLE company_staff
RENAME COLUMN joining_date TO date_of_joining;
```

```
-- 11. Drop bonus column
ALTER TABLE company_staff
DROP COLUMN bonus;
```

```
-- 12. Drop phone_number column
ALTER TABLE company_staff
DROP COLUMN phone_number;
```

```
-- 13. Rename table company_staff to staff_records
ALTER TABLE company_staff
RENAME TO staff_records;
```

```
-- 14. Rename table staff_records back to company_staff
ALTER TABLE staff_records
RENAME TO company_staff;
```

```
-- 15. Add last_updated column before joining_date
ALTER TABLE company_staff
ADD last_updated DATE after skills;
mysql> notee
```


Day-5

TRUNCATE TABLE PRODUCTS;

Day-5

=====

approach-1

INSERT INTO TABLE-NAME(column1,column2.....column-n) values(value1,value2....valuen);

mysql> INSERT INTO

EMPLOYEES(EMPID,FIRSTNAME,LASTNAME,AGE,DOJ,ADDRESS,DEPT)

-> VALUES

-> ('E001','MALLI','ARJUN',27,'2024-03-20','HYD','JAVA');

Query OK, 1 row affected (0.13 sec)

mysql> SELECT*FROM EMPLOYEES;

EMPID	FIRSTNAME	LASTNAME	AGE	DOJ	ADDRESS	DEPT
E001	MALLI	ARJUN	27	2024-03-20	HYD	JAVA

1 row in set (0.02 sec)

mysql> INSERT INTO

EMPLOYEES(EMPID,FIRSTNAME,LASTNAME,AGE,DOJ,ADDRESS,DEPT)

-> VALUES

-> ('E002','ARJUN','SARKAR',37,'2022-02-20','VJY','SALES'),

-> ('E003','SAI','VARDAN',22,'2024-03-18','HYD','PYTHON');

Query OK, 2 rows affected (0.01 sec)

Records: 2 Duplicates: 0 Warnings: 0

mysql> SELECT*FROM EMPLOYEES;

EMPID	FIRSTNAME	LASTNAME	AGE	DOJ	ADDRESS	DEPT
E001	MALLI	ARJUN	27	2024-03-20	HYD	JAVA
E002	ARJUN	SARKAR	37	2022-02-20	VJY	SALES
E003	SAI	VARDAN	22	2024-03-18	HYD	PYTHON

3 rows in set (0.00 sec)

Approac-2

insert into table-name values(value1,value2....value-n);
mysql> INSERT INTO EMPLOYEES VALUES
-> ('E004','SUMA','SRI',24,'2025-10-10','BNG','DEVELOPMENT');
Query OK, 1 row affected (0.03 sec)

mysql> INSERT INTO EMPLOYEES VALUES
-> ('E005','MANI','SHARMA',17,'2020-09-09','MUMBAI','MARKETING'),
-> ('E006','MOUNIKA','KOPPU',27,'2019-11-11','PUNE','FINANCE');
Query OK, 2 rows affected (0.03 sec)
Records: 2 Duplicates: 0 Warnings: 0

mysql> SELECT*FROM EMPLOYEES;
+-----+-----+-----+-----+-----+-----+
| EMPID | FIRSTNAME | LASTNAME | AGE | DOJ | ADDRESS | DEPT |
+-----+-----+-----+-----+-----+-----+
E001	MALLI	ARJUN	27	2024-03-20	HYD	JAVA
E002	ARJUN	SARKAR	37	2022-02-20	VJY	SALES
E003	SAI	VARDAN	22	2024-03-18	HYD	PYTHON
E004	SUMA	SRI	24	2025-10-10	BNG	DEVELOPMENT
E005	MANI	SHARMA	17	2020-09-09	MUMBAI	MARKETING
E006	MOUNIKA	KOPPU	27	2019-11-11	PUNE	FINANCE
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)

```
mysql> INSERT INTO EMPLOYEES VALUES
-> ('E004','SUMA','SRI',24,'2025-10-10','BNG','DEVELOPMENT');
Query OK, 1 row affected (0.03 sec)
```

```
mysql> INSERT INTO EMPLOYEES VALUES
-> ('E005','MANI','SHARMA',17,'2020-09-09','MUMBAI','MARKETING'),
-> ('E006','MOUNIKA','KOPPU',27,'2019-11-11','PUNE','FINANCE');
Query OK, 2 rows affected (0.03 sec)
Records: 2 Duplicates: 0 Warnings: 0
```

```
mysql> SELECT*FROM EMPLOYEES;
+-----+-----+-----+-----+-----+-----+-----+
| EMPID | FIRSTNAME | LASTNAME | AGE | DOJ      | ADDRESS | DEPT      |
+-----+-----+-----+-----+-----+-----+-----+
| E001 | MALLI     | ARJUN    | 27 | 2024-03-20 | HYD     | JAVA      |
| E002 | ARJUN     | SARKAR   | 37 | 2022-02-20 | VJY     | SALES     |
| E003 | SAI       | VARDAN   | 22 | 2024-03-18 | HYD     | PYTHON    |
| E004 | SUMA      | SRI      | 24 | 2025-10-10 | BNG     | DEVELOPMENT |
| E005 | MANI      | SHARMA   | 17 | 2020-09-09 | MUMBAI  | MARKETING |
| E006 | MOUNIKA   | KOPPU    | 27 | 2019-11-11 | PUNE    | FINANCE   |
+-----+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)
```

```
mysql> note
```

Approach-3

insert into table-name(particular columns you required) values(particular values);

```
INSERT INTO EMPLOYEES(FIRSTNAME,DOJ,LASTNAME,EMPID,ADDRESS)
VALUES
('SAI','2017-07-07','RAM','E008','VIJAYAWADA');
```

Approch-4

Insert into taable-name set Column-name=value,column-name=value,..... column-n=valuen;

```
mysql> INSERT INTO EMPLOYEES SET
```

```
EMPID='E010',FIRSTNAME='SUDHEER',LASTNAME='KUMAR',AGE=45,DOJ='2024-06-01',ADDRESS='KOLKATTA',DEPT='DEVELOPMENT';
```

Query OK, 1 row affected (0.03 sec)

```
mysql> SELECT*FROM EMPLOYEES;
```

```
+-----+-----+-----+-----+-----+-----+-----+
| EMPID | FIRSTNAME | LASTNAME | AGE | DOJ      | ADDRESS  | DEPT      |
+-----+-----+-----+-----+-----+-----+-----+
| E001  | MALLI     | ARJUN    | 27  | 2024-03-20 | HYD      | JAVA      |
| E002  | ARJUN     | SARKAR   | 37  | 2022-02-20 | VJY      | SALES     |
| E003  | SAI       | VARDAN   | 22  | 2024-03-18 | HYD      | PYTHON    |
| E004  | SUMA      | SRI      | 24  | 2025-10-10 | BNG      | DEVELOPMENT |
| E005  | MANI      | SHARMA   | 17  | 2020-09-09 | MUMBAI   | MARKETING |
| E006  | MOUNIKA   | KOPPU    | 27  | 2019-11-11 | PUNE     | FINANCE   |
| e007  | sai       | kiran    | 32  | 2018-09-12 | pune     | admin     |
| E008  | SAI       | RAM      | NULL | 2017-07-07 | VIJAYAWADA | NULL      |
| E009  | ARUN      | NULL     | 15  | NULL       | DELHI    | TRAINEE   |
| E010  | SUDHEER   | KUMAR    | 45  | 2024-06-01 | KOLKATTA | DEVELOPMENT |
+-----+-----+-----+-----+-----+-----+-----+
```

10 rows in set (0.00 sec)

UPDATE

=====

SYNTAX

UPDATE TABLE-NAME SET COLUMN-NAME=VALUE WHERE CONDITION;

```
mysql> UPDATE EMPLOYEES SET LASTNAME='KUMAR' WHERE
FIRSTNAME='ARUN';
```

Query OK, 1 row affected (0.03 sec)

Rows matched: 1 Changed: 1 Warnings: 0

```
mysql> SELECT*FROM EMPLOYEES;
```

```
+-----+-----+-----+-----+-----+-----+-----+
```



```
| EMPID | FIRSTNAME | LASTNAME | AGE | DOJ | ADDRESS | DEPT |
+-----+-----+-----+-----+-----+-----+
| E001 | MALLI | ARJUN | 27 | 2024-03-20 | HYD | JAVA |
| E002 | ARJUN | SARKAR | 37 | 2022-02-20 | VJY | SALES |
| E003 | SAI | VARDAN | 22 | 2024-03-18 | HYD | PYTHON |
| E004 | SUMA | SRI | 24 | 2025-10-10 | BNG | DEVELOPMENT |
| E005 | MANI | SHARMA | 17 | 2020-09-09 | MUMBAI | MARKETING |
| E006 | MOUNIKA | KOPPU | 27 | 2019-11-11 | PUNE | FINANCE |
| e007 | sai | kiran | 32 | 2018-09-12 | pune | admin |
| E008 | SAI | RAM | NULL | 2017-07-07 | VIJAYAWADA | NULL |
| E009 | ARUN | KUMAR | 15 | NULL | DELHI | TRAINEE |
| E010 | SUDHEER | KUMAR | 45 | 2024-06-01 | KOLKATTA | DEVELOPMENT |
+-----+-----+-----+-----+-----+-----+
10 rows in set (0.00 sec)
```

```
mysql> UPDATE EMPLOYEES SET AGE=47 WHERE EMPID='E008';
Query OK, 1 row affected (0.02 sec)
Rows matched: 1 Changed: 1 Warnings: 0
```

```
mysql> SELECT*FROM EMPLOYEES;
+-----+-----+-----+-----+-----+-----+
| EMPID | FIRSTNAME | LASTNAME | AGE | DOJ | ADDRESS | DEPT |
+-----+-----+-----+-----+-----+-----+
| E001 | MALLI | ARJUN | 27 | 2024-03-20 | HYD | JAVA |
| E002 | ARJUN | SARKAR | 37 | 2022-02-20 | VJY | SALES |
| E003 | SAI | VARDAN | 22 | 2024-03-18 | HYD | PYTHON |
| E004 | SUMA | SRI | 24 | 2025-10-10 | BNG | DEVELOPMENT |
| E005 | MANI | SHARMA | 17 | 2020-09-09 | MUMBAI | MARKETING |
| E006 | MOUNIKA | KOPPU | 27 | 2019-11-11 | PUNE | FINANCE |
| e007 | sai | kiran | 32 | 2018-09-12 | pune | admin |
| E008 | SAI | RAM | 47 | 2017-07-07 | VIJAYAWADA | NULL |
| E009 | ARUN | KUMAR | 15 | NULL | DELHI | TRAINEE |
| E010 | SUDHEER | KUMAR | 45 | 2024-06-01 | KOLKATTA | DEVELOPMENT |
+-----+-----+-----+-----+-----+-----+
10 rows in set (0.02 sec)
```

```
mysql> UPDATE EMPLOYEES SET DEPT='IT' WHERE ADDRESS='VIJAYAWADA';
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0
```

```
mysql> SELECT*FROM EMPLOYEES;
```

```
+-----+-----+-----+-----+-----+-----+
| EMPID | FIRSTNAME | LASTNAME | AGE | DOJ      | ADDRESS | DEPT      |
+-----+-----+-----+-----+-----+-----+
| E001 | MALLI     | ARJUN    | 27 | 2024-03-20 | HYD      | JAVA      |
| E002 | ARJUN     | SARKAR   | 37 | 2022-02-20 | VJY      | SALES     |
| E003 | SAI       | VARDAN   | 22 | 2024-03-18 | HYD      | PYTHON    |
| E004 | SUMA      | SRI      | 24 | 2025-10-10 | BNG      | DEVELOPMENT |
| E005 | MANI      | SHARMA   | 17 | 2020-09-09 | MUMBAI   | MARKETING |
| E006 | MOUNIKA   | KOPPU    | 27 | 2019-11-11 | PUNE     | FINANCE   |
| e007 | sai       | kiran    | 32 | 2018-09-12 | pune     | admin     |
| E008 | SAI       | RAM      | 47 | 2017-07-07 | VIJAYAWADA | IT        |
| E009 | ARUN      | KUMAR    | 15 | NULL      | DELHI    | TRAINEE   |
| E010 | SUDHEER   | KUMAR    | 45 | 2024-06-01 | KOLKATTA | DEVELOPMENT |
+-----+-----+-----+-----+-----+-----+
10 rows in set (0.00 sec)
```

```
mysql> UPDATE EMPLOYEES SET DOJ='2026-06-27' WHERE EMPID='E009';
```

```
Query OK, 1 row affected (0.03 sec)
```

```
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql> SELECT*FROM EMPLOYEES;
```

```
+-----+-----+-----+-----+-----+-----+
| EMPID | FIRSTNAME | LASTNAME | AGE | DOJ      | ADDRESS | DEPT      |
+-----+-----+-----+-----+-----+-----+
| E001 | MALLI     | ARJUN    | 27 | 2024-03-20 | HYD      | JAVA      |
| E002 | ARJUN     | SARKAR   | 37 | 2022-02-20 | VJY      | SALES     |
| E003 | SAI       | VARDAN   | 22 | 2024-03-18 | HYD      | PYTHON    |
| E004 | SUMA      | SRI      | 24 | 2025-10-10 | BNG      | DEVELOPMENT |
| E005 | MANI      | SHARMA   | 17 | 2020-09-09 | MUMBAI   | MARKETING |
| E006 | MOUNIKA   | KOPPU    | 27 | 2019-11-11 | PUNE     | FINANCE   |
| e007 | sai       | kiran    | 32 | 2018-09-12 | pune     | admin     |
| E008 | SAI       | RAM      | 47 | 2017-07-07 | VIJAYAWADA | IT        |
| E009 | ARUN      | KUMAR    | 15 | 2026-06-27 | DELHI    | TRAINEE   |
| E010 | SUDHEER   | KUMAR    | 45 | 2024-06-01 | KOLKATTA | DEVELOPMENT |
+-----+-----+-----+-----+-----+-----+
10 rows in set (0.00 sec)
```

```
mysql>
```

```
mysql> mysql> INSERT INTO EMPLOYEES VALUES
```

```
-> -> ('E004','SUMA','SRI',24,'2025-10-10','BNG','DEVELOPMENT');
```

```
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that  
corresponds to your MySQL server version for the right syntax to use near 'mysql> INSERT  
INTO EMPLOYEES VALUES
```

```
-> ('E004','SUMA','SRI',24,'2025-10-10','' at line 1
```

```
mysql> Query OK, 1 row affected (0.03 sec)
```

```
->
```

```
-> mysql> INSERT INTO EMPLOYEES VALUES
```

```
-> -> ('E005','MANI','SHARMA',17,'2020-09-09','MUMBAI','MARKETING'),
```

```
-> -> ('E006','MOUNIKA','KOPPU',27,'2019-11-11','PUNE','FINANCE');
```

```
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that  
corresponds to your MySQL server version for the right syntax to use near 'Query OK, 1 row  
affected (0.03 sec)
```

```
mysql> INSERT INTO EMPLOYEES VALUES
```

```
-> ' at line 1
```

```
mysql> Query OK, 2 rows affected (0.03 sec)
```

```
-> Records: 2 Duplicates: 0 Warnings: 0
```

```
->
```

```
-> mysql> SELECT*FROM EMPLOYEES;
```

ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'Query OK, 2 rows affected (0.03 sec)

Records: 2 Duplicates: 0 Warnings: 0

mys' at line 1

```
mysql> +-----+-----+-----+-----+-----+-----+
-> | EMPID | FIRSTNAME | LASTNAME | AGE | DOJ | ADDRESS | DEPT |
-> +-----+-----+-----+-----+-----+-----+
-> | E001 | MALLI | ARJUN | 27 | 2024-03-20 | HYD | JAVA |
-> | E002 | ARJUN | SARKAR | 37 | 2022-02-20 | VJY | SALES |
-> | E003 | SAI | VARDAN | 22 | 2024-03-18 | HYD | PYTHON |
-> | E004 | SUMA | SRI | 24 | 2025-10-10 | BNG | DEVELOPMENT |
-> | E005 | MANI | SHARMA | 17 | 2020-09-09 | MUMBAI | MARKETING |
-> | E006 | MOUNIKA | KOPPU | 27 | 2019-11-11 | PUNE | FINANCE |
-> +-----+-----+-----+-----+-----+-----+
-> ^C
```

mysql> INSERT INTO EMPLOYEES SET

EMPID='E010',FIRSTNAME='SUDHEER',LASTNAME='KUMAR',AGE=45,DOJ='2024-06-01',ADDRESS='KOLKATTA',DEPT='DEVELOPMENT';

Query OK, 1 row affected (0.03 sec)

mysql> SELECT*FROM EMPLOYEES;

```
+-----+-----+-----+-----+-----+-----+
| EMPID | FIRSTNAME | LASTNAME | AGE | DOJ | ADDRESS | DEPT |
+-----+-----+-----+-----+-----+-----+
| E001 | MALLI | ARJUN | 27 | 2024-03-20 | HYD | JAVA |
| E002 | ARJUN | SARKAR | 37 | 2022-02-20 | VJY | SALES |
| E003 | SAI | VARDAN | 22 | 2024-03-18 | HYD | PYTHON |
| E004 | SUMA | SRI | 24 | 2025-10-10 | BNG | DEVELOPMENT |
| E005 | MANI | SHARMA | 17 | 2020-09-09 | MUMBAI | MARKETING |
| E006 | MOUNIKA | KOPPU | 27 | 2019-11-11 | PUNE | FINANCE |
| e007 | sai | kiran | 32 | 2018-09-12 | pune | admin |
| E008 | SAI | RAM | NULL | 2017-07-07 | VIJAYAWADA | NULL |
| E009 | ARUN | NULL | 15 | NULL | DELHI | TRAINEE |
| E010 | SUDHEER | KUMAR | 45 | 2024-06-01 | KOLKATTA | DEVELOPMENT |
+-----+-----+-----+-----+-----+-----+
```

10 rows in set (0.00 sec)

mysql> delete from employees where empid='e010';

Query OK, 1 row affected (0.03 sec)

mysql> select*from employees;

EMPID	FIRSTNAME	LASTNAME	AGE	DOJ	ADDRESS	DEPT
E001	MALLI	ARJUN	27	2024-03-20	HYD	JAVA
E002	ARJUN	SARKAR	37	2022-02-20	VJY	SALES
E003	SAI	VARDAN	22	2024-03-18	HYD	PYTHON
E004	SUMA	SRI	24	2025-10-10	BNG	DEVELOPMENT
E005	MANI	SHARMA	17	2020-09-09	MUMBAI	MARKETING
E006	MOUNIKA	KOPPU	27	2019-11-11	PUNE	FINANCE
e007	sai	kiran	32	2018-09-12	pune	admin
E008	SAI	RAM	47	2017-07-07	VIJAYAWADA	IT
E009	ARUN	KUMAR	15	2026-06-27	DELHI	TRAINEE

9 rows in set (0.00 sec)

mysql> delete from employees where firstname='sai';

Query OK, 3 rows affected (0.03 sec)

mysql> select*from employees;

EMPID	FIRSTNAME	LASTNAME	AGE	DOJ	ADDRESS	DEPT
E001	MALLI	ARJUN	27	2024-03-20	HYD	JAVA
E002	ARJUN	SARKAR	37	2022-02-20	VJY	SALES
E004	SUMA	SRI	24	2025-10-10	BNG	DEVELOPMENT
E005	MANI	SHARMA	17	2020-09-09	MUMBAI	MARKETING
E006	MOUNIKA	KOPPU	27	2019-11-11	PUNE	FINANCE
E009	ARUN	KUMAR	15	2026-06-27	DELHI	TRAINEE

6 rows in set (0.00 sec)

mysql> truncate table employees;

Query OK, 0 rows affected (0.24 sec)

mysql> select*from employees;

Empty set (0.03 sec)

```
mysql> drop table employees;
Query OK, 0 rows affected (0.08 sec)
```

```
mysql> CREATE TABLE EMPLOYEES(
-> EMPID INT AUTO_INCREMENT,
-> NAME VARCHAR(100),
-> DEPARTMENT VARCHAR(50)
-> );
```

ERROR 1075 (42000): Incorrect table definition; there can be only one auto column and it must be defined as a key

```
mysql> EMPID INT AUTO_INCREMENT PRIMARY KEY,
-> NAME VARCHAR(100),
-> DEPARTMENT VARCHAR(50)
-> );
```

ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'EMPID INT AUTO_INCREMENT PRIMARY KEY, NAME VARCHAR(100), DEPARTMENT VARCHAR(50) ' at line 1

```
mysql> CREATE TABLE EMPLOYEES(
-> EMPID INT AUTO_INCREMENT PRIMARY KEY,
-> NAME VARCHAR(100),
-> DEPARTMENT VARCHAR(50)
-> );
```

Query OK, 0 rows affected (0.06 sec)

```
mysql> INSERT INTO EMPLOYEES(NAME,DEPARTMENT)
-> VALUES
-> ('MALLI','IT'),
-> ('INDU','HR');
```

Query OK, 2 rows affected (0.03 sec)

Records: 2 Duplicates: 0 Warnings: 0

```
mysql> SELECT*FROM EMPLOYEES;
```

```
+-----+-----+-----+
| EMPID | NAME  | DEPARTMENT |
+-----+-----+-----+
| 1     | MALLI | IT         |
| 2     | INDU  | HR         |
```

```
+-----+-----+-----+
```

2 rows in set (0.00 sec)

```
mysql> CREATE TABLE BACKUPEMPOYEEES AS
```

```
-> SELECT*FROM EMPLOYEES;
```

Query OK, 2 rows affected (0.06 sec)

Records: 2 Duplicates: 0 Warnings: 0

```
mysql> DESC BACKUPEMPOYEEES;
```

```
+-----+-----+-----+-----+-----+-----+
```

```
| Field      | Type      | Null | Key | Default | Extra |
```

```
+-----+-----+-----+-----+-----+-----+
```

```
| EMPID      | int       | NO   |     | 0       |       |
```

```
| NAME       | varchar(100) | YES  |     | NULL    |       |
```

```
| DEPARTMENT | varchar(50) | YES  |     | NULL    |       |
```

```
+-----+-----+-----+-----+-----+-----+
```

3 rows in set (0.03 sec)

```
mysql> SELECT*FROM BACKUPEMPOYEEES;
```

```
+-----+-----+-----+
```

```
| EMPID | NAME | DEPARTMENT |
```

```
+-----+-----+-----+
```

```
| 1 | MALLI | IT |
```

```
| 2 | INDU  | HR |
```

```
+-----+-----+-----+
```

2 rows in set (0.00 sec)

```
mysql> CREATE TABLE PRODUCTS(
```

```
-> PRODUCTID INT PRIMARY KEY,
```

```
-> PRICE DECIMAL(10,2),
```

```
-> QUANTITY INT,
```

```
-> TOTALAMOUNT DECIMAL(10,2) GENERATED ALWAYS AS(PRICE*QUANTITY)
```

```
STORED
```

```
-> );
```

Query OK, 0 rows affected (0.08 sec)

```
mysql> INSERT INTO PRODUCTS(PRODUCTID,PRICE,QUANTITY)
```

```
-> VALUES
```

```
-> (1,50,3),
```

```
-> (2,100.25,6);
```

Query OK, 2 rows affected (0.03 sec)
Records: 2 Duplicates: 0 Warnings: 0

```
mysql> SELECT*FROM PRODUCTS;
```

PRODUCTID	PRICE	QUANTITY	TOTALAMOUNT
1	50.00	3	150.00
2	100.25	6	601.50

2 rows in set (0.00 sec)

```
mysql> UPDATE PRODUCTS SET PRICE=500 WHERE PRODUCTID=1;
```

Query OK, 1 row affected (0.02 sec)
Rows matched: 1 Changed: 1 Warnings: 0

```
mysql> SELECT*FROM PRODUCTS;
```

PRODUCTID	PRICE	QUANTITY	TOTALAMOUNT
1	500.00	3	1500.00
2	100.25	6	601.50

2 rows in set (0.00 sec)

Drop,delete,truncate

Delete

=====

Step-by-Step Example in MySQL:

Step 1: Create the Table and Insert Data

First, create the employees table and insert sample data.

-- Drop table if it already exists

```
DROP TABLE IF EXISTS employees;
```

-- Create employees table with AUTO_INCREMENT primary key


```
CREATE TABLE employees (  
    employee_id INT PRIMARY KEY AUTO_INCREMENT,  
    first_name VARCHAR(50),  
    last_name VARCHAR(50),  
    department VARCHAR(50),  
    salary DECIMAL(10, 2)  
);  
  
INSERT INTO employees (first_name, last_name, department, salary)  
VALUES  
('Ravi', 'Kumar', 'HR', 50000.00),  
('Anjali', 'Sharma', 'Sales', 60000.00),  
('Amit', 'Patel', 'HR', 55000.00),  
('Sneha', 'Reddy', 'Engineering', 70000.00),  
('Vikram', 'Nair', 'Sales', 65000.00);
```

Step 2: Use DELETE with Rollback

The DELETE command can be used with a transaction, and changes can be rolled back if needed.

```
-- Start a transaction  
START TRANSACTION;  
  
-- Perform DELETE operation (delete employees from the Sales department)  
DELETE FROM employees WHERE department = 'Sales';  
  
-- Check the current state of the table (after DELETE but before commit or rollback)  
SELECT * FROM employees;  
  
-- Rollback the transaction (undo the DELETE)  
ROLLBACK;  
  
-- Check the data again (after rollback)  
SELECT * FROM employees;
```

- Expected Outcome: After ROLLBACK, no rows should be deleted from the table, and the Sales department employees will be restored.

If you had used COMMIT instead of ROLLBACK, the changes would be permanent.

-- COMMIT the changes if you want the DELETE to be permanent
COMMIT;

-- Check the data after commit
SELECT * FROM employees;

- Expected Outcome: The employees from the Sales department would be deleted permanently.

Step 3: Use TRUNCATE (Not Rollbackable in MySQL)

In MySQL, TRUNCATE is a DDL (Data Definition Language) operation, not a DML (Data Manipulation Language) operation. This means it cannot be rolled back in MySQL.

-- Start a new transaction
START TRANSACTION;

-- Perform TRUNCATE operation (remove all rows from the table)
TRUNCATE TABLE employees;

-- Check the data after TRUNCATE (before commit or rollback)
SELECT * FROM employees;

-- Rollback will not work here, because TRUNCATE cannot be rolled back in MySQL.
-- Try ROLLBACK (it won't undo TRUNCATE in MySQL)
ROLLBACK;

-- Check the data again (it won't restore the data)
SELECT * FROM employees;

- Expected Outcome: After TRUNCATE, the table will be empty, and ROLLBACK will have no effect because TRUNCATE in MySQL cannot be rolled back.

If you had used COMMIT, the data would be permanently deleted.

-- COMMIT the changes if you want the TRUNCATE to be permanent
COMMIT;

-- Check the data after commit
SELECT * FROM employees;

- Expected Outcome: The table will remain empty after a successful commit of the TRUNCATE.

Step 4: Use DROP (Not Rollbackable in MySQL)

In MySQL, DROP is also a DDL command and cannot be rolled back once executed. It will completely remove the table from the database.

-- Start a new transaction
START TRANSACTION;

-- Perform DROP operation (remove the employees table)
DROP TABLE employees;

-- Check if the table still exists (before commit or rollback)
SELECT * FROM employees; -- This will give an error because the table no longer exists

-- Rollback will not work here, because DROP cannot be rolled back in MySQL.
ROLLBACK;

-- Try checking the data again (it won't restore the table)
SELECT * FROM employees; -- This will give an error again

- Expected Outcome: After DROP, the table is removed from the database, and ROLLBACK will not restore the table, because DROP is not rollbackable in MySQL.

If you had used COMMIT, the table would be permanently removed.

-- COMMIT the changes if you want the DROP to be permanent
COMMIT;

-- Attempting to check the table will result in an error, as it no longer exists.
SELECT * FROM employees; -- Error: Table doesn't exist

Summary:

- DELETE: Can be rolled back within a transaction.
- TRUNCATE: Cannot be rolled back in MySQL (it's treated as a DDL command).
- DROP: Cannot be rolled back in MySQL (it's a DDL command that completely removes the table).

To safely perform actions that can be rolled back in MySQL, you should rely on DELETE within transactions. TRUNCATE and DROP are permanent changes in MySQL and cannot be undone unless you have backups or restore points.

Key Differences Summary:

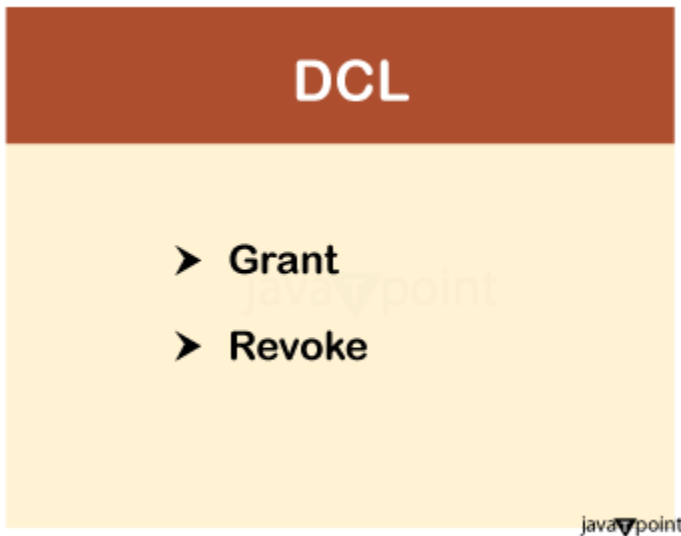
Feature	DROP	DELETE	TRUNCATE
Action	Removes the table or object	Removes specific rows from a table	Removes all rows, keeps structure
Data Removed	Table structure + data	Specific rows (based on WHERE)	All rows
Table Structure	Removed	Intact	Intact
Performance	Slow (due to full removal)	Slower (row-by-row)	Fast (bulk deletion)
Rollback	No (unless in transaction)	Yes	Yes (if in transaction, varies by DB)
Auto-Increment Reset	N/A	No	Yes (depending on DB)

Day-6

3. Data Control Language

DCL commands are used to grant and take back authority from any database user.

Following are the some commands that come under DCL:



Syntax:

```
GRANT <obj_priv> ON <obj_name> To <username>;
```

In the above syntax, <obj_priv> is the DML statement like Insert, Delete , update and Select and <obj_name> is a table, view etc. and username is the name of the authorized user.

Example

```
GRANT SELECT, UPDATE ON MY_TABLE TO SOME_USER,  
ANOTHER_USER;
```

b. Revoke: It is used to take back permissions from the user.

Syntax:

```
REVOKE <obj_priv> ON <obj_name> FROM <username>;
```

In the above syntax, <obj_priv> is the DML statement like Insert, Delete , update and Select and <obj_name> is a table, view etc. and username is the name of the user from whom the permission is revoked.

Example

```
REVOKE SELECT, UPDATE ON MY_TABLE FROM USER1, USER2;
```

Grant and Revoke

```
=====
```

```
mysql> use codegnan;
```

```
Database changed
```

```
mysql> show tables;
```

```
+-----+  
| Tables_in_codegnan |  
+-----+  
| codegnan_emp      |  
+-----+
```

```
1 row in set (0.23 sec)
```

```
mysql> create table students(  
  -> name varchar(20),  
  -> age int,  
  -> mobile varchar(12)  
  -> );
```

```
Query OK, 0 rows affected (0.15 sec)
```

```
mysql> insert into students values('student 1',23,'9182454984');
```

```
Query OK, 1 row affected (0.02 sec)
```

```
mysql> select*from students;
```

```
+-----+-----+-----+  
| name   | age | mobile   |  
+-----+-----+-----+  
| student 1 | 23 | 9182454984 |  
+-----+-----+-----+
```

```
1 row in set (0.00 sec)
```

```
mysql> create user user1 identified by 'user1';
```

```
Query OK, 0 rows affected (0.10 sec)
```

```
mysql> grant select,insert,update on students to user1;
```

```
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> create user user2 identified by 'user2';
```

Query OK, 0 rows affected (0.01 sec)

mysql> grant select,update on students to user2;

Query OK, 0 rows affected (0.01 sec)

user1 previligious

C:\Program Files\MySQL\MySQL Server 8.0\bin>mysql -u USER1 -p

Enter password: *****

Welcome to the MySQL monitor. Commands end with ; or \g.

Your MySQL connection id is 75

Server version: 8.0.39 MySQL Community Server - GPL

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> use codegnan;

Database changed

mysql> select*from user1;

ERROR 1142 (42000): SELECT command denied to user 'user1'@'localhost' for table 'user1'

mysql> select*from students;

```
+-----+-----+-----+
| name   | age  | mobile   |
+-----+-----+-----+
| student 1 | 23 | 9182454984 |
+-----+-----+-----+
```

1 row in set (0.00 sec)

mysql> insert into students values('student 2',32,'8834248768');

Query OK, 1 row affected (0.01 sec)

mysql> insert into student(name,mobile) values('student 3','874772372'),('student 4','689648797');

ERROR 1142 (42000): INSERT command denied to user 'user1'@'localhost' for table 'student'

```
mysql> insert into students(name,mobile) values('student
3','874772372'),('student 4','689648797');
```

Query OK, 2 rows affected (0.01 sec)

Records: 2 Duplicates: 0 Warnings: 0

```
mysql> update students set age=37 where name='student4';
```

Query OK, 0 rows affected (0.01 sec)

Rows matched: 0 Changed: 0 Warnings: 0

```
mysql> drop table student;
```

ERROR 1142 (42000): DROP command denied to user 'user1'@'localhost' for table 'student'

```
mysql> delete table students;
```

ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'table students' at line 1

```
mysql> delete from students where age=37;;
```

ERROR 1142 (42000): DELETE command denied to user 'user1'@'localhost' for table 'students'

ERROR:

No query specified

```
mysql>
```

user2 previloges

=====

```
C:\Program Files\MySQL\MySQL Server 8.0\bin>mysql -u USER2 -p
```

Enter password: *****

Welcome to the MySQL monitor. Commands end with ; or \g.

Your MySQL connection id is 79

Server version: 8.0.39 MySQL Community Server - GPL

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

```
mysql> select*from students;  
ERROR 1046 (3D000): No database selected
```

```
mysql> use codegnan;
```

```
Database changed
```

```
mysql> select*from students;
```

```
+-----+-----+-----+  
| name   | age | mobile   |  
+-----+-----+-----+  
| student 1 | 23 | 9182454984 |  
| student 2 | 32 | 8834248768 |  
| student 3 | NULL | 874772372 |  
| student 4 | NULL | 689648797 |  
+-----+-----+-----+
```

```
4 rows in set (0.01 sec)
```

```
mysql> insert into students values('djs','09','9889');  
ERROR 1142 (42000): INSERT command denied to user 'user2'@'localhost' for  
table 'students'
```

```
mysql> drop table students;
```

```
ERROR 1142 (42000): DROP command denied to user 'user2'@'localhost' for  
table 'students'
```

```
mysql> update students age=40 where name='student 4';
```

```
ERROR 1064 (42000): You have an error in your SQL syntax; check the  
manual that corresponds to your MySQL server version for the right  
syntax to use near '=40 where name='student 4"' at line 1
```

```
mysql> update students set age=40 where name='student 4';
```

```
Query OK, 1 row affected (0.01 sec)
```

```
Rows matched: 1 Changed: 1 Warnings: 0
```

```
mysql> select*from students;
```

```
+-----+-----+-----+  
| name   | age | mobile   |  
+-----+-----+-----+  
| student 1 | 23 | 9182454984 |  
| student 2 | 32 | 8834248768 |  
| student 3 | NULL | 874772372 |  
| student 4 | 40 | 689648797 |  
+-----+-----+-----+
```

```
4 rows in set (0.00 sec)
```

```
mysql>
```

=====

===

Revoke

```
mysql> revoke update on students from user2;  
Query OK, 0 rows affected (0.01 sec)
```

```
update students set age=40 where name='student 4';  
ERROR 1142 (42000): UPDATE command denied to user 'user2'@'localhost' for  
table 'students'
```

LOCK

-- Set transaction isolation level to READ COMMITTED

```
SET TRANSACTION ISOLATION LEVEL READ COMMITTED;
```

5.

```
START TRANSACTION;
```

-- Select the record for EMP_ID 'E001' and lock it for update

```
SELECT * FROM CODEGNAN_EMP WHERE EMP_ID = 'E001' FOR UPDATE;
```

-- At this point, the record with EMP_ID 'E001' is locked for update within this transaction.

-- Other transactions will be blocked from updating this record until the current transaction is committed or rolled back.

-- Update the AGE column for EMP_ID 'E001'

```
UPDATE CODEGNAN_EMP SET AGE = 31 WHERE EMP_ID = 'E001';
```

```
-- Commit the transaction to apply changes
```

```
COMMIT;
```

ROLLBACK

```
-- Start a new transaction
```

```
START TRANSACTION;
```

```
-- Attempt to update the AGE column (example update that might fail)
```

```
UPDATE CODEGNAN_EMP SET AGE = 32 WHERE EMP_ID = 'E001';
```

```
-- Simulate an error that prevents the update from completing
```

```
-- For example, an error might occur due to a constraint violation or data  
validation failure.
```

```
-- Select all records from CODEGNAN_EMP to observe the state before  
rollback
```

```
SELECT * FROM CODEGNAN_EMP;
```

```
-- Rollback the transaction to undo changes
```

```
ROLLBACK;
```

-- Select all records from CODEGNAN_EMP after rollback to verify changes were undone

```
SELECT * FROM CODEGNAN_EMP;
```

EXAMPLE2

-- Start a new transaction

```
START TRANSACTION;
```

-- Update the ADDRESS column for EMP_ID 'E002'

```
UPDATE CODEGNAN_EMP SET ADDRESS = '789 Elm St' WHERE EMP_ID = 'E002';
```

-- Select all records from CODEGNAN_EMP to observe changes after first update

```
SELECT * FROM CODEGNAN_EMP;
```

-- Update the DEPT column for EMP_ID 'E002'

```
UPDATE CODEGNAN_EMP SET DEPT = 'Finance' WHERE EMP_ID = 'E002';
```

-- Select all records from CODEGNAN_EMP to observe changes after second update

```
SELECT * FROM CODEGNAN_EMP;
```

-- Rollback the transaction to undo changes

ROLLBACK;

-- Select all records from CODEGNAN_EMP to verify changes were rolled back

SELECT * FROM CODEGNAN_EMP WHERE EMP_ID = 'E002';

SAVEPOINT

-- Start a new transaction

START TRANSACTION;

-- Update the AGE column for EMP_ID 'E001'

UPDATE CODEGNAN_EMP SET AGE = 32 WHERE EMP_ID = 'E001';

-- Set a SAVEPOINT named 'update_savepoint'

SAVEPOINT update_savepoint;

-- Update the ADDRESS column for EMP_ID 'E001'

UPDATE CODEGNAN_EMP SET ADDRESS = '456 Main St' WHERE EMP_ID = 'E001';

-- Select all records to observe changes

SELECT * FROM CODEGNAN_EMP;

-- Rollback to the SAVEPOINT 'update_savepoint'

ROLLBACK TO SAVEPOINT update_savepoint;

-- Select all records again to verify changes were rolled back

SELECT * FROM CODEGNAN_EMP WHERE EMP_ID = 'E001';

-- Commit the transaction

COMMIT;

Day-7

Tcl

=====

Definition

The process of combining all related operations into a single unit and executing them according to the rule "either all operations should succeed or none should succeed" is called Transaction Management.

A transaction is considered a single unit of work and follows the principle:

"Either all or none."

Example: Funds Transfer

Consider a money transfer process between two accounts:

- 1. Debit amount from the sender's account**
- 2. Credit amount into the receiver's account**

Both operations should be executed as a single transaction.

If the amount is debited from the sender's account successfully, but crediting the receiver's account fails, it may lead to data inconsistency problems.

Types of Transactions

There are two types of transactions:

- 1. Local Transactions**
- 2. Global Transactions**

1. Local Transactions

In local transactions, all operations of a transaction are executed on the same database.

Example:

Funds transfer from one account to another account where both accounts belong to the same bank.

2. Global Transactions

In global transactions, operations of a transaction are executed on different databases.

Example:

Funds transfer from one account to another account where both accounts belong to different banks.

Acid Properties

=====

In the context of **Database Management Systems (DBMS)**, the term **ACID** refers to a set of properties that ensure database transactions are processed reliably. These properties are crucial for maintaining the integrity and consistency of data, especially when there are system failures, power outages, or multiple transactions occurring simultaneously. **ACID** is an acronym that stands for:

1. **Atomicity**
2. **Consistency**
3. **Isolation**
4. **Durability**

1. Atomicity:

Atomicity ensures that a transaction is treated as a single, indivisible unit. This means that either all the operations of the transaction are successfully completed, or none of them are applied. If any part of the transaction fails, the whole transaction is rolled back.

- **Example Scenario:**

- Suppose you are transferring money from **Account A** to **Account B** in a banking system.
- The transaction involves two operations:
 1. Subtracting \$100 from **Account A**.
 2. Adding \$100 to **Account B**.
- If the system crashes after subtracting from **Account A** but before adding to **Account B**, the transfer would be incomplete. **Atomicity** ensures that the

transaction is fully executed, or not executed at all, and both accounts will remain in a consistent state.

- If the transaction fails, both operations will be rolled back, and no money will be deducted from **Account A** or added to **Account B**.

2. Consistency:

Consistency ensures that a transaction brings the database from one valid state to another, maintaining all predefined rules, constraints, and data integrity. After a transaction, the database must remain in a valid state.

- **Example Scenario:**

- Consider a rule that says a bank account balance cannot go below \$0.
- If a transaction tries to withdraw more money than the account holds, the database would be in an inconsistent state. **Consistency** ensures that the transaction is not allowed to proceed unless the integrity of the database (like account balance) is maintained.
- If the account balance would become negative as a result of the transaction, the database would reject the transaction and maintain consistency.

3. Isolation:

Isolation ensures that the operations of a transaction are isolated from the operations of other concurrent transactions. The effects of one transaction are not visible to others until the transaction is complete. This prevents "dirty reads" or inconsistent data from being accessed by other transactions during the process.

- **Example Scenario:**

- Imagine two users are accessing the same bank account at the same time:
 - User 1 wants to transfer money from **Account A** to **Account B**.
 - User 2 is trying to check the balance of **Account A** at the same time.
- If the transactions are not isolated, User 2 could see an intermediate state of **Account A** during the transfer, which could be inaccurate (e.g., the balance may appear to be higher than it should be).
- **Isolation** ensures that User 2 only sees the final result of the transaction after User 1's transaction is fully completed, maintaining the integrity of the data.

4. Durability:

Durability guarantees that once a transaction has been committed, it will persist, even in the event of a system crash or power failure. The changes made by the transaction are permanently recorded in the database.

- **Example Scenario:**

- After a successful transfer of \$100 from **Account A** to **Account B**, the transaction is committed.
- Even if the system crashes immediately after the commit, when the system restarts, the changes (the money transfer) will still be present in the database.
- **Durability** ensures that once a transaction is confirmed, its effects are permanent and not lost.

Summary of ACID Properties in DBMS:

- **Atomicity:** All or nothing.
- **Consistency:** The database remains in a valid state.
- **Isolation:** Transactions do not interfere with each other.
- **Durability:** Committed changes are permanent.

These properties ensure the database maintains its integrity, even when facing issues like system failures or concurrent transactions.

1. COMMIT

Definition

COMMIT is used to **permanently save all changes** made during the current transaction into the database.

Once committed:

- Changes become permanent.
- Other users can see the changes.
- You cannot undo the changes using **ROLLBACK**.

Syntax

COMMIT;

Example

– Step 1: Create Table

```
CREATE TABLE employees (  
    emp_id INT PRIMARY KEY,
```

```
emp_name VARCHAR(50),  
salary DECIMAL(10,2)  
);
```

– Step 2: Start Transaction
START TRANSACTION;

– **Step 3: Insert Records**

```
INSERT INTO employees VALUES (101,'Rahul',45000);  
INSERT INTO employees VALUES (102,'Priya',50000);
```

– Step 4: Commit
COMMIT;

Real-Life Example

Imagine you transfer ₹10,000 from your bank account.

- Money deducted from your account
- Money added to receiver's account

Only after both operations succeed, the bank executes:

2. ROLLBACK

Definition

ROLLBACK is used to **undo all changes** made in the current transaction before a COMMIT.

If something goes wrong, rollback restores the database to its previous state.

Syntax

Rollback;

Example

– Step 1

Start transaction;

– step2

INSERT INTO employees VALUES
(103,'Kiran',60000);

INSERT INTO employees VALUES
(104,'Sneha',70000);

– Step 3

Suppose you inserted wrong data.

Execute

Rollback;

Real-Life Example

Online shopping:

- Product reserved
- Payment failed

Database executes

Rollback;

3. SAVEPOINT

Definition

A **SAVEPOINT** creates a checkpoint inside a transaction.

Instead of rolling back the entire transaction, you can rollback only to a specific savepoint.

Syntax

Savepoint save-point-name;

Rollback to savepoint

ROLLBACK TO savepoint_name;

Commit

Commit;

Example

START TRANSACTION;

-- Increase Rahul's salary

UPDATE employees

SET salary = 48000

WHERE emp_id = 101;

SAVEPOINT s1;

-- Increase Priya's salary

UPDATE employees

SET salary = 53000

```
WHERE emp_id = 102;
```

```
-- Delete Ajay's record
```

```
DELETE FROM employees
```

```
WHERE emp_id = 105;
```

```
SAVEPOINT s2;
```

```
-- Update Sneha's salary
```

```
UPDATE employees
```

```
SET salary = 70000
```

```
WHERE emp_id = 104;
```

```
SELECT * FROM employees;
```

```
ROLLBACK TO s2;
```

```
SELECT * FROM employees;
```

```
COMMIT;
```

```
WHAT IS LOCKING
```

```
=====
```

locking is a mechanism used by the database to control concurrent access data.

it ensures that multiple users access the same data simultaneously the database maintains consistency, integrity, and prevent conflicts.

1. Shared Lock

Definition

A Shared Lock allows a transaction to read data.

Multiple transactions can hold a Shared Lock on the same row.

No transaction can modify the row until all Shared Locks are released.

Characteristics

Read only

Multiple users can read simultaneously

Blocks updates and deletes

2. Exclusive Lock

Definition

An Exclusive Lock allows a transaction to modify data.

While an Exclusive Lock exists:

No other transaction can read using a locking read or modify that row.

Only the transaction holding the lock can update/delete the data.

Characteristics

Used for INSERT, UPDATE, DELETE

Only one Exclusive Lock at a time

Blocks both reads (locking reads) and writes

3. Intent Lock

An Intent Lock is a table-level lock that indicates a transaction intends to place row-level locks.

It helps MySQL quickly determine whether a table-level lock conflicts with existing row locks.

Users do not usually create Intent Locks manually. InnoDB manages them automatically.

Real-Life Example

Imagine a library:

Shared Lock → Many students can read the same book at the same time.

Exclusive Lock → One student borrows the book, so nobody else can read or borrow it until it is returned.

Intent Lock → The librarian marks that someone is about to read or borrow books from a shelf, helping coordinate access without checking every book individually.

Table Lock

Entire table is locked.

Nobody can modify any row until the lock is released.

Create Table

None

```
CREATE TABLE student(  
  
  id INT,  
  
  name VARCHAR(30),  
  
  marks INT  
  
);  
  
INSERT INTO student VALUES  
  
(1, 'Ravi', 80),  
  
(2, 'Ram', 70),  
  
(3, 'Ajay', 90);
```

Session 1

Lock the table.

None

```
LOCK TABLE student WRITE;
```

Now Session 1 can

None

```
UPDATE student  
  
SET marks=95
```

```
WHERE id=1;
```

Works.

Session 2

None

```
SELECT * FROM student;
```

Output

None

```
Waiting...
```

or

None

```
Table locked
```

Session 2 cannot read or update because the entire table is locked.

Unlock

None

```
UNLOCK TABLES;
```

Now everyone can access the table.

Diagram

None

Student Table

Row1

Row2

Row3

Entire table locked

Session1 ✓

Session2 ✗

Session3 ✗

Read Lock

None

LOCK TABLE student READ;

Session 1

None

SELECT * FROM student;

Allowed

Session 2

None

```
SELECT * FROM student;
```

Allowed

Session 2

None

```
UPDATE student
```

```
SET marks=50
```

```
WHERE id=2;
```

Waiting.

Write Lock

None

```
LOCK TABLE student WRITE;
```

Session 1

Can Read

Can Update

Can Delete

Session 2

Cannot Read

Cannot Update

Cannot Delete

Table Lock Summary

Lock Type	Read	Write
READ	✓	✗
WRITE	✓	✓ (only lock owner)

Row Lock

Instead of locking the entire table, only the selected row is locked.

Session 1

None

```
START TRANSACTION;
```

```
SELECT *
```

```
FROM student
```

```
WHERE id=1
```

```
FOR UPDATE;
```

Row 1 is locked.

Session 2

None

```
UPDATE student
```

```
SET marks=100
```

```
WHERE id=1;
```

Output

None

```
Waiting...
```

Because row 1 is locked.

Session 2

None

```
UPDATE student
```

```
SET marks=65
```

```
WHERE id=2;
```

Works.

Only row 1 is locked.

Commit Session 1

None

COMMIT;

Now Session 2 continues.

Diagram

None

Student Table

Row1 

Row2

Row3

Session1 → Row1

Session2 → Row2 ✓

Session2 → Row1 ✗ Waiting

Shared Lock

Used for reading.

None

```
START TRANSACTION;
```

```
SELECT *
```

```
FROM student
```

```
WHERE id=1
```

```
FOR SHARE;
```

(Older MySQL versions use `LOCK IN SHARE MODE`.)

Many users can read.

Nobody can update.

Exclusive Lock

Used for updating.

None

```
START TRANSACTION;
```

```
SELECT *
```

```
FROM student
```

```
WHERE id=1  
FOR UPDATE;
```

Only one transaction owns the row.

Everyone else waits.

Intent Lock

Automatically managed by InnoDB.

When a row lock is placed,

MySQL internally creates:

None

Table

↓

Intent Lock

↓

Row Lock

You never manually create an Intent Lock.

Row Lock vs Table Lock

Feature	Table Lock	Row Lock
Locks	Entire Table	Single Row
Performance	Faster to acquire	More overhead
Concurrency	Low	High
Used In	MyISAM and explicit locks	InnoDB
Multiple Users	Mostly blocked	Allowed on different rows

Isolation Levels Summary

Isolation Level	Dirty Read	Non-repeatable Read	Phantom Read
Read Uncommitted	✓	✓	✓
Read Committed	✗	✓	✓

Repeatable Read	×	×	Mostly prevented in MySQL with MVCC/next-key locking
Serializable	×	×	×

Classroom Interview Questions

1. Why do we need locking?

To prevent multiple transactions from causing data inconsistency.

2. What is Dirty Read?

Reading data that another transaction has modified but not yet committed.

3. What is Non-repeatable Read?

Reading the same row twice within a transaction and getting different values because another transaction committed an update.

4. What is the default isolation level in MySQL?

REPEATABLE READ.

5. What is the difference between a Table Lock and a Row Lock?

A Table Lock locks the entire table, while a Row Lock locks only the selected row, allowing higher concurrency.

6. What is the difference between **FOR UPDATE** and **FOR SHARE**?

- **FOR UPDATE** acquires an exclusive row lock for updates.
- **FOR SHARE** acquires a shared row lock for reading while preventing conflicting updates.

Day-8

Constraints in MySQL

What are Constraints?

Constraints are rules applied to table columns in MySQL to ensure that only **valid and accurate data** is stored in the database.

In simple words:

Constraints = Rules that control the data entered into a table.

Why do we use constraints?

Constraints are used to:

- Maintain data integrity
- Prevent invalid data entry
- Avoid duplicate records
- Ensure relationships between tables
- Make the database reliable and consistent

integrity constraints:-

- 1) NOT NULL
- 2) UNIQUE
- 3) PRIMARY KEY
- 4) composite key
- 5) FOREIGN KEY or REFERENCIAL INTEGRITY
- 6) CHECK
- 7) default constraint

Integrity Constraints in MySQL

Integrity constraints are rules that ensure the accuracy, consistency, and reliability of data stored in a database.

1) NOT NULL Constraint

Definition

The NOT NULL constraint ensures that a column cannot contain NULL (empty) values. Every row must have a value for that column.

Example

```
CREATE TABLE employees (  
    emp_id INT,  
    emp_name VARCHAR(50) NOT NULL,  
    salary DECIMAL(10,2)  
);
```

Insert Valid Record

```
INSERT INTO employees VALUES  
(101, 'Ravi', 35000);
```

Insert Invalid Record

```
INSERT INTO employees VALUES  
(102, NULL, 40000);
```

Output

ERROR:

Column 'emp_name' cannot be null

2) UNIQUE Constraint

Definition

The UNIQUE constraint ensures that all values in a column are different. Duplicate values are not allowed.

Example

```
CREATE TABLE students (  
    student_id INT,  
    email VARCHAR(100) UNIQUE,  
    name VARCHAR(50)  
);
```

Insert Records

```
INSERT INTO students VALUES  
(1, 'ravi@gmail.com', 'Ravi');
```

```
INSERT INTO students VALUES  
(2, 'priya@gmail.com', 'Priya');
```

Duplicate Value

```
INSERT INTO students VALUES  
(3, 'ravi@gmail.com', 'Mahesh');
```

Output

ERROR:

Duplicate entry 'ravi@gmail.com' for key

3) PRIMARY KEY Constraint

Definition

A PRIMARY KEY uniquely identifies every row in a table.

Rules

- Cannot contain NULL values.
- Cannot contain duplicate values.
- Only one Primary Key per table.

Example

```
CREATE TABLE products (  
    product_id INT PRIMARY KEY,  
    product_name VARCHAR(50),  
    price DECIMAL(10,2)  
);
```

Insert Records

```
INSERT INTO products VALUES  
(101, 'Laptop', 55000);
```

```
INSERT INTO products VALUES  
(102, 'Mouse', 800);
```

Duplicate Primary Key

```
INSERT INTO products VALUES  
(101, 'Keyboard', 1200);
```

Output

ERROR:

Duplicate entry '101' for key 'PRIMARY'

4) Composite Key

Definition

A Composite Key is a Primary Key formed using two or more columns together.

Example

```
CREATE TABLE enrollment (  
    student_id INT,  
    course_id INT,  
    enroll_date DATE,  
    PRIMARY KEY(student_id, course_id)  
);
```

Insert Records

```
INSERT INTO enrollment VALUES  
(101, 1, '2026-07-21');
```

```
INSERT INTO enrollment VALUES  
(101, 2, '2026-07-21');
```

```
INSERT INTO enrollment VALUES  
(102, 1, '2026-07-21');
```

Duplicate Composite Key

```
INSERT INTO enrollment VALUES
```

(101, 1, '2026-07-22');

Output

ERROR:

Duplicate entry '101-1' for key 'PRIMARY'

5) FOREIGN KEY (Referential Integrity)

Definition

A Foreign Key creates a relationship between two tables. It ensures that a value in the child table must already exist in the parent table.

Parent Table

```
CREATE TABLE department (  
    dept_id INT PRIMARY KEY,  
    dept_name VARCHAR(50)  
);
```

Insert Records

```
INSERT INTO department VALUES  
(101,'HR'),  
(102,'IT'),  
(103,'Finance');
```

Child Table

```
CREATE TABLE employee (  
    emp_id INT PRIMARY KEY,  
    emp_name VARCHAR(50),  
    dept_id INT,  
    FOREIGN KEY(dept_id)  
    REFERENCES department(dept_id)  
);
```

Valid Insert

```
INSERT INTO employee VALUES  
(1,'Ravi',101);
```

```
INSERT INTO employee VALUES  
(2,'Priya',102);
```

Invalid Insert

```
INSERT INTO employee VALUES  
(3,'Mahesh',110);
```

Output

ERROR:

Cannot add or update a child row:

a foreign key constraint fails

6) CHECK Constraint

Definition

The CHECK constraint ensures that values satisfy a specified condition before insertion or update.

Example

```
CREATE TABLE exams (  
    student_id INT,  
    marks INT CHECK (marks BETWEEN 0 AND 100)  
);
```

Valid Records

```
INSERT INTO exams VALUES  
(101,85);
```

```
INSERT INTO exams VALUES  
(102,65);
```

Invalid Record

```
INSERT INTO exams VALUES  
(103,120);
```

Output

ERROR:

Check constraint violated

7) DEFAULT Constraint

Definition

The DEFAULT constraint automatically assigns a predefined value if no value is provided during insertion.

Example

```
CREATE TABLE customers (  
    customer_id INT PRIMARY KEY,  
    customer_name VARCHAR(50),  
    city VARCHAR(50) DEFAULT 'Hyderabad'  
);
```

Insert Without City

```
INSERT INTO customers(customer_id, customer_name)  
VALUES  
(101,'Ravi');
```

View Table

```
SELECT * FROM customers;
```

Output

customer_id	customer_name	city
101	Ravi	Hyderabad

Insert With City

```
INSERT INTO customers VALUES  
(102,'Priya','Bangalore');
```


Output

customer_id	customer_name	city
101	Ravi	Hyderabad
102	Priya	Bangalore

Summary Table

Constraint	Purpose	Duplicate Allowed	NULL Allowed
NOT NULL	Prevents NULL values	Yes	No
UNIQUE	Prevents duplicate values	No	Yes (one or more NULLs in MySQL, depending on version/usage)
PRIMARY KEY	Uniquely identifies each row	No	No
Composite Key	Combination of columns uniquely identifies a row	No (for the combination)	No (as part of the primary key)
FOREIGN KEY	Maintains parent-child relationship	Yes	Yes (unless also declared NOT NULL)

CHECK	Restricts values based on a condition	Depends on the condition	Depends on the column definition
DEFAULT	Assigns a default value when none is provided	Depends on the column	Depends on the column definition

What are Referential Actions?

Referential actions define what happens to child table data when the parent table data is updated or deleted.

They are used with FOREIGN KEY constraints.

1. ON DELETE CASCADE

Concept

When a parent row is deleted, all matching child rows are automatically deleted.

What it does

Keeps data consistent by removing dependent records automatically.

Example

-- Parent Table

```
CREATE TABLE Departments (
    dept_id INT PRIMARY KEY,
    dept_name VARCHAR(50)
);
```

-- Child Table

```
CREATE TABLE Employees (  
    emp_id INT PRIMARY KEY,  
    name VARCHAR(100),  
    dept_id INT,  
    FOREIGN KEY (dept_id)  
    REFERENCES Departments(dept_id)  
    ON DELETE CASCADE  
);
```

Insert Records

```
INSERT INTO Departments VALUES  
(1, 'IT'),  
(2, 'HR');
```

```
INSERT INTO Employees VALUES  
(101, 'Rahul', 1),  
(102, 'Sneha', 1),  
(103, 'Amit', 2);
```

Operation

```
DELETE FROM Departments WHERE dept_id = 1;
```

Result

- Dept 1 deleted
- Employees with dept_id = 1 also deleted automatically

♦ **2. ON UPDATE CASCADE**

Concept

When a parent key is updated, all matching child foreign keys are updated automatically.

What it does

Maintains consistency when primary key changes.

Example

```
CREATE TABLE Employees (  
    emp_id INT PRIMARY KEY,  
    name VARCHAR(100),  
    dept_id INT,  
    FOREIGN KEY (dept_id)  
    REFERENCES Departments(dept_id)  
    ON UPDATE CASCADE  
);
```

Insert Records

```
INSERT INTO Departments VALUES  
(1, 'IT'),  
(2, 'HR');
```

```
INSERT INTO Employees VALUES  
(101, 'Rahul', 1),  
(102, 'Sneha', 1),  
(103, 'Amit', 2);
```

Operation

```
UPDATE Departments SET dept_id = 10 WHERE dept_id = 1;
```

Result

- dept_id changed from 1 → 10
 - Employees dept_id also updated automatically to 10
-

3. ON DELETE SET NULL

Concept

When a parent row is deleted, the child foreign key becomes NULL.

What it does

Keeps the child record but removes the relationship.

Column must allow NULL

Example

```
CREATE TABLE Employees (  
    emp_id INT PRIMARY KEY,  
    name VARCHAR(100),  
    dept_id INT NULL,  
    FOREIGN KEY (dept_id)  
    REFERENCES Departments(dept_id)  
    ON DELETE SET NULL  
);
```

Insert Records

```
INSERT INTO Departments VALUES  
(1, 'IT'),  
(2, 'HR');
```

INSERT INTO Employees VALUES

(101, 'Rahul', 1),

(102, 'Sneha', 1),

(103, 'Amit', 2);

Operation

DELETE FROM Departments WHERE dept_id = 1;

Result

- **Dept 1 deleted**
 - **Employees remain, but dept_id becomes NULL**
-

4. ON UPDATE SET NULL

Concept

When a parent key is updated, the child foreign key becomes NULL.

What it does

Breaks relationships instead of updating value.

Column must allow NULL

Example

CREATE TABLE Employees (

emp_id INT PRIMARY KEY,

name VARCHAR(100),

dept_id INT NULL,

FOREIGN KEY (dept_id)

REFERENCES Departments(dept_id)

ON UPDATE SET NULL

);

Insert Records

INSERT INTO Departments VALUES

(1, 'IT'),

(2, 'HR');

INSERT INTO Employees VALUES

(101, 'Rahul', 1),

(102, 'Sneha', 1),

(103, 'Amit', 2);

Operation

UPDATE Departments SET dept_id = 10 WHERE dept_id = 1;

Result

- Parent updated
- Child dept_id becomes NULL (not 10)

Final Summary

Action	When Parent Changes	Child Table Result
ON DELETE CASCADE	Delete parent	Child rows deleted
ON UPDATE CASCADE	Update parent	Child updated

ON DELETE SET NULL

Delete parent

Child set to NULL

ON UPDATE SET NULL

Update parent

Child set to NULL

Day-9

What is an operator ?

operator is a symbol that performs a specific task.

syntax

operand operator operand

a + b

MySQL Operators and Their Usage

What are Operators?

Operators are special symbols or keywords used in MySQL to perform operations on data. They help in calculations, comparing values, filtering records, combining conditions, manipulating strings, and performing bitwise operations.

Types of Operators

1. Arithmetic Operators
2. Comparison Operators
3. Assignment Operators
4. Logical Operators
5. String Operators
6. Bitwise Operators
7. NULL Operators
8. Special Operators

1. Arithmetic Operators

Definition

Arithmetic operators are used to perform mathematical calculations on numeric values.

Operator	Description	Example	Result
+	Addition	10 + 5	15
-	Subtraction	10 - 5	5
*	Multiplication	10 * 5	50
/	Division	10 / 5	2
%	Modulus (Remainder)	10 % 3	1

STEP 1: Create Table

```
CREATE TABLE Products (
  id INT,
  name VARCHAR(100),
  price DECIMAL(10,2),
  qty INT
);
```

STEP 2: Insert Records

```
INSERT INTO Products VALUES
(1, 'Pen', 10.00, 100),
(2, 'Notebook', 25.00, 50),
(3, 'Pencil', 5.00, 200),
(4, 'Eraser', 3.00, 300),
(5, 'Marker', 15.00, 80),
(6, 'Scale', 20.00, 150),
(7, 'Sharpener', 8.00, 120),
```

```
(8, 'Geometry Box', 120.00, 40),  
(9, 'Sketch Pen', 60.00, 75),  
(10, 'Crayons', 90.00, 55);
```

SELECT Queries

Q1. Calculate the price after adding 12% GST.

```
SELECT name, price, price + (price * 0.12) AS price_with_tax  
FROM Products;
```

Q2. Calculate the discounted price after reducing ₹5.

```
SELECT name, price, price - 5 AS discounted_price  
FROM Products;
```

Q3. Find the total inventory value of each product.

```
SELECT name, price, qty, price * qty AS total_inventory_value  
FROM Products;
```

Q4. Calculate the cost of purchasing 10 units.

```
SELECT name, price, price * 10 AS price_for_10_units  
FROM Products;
```

Q5. Display products whose quantity is not divisible by 50.

```
SELECT *  
FROM Products  
WHERE qty % 50 <> 0;
```

Q6. Add ₹20 to the price of every product.

```
SELECT name, price, price + 20 AS increased_price  
FROM Products;
```

Q7. Display half of each product's quantity.

```
SELECT name, qty, qty / 2 AS half_quantity  
FROM Products;
```

Q8. Display double the quantity of every product.

```
SELECT name, qty, qty * 2 AS double_quantity  
FROM Products;
```

Q9. Find the remaining quantity after selling 10 units.

```
SELECT name, qty, qty - 10 AS remaining_quantity  
FROM Products;
```

Q10. Divide the price equally among 5 people.

```
SELECT name, price, price / 5 AS share_price  
FROM Products;
```

Q11. Display products whose price is an even number.

```
SELECT *  
FROM Products  
WHERE price % 2 = 0;
```

Q12. Display products whose quantity is divisible by 5.

```
SELECT *  
FROM Products  
WHERE qty % 5 = 0;
```

Q13. Calculate the price after giving a 10% discount.

```
SELECT name, price, price - (price * 0.10) AS discounted_price  
FROM Products;
```

Q14. Calculate the price after adding ₹15 and then 18% GST.

```
SELECT name, price, (price + 15) * 1.18 AS final_price  
FROM Products;
```

Q15. Display the remaining quantity after selling half the stock.

```
SELECT name, qty, qty - (qty / 2) AS remaining_stock  
FROM Products;
```

UPDATE Queries

Q1. Add 2 units to every product.

```
UPDATE Products  
SET qty = qty + 2;
```

Q2. Reduce the price of every product by ₹5.

```
UPDATE Products  
SET price = price - 5;
```

Q3. Double the quantity of Eraser.

```
UPDATE Products
SET qty = qty * 2
WHERE name = 'Eraser';
```

Q4. Reduce the quantity of Marker to half.

```
UPDATE Products
SET qty = qty / 2
WHERE name = 'Marker';
```

Q5. Increase the price of Notebook by ₹20.

```
UPDATE Products
SET price = price + 20
WHERE name = 'Notebook';
```

Q6. Increase the quantity of Pen by 50.

```
UPDATE Products
SET qty = qty + 50
WHERE name = 'Pen';
```

Q7. Reduce the quantity of Pencils by 25.

```
UPDATE Products
SET qty = qty - 25
WHERE name = 'Pencil';
```

Q8. Increase the price of Scale by 10%.

```
UPDATE Products
SET price = price + (price * 0.10)
WHERE name = 'Scale';
```

Q9. Double the quantity of Sharpener.

```
UPDATE Products  
SET qty = qty * 2  
WHERE name = 'Sharpener';
```

Q10. Reduce the price of Crayons by ₹15.

```
UPDATE Products  
SET price = price - 15  
WHERE name = 'Crayons';
```

DELETE Queries

Q1. Delete products whose quantity is less than 100.

```
DELETE FROM Products  
WHERE qty < 100;
```

Q2. Delete products whose price is less than ₹10.

```
DELETE FROM Products  
WHERE price < 10;
```

Q3. Delete the Marker record.

```
DELETE FROM Products  
WHERE name = 'Marker';
```

Q4. Delete products whose quantity is greater than 300.

```
DELETE FROM Products  
WHERE qty > 300;
```

Q5. Delete products whose price is greater than ₹100.

```
DELETE FROM Products  
WHERE price > 100;
```

This is plain text without any

Comparison Operators in MySQL

What are Comparison Operators?

Comparison Operators are used to compare two values or expressions. They return TRUE (1) or FALSE (0) based on whether the comparison condition is satisfied.

These operators are mainly used with the WHERE clause to filter records.

Types of Comparison Operators

Operator	Description	Example
=	Equal to	salary = 50000
!= or <>	Not Equal to	department != 'HR'
>	Greater Than	salary > 50000
<	Less Than	age < 25
>=	Greater Than or Equal To	age >= 18

<= Less Than or Equal To **salary <= 40000**

Note: In MySQL, both != and <> represent Not Equal To.

STEP 1: Create Table

```
CREATE TABLE StudentScores (  
  id INT,  
  name VARCHAR(100),  
  subject VARCHAR(50),  
  marks INT  
);
```

STEP 2: Insert Records

```
INSERT INTO StudentScores VALUES  
(1, 'Alice', 'Math', 95),  
(2, 'Bob', 'Math', 76),  
(3, 'Charlie', 'Math', 59),  
(4, 'Diana', 'Math', 88),  
(5, 'Eve', 'Math', 40),  
(6, 'Frank', 'Math', 95),  
(7, 'Grace', 'Science', 67),  
(8, 'Henry', 'Science', 82),  
(9, 'Ivy', 'English', 91),  
(10, 'Jack', 'English', 55),  
(11, 'Kevin', 'Math', 73),  
(12, 'Lily', 'Science', 48),  
(13, 'Mia', 'English', 100),  
(14, 'Noah', 'Math', 64),  
(15, 'Olivia', 'Science', 79);
```

SELECT Queries

Q1. Display students who scored more than 80.

```
SELECT *  
FROM StudentScores  
WHERE marks > 80;
```

Q2. Display students who scored 60 or more.

```
SELECT *  
FROM StudentScores  
WHERE marks >= 60;
```

Q3. Display students who scored less than 50.

```
SELECT *  
FROM StudentScores  
WHERE marks < 50;
```

Q4. Display students who scored exactly 95.

```
SELECT *  
FROM StudentScores  
WHERE marks = 95;
```

Q5. Display students who did not score 95.

```
SELECT *  
FROM StudentScores  
WHERE marks != 95;
```

Q6. Display students who scored 60 or less.

```
SELECT *  
FROM StudentScores  
WHERE marks <= 60;
```

Q7. Display students who scored more than 70.

```
SELECT *  
FROM StudentScores  
WHERE marks > 70;
```

Q8. Display students who scored less than or equal to 75.

```
SELECT *  
FROM StudentScores  
WHERE marks <= 75;
```

Q9. Display students who scored exactly 100.

```
SELECT *  
FROM StudentScores  
WHERE marks = 100;
```

Q10. Display students who did not score 40.

```
SELECT *  
FROM StudentScores  
WHERE marks <> 40;
```

Q11. Display students who scored more than or equal to 90.

```
SELECT *  
FROM StudentScores  
WHERE marks >= 90;
```

Q12. Display students who scored less than 65.

```
SELECT *  
FROM StudentScores  
WHERE marks < 65;
```

UPDATE Queries

Q1. Increase marks by 10 for students scoring below 50.

```
UPDATE StudentScores  
SET marks = marks + 10  
WHERE marks < 50;
```

Q2. Increase marks by 5 for students scoring 60 or below.

```
UPDATE StudentScores  
SET marks = marks + 5  
WHERE marks <= 60;
```

Q3. Add 2 bonus marks to students scoring above 90.

```
UPDATE StudentScores  
SET marks = marks + 2  
WHERE marks > 90;
```

Q4. Add 3 bonus marks to students scoring 80 or more.

```
UPDATE StudentScores  
SET marks = marks + 3  
WHERE marks >= 80;
```

Q5. Reduce 1 mark for students who did not score exactly 95.

```
UPDATE StudentScores  
SET marks = marks - 1  
WHERE marks != 95;
```

Q6. Set marks to 100 for students who scored exactly 95.

```
UPDATE StudentScores  
SET marks = 100  
WHERE marks = 95;
```

Q7. Increase marks by 15 for students scoring below 60.

```
UPDATE StudentScores  
SET marks = marks + 15  
WHERE marks < 60;
```

Q8. Add 5 marks for students scoring above 75.

```
UPDATE StudentScores  
SET marks = marks + 5  
WHERE marks > 75;
```

Q9. Reduce 3 marks for students scoring above 95.

```
UPDATE StudentScores  
SET marks = marks - 3  
WHERE marks > 95;
```

Q10. Set marks to 75 for students scoring below 50.

```
UPDATE StudentScores  
SET marks = 75  
WHERE marks < 50;
```

Q11. Increase marks by 8 for students scoring exactly 64.

```
UPDATE StudentScores  
SET marks = marks + 8  
WHERE marks = 64;
```

Q12. Reduce 5 marks for students scoring 100.

```
UPDATE StudentScores  
SET marks = marks - 5  
WHERE marks = 100;
```

DELETE Queries

Q1. Delete students who scored below 50.

```
DELETE FROM StudentScores  
WHERE marks < 50;
```

Q2. Delete students who scored 60 or below.

```
DELETE FROM StudentScores  
WHERE marks <= 60;
```

Q3. Delete students who scored above 90.

```
DELETE FROM StudentScores  
WHERE marks > 90;
```

Q4. Delete students who scored 80 or more.

```
DELETE FROM StudentScores  
WHERE marks >= 80;
```

Q5. Delete students whose marks are not equal to 95.

```
DELETE FROM StudentScores  
WHERE marks != 95;
```

Q6. Delete students who scored exactly 76.

```
DELETE FROM StudentScores  
WHERE marks = 76;
```

Q7. Delete students who scored exactly 100.

```
DELETE FROM StudentScores  
WHERE marks = 100;
```

Q8. Delete students who scored less than 65.

```
DELETE FROM StudentScores  
WHERE marks < 65;
```

Q9. Delete students who scored greater than or equal to 90.

```
DELETE FROM StudentScores  
WHERE marks >= 90;
```

Q10. Delete students who scored exactly 73.

```
DELETE FROM StudentScores  
WHERE marks = 73;
```

Q11. Delete students who scored exactly 48.


```
DELETE FROM StudentScores  
WHERE marks = 48;
```

Q12. Delete students who scored between 70 and 80 (using comparison operators).

```
DELETE FROM StudentScores  
WHERE marks >= 70 AND marks <= 80;
```

Logical Operators in MySQL

What are Logical Operators?

Logical Operators are used to combine multiple conditions in a SQL query. They help filter data based on one or more conditions and are commonly used with the **WHERE** clause.

Types of Logical Operators

- **AND** – Returns records only if all conditions are **TRUE**.
- **OR** – Returns records if at least one condition is **TRUE**.
- **NOT** – Reverses the result of a condition (**TRUE** becomes **FALSE** and vice versa).

STEP 1: Create Table

STEP 1: Create Table

```
CREATE TABLE Employees (  
  id INT,  
  name VARCHAR(100),  
  department VARCHAR(100),  
  salary INT,  
  isactive BOOLEAN  
);
```

STEP 2: Insert Records

INSERT INTO Employees VALUES

(1, 'Rahul Sharma', 'IT', 50000, TRUE),
(2, 'Priya Reddy', 'HR', 30000, TRUE),
(3, 'Amit Kumar', 'Finance', 45000, FALSE),
(4, 'Sneha Patel', 'IT', 35000, TRUE),
(5, 'Vikram Singh', 'Finance', 28000, TRUE),
(6, 'Anjali Mehta', 'HR', 32000, FALSE),
(7, 'Kiran Rao', 'IT', 60000, TRUE),
(8, 'Pooja Verma', 'Marketing', 27000, TRUE),
(9, 'Ramesh Naidu', 'Sales', 38000, TRUE),
(10, 'Deepika Joshi', 'Finance', 52000, TRUE),
(11, 'Suresh Babu', 'Marketing', 29000, FALSE),
(12, 'Neha Kapoor', 'HR', 41000, TRUE),
(13, 'Arjun Reddy', 'IT', 55000, FALSE),
(14, 'Lakshmi Devi', 'Sales', 33000, TRUE),
(15, 'Nikhil Jain', 'IT', 47000, TRUE);

SELECT Queries

Q1. Display IT department employees whose salary is greater than 40000.

```
SELECT *  
FROM Employees  
WHERE department = 'IT' AND salary > 40000;
```

Q2. Display employees who belong to the HR department or whose salary is less than 35000.

```
SELECT *  
FROM Employees  
WHERE department = 'HR' OR salary < 35000;
```

Q3. Display employees who are not in the HR department.

```
SELECT *  
FROM Employees  
WHERE department != 'HR';
```

Q4. Display employees whose department is Finance or whose salary is greater than 30000 and are active.

```
SELECT *  
FROM Employees  
WHERE department = 'Finance'  
OR (salary > 30000 AND isactive = TRUE);
```

Q5. Display active IT employees.

```
SELECT *  
FROM Employees  
WHERE department = 'IT' AND isactive = TRUE;
```

Q6. Display inactive employees or employees earning less than 30000.

```
SELECT *  
FROM Employees  
WHERE isactive = FALSE OR salary < 30000;
```

Q7. Display Marketing employees with salary greater than 25000.

```
SELECT *  
FROM Employees  
WHERE department = 'Marketing' AND salary > 25000;
```

Q8. Display employees who are not in the Finance department.

```
SELECT *  
FROM Employees  
WHERE department != 'Finance';
```

Q9. Display active employees earning more than 50000.

```
SELECT *  
FROM Employees  
WHERE salary > 50000 AND isactive = TRUE;
```

Q10. Display employees from Sales or HR departments.

```
SELECT *  
FROM Employees  
WHERE department = 'Sales' OR department = 'HR';
```

Q11. Display employees who are inactive and belong to the IT department.

```
SELECT *  
FROM Employees  
WHERE department = 'IT' AND isactive = FALSE;
```

Q12. Display employees whose salary is less than 40000 and are active.

```
SELECT *  
FROM Employees  
WHERE salary < 40000 AND isactive = TRUE;
```

UPDATE Queries

Q1. Increase salary by 2000 for active IT employees earning less than 60000.

```
UPDATE Employees  
SET salary = salary + 2000  
WHERE department = 'IT'  
AND isactive = TRUE  
AND salary < 60000;
```

Q2. Reduce salary by 10% for Finance department employees or inactive employees.

```
UPDATE Employees
SET salary = salary - (salary * 0.10)
WHERE department = 'Finance'
OR isactive = FALSE;
```

Q3. Make all non-IT employees inactive.

```
UPDATE Employees
SET isactive = FALSE
WHERE department != 'IT';
```

Q4. Increase salary by 3000 for active HR employees.

```
UPDATE Employees
SET salary = salary + 3000
WHERE department = 'HR'
AND isactive = TRUE;
```

Q5. Increase salary by 5000 for Sales employees earning less than 40000.

```
UPDATE Employees
SET salary = salary + 5000
WHERE department = 'Sales'
AND salary < 40000;
```

Q6. Activate all inactive Marketing employees.

```
UPDATE Employees
SET isactive = TRUE
WHERE department = 'Marketing'
AND isactive = FALSE;
```

Q7. Reduce salary by 2000 for inactive employees.

```
UPDATE Employees  
SET salary = salary - 2000  
WHERE isactive = FALSE;
```

Q8. Increase salary by 15% for active employees earning more than 50000.

```
UPDATE Employees  
SET salary = salary + (salary * 0.15)  
WHERE salary > 50000  
AND isactive = TRUE;
```

Q9. Make all Finance employees active.

```
UPDATE Employees  
SET isactive = TRUE  
WHERE department = 'Finance';
```

Q10. Increase salary by 1000 for IT or Sales employees.

```
UPDATE Employees  
SET salary = salary + 1000  
WHERE department = 'IT'  
OR department = 'Sales';
```

DELETE Queries

Q1. Delete inactive employees from the HR department.

```
DELETE FROM Employees  
WHERE department = 'HR'  
AND isactive = FALSE;
```

Q2. Delete employees from Finance or employees earning less than 35000.

```
DELETE FROM Employees  
WHERE department = 'Finance'  
OR salary < 35000;
```

Q3. Delete employees who are not in the IT department.

```
DELETE FROM Employees  
WHERE department != 'IT';
```

Q4. Delete inactive employees.

```
DELETE FROM Employees  
WHERE isactive = FALSE;
```

Q5. Delete Marketing employees with salary less than 30000.

```
DELETE FROM Employees  
WHERE department = 'Marketing'  
AND salary < 30000;
```

Q6. Delete Sales employees or inactive employees.

```
DELETE FROM Employees  
WHERE department = 'Sales'  
OR isactive = FALSE;
```

Q7. Delete active employees earning more than 55000.

```
DELETE FROM Employees  
WHERE isactive = TRUE  
AND salary > 55000;
```

Q8. Delete employees who are not in the IT or HR departments.

```
DELETE FROM Employees  
WHERE department != 'IT'  
AND department != 'HR';
```

Q9. Delete employees whose salary is below 30000 and are inactive.

```
DELETE FROM Employees  
WHERE salary < 30000  
AND isactive = FALSE;
```

Q10. Delete employees from the Marketing department or whose salary is greater than 50000.

```
DELETE FROM Employees  
WHERE department = 'Marketing'  
OR salary > 50000;
```

NULL Operators in MySQL

What are NULL Operators?

NULL Operators are used to check whether a column contains a **NULL** value (missing or unknown data).

Note: **NULL** is **not** the same as 0 (zero) or an empty string (' '). It represents the absence of a value.

MySQL provides two NULL operators:

- **IS NULL** – Checks whether a column contains a NULL value.
- **IS NOT NULL** – Checks whether a column contains a non-NULL value.

STEP 1: Create Table


```
CREATE TABLE Projects (  
  id INT,  
  name VARCHAR(100),  
  assignedto VARCHAR(100),  
  start DATE,  
  end DATE  
);
```

STEP 2: Insert Records

```
INSERT INTO Projects VALUES  
(1, 'Website Revamp', 'Alice', '2024-01-15', '2024-03-01'),  
(2, 'Mobile App', 'Bob', '2024-02-01', NULL),  
(3, 'Data Migration', NULL, NULL, NULL),  
(4, 'SEO Optimization', 'Eve', '2024-03-10', '2024-04-20'),  
(5, 'Cloud Migration', NULL, '2024-03-15', NULL),  
(6, 'ERP Development', 'John', '2024-04-01', NULL),  
(7, 'CRM Upgrade', NULL, NULL, NULL),  
(8, 'Inventory System', 'David', '2024-05-10', '2024-06-25'),  
(9, 'Online Portal', 'Sophia', '2024-06-15', NULL),  
(10, 'Security Audit', NULL, '2024-07-01', NULL),  
(11, 'AI Chatbot', 'James', NULL, NULL),  
(12, 'Payroll System', 'Emma', '2024-08-01', '2024-09-10'),  
(13, 'Hospital Management', NULL, NULL, NULL),  
(14, 'Student Portal', 'Michael', '2024-09-05', NULL),  
(15, 'Banking Application', NULL, '2024-10-01', NULL);
```

SELECT Queries

Q1. Display all projects that are not assigned to anyone.

```
SELECT *  
FROM Projects  
WHERE assignedto IS NULL;
```

Q2. Display projects that have not started yet.

```
SELECT *  
FROM Projects  
WHERE start IS NULL;
```

Q3. Display all completed projects.

```
SELECT *  
FROM Projects  
WHERE end IS NOT NULL;
```

Q4. Display all projects with an assigned employee.

```
SELECT *  
FROM Projects  
WHERE assignedto IS NOT NULL;
```

Q5. Display projects that have not yet been completed.

```
SELECT *  
FROM Projects  
WHERE end IS NULL;
```

Q6. Display projects that have both a start date and an end date.

```
SELECT *  
FROM Projects  
WHERE start IS NOT NULL  
AND end IS NOT NULL;
```

Q7. Display projects where both assigned employee and start date are NULL.

```
SELECT *  
FROM Projects  
WHERE assignedto IS NULL  
AND start IS NULL;
```

Q8. Display projects that have a start date but no end date.

```
SELECT *  
FROM Projects  
WHERE start IS NOT NULL  
AND end IS NULL;
```

Q9. Display projects whose assigned employee is available.

```
SELECT id, name, assignedto  
FROM Projects  
WHERE assignedto IS NOT NULL;
```

Q10. Display projects with all NULL values in start and end dates.

```
SELECT *  
FROM Projects  
WHERE start IS NULL  
AND end IS NULL;
```

UPDATE Queries

Q1. Assign all unassigned projects to Frank.

```
UPDATE Projects  
SET assignedto = 'Frank'  
WHERE assignedto IS NULL;
```

Q2. Update today's date as the start date for projects that have not started.

```
UPDATE Projects  
SET start = CURDATE()  
WHERE start IS NULL;
```

Q3. Rename all completed projects as 'Completed'.

```
UPDATE Projects
SET name = 'Completed'
WHERE end IS NOT NULL;
```

Q4. Set today's date as the end date for projects that have no end date.

```
UPDATE Projects
SET end = CURDATE()
WHERE end IS NULL;
```

Q5. Assign 'Admin' to projects without an assigned employee.

```
UPDATE Projects
SET assignedto = 'Admin'
WHERE assignedto IS NULL;
```

Q6. Rename projects without a start date as 'Pending Project'.

```
UPDATE Projects
SET name = 'Pending Project'
WHERE start IS NULL;
```

Q7. Assign 'Manager' to projects having both start and end dates.

```
UPDATE Projects
SET assignedto = 'Manager'
WHERE start IS NOT NULL
AND end IS NOT NULL;
```

Q8. Change the project name to 'In Progress' for projects with a start date but no end date.

```
UPDATE Projects
SET name = 'In Progress'
WHERE start IS NOT NULL
AND end IS NULL;
```

Q9. Set the start date to '2024-12-01' for projects with NULL start date.

```
UPDATE Projects
SET start = '2024-12-01'
WHERE start IS NULL;
```

Q10. Set the end date to '2025-01-31' for projects with NULL end date.

```
UPDATE Projects
SET end = '2025-01-31'
WHERE end IS NULL;
```

DELETE Queries

Q1. Delete all unassigned projects.

```
DELETE FROM Projects
WHERE assignedto IS NULL;
```

Q2. Delete projects that have not started.

```
DELETE FROM Projects
WHERE start IS NULL;
```

Q3. Delete all completed projects.

```
DELETE FROM Projects
WHERE end IS NOT NULL;
```

Q4. Delete projects that do not have an end date.

```
DELETE FROM Projects
WHERE end IS NULL;
```

Q5. Delete projects where both assigned employee and start date are NULL.

```
DELETE FROM Projects
WHERE assignedto IS NULL
AND start IS NULL;
```

Q6. Delete projects with no assigned employee and no end date.

```
DELETE FROM Projects
WHERE assignedto IS NULL
AND end IS NULL;
```

Q7. Delete projects that have both start and end dates.

```
DELETE FROM Projects
WHERE start IS NOT NULL
AND end IS NOT NULL;
```

Q8. Delete projects that have a start date but no end date.

```
DELETE FROM Projects
WHERE start IS NOT NULL
AND end IS NULL;
```

Q9. Delete projects assigned to Frank.

```
DELETE FROM Projects
WHERE assignedto = 'Frank';
```

Q10. Delete projects named 'Completed'.

```
DELETE FROM Projects  
WHERE name = 'Completed';
```

Day-10

Other Operators

MySQL offers various additional operators for specific purposes.

BETWEEN: Checks if a value falls within a specific range.

IN: Checks if a value is present within a list of specified values.

CASE: Used for conditional expressions and assigning values based on conditions

BETWEEN Operator

```
SELECT * FROM orders
WHERE order_date BETWEEN '2024-03-15' AND '2024-03-20';
```

±This retrieves all orders placed between March 15th, 2024 (inclusive) and March 20th, 2024 (inclusive).

±BETWEEN:

Checks if a value falls within a specific range.

STEP 1: Create Table

```
CREATE TABLE Orders (
  id INT,
  customer_name VARCHAR(100),
  order_date DATE,
  total_amount DECIMAL(10,2)
);
```

STEP 2: Insert Records

```
INSERT INTO Orders VALUES
(1, 'Alice', '2024-03-01', 1500.00),
(2, 'Bob', '2024-03-05', 450.00),
(3, 'Charlie', '2024-03-10', 2500.00),
(4, 'Diana', '2024-03-15', 800.00),
(5, 'Eve', '2024-03-20', 1200.00),
(6, 'Frank', '2024-03-25', 3000.00),
(7, 'Grace', '2024-04-02', 1800.00),
(8, 'Henry', '2024-04-08', 950.00),
```

(9, 'Ivy', '2024-04-12', 2750.00),
(10, 'Jack', '2024-04-18', 650.00),
(11, 'Kevin', '2024-04-22', 4200.00),
(12, 'Lily', '2024-05-01', 1350.00),
(13, 'Michael', '2024-05-08', 5000.00),
(14, 'Nancy', '2024-05-15', 2250.00),
(15, 'Oliver', '2024-05-25', 900.00);

SELECT Queries

Q1. Display orders placed between '2024-03-03' and '2024-03-10'.

```
SELECT *  
FROM Orders  
WHERE order_date BETWEEN '2024-03-03' AND '2024-03-10';
```

Q2. Display orders placed between '2024-04-01' and '2024-04-30'.

```
SELECT *  
FROM Orders  
WHERE order_date BETWEEN '2024-04-01' AND '2024-04-30';
```

Q3. Display orders whose total amount is between 1000 and 3000.

```
SELECT *  
FROM Orders  
WHERE total_amount BETWEEN 1000 AND 3000;
```

Q4. Display orders whose total amount is between 500 and 1500.

```
SELECT *  
FROM Orders  
WHERE total_amount BETWEEN 500 AND 1500;
```

Q5. Display orders placed between '2024-05-01' and '2024-05-31'.

```
SELECT *  
FROM Orders  
WHERE order_date BETWEEN '2024-05-01' AND '2024-05-31';
```

Q6. Display customer names whose order amount is between 2000 and 5000.

```
SELECT customer_name, total_amount  
FROM Orders  
WHERE total_amount BETWEEN 2000 AND 5000;
```

Q7. Display orders placed between '2024-03-15' and '2024-04-15'.

```
SELECT *  
FROM Orders  
WHERE order_date BETWEEN '2024-03-15' AND '2024-04-15';
```

Q8. Display orders whose amount is between 800 and 1800.

```
SELECT *  
FROM Orders  
WHERE total_amount BETWEEN 800 AND 1800;
```

Q9. Display orders placed between '2024-04-10' and '2024-05-10'.

```
SELECT *  
FROM Orders  
WHERE order_date BETWEEN '2024-04-10' AND '2024-05-10';
```

Q10. Display orders whose amount is between 2500 and 5000.

```
SELECT *  
FROM Orders  
WHERE total_amount BETWEEN 2500 AND 5000;
```

UPDATE Queries

Q1. Increase the total amount by 1000 for orders placed between '2024-03-04' and '2024-03-10'.

```
UPDATE Orders
SET total_amount = total_amount + 1000
WHERE order_date BETWEEN '2024-03-04' AND '2024-03-10';
```

Q2. Increase the total amount by 500 for orders placed between '2024-04-01' and '2024-04-15'.

```
UPDATE Orders
SET total_amount = total_amount + 500
WHERE order_date BETWEEN '2024-04-01' AND '2024-04-15';
```

Q3. Reduce the total amount by 200 for orders between 1000 and 2000.

```
UPDATE Orders
SET total_amount = total_amount - 200
WHERE total_amount BETWEEN 1000 AND 2000;
```

Q4. Increase the total amount by 10% for orders between 2500 and 5000.

```
UPDATE Orders
SET total_amount = total_amount + (total_amount * 0.10)
WHERE total_amount BETWEEN 2500 AND 5000;
```

Q5. Add 300 to orders placed between '2024-05-01' and '2024-05-31'.

```
UPDATE Orders
SET total_amount = total_amount + 300
WHERE order_date BETWEEN '2024-05-01' AND '2024-05-31';
```

Q6. Reduce the total amount by 100 for orders placed between '2024-03-15' and '2024-03-31'.

UPDATE Orders
SET total_amount = total_amount - 100
WHERE order_date BETWEEN '2024-03-15' AND '2024-03-31';

Q7. Increase the total amount by 750 for orders between 800 and 1500.

UPDATE Orders
SET total_amount = total_amount + 750
WHERE total_amount BETWEEN 800 AND 1500;

Q8. Reduce the total amount by 500 for orders between 3000 and 5000.

UPDATE Orders
SET total_amount = total_amount - 500
WHERE total_amount BETWEEN 3000 AND 5000;

Q9. Increase the total amount by 250 for orders placed between '2024-04-20' and '2024-05-10'.

UPDATE Orders
SET total_amount = total_amount + 250
WHERE order_date BETWEEN '2024-04-20' AND '2024-05-10';

Q10. Add 100 to all orders placed between '2024-03-01' and '2024-05-31'.

UPDATE Orders
SET total_amount = total_amount + 100
WHERE order_date BETWEEN '2024-03-01' AND '2024-05-31';

DELETE Queries

Q1. Delete orders placed between '2024-03-10' and '2024-03-20'.

```
DELETE FROM Orders  
WHERE order_date BETWEEN '2024-03-10' AND '2024-03-20';
```

Q2. Delete orders placed between '2024-04-01' and '2024-04-15'.

```
DELETE FROM Orders  
WHERE order_date BETWEEN '2024-04-01' AND '2024-04-15';
```

Q3. Delete orders whose total amount is between 500 and 1000.

```
DELETE FROM Orders  
WHERE total_amount BETWEEN 500 AND 1000;
```

Q4. Delete orders whose total amount is between 2500 and 5000.

```
DELETE FROM Orders  
WHERE total_amount BETWEEN 2500 AND 5000;
```

Q5. Delete orders placed between '2024-05-01' and '2024-05-31'.

```
DELETE FROM Orders  
WHERE order_date BETWEEN '2024-05-01' AND '2024-05-31';
```

Q6. Delete orders whose total amount is between 1000 and 2000.

```
DELETE FROM Orders  
WHERE total_amount BETWEEN 1000 AND 2000;
```

Q7. Delete orders placed between '2024-03-01' and '2024-03-05'.

```
DELETE FROM Orders  
WHERE order_date BETWEEN '2024-03-01' AND '2024-03-05';
```

Q8. Delete orders placed between '2024-04-18' and '2024-05-08'.

```
DELETE FROM Orders  
WHERE order_date BETWEEN '2024-04-18' AND '2024-05-08';
```

Q9. Delete orders whose total amount is between 1800 and 3000.

```
DELETE FROM Orders  
WHERE total_amount BETWEEN 1800 AND 3000;
```

Q10. Delete orders placed between '2024-03-25' and '2024-04-08'.

```
DELETE FROM Orders  
WHERE order_date BETWEEN '2024-03-25' AND '2024-04-08';
```

IN Operator

This operator checks if a value exists within a specified list of values

```
SELECT * FROM customers  
WHERE country IN ('USA', 'Canada', 'UK');
```

±This retrieves customer records where the country is either 'USA', 'Canada', or 'UK'.

STEP 1: Create Table

```
CREATE TABLE Emp (  
  id INT,  
  name VARCHAR(100),
```

```
dept VARCHAR(20),  
salary INT  
);
```

STEP 2: Insert Records

```
INSERT INTO Emp VALUES  
(1, 'Rahul', 'IT', 50000),  
(2, 'Ramesh', 'HR', 35000),  
(3, 'Karan', 'Finance', 45000),  
(4, 'Mahesh', 'Marketing', 40000),  
(5, 'Suresh', 'IT', 55000),  
(6, 'Arjun', 'Sales', 38000),  
(7, 'Manish', 'Admin', 42000),  
(8, 'Ganesh', 'Finance', 47000),  
(9, 'Naresh', 'HR', 39000),  
(10, 'Rohan', 'IT', 52000),  
(11, 'Mohan', 'Sales', 41000),  
(12, 'Kishan', 'Marketing', 43000),  
(13, 'Dinesh', 'Admin', 46000),  
(14, 'Rajesh', 'Finance', 51000),  
(15, 'Kiran', 'IT', 48000);
```

SELECT Queries

Q1. Display employees from the IT and Finance departments.

```
SELECT *  
FROM Emp  
WHERE dept IN ('IT', 'Finance');
```

Q2. Display employees from the HR and Sales departments.


```
SELECT *  
FROM Emp  
WHERE dept IN ('HR', 'Sales');
```

Q3. Display employees from the Marketing and Admin departments.

```
SELECT *  
FROM Emp  
WHERE dept IN ('Marketing', 'Admin');
```

Q4. Display employees from the HR, IT, and Finance departments.

```
SELECT *  
FROM Emp  
WHERE dept IN ('HR', 'IT', 'Finance');
```

Q5. Display employees whose salary is 35000, 42000, or 60000.

```
SELECT *  
FROM Emp  
WHERE salary IN (35000, 42000, 60000);
```

Q6. Display employees from the Sales, Marketing, and Finance departments.

```
SELECT *  
FROM Emp  
WHERE dept IN ('Sales', 'Marketing', 'Finance');
```

Q7. Display employees from the Admin department.

```
SELECT *  
FROM Emp  
WHERE dept IN ('Admin');
```

Q8. Display employees whose salary is 39000, 43000, or 52000.

```
SELECT *  
FROM Emp  
WHERE salary IN (39000, 43000, 52000);
```

Q9. Display employees from the IT, Admin, and Sales departments.

```
SELECT *  
FROM Emp  
WHERE dept IN ('IT', 'Admin', 'Sales');
```

Q10. Display employees whose salary is 40000, 45000, or 50000.

```
SELECT *  
FROM Emp  
WHERE salary IN (40000, 45000, 50000);
```

UPDATE Queries

Q1. Increase salary by 2000 for employees in the HR or Marketing departments.

```
UPDATE Emp  
SET salary = salary + 2000  
WHERE dept IN ('HR', 'Marketing');
```

Q2. Increase salary by 3000 for employees in the IT and Finance departments.

```
UPDATE Emp  
SET salary = salary + 3000  
WHERE dept IN ('IT', 'Finance');
```

Q3. Increase salary by 1500 for employees in the Sales and Admin departments.

UPDATE Emp
SET salary = salary + 1500
WHERE dept IN ('Sales', 'Admin');

Q4. Increase salary by 5000 for employees earning 35000, 39000, or 43000.

UPDATE Emp
SET salary = salary + 5000
WHERE salary IN (35000, 39000, 43000);

Q5. Increase salary by 1000 for employees in the HR, IT, and Sales departments.

UPDATE Emp
SET salary = salary + 1000
WHERE dept IN ('HR', 'IT', 'Sales');

Q6. Increase salary by 2500 for employees earning 45000, 48000, or 52000.

UPDATE Emp
SET salary = salary + 2500
WHERE salary IN (45000, 48000, 52000);

Q7. Increase salary by 4000 for employees in the Admin department.

UPDATE Emp
SET salary = salary + 4000
WHERE dept IN ('Admin');

Q8. Increase salary by 3500 for employees in the Marketing and Finance departments.

UPDATE Emp
SET salary = salary + 3500
WHERE dept IN ('Marketing', 'Finance');

Q9. Increase salary by 500 for employees earning 40000, 41000, or 42000.

```
UPDATE Emp
SET salary = salary + 500
WHERE salary IN (40000, 41000, 42000);
```

Q10. Increase salary by 2000 for employees in the IT, Sales, and Admin departments.

```
UPDATE Emp
SET salary = salary + 2000
WHERE dept IN ('IT', 'Sales', 'Admin');
```

DELETE Queries

Q1. Delete employees from the Finance or Marketing departments.

```
DELETE FROM Emp
WHERE dept IN ('Finance', 'Marketing');
```

Q2. Delete employees from the HR department.

```
DELETE FROM Emp
WHERE dept IN ('HR');
```

Q3. Delete employees from the Sales and Admin departments.

```
DELETE FROM Emp
WHERE dept IN ('Sales', 'Admin');
```

Q4. Delete employees whose salary is 35000, 39000, or 43000.

```
DELETE FROM Emp
WHERE salary IN (35000, 39000, 43000);
```

Q5. Delete employees from the IT department.

**DELETE FROM Emp
WHERE dept IN ('IT');**

Q6. Delete employees whose salary is 45000, 50000, or 55000.

**DELETE FROM Emp
WHERE salary IN (45000, 50000, 55000);**

Q7. Delete employees from the Marketing and HR departments.

**DELETE FROM Emp
WHERE dept IN ('Marketing', 'HR');**

Q8. Delete employees from the Admin department.

**DELETE FROM Emp
WHERE dept IN ('Admin');**

Q9. Delete employees whose salary is 37000, 38000, or 41000.

**DELETE FROM Emp
WHERE salary IN (37000, 38000, 41000);**

Q10. Delete employees from the IT, Finance, and Sales departments.

**DELETE FROM Emp
WHERE dept IN ('IT', 'Finance', 'Sales');**

LIKE Operator Practice Questions

SELECT Queries (LIKE Operator)

Q1. Display employees whose names start with 'R'.

```
SELECT *  
FROM Emp  
WHERE name LIKE 'R%';
```

Q2. Display employees whose names end with 'sh'.

```
SELECT *  
FROM Emp  
WHERE name LIKE '%sh';
```

Q3. Display employees whose names contain 'ar'.

```
SELECT *  
FROM Emp  
WHERE name LIKE '%ar%';
```

Q4. Display employees whose names have exactly 5 characters.

```
SELECT *  
FROM Emp  
WHERE name LIKE '_____';
```

– Q5. Display employees whose names start with 'K' and end with 'n'.

```
SELECT
```

```
SELECT *  
FROM Emp  
WHERE name LIKE 'K%n';
```

Q6. Display employees whose names have 'a' as the second character.

```
SELECT *  
FROM Emp  
WHERE name LIKE '_a%';
```

Q7. Display employees whose names contain 'es'.

```
SELECT *  
FROM Emp  
WHERE name LIKE '%es%';
```

Q8. Display employees whose names end with 'n'.

```
SELECT *  
FROM Emp  
WHERE name LIKE '%n';
```

Q9. Display employees whose names start with 'M' and contain 'e'.

```
SELECT *  
FROM Emp  
WHERE name LIKE 'M%e%';
```

Q10. Display employees whose names have exactly 6 characters and end with 'h'.

```
SELECT *  
FROM Emp  
WHERE name LIKE '_____h';
```

Q11. Display employees whose names start with 'A'.

```
SELECT *  
FROM Emp  
WHERE name LIKE 'A%';
```

Q12. Display employees whose names end with 'an'.

```
SELECT *  
FROM Emp  
WHERE name LIKE '%an';
```

Q13. Display employees whose names contain 'oh'.

```
SELECT *  
FROM Emp  
WHERE name LIKE '%oh%';
```

Q14. Display employees whose names have exactly 7 characters.

```
SELECT *  
FROM Emp  
WHERE name LIKE '_____';
```

Q15. Display employees whose names start with 'S'.

```
SELECT *  
FROM Emp  
WHERE name LIKE 'S%';
```

UPDATE Queries

Q1. Increase salary by 2000 for employees whose names start with 'R'.

UPDATE Emp
SET salary = salary + 2000
WHERE name LIKE 'R%';

Q2. Increase salary by 3000 for employees whose names end with 'sh'.

UPDATE Emp
SET salary = salary + 3000
WHERE name LIKE '%sh';

Q3. Increase salary by 1000 for employees whose names contain 'ar'.

UPDATE Emp
SET salary = salary + 1000
WHERE name LIKE '%ar%';

Q4. Increase salary by 500 for employees whose names have exactly 5 characters.

UPDATE Emp
SET salary = salary + 500
WHERE name LIKE '_____';

Q5. Increase salary by 2500 for employees whose names start with 'K' and end with 'n'.

UPDATE Emp
SET salary = salary + 2500
WHERE name LIKE 'K%n';

Q6. Increase salary by 1500 for employees whose names have 'a' as the second character.

UPDATE Emp
SET salary = salary + 1500
WHERE name LIKE '_a%';

Q7. Increase salary by 4000 for employees whose names contain 'es'.

```
UPDATE Emp
SET salary = salary + 4000
WHERE name LIKE '%es%';
```

Q8. Increase salary by 1000 for employees whose names end with 'n'.

```
UPDATE Emp
SET salary = salary + 1000
WHERE name LIKE '%n';
```

Q9. Increase salary by 5000 for employees whose names start with 'M' and contain 'e'.

```
UPDATE Emp
SET salary = salary + 5000
WHERE name LIKE 'M%e%';
```

Q10. Increase salary by 2000 for employees whose names have exactly 6 characters and end with 'h'.

```
UPDATE Emp
SET salary = salary + 2000
WHERE name LIKE '_____h';
```

DELETE Queries

Q1. Delete employees whose names start with 'R'.

```
DELETE FROM Emp
WHERE name LIKE 'R%';
```

Q2. Delete employees whose names end with 'sh'.

DELETE FROM Emp
WHERE name LIKE '%sh';

Q3. Delete employees whose names contain 'ar'.

DELETE FROM Emp
WHERE name LIKE '%ar%';

Q4. Delete employees whose names have exactly 5 characters.

DELETE FROM Emp
WHERE name LIKE '_____';

Q5. Delete employees whose names start with 'K' and end with 'n'.

DELETE FROM Emp
WHERE name LIKE 'K%n';

Q6. Delete employees whose names have 'a' as the second character.

DELETE FROM Emp
WHERE name LIKE '_a%';

Q7. Delete employees whose names contain 'es'.

DELETE FROM Emp
WHERE name LIKE '%es%';

Q8. Delete employees whose names end with 'n'.

DELETE FROM Emp
WHERE name LIKE '%n';

Q9. Delete employees whose names start with 'M' and contain 'e'.

DELETE FROM Emp
WHERE name LIKE 'M%e%';

Q10. Delete employees whose names have exactly 6 characters and end with 'h'.

DELETE FROM Emp
WHERE name LIKE '_____h';

CASE Statement in MySQL

The CASE statement is used to apply conditional logic in SQL. It works similarly to the if-else statement in programming languages.

Syntax

```
CASE
  WHEN condition1 THEN result1
  WHEN condition2 THEN result2
  ...
  ELSE result
END
```

Sample Table

```
CREATE TABLE employees (
  emp_id INT PRIMARY KEY,
  emp_name VARCHAR(50),
  department VARCHAR(30),
  salary DECIMAL(10,2),
  experience INT,
  age INT
);
```

INSERT INTO employees VALUES

```
(101,'Ravi','IT',65000,5,28),
(102,'Priya','HR',45000,2,24),
(103,'Amit','Finance',75000,8,35),
(104,'Sneha','IT',55000,3,27),
(105,'Rahul','Sales',40000,1,22),
(106,'Kiran','Finance',90000,12,40),
(107,'Anjali','HR',52000,4,29),
(108,'Suresh','Sales',68000,7,33);
```

CASE Examples

Example 1: Employee Salary Category

```
SELECT emp_name,
salary,
CASE
    WHEN salary >= 80000 THEN 'High Salary'
    WHEN salary >= 60000 THEN 'Medium Salary'
    ELSE 'Low Salary'
END AS salary_category
FROM employees;
```

Output

Employee	Salary	Category
Ravi	65000	Medium Salary
Priya	45000	Low Salary
Amit	75000	Medium Salary
Sneha	55000	Low Salary
Rahul	40000	Low Salary
Kiran	90000	High Salary
Anjali	52000	Low Salary
Suresh	68000	Medium Salary

Example 2: Experience Level

```
SELECT emp_name,  
experience,  
CASE  
    WHEN experience >= 10 THEN 'Expert'  
    WHEN experience >= 5 THEN 'Senior'  
    WHEN experience >= 2 THEN 'Junior'  
    ELSE 'Fresher'  
END AS experience_level  
FROM employees;
```

Example 3: Department Bonus

```
SELECT emp_name,  
department,  
salary,  
CASE  
    WHEN department='IT' THEN salary*0.20  
    WHEN department='Finance' THEN salary*0.15  
    WHEN department='HR' THEN salary*0.10  
    ELSE salary*0.05  
END AS bonus  
FROM employees;
```

Example 4: Voting Eligibility

```
SELECT emp_name,  
age,  
CASE  
    WHEN age >= 18 THEN 'Eligible'  
    ELSE 'Not Eligible'  
END AS voting_status  
FROM employees;
```

Example 5: Grade Employees

```
SELECT emp_name,  
salary,  
CASE  
    WHEN salary >= 80000 THEN 'A'  
    WHEN salary >= 60000 THEN 'B'  
    WHEN salary >= 50000 THEN 'C'  
    ELSE 'D'  
END AS grade  
FROM employees;
```

Example 6: CASE in ORDER BY

Display IT employees first.

```
SELECT *  
  
FROM employees  
ORDER BY  
CASE  
    WHEN department='IT' THEN 1  
    ELSE 2  
END;
```

Example 7: CASE in UPDATE

Increase salary based on experience.

```
UPDATE employees  
SET salary =  
CASE  
    WHEN experience >= 10 THEN salary + 10000  
    WHEN experience >= 5 THEN salary + 5000  
    ELSE salary + 2000  
END;
```

Example 8: CASE with Aggregate Function

```
SELECT  
SUM(  
CASE  
    WHEN department='IT'  
    THEN salary  
    ELSE 0  
END  
) AS total_it_salary  
FROM employees;
```

Interview Questions (CASE)

Q1. Display employee name and salary status.

```
SELECT emp_name,  
salary,  
CASE  
    WHEN salary>60000 THEN 'Good Salary'  
    ELSE 'Average Salary'  
END AS status  
FROM employees;
```

Q2. Display "Experienced" if experience is more than 5 years.

```
SELECT emp_name,  
CASE  
    WHEN experience>5 THEN 'Experienced'  
    ELSE 'Less Experienced'  
END AS experience_status  
FROM employees;
```

SYSTEM INFORMATION FUNCTIONS

System Information Functions

These functions return information about the MySQL server, current user, current database, current date, current time, etc.

Q1. Display the current date.

```
SELECT CURDATE();
```

Q2. Display the current time.

```
SELECT CURTIME();
```

Q3. Display the current date and time.

```
SELECT NOW();
```

Q4. Display the current database.

```
SELECT DATABASE();
```

Q5. Display the current logged-in user.

```
SELECT USER();
```

Q6. Display the current MySQL version.

```
SELECT VERSION();
```

Q7. Display the current system date.

```
SELECT SYSDATE();
```

Q8. Display the current connection ID.

```
SELECT CONNECTION_ID();
```

Q9. Display the current database user.

```
SELECT CURRENT_USER();
```

Q10. Display the current UTC date.

```
SELECT UTC_DATE();
```

Q11. Display the current UTC time.

```
SELECT UTC_TIME();
```

Q12. Display the current UTC date and time.

```
SELECT UTC_TIMESTAMP();
```

Day-11

AGGREGATE FUNCTIONS

Aggregate Functions

Aggregate functions perform calculations on multiple rows and return a single value.

STEP 1: Create Table

```
CREATE TABLE Employee (  
  id INT,  
  name VARCHAR(100),  
  department VARCHAR(30),  
  salary DECIMAL(10,2)  
);
```

STEP 2: Insert Records

```
INSERT INTO Employee VALUES  
(1,'Rahul','IT',50000),  
(2,'Anjali','HR',35000),  
(3,'Kiran','Finance',45000),  
(4,'Bhanu','IT',60000),  
(5,'Ramesh','Sales',40000),  
(6,'Priya','HR',38000),  
(7,'Arjun','Finance',52000),  
(8,'Mounika','Marketing',47000),  
(9,'Sunil','IT',55000),  
(10,'Keerthi','Sales',42000);
```

SELECT Queries

Q1. Find the total salary of all employees.

```
SELECT SUM(salary) AS Total_Salary  
FROM Employee;
```

Q2. Find the average salary.

```
SELECT AVG(salary) AS Average_Salary  
FROM Employee;
```

Q3. Find the highest salary.

```
SELECT MAX(salary) AS Highest_Salary  
FROM Employee;
```

Q4. Find the lowest salary.

```
SELECT MIN(salary) AS Lowest_Salary  
FROM Employee;
```

Q5. Find the total number of employees.

```
SELECT COUNT(*) AS Total_Employees  
FROM Employee;
```

Q6. Find the number of employees in the IT department.

```
SELECT COUNT(*) AS IT_Employees  
FROM Employee  
WHERE department='IT';
```

Q7. Find the total salary of IT employees.

```
SELECT SUM(salary)  
FROM Employee  
WHERE department='IT';
```

Q8. Find the average salary of HR employees.

```
SELECT AVG(salary)
FROM Employee
WHERE department='HR';
```

Q9. Find the maximum salary in the Finance department.

```
SELECT MAX(salary)
FROM Employee
WHERE department='Finance';
```

Q10. Find the minimum salary in the Sales department.

```
SELECT MIN(salary)
FROM Employee
WHERE department='Sales';
```

Q11. Count employees whose salary is greater than 45000.

```
SELECT COUNT(*)
FROM Employee
WHERE salary>45000;
```

Q12. Find the total salary of HR and IT employees.

```
SELECT SUM(salary)
FROM Employee
WHERE department IN ('HR','IT');
```

Difference Between COUNT(*) and COUNT(column_name) in MySQL

Feature	COUNT(*)	COUNT(column_name)
Counts	All rows in the table	Only rows where the specified column is NOT NULL
Includes NULL values	Yes (counts every row)	No (ignores NULL values)
Arguments	*	Column name
Purpose	Count total number of records	Count non-NULL values in a specific column

Clauses in MYSQL

In MySQL, **clauses** are used to control **how data is selected, filtered, grouped, sorted, or modified** in SQL queries.

They make SQL statements more meaningful and powerful.

Think of clauses like **instructions** given to the database.

where clause

filter records based on specified conditions where clauses can be used as comparison operators, logical operators and ..etc operators.

syntax

select column1,column2....

from table

where condition.

Group by clause

=====

The group by clause is used to group rows having the same values in one or more columns. It is commonly used with aggregate functions such as sum(),min(),max(),avg(),count().

syntax

select column-name,aggregate_function(column-name)

from table-name

group by column_name.

Having Clause

The having clause is used to filter grouped records after group by clause. it is similar to where but it works on grouped instead of individual rows.

syntax

select column-name,

aggregate_function(column-name)

from table-name

group by column-name

having condition.

Order By Clause

the order by clause is used to sort the result set in ascending(ASC)or descending(Desc)order.

asc-->Ascending order by default.

Desc-->Descending order

syntax:

select column-name

from table-name

order by column-name asc;

Example

Step 1: Create Table

```
CREATE TABLE STUDENTS (  
    StudentID INT AUTO_INCREMENT PRIMARY KEY,  
    Name VARCHAR(50),  
    Age INT,  
    Department VARCHAR(30),  
    Marks INT,  
    City VARCHAR(30)  
);
```

Insert Sample Data (without StudentID)

```
INSERT INTO STUDENTS (Name, Age, Department, Marks, City) VALUES  
( 'Anil', 20, 'CSE', 85, 'Hyderabad'),  
( 'Sneha', 22, 'ECE', 75, 'Mumbai'),  
( 'Ravi', 20, 'CSE', NULL, 'Hyderabad'),  
( 'Divya', 23, 'MECH', 65, NULL),  
( 'Kiran', 22, 'ECE', 75, 'Pune'),  
( 'Varun', 21, 'CSE', 92, 'Chennai'),  
( 'Priya', 20, 'EEE', 88, 'Delhi'),  
( 'John', 22, 'MECH', 55, 'Mumbai'),  
( 'Rekha', 21, 'ECE', 75, 'Pune'),  
( 'Ramesh', 23, 'EEE', NULL, 'Delhi');
```

```
select*from students;
```

WHERE Clause Exercises

Question 1

Display all students who belong to the CSE department.

Query

```
SELECT *  
FROM students  
WHERE Department = 'CSE';
```

Question 2

Display all students whose marks are greater than 80.

Query

```
SELECT *  
FROM students  
WHERE Marks > 80;
```

Question 3

Display all students who are 22 years old.

Query

```
SELECT *  
FROM students  
WHERE Age = 22;
```

Question 4

Display all students who belong to Mumbai.

Query

```
SELECT *  
FROM students  
WHERE City = 'Mumbai';
```

Question 5

Display all students whose marks are less than 70.

Query

```
SELECT *  
FROM students  
WHERE Marks < 70;
```

Question 6

Display all students whose marks are between 70 and 90.

Query

```
SELECT *  
FROM students  
WHERE Marks BETWEEN 70 AND 90;
```

Question 7

Display all students whose age is not equal to 20.

Query

```
SELECT *  
FROM students  
WHERE Age <> 20;
```

Question 8

Display all students who belong to CSE and have marks greater than 80.

Query

```
SELECT *  
FROM students  
WHERE Department = 'CSE'  
AND Marks > 80;
```

Question 9

Display all students who belong to ECE or EEE department.

Query

```
SELECT *  
FROM students  
WHERE Department = 'ECE'  
OR Department = 'EEE';
```

Question 10

Display all students whose city is not Delhi.

Query

```
SELECT *  
FROM students  
WHERE City <> 'Delhi';
```

GROUP BY Clause

Question 1

Display the total number of students in each department.

Query

```
SELECT Department, COUNT(*) AS Total_Students  
FROM students  
GROUP BY Department;
```

Question 2

Display the average marks of students in each department.

Query

```
SELECT Department, AVG(Marks) AS Average_Marks  
FROM students  
GROUP BY Department;
```

Question 3

Display the highest marks scored in each department.

Query

**SELECT Department, MAX(Marks) AS Highest_Marks
FROM students
GROUP BY Department;**

Question 4

Display the lowest marks scored in each department.

Query

**SELECT Department, MIN(Marks) AS Lowest_Marks
FROM students
GROUP BY Department;**

Question 5

Display the total marks obtained by students in each department.

Query

**SELECT Department, SUM(Marks) AS Total_Marks
FROM students
GROUP BY Department;**

Question 6

Display the number of students in each city.

Query

**SELECT City, COUNT(*) AS Total_Students
FROM students
GROUP BY City;**

Question 7

Display the average age of students in each department.

Query

```
SELECT Department, AVG(Age) AS Average_Age  
FROM students  
GROUP BY Department;
```

Question 8

Display the total marks obtained by students in each city.

Query

```
SELECT City, SUM(Marks) AS Total_Marks  
FROM students  
GROUP BY City;
```

Question 9

Display the highest marks in each city.

Query

```
SELECT City, MAX(Marks) AS Highest_Marks  
FROM students  
GROUP BY City;
```

Question 10

Display the total number of students according to age.

Query

```
SELECT Age, COUNT(*) AS Total_Students  
FROM students  
GROUP BY Age;
```

HAVING Clause

Question 1

Display departments having more than 2 students.

Query

```
SELECT Department, COUNT() AS Total_Students  
FROM students  
GROUP BY Department  
HAVING COUNT() > 2;
```

Question 2

Display departments whose average marks are greater than 75.

Query

```
SELECT Department, AVG(Marks) AS Average_Marks  
FROM students  
GROUP BY Department  
HAVING AVG(Marks) > 75;
```

Question 3

Display cities having more than one student.

Query

```
SELECT City, COUNT() AS Total_Students  
FROM students  
GROUP BY City  
HAVING COUNT() > 1;
```

Question 4

Display departments whose total marks are greater than 150.

Query

```
SELECT Department, SUM(Marks) AS Total_Marks
FROM students
GROUP BY Department
HAVING SUM(Marks) > 150;
```

Question 5

Display age groups having more than two students.

Query

```
SELECT Age, COUNT() AS Total_Students
FROM students
GROUP BY Age
HAVING COUNT() > 2;
```

Question 6

Display cities whose average marks are greater than 70.

Query

```
SELECT City, AVG(Marks) AS Average_Marks
FROM students
GROUP BY City
HAVING AVG(Marks) > 70;
```

Question 7

Display departments whose highest marks are greater than 90.

Query

```
SELECT Department, MAX(Marks) AS Highest_Marks
FROM students
GROUP BY Department
HAVING MAX(Marks) > 90;
```

Question 8

Display departments whose lowest marks are less than 60.

Query

```
SELECT Department, MIN(Marks) AS Lowest_Marks
FROM students
GROUP BY Department
HAVING MIN(Marks) < 60;
```

Question 9

Display cities whose total marks are greater than 100.

Query

```
SELECT City, SUM(Marks) AS Total_Marks
FROM students
GROUP BY City
HAVING SUM(Marks) > 100;
```

Question 10

Display departments having exactly 2 students.

Query

```
SELECT Department, COUNT() AS Total_Students
FROM students
```

GROUP BY Department
HAVING COUNT() = 2;

ORDER BY Clause

Question 1

Display all students ordered by name.

Query

```
SELECT *  
FROM students  
ORDER BY Name;
```

Question 2

Display all students ordered by marks in descending order.

Query

```
SELECT *  
FROM students  
ORDER BY Marks DESC;
```

Question 3

Display all students ordered by age.

Query

```
SELECT *  
FROM students  
ORDER BY Age;
```

Question 4

Display all students ordered by department.

Query

```
SELECT *  
FROM students  
ORDER BY Department;
```

Question 5

Display all students ordered by city.

Query

```
SELECT *  
FROM students  
ORDER BY City;
```

Question 6

Display all students ordered by department and marks in descending order.

Query

```
SELECT *  
FROM students  
ORDER BY Department, Marks DESC;
```

Question 7

Display all students ordered by age in descending order.

Query

```
SELECT *  
FROM students  
ORDER BY Age DESC;
```

Question 8

Display all students ordered by city and name.

Query

```
SELECT *  
FROM students  
ORDER BY City, Name;
```

Question 9

Display all students ordered by marks and age.

Query

```
SELECT *  
FROM students  
ORDER BY Marks DESC, Age ASC;
```

Question 10

Display all students ordered by department, city, and name.

Query

```
SELECT *  
FROM students  
ORDER BY Department, City, Name;
```

WHERE + GROUP BY + HAVING + ORDER BY (Order of Execution)

SQL Execution Order

1. FROM
 2. WHERE
 3. GROUP BY
 4. HAVING
 5. SELECT
 6. ORDER BY
-

Question 1

Display the average marks of each department considering only students older than 20 years. Show departments whose average marks are greater than 70 and sort the results by average marks in descending order.

Query

```
SELECT Department,  
AVG(Marks) AS Average_Marks  
FROM students  
WHERE Age > 20  
GROUP BY Department  
HAVING AVG(Marks) > 70  
ORDER BY Average_Marks DESC;
```

Question 2

Display the total number of students in each city considering only students whose marks are greater than 70. Show cities having more than one student and sort by total students.

Query

```
SELECT City,  
COUNT() AS Total_Students
```

FROM students
WHERE Marks > 70
GROUP BY City
HAVING COUNT() > 1
ORDER BY Total_Students DESC;

Question 3

Display the total marks of each department considering only students from Hyderabad and Pune. Show departments whose total marks are greater than 100 and order by total marks.

Query

```
SELECT Department,  
SUM(Marks) AS Total_Marks  
FROM students  
WHERE City IN ('Hyderabad','Pune')  
GROUP BY Department  
HAVING SUM(Marks) > 100  
ORDER BY Total_Marks DESC;
```

Question 4

Display the highest marks obtained in each city considering only students from the ECE department. Show cities whose highest marks are greater than 70 and order by highest marks.

Query

```
SELECT City,  
MAX(Marks) AS Highest_Marks  
FROM students  
WHERE Department='ECE'  
GROUP BY City  
HAVING MAX(Marks) > 70  
ORDER BY Highest_Marks DESC;
```

Question 5

Display the average age of students in each department considering only students whose city is not NULL. Show departments whose average age is greater than 20 and sort alphabetically.

Query

```
SELECT Department,  
AVG(Age) AS Average_Age  
FROM students  
WHERE City IS NOT NULL  
GROUP BY Department  
HAVING AVG(Age) > 20  
ORDER BY Department;
```

Example

-- Create the Student table

```
CREATE TABLE Student (  
    student_id INT AUTO_INCREMENT PRIMARY KEY,  
    first_name VARCHAR(50),  
    last_name VARCHAR(50),  
    email VARCHAR(100),  
    grade DECIMAL(5, 2),  
    enrollment_date DATE,  
    phone_number VARCHAR(15)  
);
```


-- Insert sample data

```
INSERT INTO Student (first_name, last_name, email, grade, enrollment_date,  
phone_number) VALUES  
  
('John', 'Doe', 'john.doe@example.com', 88.50, '2024-01-10', '555-0101'),  
  
('Jane', 'Smith', 'jane.smith@example.com', 92.00, '2024-01-15', '555-0102'),  
  
('Emily', 'Johnson', 'emily.johnson@example.com', 78.00, '2024-02-01', NULL),  
  
('Michael', 'Williams', 'michael.williams@example.com', 85.00, '2024-01-20', '555-0103'),  
  
('Anna', 'Brown', 'anna.brown@example.com', 95.00, '2024-02-05', '555-0104');
```

-- Perform WHERE clause operations

-- 1. Basic Filtering by Grade

```
SELECT * FROM Student WHERE grade > 85;
```

-- 2. Filtering by Enrollment Date

```
SELECT * FROM Student WHERE enrollment_date >= '2024-01-15';
```

-- 3. Filtering by Email Domain

```
SELECT * FROM Student WHERE email LIKE '%example.com';
```

-- 4. Filtering by NULL Values in Phone Number

```
SELECT * FROM Student WHERE phone_number IS NULL;
```

-- 5. Filtering with Multiple Conditions

```
SELECT * FROM Student WHERE grade < 90 AND enrollment_date < '2024-02-01';
```

-- Perform GROUP BY operations

-- 1. Group by Enrollment Date

```
SELECT enrollment_date, COUNT(*) AS student_count FROM Student GROUP BY enrollment_date;
```

-- 3. Group by Enrollment Month

```
SELECT MONTH(enrollment_date) AS month, COUNT(*) AS student_count FROM Student GROUP BY MONTH(enrollment_date);
```

-- 4. Group by Grade and Enrollment Date

```
SELECT grade, enrollment_date, COUNT(*) AS student_count  
FROM Student  
GROUP BY grade, enrollment_date;
```

-- 5. Group by Email Domain

```
SELECT SUBSTRING_INDEX(email, '@', -1) AS email_domain, COUNT(*) AS student_count  
FROM Student  
GROUP BY email_domain;
```

-- Perform HAVING operations

-- 1. Filtering Groups with Student Count Greater Than 1

```
SELECT enrollment_date, COUNT(*) AS student_count
```

FROM Student

GROUP BY enrollment_date

HAVING COUNT(*) > 1;

-- 2. Filtering by Average Grade

SELECT enrollment_date, AVG(grade) AS average_grade

FROM Student

GROUP BY enrollment_date

HAVING AVG(grade) > 85;

-- 3. Filtering by Total Grade

SELECT enrollment_date, SUM(grade) AS total_grade

FROM Student

GROUP BY enrollment_date

HAVING SUM(grade) > 400;

-- Perform ORDER BY operations

-- 1. Basic Sorting by Grade

SELECT * FROM Student ORDER BY grade;

-- 2. Descending Order by Grade

SELECT * FROM Student ORDER BY grade DESC;

-- 3. Sorting by Enrollment Date and Grade

```
SELECT * FROM Student ORDER BY enrollment_date, grade DESC;
```

-- 4. Sorting by Multiple Columns

```
SELECT * FROM Student ORDER BY
```

Order of execution

- **Where-> group by-> having-> order by**

-- 4. Students with Grades Between 80 and 90, Grouped by Email Domain, Having More Than 1 Student per Domain, Ordered by Student Count

```
SELECT SUBSTRING_INDEX(email, '@', -1) AS email_domain, COUNT(*) AS  
student_count
```

```
FROM Student
```

```
WHERE grade BETWEEN 80 AND 90
```

```
GROUP BY email_domain
```

```
HAVING COUNT(*) > 1
```

```
ORDER BY student_count DESC;
```

-- 5. Students with Grades Above 85, Grouped by Grade, Having an Average Grade Above 85, Ordered by Grade in Ascending Order

```
SELECT grade, AVG(grade) AS average_grade, COUNT(*) AS student_count
```

```
FROM Student
```

```
WHERE grade > 85
```

```
GROUP BY grade
```

HAVING AVG(grade) > 85

ORDER BY grade ASC;

-- 7. Students with Enrollment Date in 2024, Grouped by Enrollment Date, Having an Average Grade Above 85, Ordered by Enrollment Date

SELECT enrollment_date, AVG(grade) AS average_grade

FROM Student

WHERE YEAR(enrollment_date) = 2024

GROUP BY enrollment_date

HAVING AVG(grade) > 85

ORDER BY enrollment_date;

Day-12

Clauses Exercises

CREATE TABLE: SALES

CREATE TABLE sales (

sale_id INT PRIMARY KEY,

product VARCHAR(50),

category VARCHAR(50),

units_sold INT,

unit_price INT,

region VARCHAR(50)

);

INSERT INTO sales (sale_id, product, category, units_sold, unit_price, region) VALUES

(1, 'Keyboard', 'Electronics', 10, 1200, 'North'),

(2, 'Monitor', 'Electronics', 5, 7000, 'South'),

(3, 'Chair', 'Furniture', 15, 2500, 'North'),

(4, 'Desk', 'Furniture', 7, 4500, 'West'),

(5, 'Mouse', 'Electronics', 20, 800, 'East'),

(6, 'Sofa', 'Furniture', 3, 15000, 'South'),

(7, 'Headphones', 'Electronics', 8, 1800, 'North'),
(8, 'Laptop', 'Electronics', 6, 55000, 'West'),
(9, 'Table', 'Furniture', 12, 6000, 'East'),
(10, 'Fan', 'HomeAppliance', 18, 3000, 'South'),
(11, 'AC', 'HomeAppliance', 4, 35000, 'North'),
(12, 'Cupboard', 'Furniture', 5, 12000, 'West'),
(13, 'Printer', 'Electronics', 9, 9000, 'East'),
(14, 'Bed', 'Furniture', 2, 25000, 'South'),
(15, 'Mobile', 'Electronics', 14, 20000, 'North');

WHERE Clause

Question 1

Display all products that belong to the Electronics category.

Query

```
SELECT *  
FROM sales  
WHERE category = 'Electronics';
```

Question 2

Display all products whose unit price is greater than 10000.

Query


```
SELECT *  
FROM sales  
WHERE unit_price > 10000;
```

Question 3

Display all products sold in the North region.

Query

```
SELECT *  
FROM sales  
WHERE region = 'North';
```

Question 4

Display all products where units sold are between 5 and 15.

Query

```
SELECT *  
FROM sales  
WHERE units_sold BETWEEN 5 AND 15;
```

Question 5

Display all Furniture products whose unit price is greater than 5000.

Query

```
SELECT *  
FROM sales  
WHERE category = 'Furniture'  
AND unit_price > 5000;
```

QUESTIONS ON GROUP BY + HAVING

Question 1

Total units sold per category where total units > 20

Query

```
SELECT category, SUM(units_sold) AS total_units
FROM sales
GROUP BY category
HAVING SUM(units_sold) > 20;
```

Answer

Category	Total Units
Electronics	72
Furniture	44
HomeAppliance	22

Question 2

Total revenue per region where revenue > 50000

Query

```
SELECT region,
SUM(units_sold * unit_price) AS revenue
FROM sales
GROUP BY region
HAVING SUM(units_sold * unit_price) > 50000;
```

Answer

Region	Revenue
North	363400
South	185000
West	435500
East	191000

Question 3

Count of products per category with more than 2 products

Query

```
SELECT category,
```

```
COUNT(product) AS product_count
FROM sales
GROUP BY category
HAVING COUNT(product) > 2;
```

Answer

Category	Product Count
Electronics	7
Furniture	5

Question 4

Average unit price per category < 5000

Query

```
SELECT category,
AVG(unit_price) AS avg_price
FROM sales
GROUP BY category
HAVING AVG(unit_price) < 5000;
```

Answer

No rows found.

Question 5

Regions with more than 1 Electronics product

Query

```
SELECT region,
COUNT(product) AS electronics_count
FROM sales
WHERE category='Electronics'
GROUP BY region
HAVING COUNT(product) > 1;
```

Answer

Region	Electronics Count
North	3
East	2

QUESTIONS ON GROUPBY + ORDER BY

Question 6

Categories and total units sold, ordered descending

Query

```
SELECT category,  
SUM(units_sold) AS total_units  
FROM sales  
GROUP BY category  
ORDER BY total_units DESC;
```

Answer

Category	Total Units
Electronics	72
Furniture	44
HomeAppliance	22

Question 7

Average unit price per region, ordered alphabetically

Query

```
SELECT region,  
AVG(unit_price) AS avg_price  
FROM sales  
GROUP BY region  
ORDER BY region ASC;
```

Answer

Region	Average Price
East	5266.67
North	11700
South	15000
West	23833.33

Question 8

Total revenue per category, highest first

Query

```
SELECT category,  
SUM(units_sold * unit_price) AS revenue  
FROM sales  
GROUP BY category  
ORDER BY revenue DESC;
```

Answer

Category	Revenue
Electronics	771400
Furniture	330000
HomeAppliance	194000

Question 9

Count of products in each region, descending

Query

```
SELECT region,  
COUNT(product) AS product_count  
FROM sales  
GROUP BY region  
ORDER BY product_count DESC;
```

Answer

Region	Product Count
North	5
South	4
East	3
West	3

Question 10

Regions and total units sold, sorted by region

Query

```
SELECT region,  
SUM(units_sold) AS total_units  
FROM sales  
GROUP BY region  
ORDER BY region;
```

Answer

Region	Total Units
East	41
North	50
South	30
West	18

GROUP BY + HAVING + ORDER BY

Question 11

Total revenue per region > 20000, order by revenue descending

Query

```
SELECT region,  
SUM(units_sold * unit_price) AS revenue  
FROM sales  
GROUP BY region  
HAVING SUM(units_sold * unit_price) > 20000  
ORDER BY revenue DESC;
```

Answer

Region	Revenue
West	435500
North	363400
East	191000
South	185000

Question 12

Total units sold per category > 10, order by category

Query

```
SELECT category,  
SUM(units_sold) AS total_units  
FROM sales  
GROUP BY category  
HAVING SUM(units_sold) > 10  
ORDER BY category;
```

Answer

Category	Total Units
Electronics	72
Furniture	44
HomeAppliance	22

Question 13

Count of products per region > 1, order alphabetically

Query

```
SELECT region,
COUNT(product) AS product_count
FROM sales
GROUP BY region
HAVING COUNT(product) > 1
ORDER BY region;
```

Answer

Region	Product Count
East	3
North	5
South	4
West	3

Question 14

Categories where avg price < 6000, order by avg price

Query

```
SELECT category,
AVG(unit_price) AS avg_price
FROM sales
GROUP BY category
HAVING AVG(unit_price) < 6000
ORDER BY avg_price;
```

Answer

No rows found.

Question 15

Avg units sold per category > 5, order by average descending

Query

```
SELECT category,  
AVG(units_sold) AS avg_units  
FROM sales  
GROUP BY category  
HAVING AVG(units_sold) > 5  
ORDER BY avg_units DESC;
```

Answer

Category	Average Units
HomeAppliance	11
Electronics	10.29
Furniture	8.8

Order of Execution

Execution Order:

WHERE → GROUP BY → HAVING → ORDER BY

Question 1

Display the total units sold for each region considering only **Electronics** products. Show only regions where total units sold are greater than **15** and display the results in descending order.

Query

```
SELECT region,  
SUM(units_sold) AS total_units  
FROM sales  
WHERE category = 'Electronics'  
GROUP BY region  
HAVING SUM(units_sold) > 15  
ORDER BY total_units DESC;
```

Question 2

Display the total revenue generated by each category considering only products whose **unit price is greater than 2000**. Show only categories with revenue greater than **100000** and order by revenue in descending order.

Query

```
SELECT category,  
SUM(units_sold * unit_price) AS revenue  
FROM sales  
WHERE unit_price > 2000  
GROUP BY category  
HAVING SUM(units_sold * unit_price) > 100000  
ORDER BY revenue DESC;
```

Question 3

Display the average unit price of each region considering only products where **units sold are greater than or equal to 5**. Show only regions where the average price is greater than **5000** and order the results alphabetically.

Query

```
SELECT region,  
AVG(unit_price) AS avg_price  
FROM sales  
WHERE units_sold >= 5  
GROUP BY region  
HAVING AVG(unit_price) > 5000  
ORDER BY region;
```

Question 4

Display the count of products in each category considering only products sold in the **North** and **South** regions. Show only categories having more than **2 products** and sort the count in descending order.

Query

```
SELECT category,  
COUNT() AS product_count  
FROM sales  
WHERE region IN ('North','South')  
GROUP BY category  
HAVING COUNT() > 2  
ORDER BY product_count DESC;
```

Question 5

Display the total revenue generated in each region considering only products whose **unit price is less than 10000**. Show only regions having revenue greater than **50000** and display the results in descending order of revenue.

Query

```
SELECT region,  
SUM(units_sold * unit_price) AS revenue  
FROM sales  
WHERE unit_price < 10000  
GROUP BY region  
HAVING SUM(units_sold * unit_price) > 50000  
ORDER BY revenue DESC;
```

These questions are ideal for teaching the SQL execution order:

WHERE → GROUP BY → HAVING → ORDER BY

They clearly demonstrate:

- **WHERE** filters rows before grouping.
- **GROUP BY** creates groups.
- **HAVING** filters the groups.
- **ORDER BY** sorts the final result.

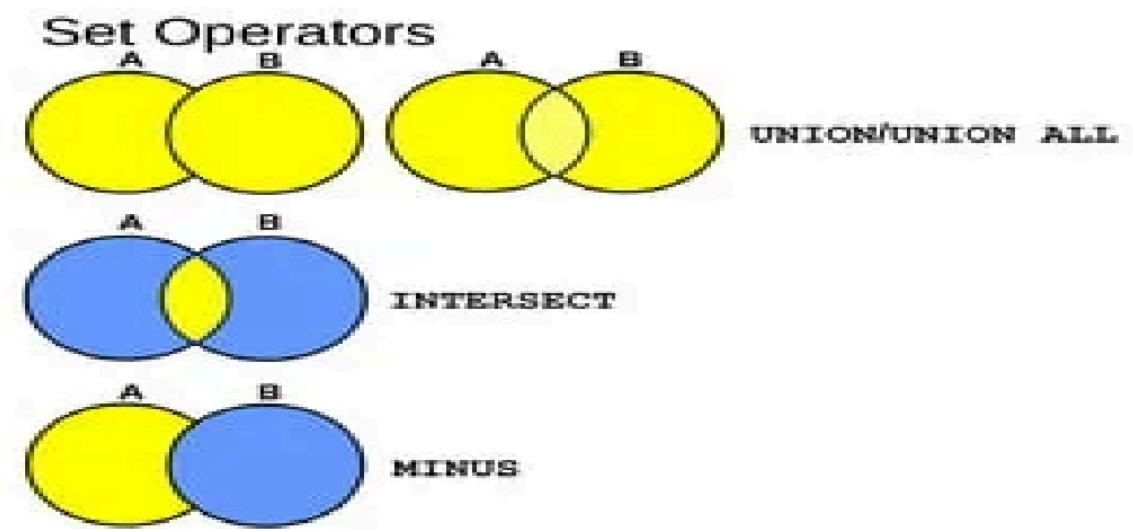
SET Operators

Union : Combines results from two queries and removes duplicate rows. Columns must have the same number and type.

Union All : Combines results from two queries and includes all rows, including duplicates. Columns must have the same number and type

Intersect : Returns rows that are common to both queries. Columns must have the same number and type. Note: MySQL does not support INTERSECT directly.

Minus : Returns rows from the first query that are not present in the second query. Columns must have the same number and type. Note: MySQL does not support MINUS directly.



-- Create the Course table

```
CREATE TABLE Course (  
    course_id INT,  
    course_name VARCHAR(100),  
    instructor VARCHAR(100)  
);
```

-- Insert sample data into Course table

```
INSERT INTO Course (course_id, course_name, instructor) VALUES  
(1, 'Mathematics', 'Dr. Smith'),  
(2, 'Physics', 'Dr. Johnson'),  
(3, 'Chemistry', 'Dr. Brown'),  
(4, 'Biology', 'Dr. Lee'),  
(5, 'History', 'Dr. Martin');
```

-- Create another table for SET operations

```
CREATE TABLE Course_Completed (  
    course_id INT,  
    course_name VARCHAR(100),  
    instructor VARCHAR(100)  
);
```

-- Insert sample data into Course_Completed table

```
INSERT INTO Course_Completed (course_id, course_name, instructor) VALUES
```

(2, 'Physics', 'Dr. Johnson'),
(3, 'Chemistry', 'Dr. Brown'),
(6, 'Computer Science', 'Dr. Walker'),
(7, 'Art', 'Dr. Scott');

-- UNION Examples

-- 1. Combine course names from both tables, removing duplicates

```
SELECT course_name FROM Course  
  
UNION  
  
SELECT course_name FROM Course_Completed;
```

-- 2. Combine instructor names from both tables, removing duplicates

```
SELECT instructor FROM Course  
  
UNION  
  
SELECT instructor FROM Course_Completed;
```

-- UNION ALL Examples

-- 1. Combine course names from both tables, including duplicates

```
SELECT course_name FROM Course  
  
UNION ALL  
  
SELECT course_name FROM Course_Completed;
```

-- 2. Combine all records from both tables, including duplicates

```
SELECT course_id, course_name, instructor FROM Course
```

```
UNION ALL
```

```
SELECT course_id, course_name, instructor FROM Course_Completed;
```

-- INTERSECT Examples (Note: MySQL does not support INTERSECT natively)

-- Using INNER JOIN to achieve similar results

-- 1. Common course names

```
SELECT DISTINCT c.course_name
```

```
FROM Course c
```

```
INNER JOIN Course_Completed cc ON c.course_name = cc.course_name;
```

-- 2. Common instructors

```
SELECT DISTINCT c.instructor
```

```
FROM Course c
```

```
INNER JOIN Course_Completed cc ON c.instructor = cc.instructor;
```

-- MINUS Examples (Note: MySQL does not support MINUS natively)

-- Using LEFT JOIN to achieve similar results

-- 1. Courses in Course table but not in Course_Completed

```
SELECT c.course_name
```

```
FROM Course c
```

```
LEFT JOIN Course_Completed cc ON c.course_id = cc.course_id
```

```
WHERE cc.course_id IS NULL;
```

-- 2. Courses in Course table but not in Course_Completed by course ID

SELECT c.course_id

FROM Course c

LEFT JOIN Course_Completed cc ON c.course_id = cc.course_id

WHERE cc.course_id IS NULL;

Day-13

--Summary of Numeric Functions:

- **ABS():** Absolute value of a number.
- **CEIL():** Round up.
- **FLOOR():** Round down.
- **ROUND():** Round to a specific number of digits.
- **MOD():** Modulo operation.
- **POW()/POWER():** Exponentiation.
- **SQRT():** Square root.
- **LN():** Natural logarithm.
- **LOG():** Logarithm to a specific base.
- **EXP():** Exponential function.
- **RADIANS():** Convert degrees to radians.
- **DEGREES():** Convert radians to degrees.
- **SIGN():** Returns the sign of a number.
- **RAND():** Generate a random number.
- **GREATEST():** Find the maximum value.
- **LEAST():** Find the minimum value.
- **PI():** Return the value of Pi.
- **TRUNCATE()** Truncate decimal places (no rounding)
- **ROUND(x, d, mode)** Round with control over mode
- **LOG2()** Log base 2
- **LOG10()** Log base 10
- **BIT_COUNT()** Count set bits in binary
- **CONV()** Convert between number bases
- **HEX()** Convert to hexadecimal
- **OCT()** Convert to octal
- **BIN()** Convert to binary

1. ABS() - Absolute Value

Description: Returns the absolute value of a number (removes the sign).

```
SELECT ABS(-100) AS absolute_value;
```

Output:

```
+-----+
| absolute_value |
+-----+
| 100           |
+-----+
```

2. CEIL() or CEILING() - Round Up

Description: Rounds a number up to the nearest integer.

SELECT CEIL(12.34) AS rounded_up;

Output:

```
+-----+
| rounded_up |
+-----+
| 13         |
+-----+
```

3. FLOOR() - Round Down

Description: Rounds a number down to the nearest integer.

SELECT FLOOR(12.34) AS rounded_down;

Output:

```
+-----+
| rounded_down |
+-----+
| 12           |
+-----+
```

4. ROUND() - Round to a Specific Number of Digits

Description: Rounds a number to a specified number of decimal places.

SELECT ROUND(12.3456, 2) AS rounded_value;

Output:

```
+-----+
| rounded_value |
+-----+
| 12.35        |
+-----+
```

5. MOD() - Modulo (Remainder of Division)

Description: Returns the remainder of the division of two numbers.

SELECT MOD(17, 5) AS remainder;

Output:

```
+-----+
| remainder |
+-----+
| 2         |
+-----+
```

6. POW() or POWER() - Exponentiation

Description: Returns the value of a number raised to the power of another number.

SELECT POW(2, 3) AS power_value;

Output:

```
+-----+
| power_value |
+-----+
| 8          |
+-----+
```

7. SQRT() - Square Root

Description: Returns the square root of a number.

SELECT SQRT(16) AS square_root;

Output:

```
+-----+
| square_root |
+-----+
| 4           |
+-----+
```

8. LN() - Natural Logarithm

Description: Returns the natural logarithm (base e) of a number.

SELECT LN(10) AS natural_log;

Output:

```
+-----+
| natural_log |
+-----+
| 2.302585    |
+-----+
```

9. LOG() - Logarithm

Description: Returns the logarithm of a number to a specified base.

SELECT LOG(10, 100) AS log_value;

Output:

```
+-----+
| log_value |
+-----+
| 2        |
+-----+
```

10. EXP() - Exponential

Description: Returns the value of e (Euler's number) raised to the power of the specified number.

SELECT EXP(1) AS exponential_value;

Output:

```
+-----+
| exponential_value |
+-----+
| 2.718282         |
+-----+
```

11. RADIANS() - Convert Degrees to Radians

Description: Converts degrees to radians.

```
SELECT RADIANS(180) AS radians_value;
```

Output:

```
+-----+
| radians_value |
+-----+
| 3.141593      |
+-----+
```

12. DEGREES() - Convert Radians to Degrees

Description: Converts radians to degrees.

```
SELECT DEGREES(PI()) AS degrees_value;
```

Output:

```
+-----+
| degrees_value |
+-----+
| 180           |
+-----+
```

13. SIGN() - Sign of a Number

Description: Returns the sign of a number:

- 1 for positive numbers
- 0 for zero
- -1 for negative numbers

```
SELECT SIGN(-25) AS sign_value;
```

Output:

```
+-----+
| sign_value |
+-----+
| -1        |
+-----+
```

14. RAND() - Generate a Random Number

Description: Returns a random floating-point number between 0 and 1.

SELECT RAND() AS random_value;

Output:

```
+-----+
| random_value |
+-----+
| 0.538953    |
+-----+
```

To generate a random number within a specific range:

SELECT FLOOR(RAND() * (100 - 1 + 1) + 1) AS random_in_range;

Output:

```
+-----+
| random_in_range |
+-----+
| 47              |
+-----+
```

15. GREATEST() - Find the Maximum Value

Description: Returns the largest value from a list of expressions.

```
SELECT GREATEST(5, 10, 15, 20) AS GREATEST_value;
```

Output:

```
+-----+
| greatest_value |
+-----+
| 20            |
+-----+
```

16. LEAST() - Find the Minimum Value

Description: Returns the smallest value from a list of expressions.

```
SELECT LEAST(5, 10, 15, 20) AS least_value;
```

Output:

```
+-----+
| least_value |
+-----+
| 5           |
+-----+
```

17. PI() - Returns the Value of Pi

Description: Returns the constant value of π (pi).

```
SELECT PI() AS pi_value;
```

Output:


```

+-----+
| pi_value |
+-----+
| 3.141593 |
+-----+

```

18. TRUNCATE() – Truncate Decimal Places

Description: Truncates a number to a specified number of decimal places without rounding.

```
SELECT TRUNCATE(12.3456, 2) AS truncated_value;
```

Output:

```

+-----+
| truncated_value|
+-----+
| 12.34         |
+-----+

```

19. ROUND(X, D [, mode]) – Round with Optional Mode

Description: Rounds a number X to D decimal places. If mode is set to 1, it truncates away from zero instead of rounding.

Output:

```

+-----+
| rounded_custom |
+-----+
| -1.5          |
+-----+

```

20. LOG2() – Logarithm Base 2

Description: Returns the base-2 logarithm of a number.

```
SELECT LOG2(8) AS log2_value;
```

Output:

```
+-----+
| log2_value |
+-----+
| 3         |
+-----+
```

21. LOG10() – Logarithm Base 10

Description: Returns the base-10 logarithm of a number.

```
SELECT LOG10(1000) AS log10_value;
```

Output:

```
+-----+
| log10_value |
+-----+
| 3           |
+-----+
```

22. BIT_COUNT() – Count of Set Bits

Description: Returns the number of bits that are set (i.e., 1s) in the binary representation of a number.

```
SELECT BIT_COUNT(7) AS set_bits_count;
```

Output:

set_bits_count
3

Because 7 in binary is 111 → 3 bits are set.

23. CONV(N, from_base, to_base) – Convert Number Between Bases

Description: Converts a number N from one base to another (between 2 and 36).

```
SELECT CONV('A', 16, 10) AS base10_value;
```

Output:

base10_value
10

24. HEX() – Convert to Hexadecimal

Description: Returns the hexadecimal value of a decimal number.

```
SELECT HEX(255) AS hex_value;
```

Output:

hex_value
FF

25. OCT() – Convert to Octal

Description: Returns the octal (base 8) representation of a number.

```
SELECT OCT(64) AS octal_value;
```

Output:

```
+-----+
| octal_value |
+-----+
| 100        |
+-----+
```

26. BIN() – Convert to Binary

Description: Converts a decimal number to its binary representation.

```
SELECT BIN(10) AS binary_value;
```

Output:

```
+-----+
| binary_value |
+-----+
| 1010        |
+-----+
```

```
CREATE TABLE employee (
    emp_id INT PRIMARY KEY,
    emp_name VARCHAR(50),
    age INT,
    department VARCHAR(30),
    salary DECIMAL(10,2),
    bonus DECIMAL(10,2),
    experience DECIMAL(5,2),
    performance_score DECIMAL(6,2)
```

```
);

INSERT INTO employee VALUES
(101, 'Rahul', 25, 'IT', 35000.75, 2500.50, 2.5, 85.45),
(102, 'Anjali', 30, 'HR', 42000.80, 3200.75, 5.8, 91.65),
(103, 'Ramesh', 27, 'Sales', 38000.60, 2100.90, 3.2, 74.55),
(104, 'Priya', 35, 'Finance', 55000.90, 5000.25, 8.7, 95.80),
(105, 'Arjun', 29, 'IT', 46000.50, 3500.60, 4.6, 82.45),
(106, 'Sneha', 24, 'HR', 32000.20, 1800.40, 1.8, 69.85),
(107, 'Kiran', 32, 'Sales', 50000.90, 4200.30, 6.5, 88.90),
(108, 'Deepika', 28, 'Finance', 41000.75, 3000.50, 4.2, 79.65),
(109, 'Mahesh', 40, 'IT', 68000.40, 6500.75, 12.5, 97.20),
(110, 'Divya', 26, 'HR', 36000.85, 2400.25, 2.9, 76.40);
```

SQL STRING FUNCTIONS

MySQL String Functions with Description

Function	Description
CONCAT()	Concatenates (joins) two or more strings into a single string without any separator unless specified.
CONCAT_WS()	Concatenates multiple strings using a specified separator (WS = With Separator).

UPPER()	Converts all characters in a string to uppercase. (UCASE() is an alias.)
LOWER()	Converts all characters in a string to lowercase. (LCASE() is an alias.)
SUBSTRING()	Extracts a specified number of characters from a string starting at a given position. (SUBSTR() is an alias.)
LEFT()	Returns the specified number of characters from the beginning (left side) of a string.
RIGHT()	Returns the specified number of characters from the end (right side) of a string.
LENGTH()	Returns the length of a string in bytes. For multi-byte characters, the byte count may be greater than the character count.
CHAR_LENGTH()	Returns the number of characters in a string, regardless of how many bytes each character occupies.
REPLACE()	Replaces all occurrences of a specified substring within a string with another substring.
TRIM()	Removes leading and trailing spaces (or specified characters) from a string.
POSITION()	Returns the position of the first occurrence of a substring within a string. Returns 0 if the substring is not found.

INSTR()	Returns the position of the first occurrence of a substring within a string. Similar to POSITION() , but uses a different syntax.
REVERSE()	Reverses the order of characters in a string.
LPAD()	Pads the left side of a string with a specified character until the string reaches the desired length.
RPAD()	Pads the right side of a string with a specified character until the string reaches the desired length.
ASCII()	Returns the ASCII value of the first character in a string.

Table Setup

A single employees table is reused for every question in this workbook — from basic single-function calls through advanced multi-function reports.

```
CREATE TABLE employees (  
  emp_id INT PRIMARY KEY,  
  first_name VARCHAR(30),  
  last_name VARCHAR(30),  
  email VARCHAR(60),  
  phone_number VARCHAR(20),  
  department VARCHAR(30),  
  city VARCHAR(30),  
  salary DECIMAL(10,2)  
);
```

Sample Data

```
INSERT INTO employees VALUES
(101, 'Rahul', 'Kumar', 'rahul.kumar@gmail.com', '987-654-3210', 'IT', 'Hyderabad', 45000),
(102, 'Anita', 'Sharma', 'anita.sharma@yahoo.com', '912-345-6789', 'HR', 'Bangalore', 38000),
(103, 'Suresh', 'Verma', 'suresh.verma@gmail.com', '998-123-4567', 'Sales', 'Chennai', 52000),
(104, 'Priya', 'Singh', 'priya.singh@outlook.com', '900-456-7890', 'Finance', 'Pune', 47000),
(105, 'Rahul', 'Joshi', 'rahul.joshi@gmail.com', '955-888-1111', 'Testing', 'Delhi', 41000),
(106, 'Kiran', 'Reddy', 'kiran.reddy@gmail.com', '901-222-3333', 'IT', 'Hyderabad', 56000),
(107, 'Deepika', 'Naidu', 'deepika.naidu@yahoo.com', '944-111-2222', 'HR', 'Vizag', 39000),
(108, 'Mahesh', 'Patel', 'mahesh.patel@gmail.com', '933-444-5555', 'Sales', 'Mumbai', 62000),
(109, 'Divya', 'Gupta', 'divya.gupta@gmail.com', '955-777-8888', 'Finance', 'Delhi', 49000),
(110, 'Arjun', 'Rao', 'arjun.rao@gmail.com', '988-999-7777', 'Developer', 'Hyderabad', 68000);
```

Basic Level

20 Questions — one string function at a time

Q1. Display full name by concatenating first_name and last_name.

```
SELECT CONCAT(first_name, ' ', last_name) AS full_name
FROM employees;
```

Q2. Convert first_name to uppercase.

```
SELECT first_name, UPPER(first_name) AS upper_name
FROM employees;
```

Q3. Convert last_name to lowercase.

```
SELECT last_name, LOWER(last_name) AS lower_name
FROM employees;
```

Q4. Extract the first 10 characters of email.

```
SELECT email, SUBSTRING(email, 1, 10) AS email_prefix
FROM employees;
```

Q5. Find the length of first_name.

```
SELECT first_name, LENGTH(first_name) AS name_length
FROM employees;
```

Q6. Remove dashes from phone_number.

```
SELECT phone_number, REPLACE(phone_number, '-', '') AS clean_phone
FROM employees;
```


Q7. Remove leading and trailing spaces from first_name.

```
SELECT TRIM(first_name) AS trimmed_name
FROM employees;
```

Q8. Concatenate first_name, last_name and email separated by a comma.

```
SELECT CONCAT_WS(', ', first_name, last_name, email) AS employee_info
FROM employees;
```

Q9. Find the position of the letter 'a' in first_name using POSITION().

```
SELECT first_name, POSITION('a' IN first_name) AS position_of_a
FROM employees;
```

Q10. Find the position of the letter 'a' in first_name using INSTR().

```
SELECT first_name, INSTR(first_name, 'a') AS position_of_a
FROM employees;
```

Q11. Display the first 3 characters of first_name.

```
SELECT first_name, LEFT(first_name, 3) AS short_name
FROM employees;
```

Q12. Display the last 3 characters of last_name.

```
SELECT last_name, RIGHT(last_name, 3) AS end_letters
FROM employees;
```

Q13. Reverse the first_name.

```
SELECT first_name, REVERSE(first_name) AS reversed_name
FROM employees;
```

Q14. Pad first_name on the left with '*' to make it 10 characters.

```
SELECT first_name, LPAD(first_name, 10, '*') AS padded_name
FROM employees;
```

Q15. Pad first_name on the right with '-' to make it 10 characters.

```
SELECT first_name, RPAD(first_name, 10, '-') AS padded_name
FROM employees;
```

Q16. Find the ASCII value of the first character of first_name.

```
SELECT first_name, ASCII(first_name) AS ascii_value
FROM employees;
```

Q17. Find the character length of first_name.

```
SELECT first_name, CHAR_LENGTH(first_name) AS char_len
FROM employees;
```

Q18. Convert email addresses to lowercase.

```
SELECT email, LOWER(email) AS lower_email
FROM employees;
```

Q19. Convert department names to uppercase.

```
SELECT department, UPPER(department) AS upper_department
FROM employees;
```

Q20. Remove leading spaces and trailing spaces from city using LTRIM() and RTRIM().

```
SELECT city, LTRIM(city) AS left_trimmed, RTRIM(city) AS right_trimmed
FROM employees;
```

Intermediate Level

20 Questions — filtering, sorting and combining string functions

Q1. Find employees whose first_name starts with 'R'.

```
SELECT first_name, last_name
FROM employees
WHERE first_name LIKE 'R%';
```

Q2. Find employees whose email contains 'yahoo'.

```
SELECT first_name, email
FROM employees
WHERE email LIKE '%yahoo%';
```

Q3. Display full name along with department as a single string.

```
SELECT CONCAT(first_name, ' ', last_name, ' - ', department) AS
name_department
FROM employees;
```

Q4. Find the length of every email address.

```
SELECT email, LENGTH(email) AS email_length
FROM employees;
```

Q5. Find employees whose first_name has more than 5 characters.

```
SELECT first_name
FROM employees
WHERE LENGTH(first_name) > 5;
```

Q6. Replace 'gmail.com' with 'company.com' in email.

```
SELECT email, REPLACE(email, 'gmail.com', 'company.com') AS updated_email
FROM employees;
```

Q7. Find the position of the '@' symbol in email.

```
SELECT email, INSTR(email, '@') AS at_position
FROM employees;
```

Q8. Extract the username part of email (before '@') using SUBSTRING_INDEX().

```
SELECT email, SUBSTRING_INDEX(email, '@', 1) AS username
FROM employees;
```

Q9. Extract the domain part of email (after '@') using SUBSTRING_INDEX().

```
SELECT email, SUBSTRING_INDEX(email, '@', -1) AS domain
FROM employees;
```

Q10. Find employees whose last_name ends with 'a'.

```
SELECT last_name
FROM employees
WHERE last_name LIKE '%a';
```

Q11. Display full name in uppercase for employees in the IT department.

```
SELECT UPPER(CONCAT(first_name, ' ', last_name)) AS full_name_upper
FROM employees
WHERE department = 'IT';
```

Q12. Find first names that are duplicated among employees.

```
SELECT first_name, COUNT(*) AS occurrences
FROM employees
GROUP BY first_name
```

```
HAVING COUNT(*) > 1;
```

Q13. Format phone numbers by removing all dashes.

```
SELECT phone_number, REPLACE(phone_number, '-', '') AS formatted_phone  
FROM employees;
```

Q14. Find employees whose first_name contains the letter 'i'.

```
SELECT first_name  
FROM employees  
WHERE first_name LIKE '%i%';
```

Q15. Sort employees by the length of their first_name.

```
SELECT first_name, LENGTH(first_name) AS name_length  
FROM employees  
ORDER BY name_length;
```

Q16. Find employees whose city name has more than 6 characters.

```
SELECT city  
FROM employees  
WHERE LENGTH(city) > 6;
```

Q17. Build a formatted string "Name - Department" for every employee.

```
SELECT CONCAT(first_name, ' ', last_name, ' - ', department) AS name_dept  
FROM employees;
```

Q18. Display the initials of each employee (first letter of first and last name).

```
SELECT CONCAT(LEFT(first_name, 1), LEFT(last_name, 1)) AS initials  
FROM employees;
```

Q19. Use REPEAT() to build a star-rating string based on salary (1 star per ₹10,000, capped at 5).

```
SELECT first_name, salary,  
       REPEAT('*', LEAST(FLOOR(salary / 10000), 5)) AS salary_rating  
FROM employees;
```

Q20. Check whether each employee's department is one of 'IT', 'HR', 'Sales' using FIND_IN_SET().

```
SELECT first_name, department,  
       FIND_IN_SET(department, 'IT,HR,Sales') AS is_core_dept
```

```
FROM employees;
```

Advanced Level

20 Questions — masking, reports, and nested string functions

Q1. Mask email addresses, showing only the first 3 characters before '@'.

```
SELECT email,  
       CONCAT(LEFT(email, 3), '***@', SUBSTRING_INDEX(email, '@', -1))  
AS masked_email  
FROM employees;
```

Q2. Check if any employee's first_name is a palindrome.

```
SELECT first_name  
FROM employees  
WHERE LOWER(first_name) = LOWER(REVERSE(first_name));
```

Q3. Generate a username using the first letter of first_name plus the full last_name, in lowercase.

```
SELECT CONCAT(LOWER(LEFT(first_name, 1)), LOWER(last_name)) AS username  
FROM employees;
```

Q4. Compare first_name and last_name alphabetically using STRCMP().

```
SELECT first_name, last_name, STRCMP(first_name, last_name) AS comparison  
FROM employees;
```

Q5. Extract the top-level domain (e.g. com) from each email using nested SUBSTRING_INDEX().

```
SELECT email,  
       SUBSTRING_INDEX(SUBSTRING_INDEX(email, '@', -1), '.', -1) AS tld  
FROM employees;
```

Q6. List all employee names grouped by department using GROUP_CONCAT().

```
SELECT department,  
       GROUP_CONCAT(CONCAT(first_name, ' ', last_name) SEPARATOR ', ') AS  
employees_list  
FROM employees  
GROUP BY department;
```

Q7. Categorize employees as 'Short Name' or 'Long Name' based on the length of first_name.

```
SELECT first_name,  
       CASE WHEN LENGTH(first_name) <= 5 THEN 'Short Name'  
            ELSE 'Long Name' END AS name_category  
FROM employees;
```

Q8. Generate a formatted employee code like 'EMP0101' by padding emp_id with zeros.

```
SELECT emp_id, CONCAT('EMP', LPAD(emp_id, 4, '0')) AS employee_code  
FROM employees;
```

Q9. Extract the area code (first 3 digits) from phone_number.

```
SELECT phone_number, SUBSTRING(phone_number, 1, 3) AS area_code  
FROM employees;
```

Q10. Find the position of the first '.' (dot) in each email using LOCATE().

```
SELECT email, LOCATE('.', email) AS dot_position  
FROM employees;
```

Q11. Mask phone numbers, showing only the last 4 digits.

```
SELECT phone_number,  
       CONCAT('XXX-XXX-', RIGHT(phone_number, 4)) AS masked_phone  
FROM employees;
```

Q12. Count the number of occurrences of the letter 'a' in each first_name using LENGTH() and REPLACE().

```
SELECT first_name,  
       (LENGTH(first_name) - LENGTH(REPLACE(LOWER(first_name), 'a', '')))  
AS a_count  
FROM employees;
```

Q13. Find employees whose full name (concatenated) exceeds 15 characters.

```
SELECT first_name, last_name,  
       LENGTH(CONCAT(first_name, ' ', last_name)) AS full_name_length  
FROM employees  
HAVING full_name_length > 15;
```

Q14. Generate an employee short code combining initials and emp_id (e.g. RK101).

```
SELECT CONCAT(LEFT(first_name, 1), LEFT(last_name, 1), emp_id) AS  
short_code
```

```
FROM employees;
```

Q15. Sort employees by last_name in descending order.

```
SELECT first_name, last_name
FROM employees
ORDER BY last_name DESC;
```

Q16. Create a 3-letter short code for each department using LEFT() and UPPER().

```
SELECT department, UPPER(LEFT(department, 3)) AS dept_code
FROM employees;
```

Q17. Display employee salary formatted with a thousands separator using FORMAT().

```
SELECT first_name, salary, FORMAT(salary, 0) AS formatted_salary
FROM employees;
```

Q18. Build a complete employee directory report combining name, masked email, and department code.

```
SELECT CONCAT(first_name, ' ', last_name) AS full_name,
       CONCAT(LEFT(email, 3), '****@', SUBSTRING_INDEX(email, '@', -1))
AS masked_email,
       UPPER(LEFT(department, 3)) AS dept_code
FROM employees;
```

Q19. Find employees whose email domain is not 'gmail.com'.

```
SELECT first_name, email
FROM employees
WHERE SUBSTRING_INDEX(email, '@', -1) <> 'gmail.com';
```

Q20. Generate a full formatted employee record: "EMP<emp_id> | <FULL NAME> | <Department> | Salary: <formatted salary>".

```
SELECT CONCAT(
    'EMP', emp_id, ' | ',
    UPPER(CONCAT(first_name, ' ', last_name)), ' | ',
    department, ' | Salary: ', FORMAT(salary, 0)
) AS employee_record
FROM employees;
```

Day-14

SQL DATE & TIME FUNCTIONS

-CURDATE(): Current date.

CURTIME(): Current time.

NOW(): Current date and time.

DATE(): Extract date part.

TIME(): Extract time part.

YEAR(): Extract year.

MONTH(): Extract month.

DAY()/DAYOFMONTH(): Extract day of the month.

WEEK(): Extract week number.

DAYOFWEEK(): Extract day of the week.

DATE_ADD(): Add time to a date.

DATE_SUB(): Subtract time from a date.

DATEDIFF(): Difference between two dates.

TIMESTAMPDIFF(): Difference between two DATETIME values.

STR_TO_DATE(): Convert string to date.

DATE_FORMAT(): Format a date.

FROM_UNIXTIME(): Convert Unix timestamp to date.

UNIX_TIMESTAMP(): Convert date to Unix timestamp.

1. CURDATE() - Current Date

Description: Returns the current date in YYYY-MM-DD format (no time).

```
SELECT CURDATE() AS current__date;
```

Output:

```
+-----+  
| current_date|
```

```
+-----+
| 2024-11-28 |
+-----+
```

2. CURTIME() - Current Time

Description: Returns the current time in HH:MM:SS format (no date).

```
SELECT CURTIME() AS current_time;
```

Output:

```
+-----+
| current_time |
+-----+
| 14:32:45    |
+-----+
```

3. NOW() - Current Date and Time

Description: Returns the current date and time in YYYY-MM-DD HH:MM:SS format.

```
SELECT NOW() AS current_datetime;
```

Output:

```
+-----+
| current_datetime |
+-----+
| 2024-11-28 14:32:45 |
+-----+
```

4. DATE() - Extract Date Part

Description: Extracts the date part from a DATETIME or TIMESTAMP value.

```
SELECT DATE('2024-11-28 14:32:45') AS extracted_date;
```

Output:

```
+-----+
| extracted_date |
+-----+
| 2024-11-28    |
+-----+
```

5. TIME() - Extract Time Part

Description: Extracts the time part from a DATETIME or TIMESTAMP value.

```
SELECT TIME('2024-11-28 14:32:45') AS extracted_time;
```

Output:

```
+-----+
| extracted_time |
+-----+
| 14:32:45      |
+-----+
```

6. YEAR() - Extract Year

Description: Returns the year part from a DATE, DATETIME, or TIMESTAMP value.

```
SELECT YEAR('2024-11-28') AS year_part;
```

Output:

+-----+
year_part
+-----+
2024
+-----+

7. MONTH() - Extract Month

Description: Returns the month part (1–12) from a DATE, DATETIME, or TIMESTAMP value.

```
SELECT MONTH('2024-11-28') AS month_part;
```

Output:

+-----+
month_part
+-----+
11
+-----+

8. DAY() or DAYOFMONTH() - Extract Day of the Month

Description: Returns the day of the month (1–31) from a DATE, DATETIME, or TIMESTAMP value.

```
SELECT DAY('2024-11-28') AS day_part;
```

Output:

+-----+
day_part
+-----+
28
+-----+

9. WEEK() - Extract Week Number

Description: Returns the week number of the year (1–53) from a DATE, DATETIME, or TIMESTAMP value.

```
SELECT WEEK('2024-11-28') AS week_number;
```

Output:

```
+-----+
| week_number |
+-----+
| 48          |
+-----+
```

10. DAYOFWEEK() - Day of the Week

Description: Returns the day of the week as an integer (1 for Sunday, 2 for Monday, ..., 7 for Saturday).

```
SELECT DAYOFWEEK('2024-11-28') AS day_of_week;
```

Output:

```
+-----+
| day_of_week |
+-----+
| 5           |
+-----+
```

11. DATE_ADD() - Add Time to a Date

Description: Adds a specified time interval to a date or time.

```
SELECT DATE_ADD('2024-11-28', INTERVAL 10 DAY) AS new_date;
```

Output:

diff

Copy code

```
+-----+
| new_date |
+-----+
| 2024-12-08 |
+-----+
```

12. DATE_SUB() - Subtract Time from a Date

Description: Subtracts a specified time interval from a date or time.

SELECT DATE_SUB('2024-11-28', INTERVAL 10 DAY) AS new_date;

Output:

```
+-----+
| new_date |
+-----+
| 2024-11-18 |
+-----+
```

13. DATEDIFF() - Difference Between Two Dates

Description: Returns the number of days between two dates.

SELECT DATEDIFF('2024-11-28', '2024-11-18') AS days_diff;

Output:

```
+-----+
| days_diff |
+-----+
| 10      |
+-----+
```

14. TIMESTAMPDIFF() - Difference Between Two Date/Time Values

Description: Returns the difference between two date or time values in the specified unit (e.g., seconds, minutes, hours, etc.).

```
SELECT TIMESTAMPDIFF(DAY, '2024-11-18', '2024-11-28') AS days_difference;
```

Output:

```
+-----+
| days_difference |
+-----+
| 10             |
+-----+
```

15. STR_TO_DATE() - Convert String to Date

Description: Converts a string to a DATE, DATETIME, or TIMESTAMP using a specified format.

```
SELECT STR_TO_DATE('2024-11-28', '%Y-%m-%d') AS formatted_date;
```

Output:

```
+-----+
| formatted_date |
+-----+
| 2024-11-28    |
+-----+
```

16. DATE_FORMAT() - Format a Date

Description: Formats a DATE, DATETIME, or TIMESTAMP value into a string according to a specified format.

```
SELECT DATE_FORMAT('2024-11-28', '%W, %M %d, %Y') AS formatted_date;
```

Output:

```
+-----+
| formatted_date |
+-----+
| Thursday, November 28, 2024 |
+-----+
```

17. FROM_UNIXTIME() - Convert Unix Timestamp to Date

Description: Converts a Unix timestamp to a DATETIME value.

```
SELECT FROM_UNIXTIME(1609459200) AS datetime_value;
```

Output:

```
+-----+
| datetime_value |
+-----+
| 2021-01-01 00:00:00 |
+-----+
```

18. UNIX_TIMESTAMP() - Convert Date to Unix Timestamp

Description: Converts a DATETIME or DATE value to a Unix timestamp.

```
SELECT UNIX_TIMESTAMP('2024-11-28 14:32:45') AS unix_timestamp;
```

Output:

+-----+
unix_timestamp
+-----+
1638111165
+-----+

19. LAST_DAY() – Last Day of the Month

Description: Returns the last day of the month for a given date.

```
SELECT LAST_DAY('2024-11-28') AS last_day_of_month;
```

Output:

+-----+
last_day_of_month
+-----+
2024-11-30
+-----+

20. MAKEDATE() – Create a Date from Year and Day Number

Description: Returns a date from a given year and day number (1–366).

```
SELECT MAKEDATE(2024, 60) AS result_date;
```

Output:

+-----+
result_date
+-----+
2024-02-29
+-----+

21. MAKETIME() – Create Time from Hour, Minute, Second

Description: Returns a time value from hour, minute, and second.

```
SELECT MAKETIME(14, 30, 15) AS result_time;
```

Output:

result_time
14:30:15

22. MONTHNAME() – Full Name of the Month

Description: Returns the name of the month from a date.

```
SELECT MONTHNAME('2024-11-28') AS month_name;
```

Output:

month_name
November

23. DAYNAME() – Name of the Day

Description: Returns the name of the weekday.

```
SELECT DAYNAME('2024-11-28') AS day_name;
```

Output:

day_name

```
+-----+
| Thursday |
+-----+
```

24. QUARTER() – Quarter of the Year

Description: Returns the quarter (1 to 4) of the year for a date.

```
SELECT QUARTER('2024-11-28') AS quarter_value;
```

Output:

```
+-----+
| quarter_value |
+-----+
| 4             |
+-----+
```

25. EXTRACT() – Extract Specific Part of Date

Description: Returns part of a date (e.g., YEAR, MONTH, DAY, etc.).

```
SELECT EXTRACT(WEEK FROM '2024-11-28') AS week_value;
```

Output:

```
+-----+
| week_value |
+-----+
| 48         |
+-----+
```

You can extract:

- YEAR

- MONTH
- DAY
- WEEK
- HOUR
- MINUTE
- SECOND
- QUARTER
- etc.

26. ADDDATE() – Same as DATE_ADD()

Description: Synonym for DATE_ADD(). Adds days or intervals.

```
SELECT ADDDATE('2024-11-28', INTERVAL 5 DAY) AS new_date;
```

Output:

```
+-----+
| new_date |
+-----+
| 2024-12-03 |
+-----+
```

27. SUBDATE() – Same as DATE_SUB()

Description: Synonym for DATE_SUB(). Subtracts days or intervals.

```
SELECT SUBDATE('2024-11-28', INTERVAL 5 DAY) AS new_date;
```

Output:

+-----+
new_date
+-----+
2024-11-23
+-----+

28. SEC_TO_TIME() – Convert Seconds to Time

Description: Converts an integer number of seconds to HH:MM:SS format.

```
SELECT SEC_TO_TIME(3661) AS time_value;
```

Output:

+-----+
time_value
+-----+
01:01:01
+-----+

29. TIME_TO_SEC() – Convert Time to Seconds

Description: Converts a time value into seconds.

```
SELECT TIME_TO_SEC('01:01:01') AS total_seconds;
```

Output:

+-----+
total_seconds
+-----+
3661
+-----+

30. SYSDATE() – Current Date and Time (Execution Time)

Description: Returns the exact time at the moment the query executes (unlike NOW() which may be cached).

```
SELECT SYSDATE() AS system_time;
```

Output:

```
+-----+
| system_time |
+-----+
| 2024-11-28 14:35:50 |
+-----+
```

Table Setup

A single employee_attendance table is reused for every question in this workbook — from basic single-function calls through advanced multi-function reports.

```
CREATE TABLE employee_attendance
(
    emp_id INT PRIMARY KEY AUTO_INCREMENT,
    emp_name VARCHAR(100),
    department VARCHAR(50),
    joining_date DATE,
    birth_date DATE,
    login_time DATETIME,
    logout_time DATETIME,
    salary DECIMAL(10,2)
);
```

Sample Data

```
INSERT INTO employee_attendance
(emp_name, department, joining_date, birth_date, login_time, logout_time, salary)
VALUES
('Rahul','IT','2021-06-15','1998-05-10','2024-11-20 09:05:10','2024-11-20 18:15:30',55000),
('Sneha','HR','2020-03-21','1996-11-18','2024-11-21 08:55:20','2024-11-21 17:45:50',48000),
('Arjun','Finance','2019-08-10','1994-07-30','2024-11-22 09:15:45','2024-11-22 18:35:20',62000),
('Priya','IT','2022-01-05','2000-01-25','2024-11-23 09:00:00','2024-11-23 17:55:40',51000),
('Kiran','Admin','2018-12-11','1992-09-15','2024-11-24 08:40:30','2024-11-24 17:30:00',45000),
('Anjali','Testing','2023-04-17','1999-12-20','2024-11-25 09:12:10','2024-11-25 18:05:00',47000),
('Vijay','IT','2021-09-14','1997-04-08','2024-11-26 09:08:15','2024-11-26 18:20:40',59000),
('Divya','HR','2019-10-19','1995-06-11','2024-11-27 08:50:30','2024-11-27 17:42:15',50000),
('Ramesh','Finance','2017-07-01','1991-08-14','2024-11-28 09:03:00','2024-11-28 18:12:00',70000),
('Pooja','Testing','2022-05-09','2001-03-19','2024-11-29 09:18:50','2024-11-29 18:25:10',53000);
```

```
select*from employee_attendance;
```

Basic Level

20 Questions — one date/time function at a time

Q1. Display today's date.

```
SELECT CURDATE() AS today_date;
```

Q2. Display the current time.

```
SELECT CURTIME() AS current_time_now;
```

Q3. Display the current date and time.

```
SELECT NOW() AS current_datetime;
```

Q4. Display only the date part from every employee's login time.

```
SELECT emp_name, DATE(login_time) AS login_date
FROM employee_attendance;
```

Q5. Display only the time part from every employee's login time.

```
SELECT emp_name, TIME(login_time) AS login_time_only
FROM employee_attendance;
```

Q6. Display employee names along with their joining year.

```
SELECT emp_name, YEAR(joining_date) AS joining_year
FROM employee_attendance;
```

Q7. Display employee names along with their birth month number.

```
SELECT emp_name, MONTH(birth_date) AS birth_month
FROM employee_attendance;
```

Q8. Display employee names and day of birth.

```
SELECT emp_name, DAY(birth_date) AS birth_day
FROM employee_attendance;
```

Q9. Display the week number of every employee's joining date.

```
SELECT emp_name, WEEK(joining_date) AS week_number
FROM employee_attendance;
```

Q10. Display the weekday number of each employee's joining date.

```
-- WEEKDAY(): 0 = Monday ... 6 = Sunday
SELECT emp_name, WEEKDAY(joining_date) AS weekday_number
FROM employee_attendance;
```

Q11. Add 30 days to every employee's joining date.

```
SELECT emp_name, DATE_ADD(joining_date, INTERVAL 30 DAY) AS new_date
FROM employee_attendance;
```

Q12. Subtract 15 days from every employee's joining date.

```
SELECT emp_name, DATE_SUB(joining_date, INTERVAL 15 DAY) AS new_date
FROM employee_attendance;
```

Q13. Find how many days each employee has worked from joining date until today.

```
SELECT emp_name, DATEDIFF(CURDATE(), joining_date) AS days_worked
FROM employee_attendance;
```

Q14. Find the number of days between login time and logout time.

```
-- Returns 0 for same-day login/logout records in the sample data
SELECT emp_name, DATEDIFF(logout_time, login_time) AS days_between
FROM employee_attendance;
```

Q15. Convert the string '2024-12-31' into a date.


```
SELECT STR_TO_DATE('2024-12-31', '%Y-%m-%d') AS converted_date;
```

Q16. Display joining dates in the format: Monday, January 01, 2024

```
SELECT emp_name,  
       DATE_FORMAT(joining_date, '%W, %M %d, %Y') AS formatted_date  
FROM employee_attendance;
```

Q17. Display the Unix timestamp of login_time.

```
SELECT emp_name, UNIX_TIMESTAMP(login_time) AS unix_ts  
FROM employee_attendance;
```

Q18. Convert Unix timestamp 1700000000 into DateTime.

```
SELECT FROM_UNIXTIME(1700000000) AS converted_datetime;
```

Q19. Display the last day of each employee's joining month.

```
SELECT emp_name, LAST_DAY(joining_date) AS last_day_of_month  
FROM employee_attendance;
```

Q20. Display the month name and day name of every employee's birth date.

```
SELECT emp_name,  
       MONTHNAME(birth_date) AS birth_month_name,  
       DAYNAME(birth_date) AS birth_day_name  
FROM employee_attendance;
```

Intermediate Level

20 Questions — filtering, calculations and formatting

Q1. Display employees who joined in the current year.

```
SELECT emp_name, joining_date  
FROM employee_attendance  
WHERE YEAR(joining_date) = YEAR(CURDATE());
```

Q2. Display employees born in November.

```
SELECT emp_name, birth_date  
FROM employee_attendance
```

```
WHERE MONTH(birth_date) = 11;
```

Q3. Display employees whose joining month is June.

```
SELECT emp_name, joining_date  
FROM employee_attendance  
WHERE MONTH(joining_date) = 6;
```

Q4. Display employees whose birthday falls on the 10th day.

```
SELECT emp_name, birth_date  
FROM employee_attendance  
WHERE DAY(birth_date) = 10;
```

Q5. Display employees who joined in Quarter 2.

```
SELECT emp_name, joining_date  
FROM employee_attendance  
WHERE QUARTER(joining_date) = 2;
```

Q6. Display employee names and their age in years (use TIMESTAMPDIFF()).

```
SELECT emp_name,  
       TIMESTAMPDIFF(YEAR, birth_date, CURDATE()) AS age_years  
FROM employee_attendance;
```

Q7. Display employees who have completed more than 3 years in the company.

```
SELECT emp_name, joining_date  
FROM employee_attendance  
WHERE TIMESTAMPDIFF(YEAR, joining_date, CURDATE()) > 3;
```

Q8. Display employee names and their total working hours.

```
SELECT emp_name,  
       TIMESTAMPDIFF(HOUR, login_time, logout_time) AS working_hours  
FROM employee_attendance;
```

Q9. Display employee names and working minutes.

```
SELECT emp_name,  
       TIMESTAMPDIFF(MINUTE, login_time, logout_time) AS working_minutes  
FROM employee_attendance;
```

Q10. Display employees whose login time is before 9 AM.

```
SELECT emp_name, login_time
FROM employee_attendance
WHERE TIME(login_time) < '09:00:00';
```

Q11. Display employees who logged in after 9:10 AM.

```
SELECT emp_name, login_time
FROM employee_attendance
WHERE TIME(login_time) > '09:10:00';
```

Q12. Display employees whose joining date falls on Monday.

```
SELECT emp_name, joining_date
FROM employee_attendance
WHERE DAYNAME(joining_date) = 'Monday';
```

Q13. Display employees whose birth month name starts with 'J'.

```
SELECT emp_name, birth_date
FROM employee_attendance
WHERE MONTHNAME(birth_date) LIKE 'J%';
```

Q14. Display employees who joined in Week Number 25.

```
SELECT emp_name, joining_date
FROM employee_attendance
WHERE WEEK(joining_date) = 25;
```

Q15. Display employee names and the last day of their birth month.

```
SELECT emp_name, LAST_DAY(birth_date) AS last_day_birth_month
FROM employee_attendance;
```

Q16. Display employee names with joining date after adding one year.

```
SELECT emp_name,
       DATE_ADD(joining_date, INTERVAL 1 YEAR) AS new_joining_date
FROM employee_attendance;
```

Q17. Display employee names with joining date before subtracting two months.

```
SELECT emp_name,
       DATE_SUB(joining_date, INTERVAL 2 MONTH) AS earlier_date
FROM employee_attendance;
```

Q18. Display login time in 12-hour format. Example: 09:15 AM

```
SELECT emp_name,  
       DATE_FORMAT(login_time, '%h:%i %p') AS login_12hr  
FROM employee_attendance;
```

Q19. Display logout time in the format: 28-Nov-2024 06:15 PM

```
SELECT emp_name,  
       DATE_FORMAT(logout_time, '%d-%b-%Y %h:%i %p') AS formatted_logout  
FROM employee_attendance;
```

Q20. Display employees whose login and logout happened on the same date.

```
SELECT emp_name  
FROM employee_attendance  
WHERE DATE(login_time) = DATE(logout_time);
```

Advanced Level

20 Questions — multi-function combinations and real-world reports

Q1. Display employee names, age, years of experience, and retirement year (assume retirement at age 60).

```
SELECT emp_name,  
       TIMESTAMPDIFF(YEAR, birth_date, CURDATE()) AS age,  
       TIMESTAMPDIFF(YEAR, joining_date, CURDATE()) AS experience,  
       YEAR(birth_date) + 60 AS retirement_year  
FROM employee_attendance;
```

Q2. Display employees who joined in the last 5 years.

```
SELECT emp_name, joining_date  
FROM employee_attendance  
WHERE joining_date >= DATE_SUB(CURDATE(), INTERVAL 5 YEAR);
```

Q3. Display employees whose birthdays are coming within the next 30 days.

```
SELECT emp_name, birth_date  
FROM employee_attendance  
WHERE DATE_ADD(birth_date,  
              INTERVAL (YEAR(CURDATE()) - YEAR(birth_date)) YEAR)  
      BETWEEN CURDATE() AND DATE_ADD(CURDATE(), INTERVAL 30 DAY);
```

Q4. Display employees whose work duration exceeded 9 hours.

```
SELECT emp_name,  
       TIMESTAMPDIFF(MINUTE, login_time, logout_time) / 60 AS work_hours  
FROM employee_attendance  
WHERE TIMESTAMPDIFF(MINUTE, login_time, logout_time) > 540;
```

Q5. Display employee names and total work duration in HH:MM:SS format.

```
SELECT emp_name,  
       SEC_TO_TIME(TIMESTAMPDIFF(SECOND, login_time, logout_time)) AS work_duration  
FROM employee_attendance;
```

Q6. Display employee names and working duration in total seconds.

```
SELECT emp_name,  
       TIMESTAMPDIFF(SECOND, login_time, logout_time) AS work_seconds  
FROM employee_attendance;
```

Q7. Display employees who joined on weekends.

```
-- DAYOFWEEK(): 1 = Sunday, 7 = Saturday  
SELECT emp_name, joining_date  
FROM employee_attendance  
WHERE DAYOFWEEK(joining_date) IN (1, 7);
```

Q8. Display employees who joined on weekdays only.

```
SELECT emp_name, joining_date  
FROM employee_attendance  
WHERE DAYOFWEEK(joining_date) NOT IN (1, 7);
```

Q9. Display employee names and salary along with the month name of joining.

```
SELECT emp_name, salary, MONTHNAME(joining_date) AS joining_month  
FROM employee_attendance;
```

Q10. Display employees who joined in leap years.

```
SELECT emp_name, joining_date  
FROM employee_attendance  
WHERE (YEAR(joining_date) % 4 = 0 AND YEAR(joining_date) % 100 <> 0)  
      OR YEAR(joining_date) % 400 = 0;
```

Q11. Display employees whose birthday falls in Quarter 4.

```
SELECT emp_name, birth_date  
FROM employee_attendance  
WHERE QUARTER(birth_date) = 4;
```

Q12. Display employee names and number of months worked in the company.

```
SELECT emp_name,  
       TIMESTAMPDIFF(MONTH, joining_date, CURDATE()) AS months_worked  
FROM employee_attendance;
```

Q13. Display employee names and number of weeks worked in the company.

```
SELECT emp_name,  
       TIMESTAMPDIFF(WEEK, joining_date, CURDATE()) AS weeks_worked  
FROM employee_attendance;
```

Q14. Display employee names along with the first day of their joining month.

```
SELECT emp_name,  
       DATE_FORMAT(joining_date, '%Y-%m-01') AS first_day_of_month  
FROM employee_attendance;
```

Q15. Display employee names whose birth year is divisible by 4.

```
SELECT emp_name, birth_date  
FROM employee_attendance  
WHERE YEAR(birth_date) % 4 = 0;
```

Q16. Display employee names along with login Unix timestamp and converted DateTime again.

```
SELECT emp_name,  
       UNIX_TIMESTAMP(login_time) AS unix_ts,  
       FROM_UNIXTIME(UNIX_TIMESTAMP(login_time)) AS converted_back  
FROM employee_attendance;
```

Q17. Display employee names whose login time is between 9 AM and 10 AM.

```
SELECT emp_name, login_time  
FROM employee_attendance  
WHERE TIME(login_time) BETWEEN '09:00:00' AND '10:00:00';
```

Q18. Display employee names with formatted joining date as: 15 June 2021

```
SELECT emp_name,  
       DATE_FORMAT(joining_date, '%d %M %Y') AS formatted_joining_date  
FROM employee_attendance;
```

Q19. Display employee names whose working duration is greater than 8 hours but less than 10 hours.

```
SELECT emp_name,  
       TIMESTAMPDIFF(MINUTE, login_time, logout_time) / 60 AS work_hours  
FROM employee_attendance  
WHERE TIMESTAMPDIFF(MINUTE, login_time, logout_time) > 480  
      AND TIMESTAMPDIFF(MINUTE, login_time, logout_time) < 600;
```

Q20. Generate a report for every employee: Employee, Age, Experience, Joining Month, Joining Quarter, Last Day of Joining Month, Work Hours.

```
SELECT emp_name          AS Employee,  
       TIMESTAMPDIFF(YEAR, birth_date, CURDATE()) AS Age,  
       TIMESTAMPDIFF(YEAR, joining_date, CURDATE()) AS Experience,  
       MONTHNAME(joining_date) AS Joining_Month,  
       QUARTER(joining_date) AS Joining_Quarter,  
       LAST_DAY(joining_date) AS Last_Day_Of_Joining_Month,  
       SEC_TO_TIME(TIMESTAMPDIFF(SECOND, login_time, logout_time)) AS Work_Hours  
FROM employee_attendance;
```

Day-15

JOINS

Joins

a join in MySQL is used to combine rows from two or more tables based on related columns.

example

employees table--->Employee details.

departments table---> department details.

if you want to see which employee belongings to which department. you need to join these two tables.

Why do we need joins?

- 1. to fetch data from multiple tables in a single Query.**
- 2. to reduce data duplication(instead of keeping everything in one table.)**
- 3. to implement relationships like(one-to-one, one-to-many, many-to-one, many-to-many) in databases.**
- 4. to generate reports and analytics combining different datasets.**

types of joins in MySQL

1.inner join

2.left join

3. right join

4. full join

5. cross join

6. self join

1. INNER JOIN

Returns only the rows that have matching values in both tables.

Syntax

SELECT columns

FROM table1

INNER JOIN table2

ON table1.common_column = table2.common_column;

2. LEFT JOIN (or LEFT OUTER JOIN)

Returns all rows from the left table and the matched rows from the right table. If no match, **NULL** values appear for right table columns.

Syntax

SELECT columns

FROM table1

LEFT JOIN table2

ON table1.common_column = table2.common_column;

3.RIGHT JOIN (or RIGHT OUTER JOIN)

Returns all rows from the right table and the matched rows from the left table. If no match, **NULL** values appear for left table columns.

SELECT columns

FROM table1

RIGHT JOIN table2

ON table1.common_column = table2.common_column;

4. FULL OUTER JOIN (Not directly supported in MySQL)

To simulate, use the **UNION** of **LEFT JOIN** and **RIGHT JOIN**.

SELECT columns

FROM table1

LEFT JOIN table2

ON table1.common_column = table2.common_column

UNION

SELECT columns

FROM table1

RIGHT JOIN table2

ON table1.common_column = table2.common_column;

5. CROSS JOIN

Returns Cartesian product — every row from the first table with every row from the second table.

SELECT columns

FROM table1

CROSS JOIN table2;

6. SELF JOIN

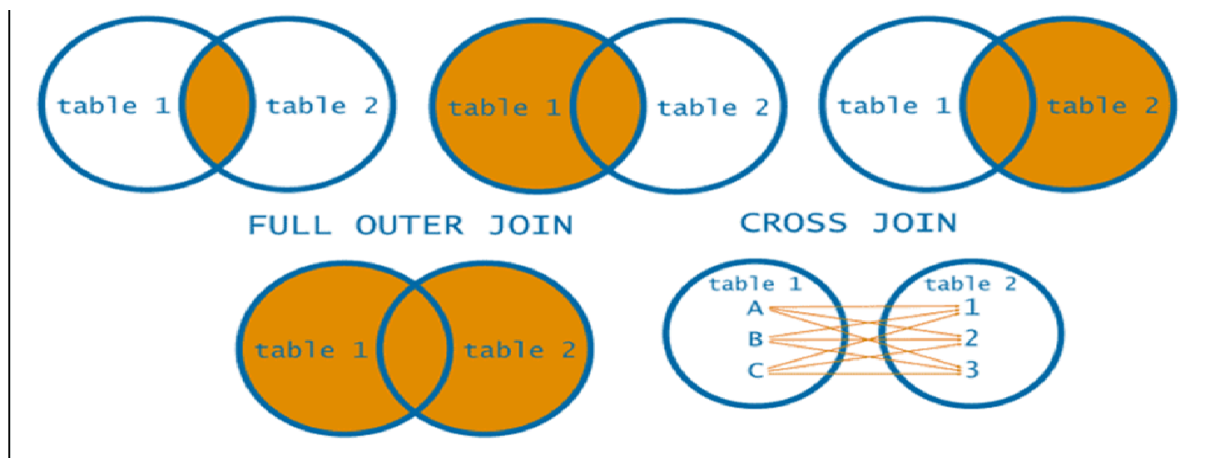
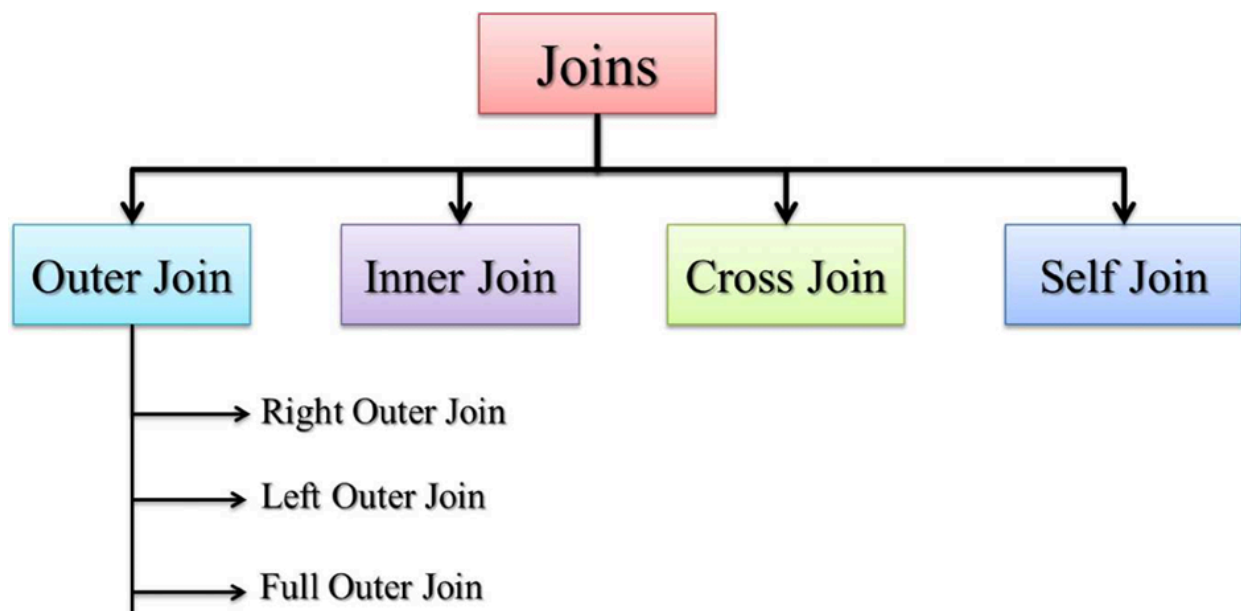
A table joined with itself.

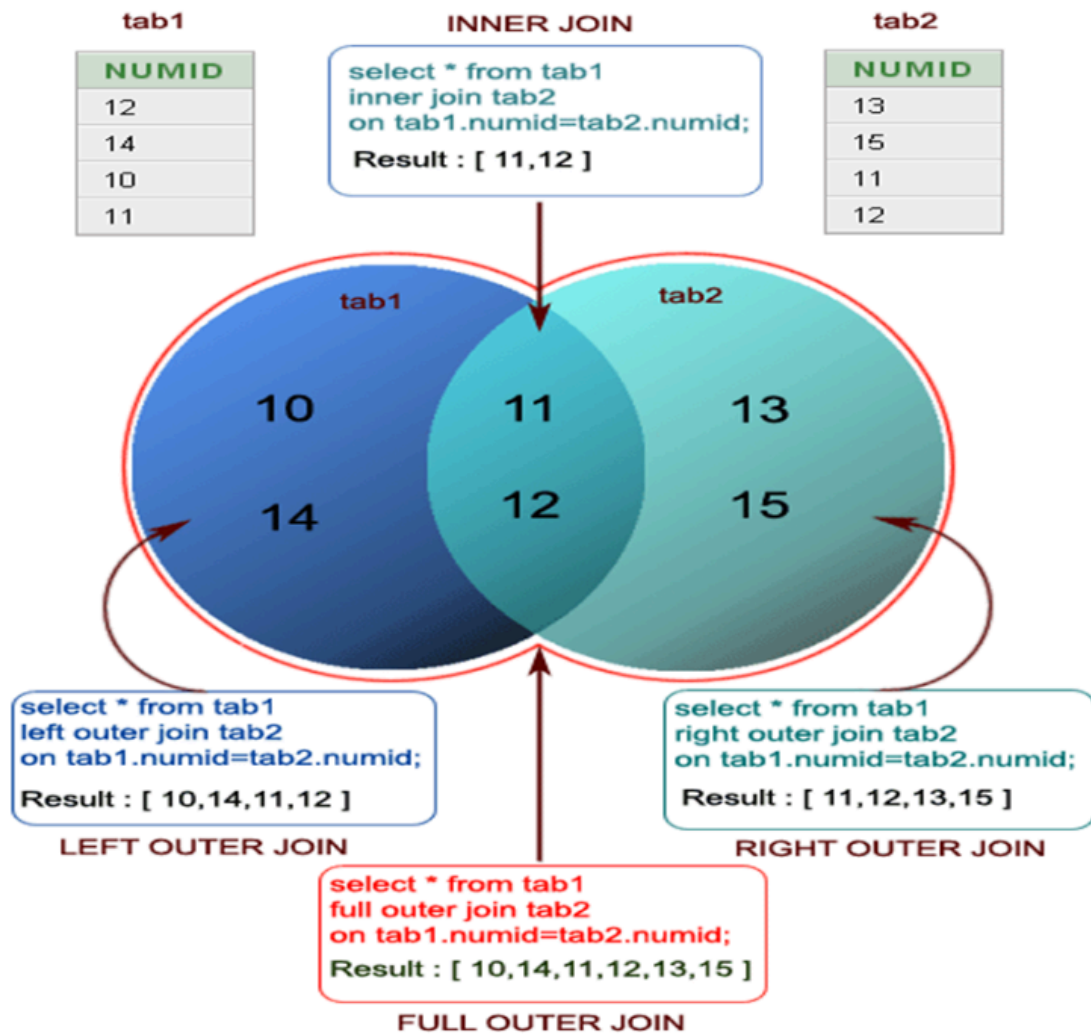
SELECT a.columns, b.columns

FROM table1 a

INNER JOIN table1 b

ON a.common_column = b.common_column;





```
CREATE TABLE TAB1(NUMID INT(3));
```

```
CREATE TABLE TAB2 (NUMID INT(3));
```

```
INSERT INTO TAB1 VALUES(10);
```

```
INSERT INTO TAB1 VALUES(11);
```

```
INSERT INTO TAB1 VALUES(12);
```

```
INSERT INTO TAB1 VALUES(14);
```

INSERT INTO TAB2 VALUES(11);

INSERT INTO TAB2 VALUES(12);

INSERT INTO TAB2 VALUES(13);

INSERT INTO TAB2 VALUES(15);

SELECT * FROM TAB1;

SELECT* FROM TAB2;

INNER JOIN:- • Only matched records in both the tables

SELECT*FROM TAB1 INNER JOIN TAB2

ON

TAB1.NUMID=TAB2.NUMID

LEFT JOIN:- Matched records+ unmatched from left table dept

SELECT*FROM TAB1 LEFT JOIN TAB2

ON

TAB1.NUMID=TAB2.NUMID

RIGHT JOIN:- Matched records+ unmatched from right table emp

SELECT*FROM TAB1 RIGHT JOIN TAB2

ON

TAB1.NUMID=TAB2.NUMID

FULL JOIN:- Matched records + unmatched records from both the tables

FULL JOIN NOT SUPPORTED mysql:

CrossJoin:- SELECT *

FROM TAB1

CROSS JOIN TAB2;

Join Exercise Questions part-1

Employees / Departments / Projects

```
CREATE TABLE Employees (  
    emp_id INT PRIMARY KEY,  
    name VARCHAR(100),  
    department_id INT,  
    salary DECIMAL(10,2)  
);
```

```
CREATE TABLE Departments (  
    department_id INT PRIMARY KEY,  
    department_name VARCHAR(100)  
);
```

```
CREATE TABLE Projects (  
    project_id INT PRIMARY KEY,  
    project_name VARCHAR(100),  
    emp_id INT  
);
```

```
INSERT INTO Departments (department_id, department_name) VALUES  
(1, 'Human Resources'), (2, 'Engineering'), (3, 'Marketing'), (4, 'Finance'), (5,  
'Operations');
```

```
INSERT INTO Employees (emp_id, name, department_id, salary) VALUES
```



```
(101, 'Arjun', 1, 50000), (102, 'Bhavana', 2, 65000), (103, 'Charan', 2, 72000),
(104, 'Divya', 3, 45000), (105, 'Eswar', NULL, 40000), (106, 'Farhan', 4, 55000),
(107, 'Gowri', 2, 62000), (108, 'Harish', 5, 48000), (109, 'Ishita', NULL, 53000),
(110, 'Jagan', 3, 60000);
```

```
INSERT INTO Projects (project_id, project_name, emp_id) VALUES
(201, 'Website Redesign', 102), (202, 'Mobile Banking App', 103), (203, 'Marketing
Campaign', 104),
(204, 'Payroll System', 106), (205, 'Backend Optimization', 107), (206, 'Cloud
Migration', 102),
(207, 'Customer Analytics', 110), (208, 'Inventory System', NULL);
```

Key facts this data is built to exercise:

- Eswar and Ishita have `department_id = NULL` — employees with no department.
- Arjun, Harish have no project. Bhavana has **two** projects.
- Project 208 "Inventory System" has `emp_id = NULL` — a project with no one assigned.
- Every department has at least one employee, so no department is ever "empty" in this dataset (unlike the Students set below — see Part 2).

INNER JOIN

Q1. Get all employees with their department names.

```
SELECT e.emp_id, e.name, d.department_name
FROM Employees e
INNER JOIN Departments d ON e.department_id = d.department_id;
```

emp_id	name	department_name
101	Arjun	Human Resources
102	Bhavana	Engineering
103	Charan	Engineering
104	Divya	Marketing
106	Farhan	Finance

107	Gowri	Engineering
108	Harish	Operations
110	Jagan	Marketing

8 rows — Eswar and Ishita are correctly dropped (NULL department, no match).

Q2. List all employees who are assigned to a project.

```
SELECT DISTINCT e.emp_id, e.name
FROM Employees e
INNER JOIN Projects p ON e.emp_id = p.emp_id;
```

emp_id	name
102	Bhavana
103	Charan
104	Divya
106	Farhan
107	Gowri
110	Jagan

Q3. Show employee names and their project names.

```
SELECT e.name, p.project_name
FROM Employees e
INNER JOIN Projects p ON e.emp_id = p.emp_id;
```

name	project_name
Bhavana	Website Redesign
Charan	Mobile Banking App
Divya	Marketing Campaign
Farhan	Payroll System
Gowri	Backend Optimization

Bhavana	Cloud Migration
Jagan	Customer Analytics

7 rows (no DISTINCT here, so Bhavana correctly appears twice for her two projects).

LEFT JOIN

Q1. All employees and their department names (even if no department).

```
SELECT e.emp_id, e.name, d.department_name
FROM Employees e
LEFT JOIN Departments d ON e.department_id = d.department_id;
```

emp_id	name	department_name
101	Arjun	Human Resources
102	Bhavana	Engineering
103	Charan	Engineering
104	Divya	Marketing
105	Eswar	<i>NULL</i>
106	Farhan	Finance
107	Gowri	Engineering
108	Harish	Operations
109	Ishita	<i>NULL</i>
110	Jagan	Marketing

Q2. All employees and their project names (if assigned).

```
SELECT e.name, p.project_name
FROM Employees e
LEFT JOIN Projects p ON e.emp_id = p.emp_id;
```

name	project_name
------	--------------

Arjun	<i>NULL</i>
Bhavana	Website Redesign
Bhavana	Cloud Migration
Charan	Mobile Banking App
Divya	Marketing Campaign
Eswar	<i>NULL</i>
Farhan	Payroll System
Gowri	Backend Optimization
Harish	<i>NULL</i>
Ishita	<i>NULL</i>
Jagan	Customer Analytics

11 rows — Arjun, Eswar, Harish, Ishita show **NULL** (no project); Bhavana still gets 2 rows.

Q3. All departments and the employees working in them.

```
SELECT d.department_name, e.name
FROM Departments d
LEFT JOIN Employees e ON d.department_id = e.department_id;
```

department_name	name
Human Resources	Arjun
Engineering	Bhavana
Engineering	Charan
Engineering	Gowri
Marketing	Divya
Marketing	Jagan
Finance	Farhan
Operations	Harish

(No NULL rows here — every department in this dataset has at least one employee.)

RIGHT JOIN

Q1. All departments and the employees in them (even if none).

```
SELECT D.DEPARTMENT_NAME,E.NAME
FROM DEPARTMENTS D
RIGHT JOIN EMPLOYEES E
ON D.DEPARTMENT_ID=E.DEPARTMENT_ID;
```

Same 8 rows as LEFT JOIN Q3 above — because every department happens to have an employee, RIGHT JOIN here produces no visible NULLs. (LEFT and RIGHT only look different when there's actually an unmatched row on the "preserved" side.)

Q2. All projects and their assigned employees (even if none).

```
SELECT P.PROJECT_NAME,E.NAME
FROM PROJECTS P
RIGHT JOIN EMPLOYEES E
-ON P.EMP_ID=E.EMP_ID;
```

8 rows — Inventory System correctly shows 'NULL' for its unassigned employee. This is the row that actually proves the RIGHT JOIN is doing something LEFT JOIN wouldn't.

Q3. Employees and departments using RIGHT JOIN.

```
SELECT e.name, d.department_name
FROM Departments d
RIGHT JOIN Employees e ON d.department_id = e.department_id;
```

name	department_name
Arjun	Human Resources
Bhavana	Engineering
Charan	Engineering

Gowri	Engineering
Divya	Marketing
Jagan	Marketing
Farhan	Finance
Harish	Operations
Eswar	<i>NULL</i>
Ishita	<i>NULL</i>

10 rows — Eswar and Ishita show up with **NULL** department, since **Employees** is on the "preserved" (right) side here.

SELF JOIN

Q1. Employees who share the same department.

```
SELECT e1.name AS Employee1, e2.name AS Employee2, e1.department_id
FROM Employees e1
JOIN Employees e2 ON e1.department_id = e2.department_id AND e1.emp_id <>
e2.emp_id;
```

Employee1	Employee2	department_id
Bhavana	Charan	2
Bhavana	Gowri	2
Charan	Bhavana	2
Charan	Gowri	2
Divya	Jagan	3
Gowri	Bhavana	2
Gowri	Charan	2
Jagan	Divya	3

Note: this returns both directions of each pair (Bhavana↔Charan appears twice). That's expected from <> — if you only want each pair once, use `e1.emp_id < e2.emp_id` instead, which would cut this to 4 rows.

Q2. Pairs of employees where one salary is more than the other.

```
SELECT e1.name AS Higher_Salary_Employee, e2.name AS Lower_Salary_Employee,
e1.salary, e2.salary
FROM Employees e1
JOIN Employees e2 ON e1.salary > e2.salary;
```

This returns 45 rows — every possible higher/lower salary pairing among all 10 employees (since all 10 salaries happen to be distinct, this is the full $C(10,2) = 45$ combinations). It's not wrong, but it's a lot of noise for an exercise answer — worth being aware the result set explodes quickly as employee count grows. (Full 45-row output available on request; omitted here for length — first few rows: Arjun > Divya, Arjun > Eswar, Arjun > Harish, Bhavana > Arjun, Bhavana > Divya...)

Q3. Employee names and colleagues from the same department.

```
SELECT e1.name AS Employee, e2.name AS Colleague, d.department_name
FROM Employees e1
JOIN Employees e2 ON e1.department_id = e2.department_id AND e1.emp_id <>
e2.emp_id
JOIN Departments d ON e1.department_id = d.department_id;
```

Employee	Colleague	department_name
Bhavana	Charan	Engineering
Bhavana	Gowri	Engineering
Charan	Bhavana	Engineering
Charan	Gowri	Engineering
Divya	Jagan	Marketing
Gowri	Bhavana	Engineering
Gowri	Charan	Engineering
Jagan	Divya	Marketing

FULL JOIN (simulated via UNION — MySQL has no native FULL JOIN)

Q1. All employees and all departments (even if no match).

```
SELECT e.name, d.department_name FROM Employees e LEFT JOIN Departments d
ON e.department_id = d.department_id
UNION
SELECT e.name, d.department_name FROM Employees e RIGHT JOIN Departments
d ON e.department_id = d.department_id;
```

name	department_name
Arjun	Human Resources
Bhavana	Engineering
Charan	Engineering
Divya	Marketing
Eswar	<i>NULL</i>
Farhan	Finance
Gowri	Engineering
Harish	Operations
Ishita	<i>NULL</i>
Jagan	Marketing

10 rows — every employee, including Eswar/Ishita with NULL department. (No department-side NULL rows appear because every department has at least one employee.)

Q2. All projects and employees (even if some don't match).

```
SELECT e.name, p.project_name FROM Employees e LEFT JOIN Projects p ON
e.emp_id = p.emp_id
UNION
SELECT e.name, p.project_name FROM Employees e RIGHT JOIN Projects p ON
e.emp_id = p.emp_id;
```

name	project_name
<i>NULL</i>	Inventory System

Arjun	<i>NULL</i>
Bhavana	Website Redesign
Bhavana	Cloud Migration
Charan	Mobile Banking App
Divya	Marketing Campaign
Eswar	<i>NULL</i>
Farhan	Payroll System
Gowri	Backend Optimization
Harish	<i>NULL</i>
Ishita	<i>NULL</i>
Jagan	Customer Analytics

12 rows — this is the one query in Part 1 that shows NULLs on both sides at once:
Inventory System has no employee, and Arjun/Eswar/Harish/Ishita have no project.

Q3. All department names and employee names, with or without a match.

```
SELECT d.department_name, e.name FROM Departments d LEFT JOIN Employees e
ON d.department_id = e.department_id
UNION
SELECT d.department_name, e.name FROM Departments d RIGHT JOIN Employees
e ON d.department_id = e.department_id;
```

department_name	name
<i>NULL</i>	Eswar
<i>NULL</i>	Ishita
Engineering	Bhavana
Engineering	Charan
Engineering	Gowri
Finance	Farhan
Human Resources	Arjun
Marketing	Divya

Marketing	Jagan
Operations	Harish

JOIN + GROUP BY

Q1. Count employees per department.

```
SELECT d.department_name, COUNT(e.emp_id) AS Total_Employees
FROM Departments d
LEFT JOIN Employees e ON d.department_id = e.department_id
GROUP BY d.department_name;
```

department_name	Total_Employees
Engineering	3
Finance	1
Human Resources	1
Marketing	2
Operations	1

Q2. Total salary per department.

```
SELECT d.department_name, SUM(e.salary) AS Total_Salary
FROM Departments d
LEFT JOIN Employees e ON d.department_id = e.department_id
GROUP BY d.department_name;
```

department_name	Total_Salary
Engineering	199000
Finance	55000
Human Resources	50000
Marketing	105000
Operations	48000

Q3. Number of projects per employee.

```
SELECT e.name, COUNT(p.project_id) AS Total_Projects
FROM Employees e
LEFT JOIN Projects p ON e.emp_id = p.emp_id
GROUP BY e.name;
```

name	Total_Projects
Arjun	0
Bhavana	2
Charan	1
Divya	1
Eswar	0
Farhan	1
Gowri	1
Harish	0
Ishita	0
Jagan	1

JOIN + ORDER BY

Q1. Employees with department names, ordered by salary (highest first).

```
SELECT e.name, d.department_name, e.salary
FROM Employees e
JOIN Departments d ON e.department_id = d.department_id
ORDER BY e.salary DESC;
```

name	department_name	salary
Charan	Engineering	72000
Bhavana	Engineering	65000

Gowri	Engineering	62000
Jagan	Marketing	60000
Farhan	Finance	55000
Arjun	Human Resources	50000
Harish	Operations	48000
Divya	Marketing	45000

(Eswar/Ishita excluded — plain JOIN behaves as INNER JOIN.)

Q2. Employee names and project names, ordered by employee name.

```
SELECT e.name, p.project_name
FROM Employees e
JOIN Projects p ON e.emp_id = p.emp_id
ORDER BY e.name;
```

name	project_name
Bhavana	Website Redesign
Bhavana	Cloud Migration
Charan	Mobile Banking App
Divya	Marketing Campaign
Farhan	Payroll System
Gowri	Backend Optimization
Jagan	Customer Analytics

Q3. Departments and employee count, ordered by count (highest first).

```
SELECT d.department_name, COUNT(e.emp_id) AS Employee_Count
FROM Departments d
LEFT JOIN Employees e ON d.department_id = e.department_id
GROUP BY d.department_name
ORDER BY Employee_Count DESC;
```

department_name	Employee_Count
Engineering	3
Marketing	2
Operations	1
Human Resources	1
Finance	1

JOIN + GROUP BY + ORDER BY + HAVING

Q1. Departments with more than 1 employee, ordered by count.

```
SELECT d.department_name, COUNT(e.emp_id) AS Employee_Count
FROM Departments d
JOIN Employees e ON d.department_id = e.department_id
GROUP BY d.department_name
HAVING COUNT(e.emp_id) > 1
ORDER BY Employee_Count DESC;
```

department_name	Employee_Count
Engineering	3
Marketing	2

Q2. Employees with more than 1 project, ordered by count.

```
SELECT e.name, COUNT(p.project_id) AS Project_Count
FROM Employees e
JOIN Projects p ON e.emp_id = p.emp_id
GROUP BY e.name
HAVING COUNT(p.project_id) > 1
ORDER BY Project_Count DESC;
```

name	Project_Count
Bhavana	2

Q3. Departments with total salary > 50000, ordered by total salary.

```
SELECT d.department_name, SUM(e.salary) AS Total_Salary
FROM Departments d
JOIN Employees e ON d.department_id = e.department_id
GROUP BY d.department_name
HAVING SUM(e.salary) > 50000
ORDER BY Total_Salary DESC;
```

department_name	Total_Salary
Engineering	199000
Marketing	105000
Finance	55000

(Human Resources = 50000 and Operations = 48000 are correctly excluded — neither is strictly greater than 50000.)

Join Exercise Part-2

=====

PART 2 — Students / Courses / Enrollments

Setup

```
CREATE TABLE Students (student_id INT PRIMARY KEY, name VARCHAR(100), age
INT);
CREATE TABLE Courses (course_id INT PRIMARY KEY, course_name
VARCHAR(100));
CREATE TABLE Enrollments (enrollment_id INT PRIMARY KEY, student_id INT,
course_id INT);

INSERT INTO Students VALUES
(1, 'Arjun', 20), (2, 'Bhavana', 21), (3, 'Charan', 22), (4, 'Divya', 20),
(5, 'Eswar', 23), (6, 'Farhan', 21), (7, 'Gowri', 22);
```

```
INSERT INTO Courses VALUES
(101, 'Mathematics'), (102, 'Physics'), (103, 'Biology'), (104, 'Chemistry'), (105, 'Computer
Science');
```

```
INSERT INTO Enrollments VALUES
(1, 1, 101), (2, 2, 101), (3, 2, 102), (4, 3, 103),
(5, 4, 101), (6, 5, 104), (7, 1, 102), (8, 6, 105);
```

Q1. List all students with the courses they are enrolled in.

```
SELECT s.name AS Student_Name, c.course_name AS Course_Name
FROM Students s
INNER JOIN Enrollments e ON s.student_id = e.student_id
INNER JOIN Courses c ON e.course_id = c.course_id;
```

Student_Name	Course_Name
Arjun	Mathematics
Bhavana	Mathematics
Bhavana	Physics
Charan	Biology
Divya	Mathematics
Eswar	Chemistry
Arjun	Physics
Farhan	Computer Science

8 rows — Gowri correctly excluded (no enrollments, INNER JOIN drops her).

Q2. Show course names along with student names.

```
SELECT c.course_name, s.name AS Student_Name
FROM Courses c
INNER JOIN Enrollments e ON c.course_id = e.course_id
INNER JOIN Students s ON e.student_id = s.student_id;
```

Same 8 pairs as Q1, just column order swapped.

Q3. Get names of students enrolled in "Physics".

```
SELECT s.name
FROM Students s
INNER JOIN Enrollments e ON s.student_id = e.student_id
INNER JOIN Courses c ON e.course_id = c.course_id
WHERE c.course_name = 'Physics';
```

| name |

|---|

| Bhavana |

| Arjun |

Q4. List all students with their courses (if enrolled).

```
SELECT s.name AS Student_Name, c.course_name
FROM Students s
LEFT JOIN Enrollments e ON s.student_id = e.student_id
LEFT JOIN Courses c ON e.course_id = c.course_id;
```

Student_Name	course_name
Arjun	Mathematics
Arjun	Physics
Bhavana	Mathematics
Bhavana	Physics
Charan	Biology
Divya	Mathematics
Eswar	Chemistry
Farhan	Computer Science
Gowri	<i>NULL</i>

9 rows — **Gowri now appears with 'NULL'** since LEFT JOIN preserves all students.

Q5. Show all courses and the students who enrolled (even if none).

```
SELECT c.course_name, s.name AS Student_Name
FROM Courses c
LEFT JOIN Enrollments e ON c.course_id = e.course_id
LEFT JOIN Students s ON e.student_id = s.student_id;
```

course_name	Student_Name
Mathematics	Arjun
Mathematics	Bhavana
Mathematics	Divya
Physics	Arjun
Physics	Bhavana
Biology	Charan
Chemistry	Eswar
Computer Science	Farhan

Only 8 rows, no NULLs — because (as noted above) every course actually has at least one enrolled student, so there's nothing for this LEFT JOIN to "rescue."

Q6. Display all students with course details, even if some courses have no enrollments.

```
SELECT s.name AS Student_Name, c.course_name
FROM Students s
LEFT JOIN Enrollments e ON s.student_id = e.student_id
LEFT JOIN Courses c ON e.course_id = c.course_id;
```

Q7. List all students and all courses, even if there is no match (simulate full join).

```
SELECT s.name AS Student_Name, c.course_name FROM Students s
LEFT JOIN Enrollments e ON s.student_id = e.student_id
LEFT JOIN Courses c ON e.course_id = c.course_id
UNION
SELECT s.name AS Student_Name, c.course_name FROM Courses c
LEFT JOIN Enrollments e ON c.course_id = e.course_id
```

```
LEFT JOIN Students s ON e.student_id = s.student_id;
```

Student_Name	course_name
Arjun	Mathematics
Arjun	Physics
Bhavana	Mathematics
Bhavana	Physics
Charan	Biology
Divya	Mathematics
Eswar	Chemistry
Farhan	Computer Science
Gowri	<i>NULL</i>

This returns exactly the same 9 rows as Q4 — the query logic is correct for simulating a FULL JOIN, but with *this* dataset every course has an enrolled student, so the "courses side" of the UNION never contributes a new NULL-course row. The query would only visibly demonstrate the FULL JOIN's extra behavior if a 6th course (e.g. (106, 'History')) were added with no enrollments — then you'd see an extra NULL | History row appear here that doesn't appear in Q4.

Q8. Show pairs of students who have the same age.

```
SELECT s1.name AS Student1, s2.name AS Student2, s1.age
FROM Students s1
JOIN Students s2 ON s1.age = s2.age AND s1.student_id < s2.student_id;
```

Student1	Student2	age
Arjun	Divya	20
Bhavana	Farhan	21
Charan	Gowri	22

Each pair appears exactly once, thanks to `student_id <` rather than `<>`.

Q9. Count how many students are enrolled in each course.

```
SELECT c.course_name, COUNT(e.student_id) AS Total_Students
FROM Courses c
LEFT JOIN Enrollments e ON c.course_id = e.course_id
GROUP BY c.course_name;
```

course_name	Total_Students
Biology	1
Chemistry	1
Computer Science	1
Mathematics	3
Physics	2

Q10. Show number of courses each student is enrolled in.

```
SELECT s.name AS Student_Name, COUNT(e.course_id) AS Total_Courses
FROM Students s
LEFT JOIN Enrollments e ON s.student_id = e.student_id
GROUP BY s.name;
```

Student_Name	Total_Courses
Arjun	2
Bhavana	2
Charan	1
Divya	1
Eswar	1
Farhan	1
Gowri	0

Gowri correctly shows 0 (LEFT JOIN + COUNT gives 0, not NULL, since COUNT ignores the NULL row but still counts the group).

Q11. List all students and courses ordered by student name.

```
SELECT s.name AS Student_Name, c.course_name
```

```

FROM Students s
INNER JOIN Enrollments e ON s.student_id = e.student_id
INNER JOIN Courses c ON e.course_id = c.course_id
ORDER BY s.name;

```

Student_Name	course_name
Arjun	Mathematics
Arjun	Physics
Bhavana	Mathematics
Bhavana	Physics
Charan	Biology
Divya	Mathematics
Eswar	Chemistry
Farhan	Computer Science

Q12. Show student names and course names ordered by course name.

```

SELECT s.name AS Student_Name, c.course_name
FROM Students s
INNER JOIN Enrollments e ON s.student_id = e.student_id
INNER JOIN Courses c ON e.course_id = c.course_id
ORDER BY c.course_name;

```

Student_Name	course_name
Charan	Biology
Eswar	Chemistry
Farhan	Computer Science
Arjun	Mathematics
Bhavana	Mathematics
Divya	Mathematics
Bhavana	Physics

Arjun	Physics
-------	---------

Q13. Show only courses with more than 1 student enrolled.

```
SELECT c.course_name, COUNT(e.student_id) AS Student_Count
FROM Courses c
JOIN Enrollments e ON c.course_id = e.course_id
GROUP BY c.course_name
HAVING COUNT(e.student_id) > 1;
```

course_name	Student_Count
Mathematics	3
Physics	2

Q14. List students enrolled in more than 1 course.

```
SELECT s.name, COUNT(e.course_id) AS Course_Count
FROM Students s
JOIN Enrollments e ON s.student_id = e.student_id
GROUP BY s.name
HAVING COUNT(e.course_id) > 1;
```

name	Course_Count
Arjun	2
Bhavana	2

Q15. Display student names and count of enrollments, ordered by highest enrollment.

```
SELECT s.name AS Student_Name, COUNT(e.enrollment_id) AS Enrollment_Count
FROM Students s
LEFT JOIN Enrollments e ON s.student_id = e.student_id
GROUP BY s.name
ORDER BY Enrollment_Count DESC;
```

Student_Name	Enrollment_Count
--------------	------------------

Bhavana	2
Arjun	2
Farhan	1
Eswar	1
Divya	1
Charan	1
Gowri	0

Join Exercise part-3

=====

SQL JOINS — Tables, Data & Verified Query Results

All queries below were actually executed against a real database (SQLite) to confirm the results — not just reasoned about.

Step 1: Create Tables

```
CREATE TABLE customers (
  cust_id INT PRIMARY KEY,
  cust_name VARCHAR(50),
  email VARCHAR(50)
);

CREATE TABLE orders (
  order_id INT PRIMARY KEY,
  cust_id INT,
  order_date DATE
);

CREATE TABLE products (
  prod_id INT PRIMARY KEY,
  prod_name VARCHAR(50),
  price DECIMAL(10,2)
);
```

```
CREATE TABLE order_items (  
    item_id INT PRIMARY KEY,  
    order_id INT,  
    prod_id INT,  
    quantity INT  
);
```

Step 2: Insert Sample Data

```
-- Customers  
INSERT INTO customers VALUES  
(1, 'Ramesh', 'ramesh@gmail.com'),  
(2, 'Suresh', 'suresh@gmail.com'),  
(3, 'Anita', 'anita@gmail.com'),  
(4, 'Priya', 'priya@gmail.com'),  
(5, 'Vikram', 'vikram@gmail.com'),  
(6, 'Neha', 'neha@gmail.com');  
  
-- Orders (order 105 has an invalid cust_id = 7, on purpose)  
INSERT INTO orders VALUES  
(101, 1, '2025-09-20'),  
(102, 2, '2025-09-21'),  
(103, 3, '2025-09-22'),  
(104, 5, '2025-09-23'),  
(105, 7, '2025-09-24');  
  
-- Products  
INSERT INTO products VALUES  
(201, 'Laptop', 55000),  
(202, 'Mobile', 22000),  
(203, 'Headphones', 3000),  
(204, 'Keyboard', 1500),  
(205, 'Mouse', 800);  
  
-- Order Items  
INSERT INTO order_items VALUES  
(1001, 101, 201, 1), -- Ramesh -> Laptop  
(1002, 101, 202, 1), -- Ramesh -> Mobile  
(1003, 102, 203, 2), -- Suresh -> Headphones
```

```
(1004, 103, 204, 1), -- Anita -> Keyboard
(1005, 104, 205, 2); -- Vikram -> Mouse
```

Key facts baked into this data on purpose:

- Priya and Neha are customers with **no orders**.
- Order 105 references **cust_id = 7**, which doesn't exist.
- All 5 products happen to have at least one sale (so LEFT JOIN from products won't show any NULLs here — noted below where the original doc assumed otherwise).

INNER JOIN — only matching rows

Q1. Show customers who have placed orders

```
SELECT c.cust_name, o.order_id
FROM customers c
INNER JOIN orders o ON c.cust_id = o.cust_id;
```

cust_name	order_id
Ramesh	101
Suresh	102
Anita	103
Vikram	104

4 rows. Priya, Neha, and order 105 are excluded (no match on either side).

Q2. Show order id with product name and quantity

```
SELECT oi.order_id, p.prod_name, oi.quantity
FROM order_items oi
INNER JOIN products p ON oi.prod_id = p.prod_id;
```

order_id	prod_name	quantity
101	Laptop	1
101	Mobile	1

102	Headphones	2
103	Keyboard	1
104	Mouse	2

Q3. Show customer name and products purchased

```
SELECT c.cust_name, p.prod_name
FROM customers c
INNER JOIN orders o ON c.cust_id = o.cust_id
INNER JOIN order_items oi ON o.order_id = oi.order_id
INNER JOIN products p ON oi.prod_id = p.prod_id;
```

cust_name	prod_name
Ramesh	Laptop
Ramesh	Mobile
Suresh	Headphones
Anita	Keyboard
Vikram	Mouse

Q4. Show total amount spent by each customer

```
SELECT c.cust_name, SUM(p.price * oi.quantity) AS total_spent
FROM customers c
INNER JOIN orders o ON c.cust_id = o.cust_id
INNER JOIN order_items oi ON o.order_id = oi.order_id
INNER JOIN products p ON oi.prod_id = p.prod_id
GROUP BY c.cust_name;
```

cust_name	total_spent
Ramesh	77000
Suresh	6000
Anita	1500
Vikram	1600

LEFT JOIN — all rows from the left table + matches

Q1. Show all customers and their orders

```
SELECT c.cust_name, o.order_id
FROM customers c
LEFT JOIN orders o ON c.cust_id = o.cust_id;
```

cust_name	order_id
Ramesh	101
Suresh	102
Anita	103
Priya	<i>NULL</i>
Vikram	104
Neha	<i>NULL</i>

6 rows — Priya and Neha appear with **NULL** since they have no orders.

Q2. Show all customers and products they purchased (if any)

```
SELECT c.cust_name, p.prod_name
FROM customers c
LEFT JOIN orders o ON c.cust_id = o.cust_id
LEFT JOIN order_items oi ON o.order_id = oi.order_id
LEFT JOIN products p ON oi.prod_id = p.prod_id;
```

cust_name	prod_name
Ramesh	Laptop
Ramesh	Mobile
Suresh	Headphones
Anita	Keyboard

Priya	<i>NULL</i>
Vikram	Mouse
Neha	<i>NULL</i>

Q3. Show all customers with total spent (including zero purchase)

```
SELECT c.cust_name, SUM(p.price * oi.quantity) AS total_spent
FROM customers c
LEFT JOIN orders o ON c.cust_id = o.cust_id
LEFT JOIN order_items oi ON o.order_id = oi.order_id
LEFT JOIN products p ON oi.prod_id = p.prod_id
GROUP BY c.cust_name;
```

cust_name	total_spent
Anita	1500
Neha	<i>NULL</i>
Priya	<i>NULL</i>
Ramesh	77000
Suresh	6000
Vikram	1600

Q4. Show all products and their sales (including unsold)

```
SELECT p.prod_name, oi.quantity
FROM products p
LEFT JOIN order_items oi ON p.prod_id = oi.prod_id;
```

prod_name	quantity
Laptop	1
Mobile	1
Headphones	2
Keyboard	1

Mouse	2
-------	---

RIGHT JOIN — all rows from the right table + matches

Q1. Show all orders with customer names

```
SELECT c.cust_name, o.order_id
FROM customers c
RIGHT JOIN orders o ON c.cust_id = o.cust_id;
```

cust_name	order_id
Ramesh	101
Suresh	102
Anita	103
Vikram	104
<i>NULL</i>	105

Order 105 still shows up — its `cust_id = 7` doesn't exist, so `cust_name` is `NULL`.

Q2. Show all order items with order details

```
SELECT o.order_id, oi.item_id
FROM orders o
RIGHT JOIN order_items oi ON o.order_id = oi.order_id;
```

order_id	item_id
101	1001
101	1002
102	1003
103	1004
104	1005

Q3. Show all products with orders

```
SELECT o.order_id, p.prod_name
FROM orders o
RIGHT JOIN order_items oi ON o.order_id = oi.order_id
RIGHT JOIN products p ON oi.prod_id = p.prod_id;
```

order_id	prod_name
101	Laptop
101	Mobile
102	Headphones
103	Keyboard
104	Mouse

Q4. Show all orders and products

```
SELECT o.order_id, p.prod_name
FROM order_items oi
RIGHT JOIN orders o ON oi.order_id = o.order_id
RIGHT JOIN products p ON oi.prod_id = p.prod_id;
```

order_id	prod_name
101	Laptop
101	Mobile
102	Headphones
103	Keyboard
104	Mouse

FULL JOIN — all rows from both tables (MySQL: simulate with UNION;
SQLite/Postgres: `FULL JOIN` works directly)

Q1. Show all customers and orders

```

SELECT c.cust_name, o.order_id
FROM customers c
LEFT JOIN orders o ON c.cust_id = o.cust_id
UNION
SELECT c.cust_name, o.order_id
FROM customers c
RIGHT JOIN orders o ON c.cust_id = o.cust_id;

```

cust_name	order_id
Ramesh	101
Suresh	102
Anita	103
Priya	<i>NULL</i>
Vikram	104
Neha	<i>NULL</i>
<i>NULL</i>	105

7 rows — all 6 customers plus the orphan order 105.

Q2. Show all products and order items

```

SELECT p.prod_name, oi.quantity
FROM products p
LEFT JOIN order_items oi ON p.prod_id = oi.prod_id
UNION
SELECT p.prod_name, oi.quantity
FROM products p
RIGHT JOIN order_items oi ON p.prod_id = oi.prod_id;

```

prod_name	quantity
Laptop	1
Mobile	1
Headphones	2
Keyboard	1

Mouse	2
-------	---

Q3. Show all customers and products

```

SELECT c.cust_name, p.prod_name
FROM customers c
LEFT JOIN orders o ON c.cust_id = o.cust_id
LEFT JOIN order_items oi ON o.order_id = oi.order_id
LEFT JOIN products p ON oi.prod_id = p.prod_id
UNION
SELECT c.cust_name, p.prod_name
FROM customers c
RIGHT JOIN orders o ON c.cust_id = o.cust_id
RIGHT JOIN order_items oi ON o.order_id = oi.order_id
RIGHT JOIN products p ON oi.prod_id = p.prod_id;

```

cust_name	prod_name
Anita	Keyboard
Neha	<i>NULL</i>
Priya	<i>NULL</i>
Ramesh	Laptop
Ramesh	Mobile
Suresh	Headphones
Vikram	Mouse

Q4. Show all orders and customers

```

SELECT c.cust_name, o.order_id
FROM customers c
LEFT JOIN orders o ON c.cust_id = o.cust_id
UNION
SELECT c.cust_name, o.order_id
FROM orders o
RIGHT JOIN customers c ON c.cust_id = o.cust_id;

```

Same 7-row result as Q1 above (all customers + the orphan order).

CROSS JOIN — Cartesian product (every combination)

Q1. Show all customers with all products → 6 customers × 5 products = **30 rows**

```
SELECT c.cust_name, p.prod_name
FROM customers c
CROSS JOIN products p;
```

Q2. Show all orders with all products → 5 orders × 5 products = **25 rows**

```
SELECT o.order_id, p.prod_name
FROM orders o
CROSS JOIN products p;
```

Q3. Show all customers with all orders → 6 customers × 5 orders = **30 rows**

```
SELECT c.cust_name, o.order_id
FROM customers c
CROSS JOIN orders o;
```

Q4. Show products with quantities 1 to 3 → 5 products × 3 quantities = **15 rows**

```
SELECT p.prod_name, q.qty
FROM products p
CROSS JOIN (SELECT 1 AS qty UNION SELECT 2 UNION SELECT 3) q;
```

Joins Exercise part-4

=====

```
CREATE TABLE departments (
  department_id INT PRIMARY KEY,
  department_name VARCHAR(100) NOT NULL,
  location VARCHAR(100),
  budget DECIMAL(10, 2),
  manager_name VARCHAR(100),
```



```

    established_date DATE,
    employee_count INT,
    headquarter BOOLEAN
);

INSERT INTO departments VALUES
(1, 'HR', 'New York', 50000.00, 'John Doe', '2000-01-01', 15, TRUE),
(2, 'Finance', 'Chicago', 75000.00, 'Jane Smith', '2005-06-15', 20, FALSE),
(3, 'Engineering', 'San Francisco', 100000.00, 'Jim Brown', '2010-09-10', 50, TRUE),
(4, 'Sales', 'Los Angeles', 60000.00, 'Jessica Green', '2012-03-20', 25, FALSE);

CREATE TABLE employees (
    employee_id INT PRIMARY KEY,
    employee_name VARCHAR(100) NOT NULL,
    department_id INT,
    job_title VARCHAR(100),
    salary DECIMAL(10, 2),
    hire_date DATE,
    status VARCHAR(50),
    email VARCHAR(100),
    FOREIGN KEY (department_id) REFERENCES departments(department_id)
);

INSERT INTO employees VALUES
(101, 'Alice', 1, 'HR Manager', 60000.00, '2015-05-01', 'Active', 'alice@example.com'),
(102, 'Bob', 2, 'Financial Analyst', 65000.00, '2016-07-15', 'Active', 'bob@example.com'),
(103, 'Charlie', 2, 'Accountant', 62000.00, '2017-08-30', 'Inactive', 'charlie@example.com'),
(104, 'David', 3, 'Software Engineer', 90000.00, '2018-01-10', 'Active',
'david@example.com'),
(105, 'Eve', NULL, 'Sales Representative', 55000.00, '2019-11-25', 'Active',
'eve@example.com');

```

The two facts this dataset is built to demonstrate:

- **Eve** (employee 105) has `department_id = NULL` → she's an employee with **no department**.
- **Sales** (department 4) has **no employees at all** → the only department with zero matching rows in `employees`.

1. INNER JOIN — only rows that match on both sides

Base query

```
SELECT * FROM departments  
INNER JOIN employees ON departments.department_id = employees.department_id;
```

→ **4 rows.** Eve is dropped (no department to match) and Sales is dropped (no employee to match).

Q1. Employees with their department names

```
SELECT e.employee_id, e.employee_name, d.department_name  
FROM employees e  
INNER JOIN departments d ON e.department_id = d.department_id;
```

employee_id	employee_name	department_name
101	Alice	HR
102	Bob	Finance
103	Charlie	Finance
104	David	Engineering

Q2. Employees with department names and budgets

```
SELECT e.employee_id, e.employee_name, d.department_name, d.budget  
FROM employees e  
INNER JOIN departments d ON e.department_id = d.department_id;
```

employee_id	employee_name	department_name	budget
101	Alice	HR	50000
102	Bob	Finance	75000
103	Charlie	Finance	75000
104	David	Engineering	100000

Q3. Employees with job titles and department location

```
SELECT e.employee_id, e.employee_name, e.job_title, d.location
FROM employees e
INNER JOIN departments d ON e.department_id = d.department_id;
```

employee_id	employee_name	job_title	location
101	Alice	HR Manager	New York
102	Bob	Financial Analyst	Chicago
103	Charlie	Accountant	Chicago
104	David	Software Engineer	San Francisco

Q4. Employees with email and department manager

```
SELECT e.employee_id, e.employee_name, e.email, d.manager_name
FROM employees e
INNER JOIN departments d ON e.department_id = d.department_id;
```

employee_id	employee_name	email	manager_name
101	Alice	alice@example.com	John Doe
102	Bob	bob@example.com	Jane Smith
103	Charlie	charlie@example.com	Jane Smith
104	David	david@example.com	Jim Brown

2. LEFT JOIN — all rows from `employees` + matches

Q1. All employees and their departments (incl. no department)

```
SELECT e.employee_id, e.employee_name, d.department_name
FROM employees e
LEFT JOIN departments d ON e.department_id = d.department_id;
```

employee_id	employee_name	department_name
-------------	---------------	-----------------

101	Alice	HR
102	Bob	Finance
103	Charlie	Finance
104	David	Engineering
105	Eve	<i>NULL</i>

Q2. Employees, salaries, department location (incl. no department)

```
SELECT e.employee_id, e.employee_name, e.salary, d.location
FROM employees e
LEFT JOIN departments d ON e.department_id = d.department_id;
```

employee_id	employee_name	salary	location
101	Alice	60000	New York
102	Bob	65000	Chicago
103	Charlie	62000	Chicago
104	David	90000	San Francisco
105	Eve	55000	<i>NULL</i>

Q3. Employees, status, department budget (incl. no department)

```
SELECT e.employee_id, e.employee_name, e.status, d.budget
FROM employees e
LEFT JOIN departments d ON e.department_id = d.department_id;
```

employee_id	employee_name	status	budget
101	Alice	Active	50000
102	Bob	Active	75000
103	Charlie	Inactive	75000
104	David	Active	100000
105	Eve	Active	<i>NULL</i>

Q4. Employees, hire date, department name (incl. no department)

```
SELECT e.employee_id, e.employee_name, e.hire_date, d.department_name
FROM employees e
LEFT JOIN departments d ON e.department_id = d.department_id;
```

employee_id	employee_name	hire_date	department_name
101	Alice	2015-05-01	HR
102	Bob	2016-07-15	Finance
103	Charlie	2017-08-30	Finance
104	David	2018-01-10	Engineering
105	Eve	2019-11-25	<i>NULL</i>

3. RIGHT JOIN — all rows from `departments` + matches

Q1. All departments and their employees (incl. no employees)

```
SELECT e.employee_id, e.employee_name, d.department_name
FROM employees e
RIGHT JOIN departments d ON e.department_id = d.department_id;
```

employee_id	employee_name	department_name
101	Alice	HR
102	Bob	Finance
103	Charlie	Finance
104	David	Engineering
<i>NULL</i>	<i>NULL</i>	Sales

Q2. Departments with manager and employee names (incl. no employees)

```
SELECT e.employee_id, e.employee_name, d.manager_name, d.department_name
FROM employees e
RIGHT JOIN departments d ON e.department_id = d.department_id;
```

employee_id	employee_name	manager_name	department_name
101	Alice	John Doe	HR
102	Bob	Jane Smith	Finance
103	Charlie	Jane Smith	Finance
104	David	Jim Brown	Engineering
<i>NULL</i>	<i>NULL</i>	Jessica Green	Sales

Q3. Departments with budgets and employees' job titles

```
SELECT e.job_title, d.budget, d.department_name
FROM employees e
RIGHT JOIN departments d ON e.department_id = d.department_id;
```

job_title	budget	department_name
HR Manager	50000	HR
Financial Analyst	75000	Finance
Accountant	75000	Finance
Software Engineer	100000	Engineering
<i>NULL</i>	60000	Sales

Q4. Departments with established dates and employee salaries

```
SELECT e.salary, d.established_date, d.department_name
FROM employees e
RIGHT JOIN departments d ON e.department_id = d.department_id;
```

salary	established_date	department_name
60000	2000-01-01	HR
65000	2005-06-15	Finance
62000	2005-06-15	Finance

90000	2010-09-10	Engineering
<i>NULL</i>	2012-03-20	Sales

4. FULL JOIN — all rows from both tables

Q1. All employees and departments, incl. no match

```
-- MySQL version:
SELECT e.employee_id, e.employee_name, d.department_name
FROM employees e
LEFT JOIN departments d ON e.department_id = d.department_id
UNION
SELECT e.employee_id, e.employee_name, d.department_name
FROM employees e
RIGHT JOIN departments d ON e.department_id = d.department_id;
```

employee_id	employee_name	department_name
101	Alice	HR
102	Bob	Finance
103	Charlie	Finance
104	David	Engineering
105	Eve	<i>NULL</i>
<i>NULL</i>	<i>NULL</i>	Sales

6 rows — every employee (including unassigned Eve) plus every department (including employee-less Sales).

Q2. Employees' job titles and departments, incl. no match

```
SELECT e.employee_id, e.employee_name, e.job_title, d.department_name
FROM employees e
LEFT JOIN departments d ON e.department_id = d.department_id
UNION
SELECT e.employee_id, e.employee_name, e.job_title, d.department_name
FROM employees e
```

RIGHT JOIN departments d ON e.department_id = d.department_id;

employee_id	employee_name	job_title	department_name
101	Alice	HR Manager	HR
102	Bob	Financial Analyst	Finance
103	Charlie	Accountant	Finance
104	David	Software Engineer	Engineering
105	Eve	Sales Representative	<i>NULL</i>
<i>NULL</i>	<i>NULL</i>	<i>NULL</i>	Sales

Q3. Employees' salary and departments' budget, incl. no match

```
SELECT e.employee_id, e.salary, d.department_name, d.budget
FROM employees e
LEFT JOIN departments d ON e.department_id = d.department_id
UNION
SELECT e.employee_id, e.salary, d.department_name, d.budget
FROM employees e
RIGHT JOIN departments d ON e.department_id = d.department_id;
```

employee_id	salary	department_name	budget
101	60000	HR	50000
102	65000	Finance	75000
103	62000	Finance	75000
104	90000	Engineering	100000
105	55000	<i>NULL</i>	<i>NULL</i>
<i>NULL</i>	<i>NULL</i>	Sales	60000

Q4. Employees' hire dates and departments' established dates, incl. no match

```
SELECT e.employee_id, e.hire_date, d.department_name, d.established_date
FROM employees e
LEFT JOIN departments d ON e.department_id = d.department_id
```


UNION

```
SELECT e.employee_id, e.hire_date, d.department_name, d.established_date
FROM employees e
RIGHT JOIN departments d ON e.department_id = d.department_id;
```

employee_id	hire_date	department_name	established_date
101	2015-05-01	HR	2000-01-01
102	2016-07-15	Finance	2005-06-15
103	2017-08-30	Finance	2005-06-15
104	2018-01-10	Engineering	2010-09-10
105	2019-11-25	<i>NULL</i>	<i>NULL</i>
<i>NULL</i>	<i>NULL</i>	Sales	2012-03-20

5. CROSS JOIN — Cartesian product, every combination

5 employees × 4 departments = **20 rows** for every query below.

Q1. All combinations of employees and departments

```
SELECT e.employee_id, e.employee_name, d.department_name
FROM employees e
CROSS JOIN departments d;
```

→ 20 rows: each of Alice/Bob/Charlie/David/Eve paired with each of HR/Finance/Engineering/Sales.

Q2. Combinations with employee salary and department budget → 20 rows

```
SELECT e.employee_id, e.employee_name, e.salary, d.department_name, d.budget
FROM employees e
CROSS JOIN departments d;
```

Q3. Combinations with job titles and department locations → 20 rows

```
SELECT e.employee_id, e.employee_name, e.job_title, d.department_name, d.location
FROM employees e
```

```
CROSS JOIN departments d;
```

Q4. Combinations with employee email and department managers → 20 rows

```
SELECT e.employee_id, e.employee_name, e.email, d.department_name, d.manager_name  
FROM employees e  
CROSS JOIN departments d;
```

Day-16

MySQL VIEWS

1. What Is a View?

A view is a virtual table in MySQL, built from the result of a stored SELECT statement. It looks and behaves like a regular table when you query it — you can SELECT from it, filter it, and join it — but it does not store any data of its own. Every time a view is queried, MySQL runs its underlying SELECT statement against the real ("base") tables and returns the result.

Think of a view as a saved, reusable query with a name. Instead of typing out a long JOIN every time, you create the view once, then simply run: `SELECT * FROM view_name;`

```
CREATE VIEW view_name AS
SELECT column1, column2, ...
FROM table_name
WHERE condition;
```

2. Why Do We Need Views?

- **Simplify complex queries:** Hide multi-table JOINS, subqueries, and calculations behind a single, simple table-like name.
- **Reuse query logic:** Write a complex SELECT once and reuse it everywhere — reports, applications, dashboards — without copy-pasting SQL.
- **Data security:** Expose only specific columns or rows to certain users, without granting them access to the full base table.
- **Different perspectives of the same data:** Different departments can have their own tailored 'view' of the same underlying tables.
- **Logical data independence:** If the underlying table structure changes slightly, you can often adjust just the view definition and keep all dependent queries working unchanged.

- **Simplified reporting:** Pre-package aggregations (totals, averages, counts) into a view so business users can query them without knowing SQL joins or GROUP BY.

3. Advantages of Views

- Security: restrict access to sensitive columns (e.g. salary) or sensitive rows (e.g. only a user's own department).
- Simplicity: end users and applications interact with a simple, named virtual table instead of complex SQL.
- Reusability: one view definition can be reused across many reports, applications, and ad-hoc queries.
- Consistency: business logic (calculations, filters) lives in one place, so results are always consistent.
- Maintainability: a business-rule change only needs to be made once, inside the view.
- Custom data presentation: rename or reformat columns, add computed columns, without touching the base table.
- No extra storage cost: a standard view stores no data — only its query definition.

4. What Happens Without Views?

Without views, every application, report, or analyst that needs the same joined/filtered/aggregated data must write out (and maintain) the full SQL logic independently. This leads to duplicated code, inconsistent results when the logic drifts between copies, harder security control, and painful maintenance whenever the business rules or schema change.

Aspect	Without Views	With Views
Repeated complex JOINS	Every report/query rewrites the same 5-table JOIN logic	JOIN logic is written once inside the view
Security	Users need direct table access; hard to hide specific columns/rows	Users query a view that exposes only approved columns/rows
Consistency	Business logic can drift between copies of the same query	Single source of truth — one definition used everywhere
Maintenance	A schema or rule change means updating every copy of the query	Change the view definition once; every consumer gets the update
Readability	Application code full of long, repeated SQL joins	Application code simply does <code>SELECT * FROM view_name</code>

Storage	N/A — no extra storage either way	No extra storage — a view is virtual, not a copy of data
---------	-----------------------------------	--

Creating Views

Basic Syntax:

```
CREATE VIEW view_name AS
```

```
SELECT columns
```

```
FROM tables
```

```
WHERE conditions;
```

Altering Views

Syntax for ALTER VIEW:

```
ALTER VIEW view_name AS
```

```
SELECT new_columns
```

```
FROM new_tables
```

```
WHERE new_conditions;
```

Note: Not all database systems support ALTER VIEW. Some require DROP and CREATE.

Dropping Views

Basic Syntax:

```
DROP VIEW view_name;
```

```
DROP VIEW IF EXISTS view_name; -- Safe drop
```

5. Real-Time Example: Database & Table Setup

The examples in this guide use a simple company database with two related tables: departments and employees.

```
CREATE TABLE departments (  
    dept_id INT PRIMARY KEY AUTO_INCREMENT,  
    dept_name VARCHAR(50),  
    location VARCHAR(50)  
);  
  
CREATE TABLE employees (  
    emp_id INT PRIMARY KEY AUTO_INCREMENT,  
    emp_name VARCHAR(100),  
    dept_id INT,  
    designation VARCHAR(50),  
    salary DECIMAL(10,2),  
    join_date DATE,  
    FOREIGN KEY (dept_id) REFERENCES departments(dept_id)  
);
```

Sample Data

```
INSERT INTO departments (dept_name, location) VALUES  
( 'IT','Hyderabad'), ( 'HR','Bangalore'), ( 'Finance','Chennai'), ( 'Sales','Mumbai');  
  
INSERT INTO employees (emp_name, dept_id, designation, salary, join_date) VALUES  
( 'Rahul',1,'Software Engineer',55000,'2021-06-15'),  
( 'Anita',2,'HR Executive',38000,'2020-03-21'),  
( 'Suresh',4,'Sales Manager',52000,'2019-08-10'),  
( 'Priya',1,'Team Lead',65000,'2022-01-05'),  
( 'Kiran',3,'Finance Analyst',48000,'2018-12-11'),  
( 'Deepika',2,'HR Manager',60000,'2019-10-19');
```

6. View Examples

6.1 Basic View — Employee-Department Join

Use case: avoid re-writing the JOIN between employees and departments every time someone needs an employee's department name.

```
CREATE VIEW vw_employee_department AS
SELECT e.emp_id, e.emp_name, e.designation, d.dept_name, d.location
FROM employees e
JOIN departments d ON e.dept_id = d.dept_id;

-- Usage
SELECT * FROM vw_employee_department;
```

6.2 Security View — Hiding Salary

Use case: HR generalists should see employee details, but salary must stay restricted to payroll staff only.

```
CREATE VIEW vw_employee_public AS
SELECT emp_id, emp_name, designation, dept_id, join_date
FROM employees;

-- Grant access to the view only, never to the base table
GRANT SELECT ON company.vw_employee_public TO 'hr_user'@'%';
```

6.3 Aggregate / Reporting View — Department Salary

Summary

Use case: management wants total and average salary per department without re-writing GROUP BY logic every time.

```
CREATE VIEW vw_department_salary_summary AS
SELECT d.dept_name,
       COUNT(e.emp_id) AS total_employees,
       SUM(e.salary)   AS total_salary,

       AVG(e.salary)   AS average_salary
FROM employees e
JOIN departments d ON e.dept_id = d.dept_id
GROUP BY d.dept_name;

-- Usage
SELECT * FROM vw_department_salary_summary;
```

Note: because this view uses GROUP BY and aggregate functions, it is read-only — it cannot be used for INSERT, UPDATE, or DELETE.

6.4 Filtered, Updatable View — WITH CHECK OPTION

Use case: an 'IT department only' view that stays consistent even when new rows are added through it.


```
CREATE VIEW vw_it_employees AS
SELECT emp_id, emp_name, designation, dept_id, salary
FROM employees
WHERE dept_id = 1
WITH CHECK OPTION;

-- This succeeds (dept_id matches the view's WHERE clause)
INSERT INTO vw_it_employees (emp_name, dept_id, designation, salary)
VALUES ('Vijay', 1, 'DevOps Engineer', 58000);

-- This fails because dept_id = 2 violates the view's WHERE condition
INSERT INTO vw_it_employees (emp_name, dept_id, designation, salary)
VALUES ('Sneha', 2, 'HR Assistant', 40000);
```

6.5 Computed Column View — Annual Salary

Use case: present a calculated column (annual salary) without altering the base table.

```
CREATE VIEW vw_employee_annual_salary AS
SELECT emp_name, salary AS monthly_salary, salary * 12 AS annual_salary
FROM employees;
```

6.6 Multi-Level View — Built on Another View

Use case: a 'high earners' view built directly on top of the department salary summary view.

```
CREATE VIEW vw_high_paying_departments AS
SELECT dept_name, average_salary
FROM vw_department_salary_summary
WHERE average_salary > 50000;
```

7. Updating Data Through Views

A view is updatable only when MySQL can unambiguously map each row of the view back to exactly one row of one base table. A view becomes non-updatable if it contains any of the following:

- Aggregate functions: SUM(), COUNT(), AVG(), MIN(), MAX()
- GROUP BY or HAVING
- DISTINCT

- UNION or UNION ALL
- Subqueries in the SELECT list that refer to the same base table
- Joins where the modified column doesn't map to a single base table

```
-- Updatable example
UPDATE vw_it_employees
SET salary = 60000
WHERE emp_name = 'Vijay';

-- Reflected immediately in the base employees table
```

8. Managing Views — ALTER, DROP, SHOW

```
-- View the exact definition of a view
SHOW CREATE VIEW vw_employee_department;

-- Modify an existing view's definition
ALTER VIEW vw_employee_department AS
SELECT e.emp_id, e.emp_name, d.dept_name
FROM employees e
JOIN departments d ON e.dept_id = d.dept_id;

-- Remove a view (base table data is untouched)
DROP VIEW vw_employee_department;

-- List all views in the current database
SELECT table_name
FROM information_schema.views
WHERE table_schema = 'jfs2';
```

9. Types of Views

- **Simple View:** Built from a single table, with no aggregate functions, GROUP BY, or joins. Usually fully updatable.
- **Complex View:** Built from multiple tables using joins, and/or includes aggregate functions, GROUP BY, or subqueries. Often read-only.
- **Updatable View:** A view (simple or a well-formed join) through which INSERT/UPDATE/DELETE can be performed.

- **Read-Only View:** Any view containing aggregates, GROUP BY, DISTINCT, or UNION — data can only be selected, never modified through it.
- **Materialized View (not native to MySQL):** A view whose results are physically stored and periodically refreshed. MySQL does not support this natively — it must be simulated using a real summary table refreshed via a scheduled EVENT or trigger.

10. Interview Questions — Basic Level

Q1. What is a view in MySQL?

A view is a virtual table based on the result of a stored SELECT query. It does not store data itself (in standard MySQL views) — it stores the query definition, and the data is fetched from the underlying base table(s) each time the view is used.

Q2. How do you create a view?

Use the CREATE VIEW statement followed by a SELECT query. Example: CREATE VIEW view_name AS SELECT column1, column2 FROM table_name WHERE condition;

Q3. How do you see the data inside a view?

You query it exactly like a table: SELECT * FROM view_name;

Q4. How do you check the underlying SQL definition of a view?

Use SHOW CREATE VIEW view_name; — this returns the exact CREATE VIEW statement used to define it.

Q5. How do you delete (remove) a view?

Use DROP VIEW view_name; This removes only the view definition, never the underlying table data.

Q6. Does a view store data physically?

No. A standard MySQL view does not store data. It stores only the query. Every time the view is accessed, MySQL re-executes the underlying query against the base table(s).

Q7. What is the main difference between a table and a view?

A table physically stores data on disk, while a view is a stored query that presents data drawn live from one or more tables. Views have no independent storage for rows.

Q8. Can a view be created using multiple tables?

Yes. A view can be based on a JOIN across two or more tables, exactly as a normal SELECT query can.

Q9. Can you use WHERE, GROUP BY, and ORDER BY inside a view definition?

Yes, a view can include WHERE, GROUP BY, HAVING, JOIN, and (with some restrictions in subqueries) ORDER BY clauses, since it is built from any valid SELECT statement.

Q10. Is it possible to rename a view?

MySQL has no direct RENAME VIEW command. The common approach is to create a new view with the desired name using the same SELECT definition, then drop the old view.

11. Interview Questions — Intermediate Level

Q1. What is an updatable view? What restrictions apply?

An updatable view allows INSERT, UPDATE, and DELETE operations that pass through to the base table. A view is updatable only if it has no aggregate functions (SUM, COUNT, AVG, etc.), no GROUP BY/HAVING/DISTINCT, no UNION, and comes from a single table (or a join where the modified columns map to one underlying table).

Q2. What is the WITH CHECK OPTION and why is it used?

WITH CHECK OPTION ensures that any INSERT or UPDATE performed through the view still satisfies the view's WHERE clause. Without it, a user could insert or update a row through a filtered view and the row would then silently disappear from that view's result set.

Q3. Can you insert data into a view? Under what conditions?

Yes, but only if the view is updatable (see above) and the INSERT provides values for all NOT NULL columns of the base table that are not included in the view (or those columns have defaults).

Q4. Does MySQL support materialized views?

Not natively. MySQL views are always virtual — they run the underlying query on every access. To get materialized-view-like behavior, you must manually create a real table and refresh it periodically (e.g. via a scheduled event or trigger), which acts as a workaround.

Q5. Can a view be created on top of another view?

Yes, MySQL allows nested views — a view can be defined using a SELECT that reads from another view. However, deeply nested views can hurt performance and make debugging harder.

Q6. What happens if you drop or alter a base table that a view depends on?

MySQL does not check view dependencies at DROP time. If you drop or rename a column/table used by a view, the view becomes invalid, and querying it will raise an error at execution time (MySQL uses a 'late binding' approach for views).

Q7. Do views improve query performance?

Not directly. A view is essentially a saved query — MySQL still executes the full underlying SELECT (join, filters, aggregation) every time the view is queried. Views improve readability, reusability, and security, not raw execution speed. In fact, deeply nested or poorly written views can slow things down.

Q8. How do you modify an existing view's definition?

Use ALTER VIEW view_name AS SELECT ...; to redefine it, or DROP VIEW followed by CREATE VIEW with the new definition.

Q9. Can indexes be created directly on a MySQL view?

No. Since a standard MySQL view has no physical storage, you cannot create an index directly on it. Performance for a view depends entirely on the indexes that exist on its underlying base tables.

Q10. Can a view include an ORDER BY clause, and does it always take effect?

A view can include ORDER BY, but if the outer query that selects from the view also has its own ORDER BY, the outer ORDER BY takes precedence, and the view's internal ordering may be ignored by the optimizer.

12. Interview Questions — Advanced Level

Q1. How do views help implement security and access control?

Views let you expose only specific columns or rows of a table to certain users (column-level and row-level security), without granting them direct access to the base table. For example, a view can omit a salary column entirely so HR staff can see employee details but not compensation data, and you grant SELECT only on the view, not on the table.

Q2. What are the key limitations of MySQL views?

Views cannot have their own indexes; they cannot use certain constructs (like variables, some subquery forms) that make them non-updatable; performance depends on the underlying query being re-run every time; and MySQL does not validate view dependencies automatically when base tables change.

Q3. Explain MySQL's ALGORITHM=MERGE vs ALGORITHM=TEMPTABLE for views.

MERGE algorithm rewrites the outer query and the view's query into a single query and pushes conditions down to the base tables — this is generally faster and allows the view to be updatable. TEMPTABLE algorithm executes the view's query first into an internal temporary table, then runs the outer query against that temp table — this is often slower and makes the view non-updatable. MySQL uses MERGE whenever possible and falls back to TEMPTABLE when the view contains aggregates, DISTINCT, GROUP BY, UNION, or subqueries in the SELECT list.

Q4. How would you find out which views depend on a specific table?

Query the `information_schema.VIEWS` table and inspect the `VIEW_DEFINITION` column (or use `information_schema.TABLES` together with view metadata) to search for references to the table name. There isn't a single built-in 'dependency' table, so text-searching `VIEW_DEFINITION` for the table name is the common approach: `SELECT table_name, view_definition FROM information_schema.views WHERE view_definition LIKE '%employees%';`

Q5. Can a view span multiple databases in MySQL?

Yes. A view's `SELECT` statement can reference tables from a different database using `db_name.table_name` syntax, as long as the user creating/using the view has the right privileges on both databases.

Q6. Describe a scenario where a view could hurt performance.

A reporting view that joins several large tables with aggregate functions, built on top of two or three other nested views, forces MySQL to use the `TEMPTABLE` algorithm and materialize large intermediate result sets in memory or on disk for every single query — even a simple `SELECT * FROM view WHERE emp_id = 5` ends up scanning and joining much more data than necessary.

Q7. What is the `DEFINER` clause in a view, and why does it matter?

When a view is created, MySQL records a `DEFINER` (the user account that created it). By default, the view executes with the privileges of that definer, not the querying user (`SQL SECURITY DEFINER`, the default). This lets you grant end users access to a view without granting them any direct privileges on the underlying base tables — the view runs with the definer's higher privileges. `SQL SECURITY INVOKER` changes this so the view runs with the querying user's own privileges instead.

Q8. Can views support aggregate functions and `GROUP BY`? Are such views updatable?

Yes, views can include `SUM()`, `COUNT()`, `AVG()`, `GROUP BY`, and `HAVING`. However, such views are never updatable — MySQL disallows `INSERT/UPDATE/DELETE` through any view containing aggregate functions, `GROUP BY`, `HAVING`, `DISTINCT`, or `UNION`, because there's no unambiguous way to map a change back to a single base-table row.

Q9. How can you achieve materialized-view-like behavior in MySQL, since it isn't supported natively?

Create a real physical 'summary' table that mirrors the view's result, and populate/refresh it periodically using a MySQL `EVENT` (scheduled job) or a trigger-driven update. Applications query this summary table directly for fast reads, while the refresh job keeps it in sync with the source tables — trading real-time accuracy for read performance.

Q10. If two views are nested (a view built on another view), how does MySQL execute the outer query?

MySQL expands nested views at parse time — the outer view's query is effectively rewritten to substitute the inner view's `SELECT` definition, then the whole thing is optimized together where possible using the `MERGE` algorithm. If any level requires `TEMPTABLE` (e.g. due to aggregation), MySQL materializes that intermediate result first, which can compound performance costs across multiple nesting levels.

13. Real-World Scenario-Based Questions

These questions are commonly asked in interviews to test whether a candidate understands practical, real-world use of views rather than just syntax.

Scenario 1. *HR staff need to see employee names, designations, and departments, but must NOT see salary information. How would you solve this using views?*

Solution: Create a view that selects only the non-sensitive columns (emp_id, emp_name, designation, dept_id) and excludes salary entirely. Grant HR staff SELECT privilege only on this view — not on the base employees table. They can never query a column that was never included in the view's definition.

Scenario 2. *Management wants one query that always shows total and average salary per department, without every analyst having to write the same JOIN + GROUP BY each time. How do views help?*

Solution: Create a reporting view (e.g. vw_department_salary_summary) that performs the JOIN between employees and departments with SUM(salary) and AVG(salary) grouped by department, once. Analysts then simply run SELECT * FROM vw_department_salary_summary; getting consistent, up-to-date figures without duplicating the underlying logic in every report or tool.

Scenario 3. *A 5-table join with complex business logic is copy-pasted into 20 different places in the application. What problem does this cause, and how do views fix it?*

Solution: Duplicating the same complex query in 20 places means every future business-rule change (a new filter, a renamed column, an extra join condition) has to be updated in 20 places, which is error-prone and inconsistent. Wrapping the logic in a single view centralizes it — the application code everywhere simply does SELECT ... FROM the_view, and any future change to the business logic is made once, inside the view definition.

Scenario 4. *The DBA wants to restrict staff to only certain columns of a sensitive table, without managing column-level GRANTS. How do views solve this?*

Solution: Create a view exposing only the approved columns, and grant privileges on the view instead of the base table. This achieves column-level security without MySQL's more cumbersome column-level GRANT syntax, and it's easy to audit since all restricted access flows through one well-named view.

Scenario 5. *A view is created with WITH CHECK OPTION and a WHERE clause restricting it to department = 'IT'. A user tries to INSERT a row through this view with department = 'HR'. What happens?*

Solution: The INSERT is rejected by MySQL with an error, because WITH CHECK OPTION enforces that any row inserted or updated through the view must still satisfy the view's WHERE condition (department = 'IT'). Without WITH CHECK OPTION, the insert would silently succeed in the base table but the new HR row would simply not appear when querying the view again.

Scenario 6. *A view was created using `ALGORITHM=TEMPTABLE` because it contains a `GROUP BY`. What's the practical impact for the team using it?*

Solution: The view will always be read-only (no INSERT/UPDATE/DELETE through it), and every SELECT against it forces MySQL to build an internal temporary table first — which is slower than the MERGE approach, especially on large base tables. For a frequently queried report, the team should consider periodically caching the results in a real summary table instead of hitting this view constantly.

Scenario 7. *A column referenced by a view gets renamed by another team during a migration. What happens to the view, and how would you catch this early?*

Solution: MySQL does not validate views against schema changes at the time the column is renamed — the view keeps its old definition text. The problem only surfaces when someone actually queries the view and gets an 'Unknown column' error. To catch this early, teams should search `information_schema.views` for references to any column/table before making a breaking schema change, or add automated tests that run `SELECT * FROM view_name;` against a staging database as part of the migration checklist.

Scenario 8. *Your reporting dashboard needs fast reads on aggregated sales data, but MySQL views don't cache or materialize results. How would you work around this?*

Solution: Build a real, physical 'materialized view' equivalent: a summary table (e.g. `daily_sales_summary`) populated by a scheduled MySQL EVENT or an ETL job that runs the aggregate query and stores the results. The dashboard reads from this fast, indexed summary table instead of recomputing the aggregation on every page load, accepting a small refresh-interval delay in exchange for speed.

What is Subquery

=====

a subquery is a query inside another query.
it is written inside parenthesis() and its result is used by the outer query.

it is also called.
inner Query
Nested Query
inner select.

general syntax

select column-list
from table
where column OPERATOR(SELECT COLUMN FROM TABLE WHERE CONDITION)

purpose of subquery

to break down complex problems into smaller queries.
to use the result of one query as input to another
to avoid multiple steps(write in a single Query insted of temp tables)

types of subqueries

-
1. Single-row-subquery
 2. multi-row subquery
 3. nested subquery.
 4. Correlated subquery.
 5. Exists/not Exists subquery

1. Single-row-subquery

1. Single Row Subqueries

Single row subqueries return exactly one row and one column. They use comparison operators like =, >, <, >=, <=, !=.

Characteristics:

- Returns only one value
- Can use single-value comparison operators
- If subquery returns no rows or multiple rows, it causes an error

Syntax:

SELECT columns

FROM table

WHERE column operator (SELECT column FROM table WHERE condition);

2. Multi-row Subquery (IN)

1. Multi-Row Subqueries with IN

A Multi-Row Subquery returns more than one row. The **IN** operator is used when you want to compare a value with multiple values returned by the subquery.

Characteristics

- Returns multiple rows
- Uses the **IN** operator
- Checks whether a value exists in the list returned by the subquery
- Equivalent to multiple **OR** conditions
- Most commonly used for matching values

Syntax

```
SELECT columns  
FROM table  
WHERE column IN  
    (SELECT column  
     FROM table  
     WHERE condition);
```

3. Multi-row Subquery (ANY / SOME)

Multi row subqueries return multiple rows. When using ANY or SOME (they are synonymous), the condition is true if it matches at least one value returned by the subquery.

Characteristics:

- Returns multiple values
- ANY and SOME are interchangeable
- Condition is true if ANY of the values satisfy the condition
- Often used with IN, >ANY, <ANY, =ANY, etc.

Syntax:

```
SELECT columns  
  
FROM table
```

WHERE column operator ANY/SOME (SELECT column FROM table WHERE condition);

4. Multi-row Subquery (ALL)

The ALL operator means the condition must be true for ALL values returned by the subquery.

Characteristics:

- Condition must be true for ALL values in the subquery result
- Often used with >ALL, <ALL, >=ALL, <=ALL
- >ALL means greater than the maximum value
- <ALL means less than the minimum value

Syntax:

SELECT columns

FROM table

WHERE column operator ALL (SELECT column FROM table WHERE condition);

5. Nested Subquery

Nested subqueries are subqueries within subqueries. They can be multiple levels deep and are executed from the innermost to the outermost.

Characteristics:

- Multiple levels of nesting
- Inner queries execute first
- Can combine different types of subqueries
- Useful for complex conditions

Syntax:

```
SELECT column_list
FROM table
WHERE column = (SELECT column
                FROM table
                WHERE column = (SELECT column
                                FROM table
                                WHERE condition));
```

6. Correlated Subquery

Correlated subqueries are subqueries that reference columns from the outer query. Unlike regular subqueries that execute once, correlated subqueries execute once for each row processed by the outer query.

Characteristics:

- References columns from the outer query
- Executes multiple times (once per outer row)
- Cannot be executed independently
- Often used with EXISTS, NOT EXISTS
- Generally slower than non-correlated subqueries

Syntax

```
SELECT columns
FROM table1 outer_alias
WHERE condition (SELECT column
                FROM table2 inner_alias
                WHERE inner_alias.column = outer_alias.column);
```

7. EXISTS / NOT EXISTS Subquery

EXISTS and NOT EXISTS

EXISTS is used to test whether a subquery returns any rows. It returns TRUE if the subquery returns at least one row, FALSE otherwise. NOT EXISTS returns TRUE if the subquery returns no rows.

Characteristics:

- Returns TRUE/FALSE (boolean result)
- More efficient than IN for large datasets
- Often used with correlated subqueries
- Stops execution as soon as first match is found
- SELECT clause in EXISTS doesn't matter (often use SELECT 1)

Syntax:

SELECT columns

FROM table1

WHERE EXISTS (SELECT 1 FROM table2 WHERE condition);

SELECT columns

FROM table1

WHERE NOT EXISTS (SELECT 1 FROM table2 WHERE condition);

Country Tables and Records

=====

```
CREATE TABLE country (  
  Code      VARCHAR(3) PRIMARY KEY,  
  Name      VARCHAR(100) NOT NULL,  
  Continent VARCHAR(30) NOT NULL,  
  Region    VARCHAR(50) NOT NULL,  
  Population BIGINT NOT NULL,  
  LifeExpectancy DECIMAL(4,1),
```

```
    GNP          DECIMAL(15,2),
    Capital      INT
);
```

```
CREATE TABLE city (
    ID           INT PRIMARY KEY AUTO_INCREMENT,
    Name         VARCHAR(100) NOT NULL,
    CountryCode  VARCHAR(3)  NOT NULL,
    District     VARCHAR(50),
    Population   BIGINT      NOT NULL
);
```

```
CREATE TABLE countrylanguage (
    CountryCode  VARCHAR(3) NOT NULL,
    Language     VARCHAR(50) NOT NULL,
    IsOfficial   ENUM('T','F') NOT NULL,
    Percentage    DECIMAL(4,1)
);
```

```
-- 2. INSERT RECORDS INTO city (95 rows) — insert BEFORE country, since
--   country.Capital references city.ID
```

```
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(1,'Berlin','DEU','Berlin',3677000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(2,'Hamburg','DEU','Hamburg',1900000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(3,'Munich','DEU','Bavaria',1500000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(4,'Paris','FRA','Ile-de-France',2140000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(5,'Marseille','FRA','PACA',870000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(6,'Lyon','FRA','Auvergne-Rhone-Alpes',520000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(7,'London','GBR','England',8980000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(8,'Birmingham','GBR','England',1145000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(9,'Manchester','GBR','England',553000);
```

```

INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(10,'Rome','ITA','Lazio',2870000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(11,'Milan','ITA','Lombardy',1370000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(12,'Naples','ITA','Campania',950000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(13,'Madrid','ESP','Madrid',3280000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(14,'Barcelona','ESP','Catalonia',1620000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(15,'Valencia','ESP','Valencia',800000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(16,'Warsaw','POL','Masovian',1860000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(17,'Krakow','POL','Lesser Poland',780000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(18,'Amsterdam','NLD','North Holland',905000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(19,'Rotterdam','NLD','South Holland',656000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(20,'Brussels','BEL','Brussels',1220000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(21,'Antwerp','BEL','Antwerp',530000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(22,'Stockholm','SWE','Stockholm',980000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(23,'Gothenburg','SWE','Vastra Gotaland',580000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(24,'Lisbon','PRT','Lisbon',545000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(25,'Porto','PRT','Porto',230000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(26,'Athens','GRC','Attica',660000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(27,'Thessaloniki','GRC','Central Macedonia',320000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(28,'Monaco-Ville','MCO','Monaco',1300);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(29,'Abuja','NGA','Federal Capital Territory',1780000);

```

INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (30,'Lagos','NGA','Lagos',15400000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (31,'Kano','NGA','Kano',4100000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (32,'Cairo','EGY','Cairo',10100000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (33,'Alexandria','EGY','Alexandria',5280000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (34,'Pretoria','ZAF','Gauteng',2470000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (35,'Johannesburg','ZAF','Gauteng',5780000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (36,'Cape Town','ZAF','Western Cape',4710000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (37,'Nairobi','KEN','Nairobi',4400000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (38,'Mombasa','KEN','Mombasa',1300000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (39,'Addis Ababa','ETH','Addis Ababa',5230000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (40,'Accra','GHA','Greater Accra',2560000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (41,'Kumasi','GHA','Ashanti',3350000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (42,'Rabat','MAR','Rabat-Sale',580000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (43,'Casablanca','MAR','Casablanca-Settat',3750000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (44,'Algiers','DZA','Algiers',2760000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (45,'Oran','DZA','Oran',850000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (46,'Beijing','CHN','Beijing',21890000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (47,'Shanghai','CHN','Shanghai',24870000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (48,'Shenzhen','CHN','Guangdong',17560000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (49,'New Delhi','IND','Delhi',34700000);


```

INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(50,'Mumbai','IND','Maharashtra',21300000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(51,'Bengaluru','IND','Karnataka',13600000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(52,'Tokyo','JPN','Tokyo',37400000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(53,'Osaka','JPN','Osaka',19100000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(54,'Jakarta','IDN','Jakarta',10900000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(55,'Surabaya','IDN','East Java',2870000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(56,'Islamabad','PAK','Islamabad',1140000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(57,'Karachi','PAK','Sindh',16800000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(58,'Lahore','PAK','Punjab',13000000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(59,'Dhaka','BGD','Dhaka',22400000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(60,'Chittagong','BGD','Chittagong',5200000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(61,'Hanoi','VNM','Hanoi',8400000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (62,'Ho Chi
Minh City','VNM','Ho Chi Minh',9300000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(63,'Bangkok','THA','Bangkok',10700000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (64,'Chiang
Mai','THA','Chiang Mai',1200000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(65,'Seoul','KOR','Seoul',9500000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(66,'Busan','KOR','Busan',3400000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(67,'Riyadh','SAU','Riyadh',7680000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(68,'Jeddah','SAU','Makkah',4780000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(69,'Washington','USA','District of Columbia',690000);

```

INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (70,'New York','USA','New York',8800000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (71,'Los Angeles','USA','California',3900000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (72,'Chicago','USA','Illinois',2700000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (73,'Ottawa','CAN','Ontario',1070000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (74,'Toronto','CAN','Ontario',2930000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (75,'Vancouver','CAN','British Columbia',675000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (76,'Mexico City','MEX','Distrito Federal',9200000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (77,'Guadalajara','MEX','Jalisco',1500000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (78,'Brasilia','BRA','Distrito Federal',3100000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (79,'Sao Paulo','BRA','Sao Paulo',12300000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (80,'Rio de Janeiro','BRA','Rio de Janeiro',6750000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (81,'Buenos Aires','ARG','Buenos Aires',3080000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (82,'Cordoba','ARG','Cordoba',1450000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (83,'Bogota','COL','Bogota',7970000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (84,'Medellin','COL','Antioquia',2530000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (85,'Lima','PER','Lima',9700000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (86,'Arequipa','PER','Arequipa',1080000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (87,'Santiago','CHL','Santiago',6270000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (88,'Valparaiso','CHL','Valparaiso',300000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (89,'Caracas','VEN','Distrito Capital',2900000);

```

INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(90,'Maracaibo','VEN','Zulia',1650000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(91,'Canberra','AUS','Australian Capital Territory',470000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(92,'Sydney','AUS','New South Wales',5310000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(93,'Melbourne','AUS','Victoria',5150000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(94,'Wellington','NZL','Wellington',215000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(95,'Auckland','NZL','Auckland',1700000);

```

-- 3. INSERT RECORDS INTO country (41 rows)

```

INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('ARG','Argentina','South America','South America',45800000,76.7,630000.00,81);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('AUS','Australia','Oceania','Australia and New Zealand',26600000,83.3,1690000.00,91);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('BEL','Belgium','Europe','Western Europe',11600000,82.0,590000.00,20);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('BGD','Bangladesh','Asia','Southern and Central Asia',172000000,72.0,460000.00,59);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('BRA','Brazil','South America','South America',216400000,75.9,1920000.00,78);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('CAN','Canada','North America','North America',38900000,82.4,2140000.00,73);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('CHL','Chile','South America','South America',19600000,80.2,317000.00,87);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('CHN','China','Asia','Eastern Asia',1412000000,78.2,17960000.00,46);

```

```

INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('COL','Colombia','South America','South America',52200000,73.0,343000.00,83);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('DEU','Germany','Europe','Western Europe',83200000,81.0,4260000.00,1);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('DZA','Algeria','Africa','Northern Africa',45400000,77.1,195000.00,44);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('EGY','Egypt','Africa','Northern Africa',112700000,71.8,435000.00,32);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('ESP','Spain','Europe','Southern Europe',47400000,83.5,1430000.00,13);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('ETH','Ethiopia','Africa','Eastern Africa',126500000,66.2,126000.00,39);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('FRA','France','Europe','Western Europe',68000000,82.5,2780000.00,4);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('GBR','United Kingdom','Europe','British Isles',67300000,81.2,3130000.00,7);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('GHA','Ghana','Africa','Western Africa',33500000,64.6,77000.00,40);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('GRC','Greece','Europe','Southern Europe',10400000,81.9,210000.00,26);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('IDN','Indonesia','Asia','Southeast Asia',277500000,68.0,1320000.00,54);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('IND','India','Asia','Southern and Central Asia',1428000000,70.2,3730000.00,49);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('ITA','Italy','Europe','Southern Europe',59000000,83.2,2010000.00,10);

```

```

INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('JPN','Japan','Asia','Eastern Asia',123300000,84.5,4230000.00,52);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('KEN','Kenya','Africa','Eastern Africa',55100000,66.7,113000.00,37);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('KOR','South Korea','Asia','Eastern Asia',51700000,83.5,1670000.00,65);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('MAR','Morocco','Africa','Northern Africa',37800000,74.0,135000.00,42);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('MCO','Monaco','Europe','Southern Europe',39000,85.9,7500.00,28);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('MEX','Mexico','North America','Central America',128900000,70.7,1420000.00,76);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('NGA','Nigeria','Africa','Western Africa',223800000,55.8,477000.00,29);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('NLD','Netherlands','Europe','Western Europe',17700000,82.3,990000.00,18);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('NZL','New Zealand','Oceania','Australia and New Zealand',5200000,82.5,250000.00,94);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('PAK','Pakistan','Asia','Southern and Central Asia',240500000,66.6,375000.00,56);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('PER','Peru','South America','South America',34400000,72.9,242000.00,85);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('POL','Poland','Europe','Eastern Europe',37700000,78.0,690000.00,16);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('PRT','Portugal','Europe','Southern Europe',10300000,82.1,250000.00,24);

```

```

INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('SAU','Saudi Arabia','Asia','Middle East',36900000,76.9,1010000.00,67);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('SWE','Sweden','Europe','Nordic Countries',10500000,83.0,590000.00,22);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('THA','Thailand','Asia','Southeast Asia',71800000,79.3,495000.00,63);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('USA','United States','North America','North America',339900000,79.3,25460000.00,69);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('VEN','Venezuela','South America','South America',28300000,72.1,90000.00,89);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('VNM','Vietnam','Asia','Southeast Asia',98900000,73.6,410000.00,61);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('ZAF','South Africa','Africa','Southern Africa',60400000,65.3,405000.00,34);

```

-- 4. INSERT RECORDS INTO countrylanguage (60 rows)

```

INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('ARG','Spanish','T',96.8);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('AUS','English','T',81.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('BEL','French','T',40.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('BEL','Dutch','T',59.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('BGD','Bengali','T',98.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('BRA','Portuguese','T',97.5);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('CAN','English','T',58.7);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('CAN','French','T',22.0);

```

```

INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('CHL','Spanish','T',99.5);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('CHN','Chinese','T',92.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('COL','Spanish','T',99.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('DEU','German','T',92.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('DZA','Berber languages','T',17.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('DZA','Arabic','T',83.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('EGY','Arabic','T',98.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('ESP','Catalan','F',17.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('ESP','Spanish','T',74.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('ETH','Oromo','F',25.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('ETH','Amharic','T',30.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('FRA','French','T',90.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('GBR','English','T',95.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('GHA','English','T',100.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('GRC','Greek','T',99.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('IDN','Indonesian','T',100.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('IND','Bengali','F',8.1);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('IND','Hindi','T',41.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('IND','English','T',12.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('ITA','Italian','T',93.0);

```

```

INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('JPN','Japanese','T',99.1);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('KEN','Swahili','T',60.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('KEN','English','T',20.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('KOR','Korean','T',100.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('MAR','Arabic','T',65.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('MAR','Berber languages','T',28.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('MCO','Monegasque','F',12.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('MCO','French','T',58.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('MEX','Spanish','T',92.7);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('NGA','English','T',100.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('NGA','Hausa','F',25.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('NGA','Yoruba','F',21.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('NLD','Dutch','T',95.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('NZL','English','T',91.2);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('NZL','Maori','T',4.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('PAK','English','T',1.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('PAK','Urdu','T',8.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('PAK','Punjabi','F',48.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('PER','Spanish','T',83.9);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('PER','Quechua','T',13.0);

```



```

INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('POL','Polish','T',97.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('PRT','Portuguese','T',99.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('SAU','Arabic','T',95.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('SWE','Swedish','T',89.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('THA','Thai','T',97.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('USA','English','T',86.2);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('USA','Spanish','F',9.7);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('VEN','Spanish','T',96.9);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('VNM','Vietnamese','T',87.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('ZAF','Afrikaans','T',13.5);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('ZAF','English','T',9.6);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('ZAF','Zulu','T',23.0);

```

-- 5. (OPTIONAL) FOREIGN KEY CONSTRAINTS — add these AFTER all data above is loaded

```

ALTER TABLE city ADD CONSTRAINT fk_city_country FOREIGN KEY
(CountryCode) REFERENCES country(Code);
ALTER TABLE country ADD CONSTRAINT fk_country_capital FOREIGN KEY
(Capital) REFERENCES city(ID);
ALTER TABLE countrylanguage ADD CONSTRAINT fk_lang_country FOREIGN KEY
(CountryCode) REFERENCES country(Code);

```

```

select*from city ;
select*from country;
select*from countrylanguage;

```

If You Want Only 1st, 2nd, 3rd Separately

-- 1st Min Salary

```
SELECT MIN(salary) AS First_Min FROM Employee;
```

-- 2nd Min Salary

```
SELECT MIN(salary) AS Second_Min  
FROM Employee  
WHERE salary > (SELECT MIN(salary) FROM Employee);
```

-- 3rd Min Salary

```
SELECT MIN(salary) AS Third_Min  
FROM Employee  
WHERE salary > (  
    SELECT MIN(salary)  
    FROM Employee  
    WHERE salary > (SELECT MIN(salary) FROM Employee)  
);
```

-- 1st Max Salary

```
SELECT MAX(salary) AS First_Max FROM Employee;
```

-- 2nd Max Salary

```
SELECT MAX(salary) AS Second_Max  
FROM Employee  
WHERE salary < (SELECT MAX(salary) FROM Employee);
```

-- 3rd Max Salary

```
SELECT MAX(salary) AS Third_Max  
FROM Employee  
WHERE salary < (  
    SELECT MAX(salary)  
    FROM Employee  
    WHERE salary < (SELECT MAX(salary) FROM Employee)  
);
```

Department-wise 2nd Min Salary

```
SELECT d.dept_id, d.dept_name, MIN(e.salary) AS Second_Min_Salary
FROM Employee e
JOIN Department d ON e.dept_id = d.dept_id
WHERE e.salary > (
    SELECT MIN(salary)
    FROM Employee e2
    WHERE e2.dept_id = e.dept_id
)
GROUP BY d.dept_id, d.dept_name;
```

Department-wise 2nd Max Salary

```
SELECT d.dept_id, d.dept_name, MAX(e.salary) AS Second_Max_Salary
FROM Employee e
JOIN Department d ON e.dept_id = d.dept_id
WHERE e.salary < (
    SELECT MAX(salary)
    FROM Employee e2
    WHERE e2.dept_id = e.dept_id
)
GROUP BY d.dept_id, d.dept_name;
```

For 3rd Min and 3rd Max

You just nest one more level:

- **3rd Min Salary**

```
SELECT d.dept_id, d.dept_name, MIN(e.salary) AS Third_Min_Salary
FROM Employee e
JOIN Department d ON e.dept_id = d.dept_id
WHERE e.salary > (
    SELECT MIN(salary)
    FROM Employee e2
    WHERE e2.dept_id = e.dept_id
)
```

```

        AND e2.salary > (
            SELECT MIN(salary)
            FROM Employee e3
            WHERE e3.dept_id = e2.dept_id
        )
    )
GROUP BY d.dept_id, d.dept_name;

```

- **3rd Max Salary**

```

SELECT d.dept_id, d.dept_name, MAX(e.salary) AS Third_Max_Salary
FROM Employee e
JOIN Department d ON e.dept_id = d.dept_id
WHERE e.salary < (
    SELECT MAX(salary)
    FROM Employee e2
    WHERE e2.dept_id = e.dept_id
    AND e2.salary < (
        SELECT MAX(salary)
        FROM Employee e3
        WHERE e3.dept_id = e2.dept_id
    )
)
GROUP BY d.dept_id, d.dept_name;

```

```

=====
=====
=====
=====

```

PART 1 - TABLE CREATION AND DATA INSERT STATEMENTS

Copy everything in this section directly into your MySQL / MariaDB client
(or Google Cloud SQL, phpMyAdmin, etc.) to recreate the exact same database.

```

=====
=====

```

-- 1. CREATE TABLES

```

CREATE TABLE country (

```

```

    Code      VARCHAR(3) PRIMARY KEY,
    Name      VARCHAR(100) NOT NULL,
    Continent  VARCHAR(30) NOT NULL,
    Region    VARCHAR(50) NOT NULL,
    Population BIGINT   NOT NULL,
    LifeExpectancy DECIMAL(4,1),
    GNP       DECIMAL(15,2),
    Capital   INT
);

```

```

CREATE TABLE city (
    ID      INT PRIMARY KEY AUTO_INCREMENT,
    Name    VARCHAR(100) NOT NULL,
    CountryCode VARCHAR(3) NOT NULL,
    District VARCHAR(50),
    Population BIGINT   NOT NULL
);

```

```

CREATE TABLE countrylanguage (
    CountryCode VARCHAR(3) NOT NULL,
    Language    VARCHAR(50) NOT NULL,
    IsOfficial  ENUM('T','F') NOT NULL,
    Percentage  DECIMAL(4,1)
);

```

```

-- 2. INSERT RECORDS INTO city (95 rows) — insert BEFORE country, since
--    country.Capital references city.ID

```

```

INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(1,'Berlin','DEU','Berlin',3677000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(2,'Hamburg','DEU','Hamburg',1900000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(3,'Munich','DEU','Bavaria',1500000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(4,'Paris','FRA','Ile-de-France',2140000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(5,'Marseille','FRA','PACA',870000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(6,'Lyon','FRA','Auvergne-Rhone-Alpes',520000);

```

INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(7,'London','GBR','England',8980000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(8,'Birmingham','GBR','England',1145000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(9,'Manchester','GBR','England',553000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(10,'Rome','ITA','Lazio',2870000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(11,'Milan','ITA','Lombardy',1370000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(12,'Naples','ITA','Campania',950000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(13,'Madrid','ESP','Madrid',3280000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(14,'Barcelona','ESP','Catalonia',1620000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(15,'Valencia','ESP','Valencia',800000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(16,'Warsaw','POL','Masovian',1860000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(17,'Krakow','POL','Lesser Poland',780000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(18,'Amsterdam','NLD','North Holland',905000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(19,'Rotterdam','NLD','South Holland',656000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(20,'Brussels','BEL','Brussels',1220000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(21,'Antwerp','BEL','Antwerp',530000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(22,'Stockholm','SWE','Stockholm',980000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(23,'Gothenburg','SWE','Vastra Gotaland',580000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(24,'Lisbon','PRT','Lisbon',545000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(25,'Porto','PRT','Porto',230000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(26,'Athens','GRC','Attica',660000);

```

INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(27,'Thessaloniki','GRC','Central Macedonia',320000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(28,'Monaco-Ville','MCO','Monaco',1300);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(29,'Abuja','NGA','Federal Capital Territory',1780000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(30,'Lagos','NGA','Lagos',15400000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(31,'Kano','NGA','Kano',4100000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(32,'Cairo','EGY','Cairo',10100000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(33,'Alexandria','EGY','Alexandria',5280000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(34,'Pretoria','ZAF','Gauteng',2470000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(35,'Johannesburg','ZAF','Gauteng',5780000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (36,'Cape
Town','ZAF','Western Cape',4710000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(37,'Nairobi','KEN','Nairobi',4400000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(38,'Mombasa','KEN','Mombasa',1300000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (39,'Addis
Ababa','ETH','Addis Ababa',5230000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(40,'Accra','GHA','Greater Accra',2560000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(41,'Kumasi','GHA','Ashanti',3350000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(42,'Rabat','MAR','Rabat-Sale',580000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(43,'Casablanca','MAR','Casablanca-Settat',3750000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(44,'Algiers','DZA','Algiers',2760000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(45,'Oran','DZA','Oran',850000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(46,'Beijing','CHN','Beijing',21890000);

```

```

INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(47,'Shanghai','CHN','Shanghai',24870000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(48,'Shenzhen','CHN','Guangdong',17560000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (49,'New
Delhi','IND','Delhi',34700000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(50,'Mumbai','IND','Maharashtra',21300000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(51,'Bengaluru','IND','Karnataka',13600000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(52,'Tokyo','JPN','Tokyo',37400000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(53,'Osaka','JPN','Osaka',19100000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(54,'Jakarta','IDN','Jakarta',10900000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(55,'Surabaya','IDN','East Java',2870000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(56,'Islamabad','PAK','Islamabad',1140000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(57,'Karachi','PAK','Sindh',16800000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(58,'Lahore','PAK','Punjab',13000000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(59,'Dhaka','BGD','Dhaka',22400000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(60,'Chittagong','BGD','Chittagong',5200000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(61,'Hanoi','VNM','Hanoi',8400000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (62,'Ho Chi
Minh City','VNM','Ho Chi Minh',9300000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(63,'Bangkok','THA','Bangkok',10700000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (64,'Chiang
Mai','THA','Chiang Mai',1200000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(65,'Seoul','KOR','Seoul',9500000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(66,'Busan','KOR','Busan',3400000);

```


INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (67,'Riyadh','SAU','Riyadh',7680000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (68,'Jeddah','SAU','Makkah',4780000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (69,'Washington','USA','District of Columbia',690000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (70,'New York','USA','New York',8800000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (71,'Los Angeles','USA','California',3900000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (72,'Chicago','USA','Illinois',2700000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (73,'Ottawa','CAN','Ontario',1070000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (74,'Toronto','CAN','Ontario',2930000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (75,'Vancouver','CAN','British Columbia',675000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (76,'Mexico City','MEX','Distrito Federal',9200000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (77,'Guadalajara','MEX','Jalisco',1500000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (78,'Brasilia','BRA','Distrito Federal',3100000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (79,'Sao Paulo','BRA','Sao Paulo',12300000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (80,'Rio de Janeiro','BRA','Rio de Janeiro',6750000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (81,'Buenos Aires','ARG','Buenos Aires',3080000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (82,'Cordoba','ARG','Cordoba',1450000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (83,'Bogota','COL','Bogota',7970000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (84,'Medellin','COL','Antioquia',2530000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (85,'Lima','PER','Lima',9700000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES (86,'Arequipa','PER','Arequipa',1080000);

```

INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(87,'Santiago','CHL','Santiago',6270000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(88,'Valparaiso','CHL','Valparaiso',300000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(89,'Caracas','VEN','Distrito Capital',2900000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(90,'Maracaibo','VEN','Zulia',1650000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(91,'Canberra','AUS','Australian Capital Territory',470000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(92,'Sydney','AUS','New South Wales',5310000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(93,'Melbourne','AUS','Victoria',5150000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(94,'Wellington','NZL','Wellington',215000);
INSERT INTO city (ID,Name,CountryCode,District,Population) VALUES
(95,'Auckland','NZL','Auckland',1700000);

```

-- 3. INSERT RECORDS INTO country (41 rows)

```

INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('ARG','Argentina','South America','South America',45800000,76.7,630000.00,81);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('AUS','Australia','Oceania','Australia and New Zealand',26600000,83.3,1690000.00,91);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('BEL','Belgium','Europe','Western Europe',11600000,82.0,590000.00,20);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('BGD','Bangladesh','Asia','Southern and Central Asia',172000000,72.0,460000.00,59);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('BRA','Brazil','South America','South America',216400000,75.9,1920000.00,78);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('CAN','Canada','North America','North America',38900000,82.4,2140000.00,73);

```

```

INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('CHL','Chile','South America','South America',19600000,80.2,317000.00,87);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('CHN','China','Asia','Eastern Asia',1412000000,78.2,17960000.00,46);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('COL','Colombia','South America','South America',52200000,73.0,343000.00,83);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('DEU','Germany','Europe','Western Europe',83200000,81.0,4260000.00,1);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('DZA','Algeria','Africa','Northern Africa',45400000,77.1,195000.00,44);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('EGY','Egypt','Africa','Northern Africa',112700000,71.8,435000.00,32);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('ESP','Spain','Europe','Southern Europe',47400000,83.5,1430000.00,13);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('ETH','Ethiopia','Africa','Eastern Africa',126500000,66.2,126000.00,39);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('FRA','France','Europe','Western Europe',68000000,82.5,2780000.00,4);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('GBR','United Kingdom','Europe','British Isles',67300000,81.2,3130000.00,7);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('GHA','Ghana','Africa','Western Africa',33500000,64.6,77000.00,40);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('GRC','Greece','Europe','Southern Europe',10400000,81.9,210000.00,26);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('IDN','Indonesia','Asia','Southeast Asia',277500000,68.0,1320000.00,54);

```

```

INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('IND','India','Asia','Southern and Central Asia',1428000000,70.2,3730000.00,49);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('ITA','Italy','Europe','Southern Europe',59000000,83.2,2010000.00,10);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('JPN','Japan','Asia','Eastern Asia',123300000,84.5,4230000.00,52);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('KEN','Kenya','Africa','Eastern Africa',55100000,66.7,113000.00,37);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('KOR','South Korea','Asia','Eastern Asia',51700000,83.5,1670000.00,65);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('MAR','Morocco','Africa','Northern Africa',37800000,74.0,135000.00,42);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('MCO','Monaco','Europe','Southern Europe',39000,85.9,7500.00,28);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('MEX','Mexico','North America','Central America',128900000,70.7,1420000.00,76);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('NGA','Nigeria','Africa','Western Africa',223800000,55.8,477000.00,29);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('NLD','Netherlands','Europe','Western Europe',17700000,82.3,990000.00,18);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('NZL','New Zealand','Oceania','Australia and New Zealand',5200000,82.5,250000.00,94);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('PAK','Pakistan','Asia','Southern and Central Asia',240500000,66.6,375000.00,56);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('PER','Peru','South America','South America',34400000,72.9,242000.00,85);

```

```

INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('POL','Poland','Europe','Eastern Europe',37700000,78.0,690000.00,16);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('PRT','Portugal','Europe','Southern Europe',10300000,82.1,250000.00,24);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('SAU','Saudi Arabia','Asia','Middle East',36900000,76.9,1010000.00,67);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('SWE','Sweden','Europe','Nordic Countries',10500000,83.0,590000.00,22);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('THA','Thailand','Asia','Southeast Asia',71800000,79.3,495000.00,63);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('USA','United States','North America','North America',339900000,79.3,25460000.00,69);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('VEN','Venezuela','South America','South America',28300000,72.1,90000.00,89);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('VNM','Vietnam','Asia','Southeast Asia',98900000,73.6,410000.00,61);
INSERT INTO country
(Code,Name,Continent,Region,Population,LifeExpectancy,GNP,Capital) VALUES
('ZAF','South Africa','Africa','Southern Africa',60400000,65.3,405000.00,34);

```

-- 4. INSERT RECORDS INTO countrylanguage (60 rows)

```

INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('ARG','Spanish','T',96.8);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('AUS','English','T',81.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('BEL','French','T',40.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('BEL','Dutch','T',59.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('BGD','Bengali','T',98.0);

```

INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES ('BRA','Portuguese','T',97.5);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES ('CAN','English','T',58.7);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES ('CAN','French','T',22.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES ('CHL','Spanish','T',99.5);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES ('CHN','Chinese','T',92.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES ('COL','Spanish','T',99.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES ('DEU','German','T',92.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES ('DZA','Berber languages','T',17.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES ('DZA','Arabic','T',83.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES ('EGY','Arabic','T',98.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES ('ESP','Catalan','F',17.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES ('ESP','Spanish','T',74.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES ('ETH','Oromo','F',25.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES ('ETH','Amharic','T',30.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES ('FRA','French','T',90.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES ('GBR','English','T',95.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES ('GHA','English','T',100.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES ('GRC','Greek','T',99.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES ('IDN','Indonesian','T',100.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES ('IND','Bengali','F',8.1);

```

INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('IND','Hindi','T',41.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('IND','English','T',12.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('ITA','Italian','T',93.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('JPN','Japanese','T',99.1);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('KEN','Swahili','T',60.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('KEN','English','T',20.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('KOR','Korean','T',100.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('MAR','Arabic','T',65.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('MAR','Berber languages','T',28.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('MCO','Monegasque','F',12.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('MCO','French','T',58.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('MEX','Spanish','T',92.7);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('NGA','English','T',100.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('NGA','Hausa','F',25.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('NGA','Yoruba','F',21.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('NLD','Dutch','T',95.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('NZL','English','T',91.2);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('NZL','Maori','T',4.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('PAK','English','T',1.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('PAK','Urdu','T',8.0);

```

```

INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('PAK','Punjabi','F',48.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('PER','Spanish','T',83.9);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('PER','Quechua','T',13.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('POL','Polish','T',97.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('PRT','Portuguese','T',99.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('SAU','Arabic','T',95.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('SWE','Swedish','T',89.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('THA','Thai','T',97.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('USA','English','T',86.2);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('USA','Spanish','F',9.7);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('VEN','Spanish','T',96.9);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('VNM','Vietnamese','T',87.0);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('ZAF','Afrikaans','T',13.5);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('ZAF','English','T',9.6);
INSERT INTO countrylanguage (CountryCode,Language,IsOfficial,Percentage) VALUES
('ZAF','Zulu','T',23.0);

```

-- 5. (OPTIONAL) FOREIGN KEY CONSTRAINTS — add these AFTER all data above is loaded

```

ALTER TABLE city ADD CONSTRAINT fk_city_country FOREIGN KEY
(CountryCode) REFERENCES country(Code);
ALTER TABLE country ADD CONSTRAINT fk_country_capital FOREIGN KEY
(Capital) REFERENCES city(ID);
ALTER TABLE countrylanguage ADD CONSTRAINT fk_lang_country FOREIGN KEY
(CountryCode) REFERENCES country(Code);

```

DAY 21: SUBQUERIES — 1. SINGLE ROW SUBQUERIES

Examples:

Example 1: Find the country with the largest population

SELECT Name, Population

FROM country

WHERE Population = (SELECT MAX(Population) FROM country);

```
mysql> SELECT Name, Population
      -> FROM country
      -> WHERE Population = (SELECT MAX(Population) FROM country);
+-----+-----+
| Name   | Population |
+-----+-----+
| China  | 1277558000 |
+-----+-----+
1 row in set (0.04 sec)
```

Example 2: Find cities in the same country as 'New York'

SELECT Name, Population

FROM city

WHERE CountryCode = (SELECT CountryCode FROM city WHERE Name = 'New York');

```
mysql> SELECT Name, Population
-> FROM city
-> WHERE CountryCode = (SELECT CountryCode FROM city WHERE Name = 'New York');
+-----+-----+
| Name | Population |
+-----+-----+
| New York | 8008278 |
| Los Angeles | 3694820 |
| Chicago | 2896016 |
| Houston | 1953631 |
| Philadelphia | 1517550 |
| Phoenix | 1321045 |
| San Diego | 1223400 |
| Dallas | 1188580 |
| San Antonio | 1144646 |
| Detroit | 951270 |
| San Jose | 894943 |
| Indianapolis | 791926 |
| San Francisco | 776733 |
+-----+-----+
```

Example 3: Find countries with population greater than India's population

SELECT Name, Population

FROM country

WHERE Population > (SELECT Population FROM country WHERE Name = 'India');

```
mysql> SELECT Name, Population
-> FROM country
-> WHERE Population > (SELECT Population FROM country WHERE Name = 'India');
+-----+-----+
| Name | Population |
+-----+-----+
| China | 1277558000 |
+-----+-----+
1 row in set (0.00 sec)
```

Example 4: Find the capital city of Brazil

SELECT c.Name as CityName, c.Population

FROM city c

WHERE c.ID = (SELECT Capital FROM country WHERE Name = 'Brazil');

```
mysql> SELECT c.Name as CityName, c.Population
-> FROM city c
-> WHERE c.ID = (SELECT Capital FROM country WHERE Name = 'Brazil');
+-----+-----+
| CityName | Population |
+-----+-----+
| Brasília | 1969868 |
+-----+-----+
1 row in set (0.00 sec)
```

Example 5: Find countries in the same continent as Japan

SELECT Name, Continent

FROM country

WHERE Continent = (SELECT Continent FROM country WHERE Name = 'Japan');

```
mysql> SELECT Name, Continent
-> FROM country
-> WHERE Continent = (SELECT Continent FROM country WHERE Name = 'Japan');
+-----+-----+
| Name          | Continent |
+-----+-----+
| Afghanistan   | Asia      |
| United Arab Emirates | Asia      |
| Armenia        | Asia      |
| Azerbaijan     | Asia      |
| Bangladesh     | Asia      |
| Bahrain        | Asia      |
| Brunei         | Asia      |
| Bhutan         | Asia      |
| China          | Asia      |
| Cyprus         | Asia      |
| Georgia        | Asia      |
+-----+-----+
```

DAY 21: SUBQUERIES — 2. MULTI ROW SUBQUERIES WITH ANY/SOME

Examples:

Example 1: Find countries with population greater than ANY European country

SELECT Name, Population, Continent

FROM country

WHERE Population > ANY (SELECT Population FROM country WHERE Continent = 'Europe')

AND Continent != 'Europe';

```
mysql> SELECT Name, Population, Continent
-> FROM country
-> WHERE Population > ANY (SELECT Population FROM country WHERE Continent = 'Europe')
-> AND Continent != 'Europe';
```

Name	Population	Continent
Aruba	103000	North America
Afghanistan	22720000	Asia
Angola	12878000	Africa
Anguilla	8000	North America
Netherlands Antilles	217000	North America
United Arab Emirates	2441000	Asia
Argentina	37032000	South America
Armenia	3520000	Asia
American Samoa	68000	Oceania
Antigua and Barbuda	68000	North America
Australia	18886000	Oceania
Azerbaijan	7734000	Asia
Burundi	6695000	Africa
Benin	6097000	Africa
Burkina Faso	11937000	Africa
Bangladesh	129155000	Asia
Bahrain	617000	Asia

Example 2: Find cities with population greater than ANY city in USA

SELECT Name, Population, CountryCode

FROM city

WHERE Population > ANY (SELECT Population FROM city WHERE CountryCode = 'USA')

AND CountryCode != 'USA';

```
mysql> SELECT Name, Population, CountryCode
-> FROM city
-> WHERE Population > ANY (SELECT Population FROM city WHERE CountryCode = 'USA')
-> AND CountryCode != 'USA';
```

Name	Population	CountryCode
Kabul	1780000	AFG
Qandahar	237500	AFG
Herat	186800	AFG
Mazar-e-Sharif	127800	AFG
Amsterdam	731200	NLD
Rotterdam	593321	NLD
Haag	440900	NLD
Utrecht	234323	NLD
Eindhoven	201843	NLD
Tilburg	193238	NLD
Groningen	172701	NLD
Breda	160398	NLD
Apeldoorn	153491	NLD
Nijmegen	152463	NLD
Enschede	149544	NLD
Haarlem	148772	NLD
Almere	142465	NLD

Example 3: Using =ANY (equivalent to IN)

SELECT Name, Population

FROM city

WHERE CountryCode = ANY (SELECT Code FROM country WHERE Continent = 'Asia');

```
mysql> SELECT Name, Population
-> FROM city
-> WHERE CountryCode = ANY (SELECT Code FROM country WHERE Continent = 'Asia');
```

Name	Population
Kabul	1780000
Qandahar	237500
Herat	186800
Mazar-e-Sharif	127800
Dubai	669181
Abu Dhabi	398695
Sharja	320095
al-Ayn	225970
Ajman	114395
Yerevan	1248700
Gjumri	211700
Vanadzor	172700
Baku	1787800
Gancä	299300

-- This is equivalent to:

SELECT Name, Population

FROM city

WHERE CountryCode IN (SELECT Code FROM country WHERE Continent = 'Asia');

```
mysql> SELECT Name, Population
-> FROM city
-> WHERE CountryCode IN (SELECT Code FROM country WHERE Continent = 'Asia');
```

Name	Population
Kabul	1780000
Qandahar	237500
Herat	186800
Mazar-e-Sharif	127800
Dubai	669181
Abu Dhabi	398695
Sharja	320095
al-Ayn	225970
Ajman	114395
Yerevan	1248700
Gjumri	211700
Vanadzor	172700

Example 4: Find countries with life expectancy greater than ANY African country

SELECT Name, LifeExpectancy, Continent

FROM country

WHERE LifeExpectancy > ANY (SELECT LifeExpectancy

FROM country

WHERE Continent = 'Africa' AND LifeExpectancy IS NOT NULL)

AND Continent != 'Africa';

```
mysql> SELECT Name, LifeExpectancy, Continent
-> FROM country
-> WHERE LifeExpectancy > ANY (SELECT LifeExpectancy
-> FROM country
-> WHERE Continent = 'Africa' AND LifeExpectancy IS NOT NULL)
-> AND Continent != 'Africa';
```

Name	LifeExpectancy	Continent
Aruba	78.4	North America
Afghanistan	45.9	Asia
Anguilla	76.1	North America
Albania	71.6	Europe
Andorra	83.5	Europe
Netherlands Antilles	74.7	North America
United Arab Emirates	74.1	Asia
Argentina	75.1	South America
Armenia	66.4	Asia
American Samoa	75.1	Oceania
Antigua and Barbuda	70.5	North America
Australia	79.8	Oceania
Austria	77.7	Europe
Azerbaijan	62.9	Asia

Example 5: Cities larger than ANY capital city

```
SELECT c.Name, c.Population, co.Name as CountryName
```

FROM city c

```
JOIN country co ON c.CountryCode = co.Code
```

WHERE c.Population > ANY (SELECT ci.Population

FROM city ci

```
JOIN country co2 ON ci.ID = co2.Capital
```

WHERE ci.Population IS NOT NULL);

```
mysql> SELECT c.Name, c.Population, co.Name as CountryName
-> FROM city c
-> JOIN country co ON c.CountryCode = co.Code
-> WHERE c.Population > ANY (SELECT ci.Population
-> FROM city ci
-> JOIN country co2 ON ci.ID = co2.Capital
-> WHERE ci.Population IS NOT NULL);
```

Name	Population	CountryName
Kabul	1780000	Afghanistan
Qandahar	237500	Afghanistan
Herat	186800	Afghanistan
Mazar-e-Sharif	127800	Afghanistan
Amsterdam	731200	Netherlands
Rotterdam	593321	Netherlands
Haag	440900	Netherlands
Utrecht	234323	Netherlands
Eindhoven	201843	Netherlands
Tilburg	193238	Netherlands
Groningen	172701	Netherlands
Breda	160398	Netherlands
Apeldoorn	153491	Netherlands
Nijmegen	152463	Netherlands
Enschede	149544	Netherlands

DAY 21: SUBQUERIES — 3. MULTI ROW SUBQUERIES WITH ALL

Examples:

Example 1: Find countries with population greater than ALL European countries

SELECT Name, Population, Continent

FROM country

WHERE Population > ALL (SELECT Population FROM country WHERE Continent = 'Europe');

```
mysql> SELECT Name, Population, Continent
-> FROM country
-> WHERE Population > ALL (SELECT Population FROM country WHERE Continent = 'Europe');
```

Name	Population	Continent
Brazil	170115000	South America
China	1277558000	Asia
Indonesia	212107000	Asia
India	1013662000	Asia
Pakistan	156483000	Asia
United States	278357000	North America

6 rows in set (0.00 sec)

Example 2: Find the smallest city (population-wise) that is still larger than ALL cities in Monaco

SELECT Name, Population, CountryCode

FROM city

WHERE Population > ALL (SELECT Population FROM city WHERE CountryCode = 'MCO')

ORDER BY Population

LIMIT 1;

```
mysql> SELECT Name, Population, CountryCode
-> FROM city
-> WHERE Population > ALL (SELECT Population FROM city WHERE CountryCode = 'MCO')
-> ORDER BY Population
-> LIMIT 1;
+-----+-----+-----+
| Name | Population | CountryCode |
+-----+-----+-----+
| Nuuk | 13445 | GRL |
+-----+-----+-----+
1 row in set (0.01 sec)
```

Example 3: Find countries with life expectancy greater than ALL African countries

SELECT Name, LifeExpectancy, Continent

FROM country

WHERE LifeExpectancy > ALL (SELECT LifeExpectancy

FROM country

WHERE Continent = 'Africa' AND LifeExpectancy IS NOT NULL);

```
mysql> SELECT Name, LifeExpectancy, Continent
-> FROM country
-> WHERE LifeExpectancy > ALL (SELECT LifeExpectancy
-> FROM country
-> WHERE Continent = 'Africa' AND LifeExpectancy IS NOT NULL);
+-----+-----+-----+
| Name | LifeExpectancy | Continent |
+-----+-----+-----+
| Aruba | 78.4 | North America |
| Andorra | 83.5 | Europe |
| Australia | 79.8 | Oceania |
| Austria | 77.7 | Europe |
| Belgium | 77.8 | Europe |
| Bermuda | 76.9 | North America |
| Canada | 79.4 | North America |
| Switzerland | 79.6 | Europe |
| Cayman Islands | 78.9 | North America |
| Germany | 77.4 | Europe |
| Spain | 78.8 | Europe |
| Finland | 77.4 | Europe |
+-----+-----+-----+
```

Example 4: Find languages spoken in more countries than ALL languages in South America

SELECT Language, COUNT(*) as CountryCount

FROM countrylanguage

GROUP BY Language

HAVING COUNT(*) > ALL (SELECT COUNT(*)

FROM countrylanguage cl

JOIN country c ON cl.CountryCode = c.Code

WHERE c.Continent = 'South America'

GROUP BY cl.Language);

```
mysql> SELECT Language, COUNT(*) as CountryCount
-> FROM countrylanguage
-> GROUP BY Language
-> HAVING COUNT(*) > ALL (SELECT COUNT(*)
-> FROM countrylanguage cl
-> JOIN country c ON cl.CountryCode = c.Code
-> WHERE c.Continent = 'South America'
-> GROUP BY cl.Language);
```

Language	CountryCount
English	60
Spanish	28
French	25
Portuguese	12
Arabic	33
Italian	15
Creole English	14
German	19
Polish	10
Turkish	12
Russian	17
Ful	12
Ukrainian	12
Chinese	19

14 rows in set (0.23 sec)

Example 5: Cities with population smaller than ALL capital cities

SELECT c.Name, c.Population

FROM city c

WHERE c.Population < ALL (SELECT ci.Population

FROM city ci

JOIN country co ON ci.ID = co.Capital

WHERE ci.Population IS NOT NULL)

ORDER BY c.Population DESC;

```
mysql> SELECT c.Name, c.Population
-> FROM city c
-> WHERE c.Population < ALL (SELECT ci.Population
-> FROM city ci
-> JOIN country co ON ci.ID = co.Capital
-> WHERE ci.Population IS NOT NULL)
-> ORDER BY c.Population DESC;
Empty set (0.01 sec)
```

DAY 21: SUBQUERIES — 4. NESTED SUBQUERIES

Examples:

Example 1: Find cities in countries that have the same continent as the country with the largest population

```
SELECT c.Name as CityName, c.Population, co.Name as CountryName
```

```
FROM city c
```

```
JOIN country co ON c.CountryCode = co.Code
```

```
WHERE co.Continent =
```

```
(SELECT Continent FROM country WHERE Population = (SELECT MAX(Population) FROM country));
```

```
mysql> SELECT c.Name as CityName, c.Population, co.Name as CountryName
-> FROM city c
-> JOIN country co ON c.CountryCode = co.Code
-> WHERE co.Continent =
-> (SELECT Continent FROM country WHERE Population = (SELECT MAX(Population) FROM country));
```

CityName	Population	CountryName
Kabul	1780000	Afghanistan
Qandahar	237500	Afghanistan
Herat	186800	Afghanistan
Mazar-e-Sharif	127800	Afghanistan
Dubai	669181	United Arab Emirates
Abu Dhabi	398695	United Arab Emirates
Sharja	320095	United Arab Emirates
al-Ayn	225970	United Arab Emirates
Ajman	114395	United Arab Emirates
Yerevan	1248700	Armenia
Gjumri	211700	Armenia
Vanadzor	172700	Armenia
Baku	1787800	Azerbaijan
Gāncā	299300	Azerbaijan
Sumqayit	283000	Azerbaijan
Mingäcevir	93900	Azerbaijan

Example 2: Find the most spoken language in the continent where the country with the highest GDP is located

```
SELECT cl.Language, SUM(c.Population * cl.Percentage / 100) as TotalSpeakers
```

```
FROM countrylanguage cl
```

```
JOIN country c ON cl.CountryCode = c.Code
```

```
WHERE c.Continent = (SELECT Continent
```

```
FROM country
```

```
WHERE GNP = (SELECT MAX(GNP) FROM country WHERE GNP IS NOT NULL))
```

```
GROUP BY cl.Language
```

```
ORDER BY TotalSpeakers DESC
```

```
LIMIT 1;
```

```
mysql> SELECT cl.Language, SUM(c.Population * cl.Percentage / 100) as TotalSpeakers
-> FROM countrylanguage cl
-> JOIN country c ON cl.CountryCode = c.Code
-> WHERE c.Continent = (SELECT Continent
-> FROM country
-> WHERE GNP = (SELECT MAX(GNP) FROM country WHERE GNP IS NOT NULL))
-> GROUP BY cl.Language
-> ORDER BY TotalSpeakers DESC
-> LIMIT 1;
+-----+-----+
| Language | TotalSpeakers |
+-----+-----+
| English  | 262122173.00000 |
+-----+-----+
1 row in set (0.12 sec)
```

Example 3: Find cities in countries that have more cities than the country with the most languages

```
SELECT c.Name, c.Population, co.Name as CountryName
```

```
FROM city c
```

```
JOIN country co ON c.CountryCode = co.Code
```

```
WHERE co.Code IN (SELECT CountryCode
```

```
FROM city
```

```
GROUP BY CountryCode
```

```
HAVING COUNT(*) > (SELECT MAX(lang_count)
```

FROM (SELECT COUNT(*) as lang_count

FROM countrylanguage

GROUP BY CountryCode) as subq));

```
mysql> SELECT c.Name, c.Population, co.Name as CountryName
-> FROM city c
-> JOIN country co ON c.CountryCode = co.Code
-> WHERE co.Code IN (SELECT CountryCode
-> FROM city
-> GROUP BY CountryCode
-> HAVING COUNT(*) > (SELECT MAX(lang_count)
-> FROM (SELECT COUNT(*) as lang_count
-> FROM countrylanguage
-> GROUP BY CountryCode) as subq));
```

Name	Population	CountryName
Buenos Aires	2982146	Argentina
La Matanza	1266461	Argentina
Córdoba	1157507	Argentina
Rosario	907718	Argentina
Lomas de Zamora	622013	Argentina
Quilmes	559249	Argentina
Almirante Brown	538918	Argentina
La Plata	521936	Argentina
Mar del Plata	512880	Argentina
San Miguel de Tucumán	470809	Argentina
Lanús	469735	Argentina
Merlo	463846	Argentina
General San Martín	422542	Argentina
Salta	367550	Argentina
Moreno	356993	Argentina

Example 5: Complex nested query - Find countries in the same region as countries that have cities larger than the largest city in Europe

SELECT DISTINCT co.Name, co.Region

FROM country co

WHERE co.Region IN (SELECT DISTINCT c2.Region

FROM country c2

WHERE c2.Code IN (SELECT DISTINCT ci.CountryCode

FROM city ci

WHERE ci.Population > (SELECT MAX(ci2.Population)

FROM city ci2

JOIN country co2 ON ci2.CountryCode = co2.Code

WHERE co2.Continent = 'Europe')));

```
mysql> SELECT DISTINCT co.Name, co.Region
-> FROM country co
-> WHERE co.Region IN (SELECT DISTINCT c2.Region
-> FROM country c2
-> WHERE c2.Code IN (SELECT DISTINCT ci.CountryCode
-> FROM city ci
-> WHERE ci.Population > (SELECT MAX(ci2.Population)
-> FROM city ci2
-> JOIN country co2 ON ci2.CountryCode = co2.Code
-> WHERE co2.Continent = 'Europe')));
```

Name	Region
Afghanistan	Southern and Central Asia
United Arab Emirates	Middle East
Argentina	South America
Armenia	Middle East
Azerbaijan	Middle East
Bangladesh	Southern and Central Asia
Bahrain	Middle East
Belize	Central America
Bolivia	South America
Brazil	South America
Brunei	Southeast Asia
Bhutan	Southern and Central Asia
Chile	South America
China	Eastern Asia
Colombia	South America
Costa Rica	Central America
Cyprus	Middle East
Ecuador	South America
Falkland Islands	South America
Georgia	Middle East

DAY 22: CORRELATED SUBQUERIES — CORRELATED SUBQUERIES

Examples:

Example 1: Find countries where the capital city has above-average population for that country

```
SELECT co.Name as CountryName, cap.Name as CapitalName, cap.Population
```

```
FROM country co
```

```
JOIN city cap ON co.Capital = cap.ID
```

```
WHERE cap.Population > (SELECT AVG(c.Population)
```

```
FROM city c
```

```
WHERE c.CountryCode = co.Code);
```

```
mysql> SELECT co.Name as CountryName, cap.Name as CapitalName, cap.Population
-> FROM country co
-> JOIN city cap ON co.Capital = cap.ID
-> WHERE cap.Population > (SELECT AVG(c.Population)
-> FROM city c
-> WHERE c.CountryCode = co.Code);
```

CountryName	CapitalName	Population
Afghanistan	Kabul	1780000
Angola	Luanda	2022000
United Arab Emirates	Abu Dhabi	398695
Argentina	Buenos Aires	2982146
Armenia	Yerevan	1248700
Austria	Wien	1608144
Azerbaijan	Baku	1787800
Burkina Faso	Ouagadougou	824000
Bangladesh	Dhaka	3612850
Bulgaria	Sofija	1122302
Bosnia and Herzegovina	Sarajevo	360000
Belarus	Minsk	1674000
Bolivia	La Paz	758141
Brazil	Brasília	1969868
Botswana	Gaborone	213017

Example 2: Find countries with above-average life expectancy for their continent

```
SELECT c1.Name, c1.LifeExpectancy, c1.Continent
FROM country c1
WHERE c1.LifeExpectancy > (SELECT AVG(c2.LifeExpectancy)
FROM country c2
WHERE c2.Continent = c1.Continent
AND c2.LifeExpectancy IS NOT NULL);
```

```
mysql> SELECT c1.Name, c1.LifeExpectancy, c1.Continent
-> FROM country c1
-> WHERE c1.LifeExpectancy > (SELECT AVG(c2.LifeExpectancy)
-> FROM country c2
-> WHERE c2.Continent = c1.Continent
-> AND c2.LifeExpectancy IS NOT NULL);
```

Name	LifeExpectancy	Continent
Aruba	78.4	North America
Anguilla	76.1	North America
Andorra	83.5	Europe
Netherlands Antilles	74.7	North America
United Arab Emirates	74.1	Asia
Argentina	75.1	South America
American Samoa	75.1	Oceania
Australia	79.8	Oceania
Austria	77.7	Europe
Belgium	77.8	Europe
Bahrain	73.0	Asia
Bermuda	76.9	North America
Barbados	73.0	North America

Example 3: Find cities that are the largest in their country

```
SELECT c1.Name, c1.Population, c1.CountryCode
FROM city c1
WHERE c1.Population = (SELECT MAX(c2.Population)
                       FROM city c2
                       WHERE c2.CountryCode = c1.CountryCode);
```

```
mysql> SELECT c1.Name, c1.Population, c1.CountryCode
-> FROM city c1
-> WHERE c1.Population = (SELECT MAX(c2.Population)
->                        FROM city c2
->                        WHERE c2.CountryCode = c1.CountryCode);
```

Name	Population	CountryCode
Kabul	1780000	AFG
Amsterdam	731200	NLD
Willemstad	2345	ANT
Tirana	270000	ALB
Alger	2168000	DZA
Tafuna	5200	ASM
Andorra la Vella	21189	AND
Luanda	2022000	AGO
South Hill	961	AIA
Saint John's	24000	ATG
Dubai	669181	ARE
Buenos Aires	2982146	ARG

Example 4: Find countries where at least one official language is spoken by more than 50% of population

```
SELECT DISTINCT c.Name, c.Continent
FROM country c
WHERE 50 < (SELECT MAX(cl.Percentage)
            FROM countrylanguage cl
            WHERE cl.CountryCode = c.Code
            AND cl.IsOfficial = 'T');
```

```
mysql> SELECT DISTINCT c.Name, c.Continent
-> FROM country c
-> WHERE 50 < (SELECT MAX(c1.Percentage)
->             FROM countrylanguage c1
->             WHERE c1.CountryCode = c.Code
->             AND c1.IsOfficial = 'T');
```

Name	Continent
Afghanistan	Asia
Albania	Europe
Netherlands Antilles	North America
Argentina	South America
Armenia	Asia
American Samoa	Oceania
Australia	Oceania
Austria	Europe
Azerbaijan	Asia
Burundi	Africa
Belgium	Europe
Bangladesh	Asia
Bulgaria	Europe
Bahrain	Asia
Bosnia and Herzegovina	Europe
Belarus	Europe

Example 5: Find cities with population greater than their country's average city population

```
SELECT c1.Name, c1.Population, co.Name as CountryName
FROM city c1
JOIN country co ON c1.CountryCode = co.Code
WHERE c1.Population > (SELECT AVG(c2.Population)
                       FROM city c2
                       WHERE c2.CountryCode = c1.CountryCode);
```



```
mysql> SELECT c1.Name, c1.Population, co.Name as CountryName
-> FROM city c1
-> JOIN country co ON c1.CountryCode = co.Code
-> WHERE c1.Population > (SELECT AVG(c2.Population)
-> FROM city c2
-> WHERE c2.CountryCode = c1.CountryCode);
```

Name	Population	CountryName
Kabul	1780000	Afghanistan
Luanda	2022000	Angola
South Hill	961	Anguilla
Dubai	669181	United Arab Emirates
Abu Dhabi	398695	United Arab Emirates
Buenos Aires	2982146	Argentina
La Matanza	1266461	Argentina
Córdoba	1157507	Argentina
Rosario	907718	Argentina
Lomas de Zamora	622013	Argentina
Quilmes	559249	Argentina
Almirante Brown	538918	Argentina
La Plata	521936	Argentina
Mar del Plata	512880	Argentina
San Miguel de Tucumán	470809	Argentina
Lanús	469735	Argentina
Merlo	463846	Argentina
General San Martín	422542	Argentina

Example 6: Countries with more cities than the average number of cities per country in their continent

```
SELECT c.Name, c.Continent,
       (SELECT COUNT(*) FROM city ci WHERE ci.CountryCode = c.Code) as CityCount
FROM country c
WHERE (SELECT COUNT(*) FROM city ci WHERE ci.CountryCode = c.Code) >
      (SELECT AVG(city_count)
FROM (SELECT COUNT(*) as city_count
FROM city ci2
JOIN country c2 ON ci2.CountryCode = c2.Code
WHERE c2.Continent = c.Continent
GROUP BY c2.Code) as avg_cities);
```

```
mysql> SELECT c.Name, c.Continent,
-> (SELECT COUNT(*) FROM city ci WHERE ci.CountryCode = c.Code) as CityCount
-> FROM country c
-> WHERE (SELECT COUNT(*) FROM city ci WHERE ci.CountryCode = c.Code) >
-> (SELECT AVG(city_count)
-> FROM (SELECT COUNT(*) as city_count
-> FROM city ci2
-> JOIN country c2 ON ci2.CountryCode = c2.Code
-> WHERE c2.Continent = c.Continent
-> GROUP BY c2.Code) as avg_cities);
```

Name	Continent	CityCount
Argentina	South America	57
Australia	Oceania	14
Brazil	South America	250
Canada	North America	49
China	Asia	363
Cameroon	Africa	7
Congo, The Democratic Republic of the	Africa	18
Colombia	South America	38
Germany	Europe	93
Algeria	Africa	18
Egypt	Africa	37
Spain	Europe	59
Ethiopia	Africa	7
France	Europe	40
United Kingdom	Europe	81
Indonesia	Asia	85
India	Asia	341
Iran	Asia	67
Italy	Europe	58
Japan	Asia	248

DAY 22: CORRELATED SUBQUERIES — EXISTS EXAMPLES

Examples:

Example 1: Find countries where the capital city has above-average population for that country

```
SELECT co.Name as CountryName, cap.Name as CapitalName, cap.Population
```

```
FROM country co
```

```
JOIN city cap ON co.Capital = cap.ID
```

```
WHERE cap.Population > (SELECT AVG(c.Population)
```

```
FROM city c
```

```
WHERE c.CountryCode = co.Code);
```

```
mysql> SELECT co.Name as CountryName, cap.Name as CapitalName, cap.Population
-> FROM country co
-> JOIN city cap ON co.Capital = cap.ID
-> WHERE cap.Population > (SELECT AVG(c.Population)
-> FROM city c
-> WHERE c.CountryCode = co.Code);
```

CountryName	CapitalName	Population
Afghanistan	Kabul	1780000
Angola	Luanda	2022000
United Arab Emirates	Abu Dhabi	398695
Argentina	Buenos Aires	2982146
Armenia	Yerevan	1248700
Austria	Wien	1608144
Azerbaijan	Baku	1787800
Burkina Faso	Ouagadougou	824000
Bangladesh	Dhaka	3612850
Bulgaria	Sofija	1122302
Bosnia and Herzegovina	Sarajevo	360000
Belarus	Minsk	1674000
Bolivia	La Paz	758141
Brazil	Brasília	1969868
Botswana	Gaborone	213017

Example 2: Find countries with above-average life expectancy for their continent

```
SELECT c1.Name, c1.LifeExpectancy, c1.Continent
FROM country c1
WHERE c1.LifeExpectancy > (SELECT AVG(c2.LifeExpectancy)
FROM country c2
WHERE c2.Continent = c1.Continent
AND c2.LifeExpectancy IS NOT NULL);
```

```
mysql> SELECT c1.Name, c1.LifeExpectancy, c1.Continent
-> FROM country c1
-> WHERE c1.LifeExpectancy > (SELECT AVG(c2.LifeExpectancy)
-> FROM country c2
-> WHERE c2.Continent = c1.Continent
-> AND c2.LifeExpectancy IS NOT NULL);
```

Name	LifeExpectancy	Continent
Aruba	78.4	North America
Anguilla	76.1	North America
Andorra	83.5	Europe
Netherlands Antilles	74.7	North America
United Arab Emirates	74.1	Asia
Argentina	75.1	South America
American Samoa	75.1	Oceania
Australia	79.8	Oceania
Austria	77.7	Europe
Belgium	77.8	Europe
Bahrain	73.0	Asia
Bermuda	76.9	North America
Barbados	73.0	North America

Example 3: Find cities that are the largest in their country

```
SELECT c1.Name, c1.Population, c1.CountryCode
FROM city c1
WHERE c1.Population = (SELECT MAX(c2.Population)
                       FROM city c2
                       WHERE c2.CountryCode = c1.CountryCode);
```

```
mysql> SELECT c1.Name, c1.Population, c1.CountryCode
-> FROM city c1
-> WHERE c1.Population = (SELECT MAX(c2.Population)
->                        FROM city c2
->                        WHERE c2.CountryCode = c1.CountryCode);
```

Name	Population	CountryCode
Kabul	1780000	AFG
Amsterdam	731200	NLD
Willemstad	2345	ANT
Tirana	270000	ALB
Alger	2168000	DZA
Tafuna	5200	ASM
Andorra la Vella	21189	AND
Luanda	2022000	AGO
South Hill	961	AIA
Saint John's	24000	ATG
Dubai	669181	ARE
Buenos Aires	2982146	ARG

Example 4: Find countries where at least one official language is spoken by more than 50% of population

```
SELECT DISTINCT c.Name, c.Continent
FROM country c
WHERE 50 < (SELECT MAX(cl.Percentage)
            FROM countrylanguage cl
            WHERE cl.CountryCode = c.Code
            AND cl.IsOfficial = 'T');
```

```
mysql> SELECT DISTINCT c.Name, c.Continent
-> FROM country c
-> WHERE 50 < (SELECT MAX(c1.Percentage)
->             FROM countrylanguage c1
->             WHERE c1.CountryCode = c.Code
->             AND c1.IsOfficial = 'T');
```

Name	Continent
Afghanistan	Asia
Albania	Europe
Netherlands Antilles	North America
Argentina	South America
Armenia	Asia
American Samoa	Oceania
Australia	Oceania
Austria	Europe
Azerbaijan	Asia
Burundi	Africa
Belgium	Europe
Bangladesh	Asia
Bulgaria	Europe
Bahrain	Asia
Bosnia and Herzegovina	Europe
Belarus	Europe

Example 5: Find cities with population greater than their country's average city population

```
SELECT c1.Name, c1.Population, co.Name as CountryName
FROM city c1
JOIN country co ON c1.CountryCode = co.Code
WHERE c1.Population > (SELECT AVG(c2.Population)
                       FROM city c2
                       WHERE c2.CountryCode = c1.CountryCode);
```

```
mysql> SELECT c1.Name, c1.Population, co.Name as CountryName
-> FROM city c1
-> JOIN country co ON c1.CountryCode = co.Code
-> WHERE c1.Population > (SELECT AVG(c2.Population)
-> FROM city c2
-> WHERE c2.CountryCode = c1.CountryCode);
```

Name	Population	CountryName
Kabul	1780000	Afghanistan
Luanda	2022000	Angola
South Hill	961	Anguilla
Dubai	669181	United Arab Emirates
Abu Dhabi	398695	United Arab Emirates
Buenos Aires	2982146	Argentina
La Matanza	1266461	Argentina
Córdoba	1157507	Argentina
Rosario	907718	Argentina
Lomas de Zamora	622013	Argentina
Quilmes	559249	Argentina
Almirante Brown	538918	Argentina
La Plata	521936	Argentina
Mar del Plata	512880	Argentina
San Miguel de Tucumán	470809	Argentina
Lanús	469735	Argentina
Merlo	463846	Argentina
General San Martín	422542	Argentina

Example 6: Countries with more cities than the average number of cities per country in their continent

```
SELECT c.Name, c.Continent,
       (SELECT COUNT(*) FROM city ci WHERE ci.CountryCode = c.Code) as CityCount
FROM country c
WHERE (SELECT COUNT(*) FROM city ci WHERE ci.CountryCode = c.Code) >
      (SELECT AVG(city_count)
FROM (SELECT COUNT(*) as city_count
FROM city ci2
JOIN country c2 ON ci2.CountryCode = c2.Code
WHERE c2.Continent = c.Continent
GROUP BY c2.Code) as avg_cities);
```

```
mysql> SELECT c.Name, c.Continent,
-> (SELECT COUNT(*) FROM city ci WHERE ci.CountryCode = c.Code) as CityCount
-> FROM country c
-> WHERE (SELECT COUNT(*) FROM city ci WHERE ci.CountryCode = c.Code) >
-> (SELECT AVG(city_count)
-> FROM (SELECT COUNT(*) as city_count
-> FROM city ci2
-> JOIN country c2 ON ci2.CountryCode = c2.Code
-> WHERE c2.Continent = c.Continent
-> GROUP BY c2.Code) as avg_cities);
```

Name	Continent	CityCount
Argentina	South America	57
Australia	Oceania	14
Brazil	South America	250
Canada	North America	49
China	Asia	363
Cameroon	Africa	7
Congo, The Democratic Republic of the	Africa	18
Colombia	South America	38
Germany	Europe	93
Algeria	Africa	18
Egypt	Africa	37
Spain	Europe	59
Ethiopia	Africa	7
France	Europe	40
United Kingdom	Europe	81
Indonesia	Asia	85
India	Asia	341
Iran	Asia	67
Italy	Europe	58
Japan	Asia	248

DAY 22: CORRELATED SUBQUERIES — NOT EXISTS EXAMPLES

EXISTS Examples:

Example 1: Find countries that have at least one city

SELECT Name, Continent, Population

FROM country c

WHERE EXISTS (SELECT 1

FROM city ci

WHERE ci.CountryCode = c.Code);

```
mysql> SELECT Name, Continent, Population
-> FROM country c
-> WHERE EXISTS (SELECT 1
-> FROM city ci
-> WHERE ci.CountryCode = c.Code);
```

Name	Continent	Population
Aruba	North America	103000
Afghanistan	Asia	22720000
Angola	Africa	12878000
Anguilla	North America	8000
Albania	Europe	3401200
Andorra	Europe	78000
Netherlands Antilles	North America	217000
United Arab Emirates	Asia	2441000
Argentina	South America	37032000
Armenia	Asia	3520000
American Samoa	Oceania	68000
Antigua and Barbuda	North America	68000
Australia	Oceania	18886000
Austria	Europe	8881800

Example 2: Find countries that have cities with population over 5 million

SELECT c.Name, c.Continent

FROM country c

WHERE EXISTS (SELECT 1

FROM city ci

WHERE ci.CountryCode = c.Code

AND ci.Population > 5000000);


```
mysql> SELECT c.Name, c.Continent
-> FROM country c
-> WHERE EXISTS (SELECT 1
->                FROM city ci
->                WHERE ci.CountryCode = c.Code
->                AND ci.Population > 5000000);
```

Name	Continent
Brazil	South America
China	Asia
Congo, The Democratic Republic of the	Africa
Colombia	South America
Egypt	Africa
United Kingdom	Europe
Indonesia	Asia
India	Asia
Iran	Asia
Japan	Asia
South Korea	Asia
Mexico	North America
Pakistan	Asia
Peru	South America
Russian Federation	Europe
Thailand	Asia
Turkey	Asia
United States	North America

```
18 rows in set (0.00 sec)
```

Find countries that have at least one official language

```
SELECT c.Name, c.Continent
FROM country c
WHERE EXISTS (SELECT 1
              FROM countrylanguage cl
              WHERE cl.CountryCode = c.Code
              AND cl.IsOfficial = 'T');
```

```
mysql> SELECT c.Name, c.Continent
-> FROM country c
-> WHERE EXISTS (SELECT 1
->                FROM countrylanguage c1
->                WHERE c1.CountryCode = c.Code
->                AND c1.IsOfficial = 'T');
```

Name	Continent
Aruba	North America
Afghanistan	Asia
Anguilla	North America
Albania	Europe
Andorra	Europe
Netherlands Antilles	North America
United Arab Emirates	Asia
Argentina	South America
Armenia	Asia
American Samoa	Oceania
Antigua and Barbuda	North America
Australia	Oceania
Austria	Europe
Azerbaijan	Asia
Burundi	Africa
Belgium	Europe
Bangladesh	Asia
Bulgaria	Europe

Example 4: Find continents that have countries with life expectancy above 80

```
SELECT DISTINCT c1.Continent
FROM country c1
WHERE EXISTS (SELECT 1
              FROM country c2
              WHERE c2.Continent = c1.Continent
              AND c2.LifeExpectancy > 80);
```



```
mysql> SELECT c.Name
-> FROM country c
-> WHERE EXISTS (SELECT 1
-> FROM city ci1
-> WHERE ci1.CountryCode = c.Code
-> AND EXISTS (SELECT 1
-> FROM city ci2
-> WHERE ci2.CountryCode = c.Code
-> AND ci2.District != ci1.District));
```

Name
Afghanistan
Angola
United Arab Emirates
Argentina
Armenia
Australia
Austria
Azerbaijan
Belgium
Benin
Burkina Faso
Bangladesh
Bulgaria
Bosnia and Herzegovina

Example 6: Find languages that are official in at least one country

```
SELECT DISTINCT cl1.Language
FROM countrylanguage cl1
WHERE EXISTS (SELECT 1
FROM countrylanguage cl2
WHERE cl2.Language = cl1.Language
AND cl2.IsOfficial = 'T');
```

```
mysql> SELECT DISTINCT c11.Language
-> FROM countrylanguage c11
-> WHERE EXISTS (SELECT 1
->                FROM countrylanguage c12
->                WHERE c12.Language = c11.Language
->                AND c12.IsOfficial = 'T');
```

Language
Dutch
English
Papiamentu
Spanish
Dari
Pashto
Turkmenian
Uzbek
Albanian
Greek
Macedonian
Catalan
French
Portuguese
Arabic
Hindi

NOT EXISTS Examples:

Example 1: Find countries that have no cities in the database

```
SELECT Name, Continent, Population
```

```
FROM country c
```

```
WHERE NOT EXISTS (SELECT 1
```

```
    FROM city ci
```

```
    WHERE ci.CountryCode = c.Code);
```

```
mysql> SELECT Name, Continent, Population
-> FROM country c
-> WHERE NOT EXISTS (SELECT 1
->                     FROM city ci
->                     WHERE ci.CountryCode = c.Code);
```

Name	Continent	Population
Antarctica	Antarctica	0
French Southern territories	Antarctica	0
Bouvet Island	Antarctica	0
Heard Island and McDonald Islands	Antarctica	0
British Indian Ocean Territory	Africa	0
South Georgia and the South Sandwich Islands	Antarctica	0
United States Minor Outlying Islands	Oceania	0

7 rows in set (0.01 sec)

Example 2: Find countries that don't have any official languages recorded

```
SELECT c.Name, c.Continent
FROM country c
WHERE NOT EXISTS (SELECT 1
                   FROM countrylanguage cl
                   WHERE cl.CountryCode = c.Code
                   AND cl.IsOfficial = 'T');
```

```
mysql> SELECT c.Name, c.Continent
-> FROM country c
-> WHERE NOT EXISTS (SELECT 1
->                     FROM countrylanguage cl
->                     WHERE cl.CountryCode = c.Code
->                     AND cl.IsOfficial = 'T');
```

Name	Continent
Angola	Africa
Antarctica	Antarctica
French Southern territories	Antarctica
Benin	Africa
Burkina Faso	Africa
Bahamas	North America
Bouvet Island	Antarctica
Botswana	Africa
Central African Republic	Africa
Côte d'Ivoire	Africa
Cameroon	Africa
Congo, The Democratic Republic of the	Africa
Congo	Africa
Dominica	North America
Ethiopia	Africa

Example 3: Find countries with no cities having population over 1 million

```
SELECT c.Name, c.Population
FROM country c
WHERE NOT EXISTS (SELECT 1
                   FROM city ci
                   WHERE ci.CountryCode = c.Code
                   AND ci.Population > 1000000);
```

```
mysql> SELECT c.Name, c.Population
-> FROM country c
-> WHERE NOT EXISTS (SELECT 1
->                     FROM city ci
->                     WHERE ci.CountryCode = c.Code
->                     AND ci.Population > 1000000);
```

Name	Population
Aruba	103000
Anguilla	8000
Albania	3401200
Andorra	78000
Netherlands Antilles	217000
United Arab Emirates	2441000
American Samoa	68000
Antarctica	0
French Southern territories	0
Antigua and Barbuda	68000
Burundi	6695000
Belgium	10239000
Benin	6097000
Burkina Faso	11937000
Bahrain	617000
Bahamas	307000

Example 4: Find countries that don't have English as any language

```
SELECT c.Name, c.Continent
FROM country c
WHERE NOT EXISTS (SELECT 1
```

```

FROM countrylanguage cl
WHERE cl.CountryCode = c.Code
AND cl.Language = 'English');

```

```
mysql> SELECT c.Name, c.Continent
-> FROM country c
-> WHERE NOT EXISTS (SELECT 1
->                     FROM countrylanguage cl
->                     WHERE cl.CountryCode = c.Code
->                     AND cl.Language = 'English');
```

Name	Continent
Afghanistan	Asia
Angola	Africa
Albania	Europe
Andorra	Europe
United Arab Emirates	Asia
Argentina	South America
Armenia	Asia
Antarctica	Antarctica
French Southern territories	Antarctica
Austria	Europe
Azerbaijan	Asia
Burundi	Africa
Belgium	Europe
Benin	Africa
Burkina Faso	Africa
Bangladesh	Asia
Bulgaria	Europe

Example 5: Find countries where no city has the same name as the country

```

SELECT c.Name
FROM country c
WHERE NOT EXISTS (SELECT 1
                    FROM city ci
                    WHERE ci.CountryCode = c.Code
                    AND ci.Name = c.Name);

```



```
mysql> SELECT c.Name
-> FROM country c
-> WHERE NOT EXISTS (SELECT 1
->                     FROM city ci
->                     WHERE ci.CountryCode = c.Code
->                     AND ci.Name = c.Name);
```

Name
Aruba
Afghanistan
Angola
Anguilla
Albania
Andorra
Netherlands Antilles
United Arab Emirates
Argentina
Armenia
American Samoa
Antarctica
French Southern territories
Antigua and Barbuda
Australia
Austria
Azerbaijan
Burundi

Example 6: Find continents with no countries having life expectancy below 50

```
SELECT DISTINCT c1.Continent
FROM country c1
WHERE NOT EXISTS (SELECT 1
                   FROM country c2
                   WHERE c2.Continent = c1.Continent
                   AND c2.LifeExpectancy < 50);
```

```
mysql> SELECT DISTINCT c1.Continent
-> FROM country c1
-> WHERE NOT EXISTS (SELECT 1
->                     FROM country c2
->                     WHERE c2.Continent = c1.Continent
->                     AND c2.LifeExpectancy < 50);
```

Continent
Europe
South America
Oceania
Antarctica

```
4 rows in set (0.00 sec)
```

Day-17

1. Introduction to Stored Procedures

A stored procedure is a named, precompiled collection of one or more SQL statements that is stored inside the database and can be executed (called) whenever required. Instead of sending a long block of SQL from an application every time, the application simply calls the procedure by name.

Stored procedures behave like functions or subroutines in a programming language — they can accept parameters, perform logic (loops, conditions, variable declarations), and return results.

Why use Stored Procedures?

- **Performance:** Precompiled and cached by the database engine, so repeated execution is faster than sending raw SQL each time.
- **Reusability:** Write the logic once, call it from many applications or scripts.
- **Security:** Users can be granted EXECUTE permission on a procedure without granting direct access to the underlying tables.
- **Reduced network traffic:** A single CALL statement replaces many lines of SQL sent over the network.
- **Maintainability:** Business logic is centralized in the database, so changes are made in one place.
- **Consistency:** Ensures the same validation/business rules are applied every time, regardless of which application calls it.

2. CREATE PROCEDURE Syntax

The general syntax (MySQL dialect) to create a stored procedure is:

```
DELIMITER //

CREATE PROCEDURE procedure_name (
    [IN | OUT | INOUT] parameter_name data_type, ...
)
BEGIN
    -- SQL statements / logic
    -- declarations, conditions, loops, etc.
END //

DELIMITER ;
```

Key points:

- DELIMITER // temporarily changes the statement terminator from ';' to '//' so that the semicolons inside the procedure body do not end the CREATE PROCEDURE statement prematurely.
- The procedure body is enclosed between BEGIN and END.
- Parameters are optional — a procedure can be created with zero, one, or many parameters.
- After the procedure is created, DELIMITER is reset back to ';'.

Advantages

Faster execution after compilation.

Reduced duplicate SQL.

Supports transactions, loops, conditions, variables, cursors, and exception handlers.

Easy maintenance.

Disadvantages

Debugging can be difficult.

Vendor-specific syntax.

Complex procedures may become hard to maintain.

3. IN, OUT, and INOUT Parameters

Parameter Type	Direction	Behaviour
IN	Caller -> Procedure	Default type. Passes a value INTO the procedure. Changes made inside do NOT reflect back to the caller.
OUT	Procedure -> Caller	Procedure sets a value that is returned OUT to the caller. Initial value passed in is ignored.
INOUT	Both directions	Caller passes a value IN, procedure can modify it, and the updated value is returned back OUT.

NoParameters

=====

1. Display All Employees

Question:

Write a stored procedure to display all employee records.

Answer:

DELIMITER //

```
CREATE PROCEDURE GetAllEmployees()
BEGIN
    SELECT * FROM Employees;
END //
```

DELIMITER ;

Call

```
CALL GetAllEmployees();
```

2. Display IT Department Employees

Question:

Write a procedure to display employees working in the IT department.

Answer

DELIMITER //

```
CREATE PROCEDURE GetITEmployees()
BEGIN
    SELECT * FROM Employees
    WHERE department='IT';
END //
```

DELIMITER ;

Call

CALL GetITEmployees();

3. Display Employees with Salary Greater Than 60000

Question:

Write a procedure to display employees whose salary is greater than 60000.

Answer

DELIMITER //

CREATE PROCEDURE HighSalaryEmployees()

BEGIN

 SELECT * FROM Employees

 WHERE salary > 60000;

END //

DELIMITER ;

Call

CALL HighSalaryEmployees();

4. Display Employees from Hyderabad

Question:

Write a procedure to display employees from Hyderabad.

Answer

DELIMITER //

CREATE PROCEDURE HyderabadEmployees()

```
BEGIN
  SELECT * FROM Employees
  WHERE city='Hyderabad';
END //
```

DELIMITER ;

Call

```
CALL HyderabadEmployees();
```

5. Display Employees with Experience Greater Than 5 Years

Question:

Write a procedure to display employees having more than 5 years of experience.

Answer

```
DELIMITER //
```

```
CREATE PROCEDURE ExperiencedEmployees()
```

```
BEGIN
```

```
  SELECT * FROM Employees
```

```
  WHERE experience > 5;
```

```
END //
```

DELIMITER ;

Call

```
CALL ExperiencedEmployees();
```

6. Display Employee Names Only

Question:

Write a procedure to display only employee names.

Answer

```
DELIMITER //
```

```
CREATE PROCEDURE EmployeeNames()
```

```
BEGIN
```

```
    SELECT emp_name
```

```
    FROM Employees;
```

```
END //
```

```
DELIMITER ;
```

Call

```
CALL EmployeeNames();
```

7. Count Total Employees

Question:

Write a procedure to count the total number of employees.

Answer

```
DELIMITER //
```

```
CREATE PROCEDURE TotalEmployees()
```

```
BEGIN
```

```
    SELECT COUNT(*) AS TotalEmployees
```

```
    FROM Employees;
```

```
END //
```

```
DELIMITER ;
```

Call

```
CALL TotalEmployees();
```

8. Find Maximum Salary

Question:

Write a procedure to display the maximum salary.

Answer

```
DELIMITER //
```

```
CREATE PROCEDURE MaximumSalary()  
BEGIN  
    SELECT MAX(salary) AS HighestSalary  
    FROM Employees;  
END //
```

```
DELIMITER ;
```

Call

```
CALL MaximumSalary();
```

9. Find Minimum Salary

Question:

Write a procedure to display the minimum salary.

Answer

```
DELIMITER //
```

```
CREATE PROCEDURE MinimumSalary()  
BEGIN  
    SELECT MIN(salary) AS LowestSalary  
    FROM Employees;  
END //
```

```
DELIMITER ;
```

Call

CALL MinimumSalary();

10. Calculate Average Salary

Question:

Write a procedure to calculate the average salary.

Answer

DELIMITER //

```
CREATE PROCEDURE AverageSalary()
BEGIN
    SELECT AVG(salary) AS AverageSalary
    FROM Employees;
END //
```

DELIMITER ;

Call

CALL AverageSalary();

11. Display Employees Ordered by Salary

Question:

Write a procedure to display employees in ascending order of salary.

Answer

DELIMITER //

```
CREATE PROCEDURE EmployeesBySalary()
BEGIN
    SELECT *
    FROM Employees
    ORDER BY salary;
```

END //

DELIMITER ;

Call

CALL EmployeesBySalary();

12. Display Employees Ordered by Experience

Question:

Write a procedure to display employees sorted by experience in descending order.

Answer

DELIMITER //

CREATE PROCEDURE EmployeesByExperience()

BEGIN

 SELECT *

 FROM Employees

 ORDER BY experience DESC;

END //

DELIMITER ;

Call

CALL EmployeesByExperience();

13. Display Finance Department Employees

Question:

Write a procedure to display Finance department employees.

Answer

DELIMITER //

```
CREATE PROCEDURE FinanceEmployees()  
BEGIN  
    SELECT *  
    FROM Employees  
    WHERE department='Finance';  
END //
```

DELIMITER ;

Cal

lCALL FinanceEmployees();

14. Display HR Employees

Question:

Write a procedure to display all HR employees.

Answer

DELIMITER //

```
CREATE PROCEDURE HREmployees()  
BEGIN  
    SELECT *  
    FROM Employees  
    WHERE department='HR';  
END //
```

DELIMITER ;

Call

CALL HREmployees();

15. Total Salary Paid

Question:

Write a procedure to calculate the total salary paid to employees.

Answer

DELIMITER //

```
CREATE PROCEDURE TotalSalary()  
BEGIN  
    SELECT SUM(salary) AS TotalSalary  
    FROM Employees;  
END //
```

DELIMITER ;

Call

CALL TotalSalary();

16. Employees with Salary Between 50000 and 80000

Question:

Write a procedure to display employees whose salary is between 50000 and 80000.

Answer

DELIMITER //

```
CREATE PROCEDURE SalaryRange()  
BEGIN  
    SELECT *  
    FROM Employees  
    WHERE salary BETWEEN 50000 AND 80000;  
END //
```

DELIMITER ;

Call

CALL SalaryRange();

17. Employees from Bangalore

Question:

Write a procedure to display employees from Bangalore.

Answer

DELIMITER //

CREATE PROCEDURE BangaloreEmployees()

BEGIN

 SELECT *

 FROM Employees

 WHERE city='Bangalore';

END //

DELIMITER ;

Call

CALL BangaloreEmployees();

18. Employees Having Salary Above Average

Question:

Write a procedure to display employees whose salary is greater than the average salary.

Answer

DELIMITER //

CREATE PROCEDURE AboveAverageSalary()

BEGIN

 SELECT *

 FROM Employees

```
WHERE salary > (SELECT AVG(salary) FROM Employees);  
END //
```

```
DELIMITER ;
```

```
Call
```

```
CALL AboveAverageSalary();
```

19. Department-wise Employee Count

Question:

Write a procedure to display the number of employees in each department.

Answer

```
DELIMITER //
```

```
CREATE PROCEDURE DepartmentWiseCount()
```

```
BEGIN
```

```
    SELECT department,  
           COUNT(*) AS TotalEmployees  
    FROM Employees  
    GROUP BY department;
```

```
END //
```

```
DELIMITER ;
```

```
Call
```

```
CALL DepartmentWiseCount();
```

20. Employees with Maximum Experience

Question:

Write a procedure to display the employee(s) having the highest experience.

Answer

DELIMITER //

CREATE PROCEDURE HighestExperience()

BEGIN

SELECT *

FROM Employees

WHERE experience = (

SELECT MAX(experience)

FROM Employees

);

END //

DELIMITER ;

Call

CALL HighestExperience();

IN Parameter Procedures

1. Employee by ID

DELIMITER //

CREATE PROCEDURE GetEmployeeById(IN eid INT)

BEGIN

SELECT * FROM Employees

WHERE emp_id = eid;

END //

DELIMITER ;

CALL GetEmployeeById(101);

2. Employees by Department

DELIMITER //

CREATE PROCEDURE GetEmployeesByDepartment(IN dept VARCHAR(30))

BEGIN

SELECT *

FROM Employees

WHERE department = dept;

END //

```
DELIMITER ;  
CALL GetEmployeesByDepartment('IT');
```

3. Employees by Salary

```
DELIMITER //
```

```
CREATE PROCEDURE GetEmployeesBySalary(IN sal DECIMAL(10,2))  
BEGIN  
    SELECT *  
    FROM Employees  
    WHERE salary >= sal;  
END //
```

```
DELIMITER ;  
CALL GetEmployeesBySalary(60000);
```

4. Employees by City

```
DELIMITER //
```

```
CREATE PROCEDURE GetEmployeesByCity(IN cityName VARCHAR(30))  
BEGIN  
    SELECT *  
    FROM Employees  
    WHERE city = cityName;  
END //
```

```
DELIMITER ;  
CALL GetEmployeesByCity('Hyderabad');
```

5. Employees by Experience

```
DELIMITER //
```

```
CREATE PROCEDURE GetExperiencedEmployees(IN exp INT)  
BEGIN  
    SELECT *  
    FROM Employees  
    WHERE experience >= exp;  
END //
```

```
DELIMITER ;  
CALL GetExperiencedEmployees(5);
```

6. Employee by Name

```
DELIMITER //
```

```
CREATE PROCEDURE GetEmployeeByName(IN ename VARCHAR(50))  
BEGIN  
    SELECT *  
    FROM Employees  
    WHERE emp_name = ename;  
END //
```

```
DELIMITER ;  
CALL GetEmployeeByName('Rahul');
```

7. Salary Range

```
DELIMITER //
```

```
CREATE PROCEDURE SalaryRange(IN minSal DECIMAL(10,2), IN maxSal  
DECIMAL(10,2))  
BEGIN  
    SELECT *  
    FROM Employees  
    WHERE salary BETWEEN minSal AND maxSal;  
END //
```

```
DELIMITER ;  
CALL SalaryRange(40000,70000);
```

8. Department and Experience

```
DELIMITER //
```

```
CREATE PROCEDURE DeptExperience(IN dept VARCHAR(30), IN exp INT)  
BEGIN  
    SELECT *  
    FROM Employees  
    WHERE department=dept  
    AND experience>=exp;  
END //
```

```
DELIMITER ;  
CALL DeptExperience('IT',5);
```

9. City and Salary

```
DELIMITER //
```

```
CREATE PROCEDURE CitySalary(IN cityName VARCHAR(30), IN sal DECIMAL(10,2))  
BEGIN  
    SELECT *  
    FROM Employees  
    WHERE city=cityName  
    AND salary>=sal;  
END //
```

```
DELIMITER ;  
CALL CitySalary('Hyderabad',60000);
```

10. Finance Employees

```
DELIMITER //
```

```
CREATE PROCEDURE FinanceEmployees()  
BEGIN  
    SELECT *  
    FROM Employees  
    WHERE department='Finance';  
END //
```

```
DELIMITER ;  
CALL FinanceEmployees();
```

OUT Parameter Procedures

1. Total Employees

```
DELIMITER //
```

```
CREATE PROCEDURE TotalEmployees(OUT total INT)  
BEGIN  
    SELECT COUNT(*) INTO total  
    FROM Employees;  
END //
```

```
DELIMITER ;  
CALL TotalEmployees(@t);  
SELECT @t;
```

2. Maximum Salary

```
DELIMITER //
```

```
CREATE PROCEDURE MaximumSalary(OUT maxSal DECIMAL(10,2))  
BEGIN  
    SELECT MAX(salary)  
    INTO maxSal  
    FROM Employees;  
END //
```

```
DELIMITER ;  
CALL MaximumSalary(@m);  
SELECT @m;
```

3. Minimum Salary

```
DELIMITER //
```

```
CREATE PROCEDURE MinimumSalary(OUT minSal DECIMAL(10,2))  
BEGIN  
    SELECT MIN(salary)  
    INTO minSal  
    FROM Employees;  
END //
```

```
DELIMITER ;
```

4. Average Salary

```
DELIMITER //
```

```
CREATE PROCEDURE AverageSalary(OUT avgSal DECIMAL(10,2))  
BEGIN  
    SELECT AVG(salary)  
    INTO avgSal  
    FROM Employees;  
END //
```

DELIMITER ;

5. Total Salary

DELIMITER //

```
CREATE PROCEDURE TotalSalary(OUT totalSal DECIMAL(10,2))
BEGIN
    SELECT SUM(salary)
    INTO totalSal
    FROM Employees;
END //
```

DELIMITER ;

6. Highest Experience

DELIMITER //

```
CREATE PROCEDURE HighestExperience(OUT exp INT)
BEGIN
    SELECT MAX(experience)
    INTO exp
    FROM Employees;
END //
```

DELIMITER ;

7. Lowest Experience

DELIMITER //

```
CREATE PROCEDURE LowestExperience(OUT exp INT)
BEGIN
    SELECT MIN(experience)
    INTO exp
    FROM Employees;
END //
```

DELIMITER ;

8. IT Employee Count

DELIMITER //

```
CREATE PROCEDURE IEmployeeCount(OUT cnt INT)
BEGIN
    SELECT COUNT(*)
    INTO cnt
    FROM Employees
    WHERE department='IT';
END //
```

DELIMITER ;

9. Hyderabad Employee Count

DELIMITER //

```
CREATE PROCEDURE HyderabadCount(OUT cnt INT)
BEGIN
    SELECT COUNT(*)
    INTO cnt
    FROM Employees
    WHERE city='Hyderabad';
END //
```

DELIMITER ;

10. Average Experience

DELIMITER //

```
CREATE PROCEDURE AverageExperience(OUT avgExp DECIMAL(5,2))
BEGIN
    SELECT AVG(experience)
    INTO avgExp
    FROM Employees;
END //
```

DELIMITER ;

INOUT Parameter Procedures

1. Add Bonus

DELIMITER //

```

CREATE PROCEDURE AddBonus(
    IN empid INT,
    INOUT bonus DECIMAL(10,2)
)
BEGIN
    UPDATE Employees
    SET salary = salary + bonus
    WHERE emp_id = empid;

    SELECT salary
    INTO bonus
    FROM Employees
    WHERE emp_id = empid;
END //

DELIMITER ;
SET @b=5000;
CALL AddBonus(101,@b);
SELECT @b;

```

2. Deduct Salary

```

DELIMITER //

CREATE PROCEDURE DeductSalary(
    IN empid INT,
    INOUT amount DECIMAL(10,2)
)
BEGIN
    UPDATE Employees
    SET salary=salary-amount
    WHERE emp_id=empid;

    SELECT salary
    INTO amount
    FROM Employees
    WHERE emp_id=empid;
END //

DELIMITER ;

```



```
SET @deduction = 5000;  
CALL DeductSalary(101,@deduction);  
SELECT @deduction;
```

3. Increase Experience

DELIMITER //

```
CREATE PROCEDURE IncreaseExperience(  
    IN empid INT,  
    INOUT exp INT  
)  
BEGIN  
    UPDATE Employees  
    SET experience=experience+exp  
    WHERE emp_id=empid;  
  
    SELECT experience  
    INTO exp  
    FROM Employees  
    WHERE emp_id=empid;  
END //
```

```
DELIMITER ;  
SET @exp = 2;  
CALL IncreaseExperience(101,@exp);  
SELECT @exp;
```

5. Increase Salary by Amount

DELIMITER //

```
CREATE PROCEDURE SalaryIncrement(  
    IN empid INT,  
    INOUT inc DECIMAL(10,2)  
)  
BEGIN  
    UPDATE Employees  
    SET salary=salary+inc  
    WHERE emp_id=empid;
```

```
SELECT salary
  INTO inc
  FROM Employees
  WHERE emp_id=empid;
END //

DELIMITER ;
```

Stored Procedures - Scenario Based Questions (MySQL)

These questions are commonly asked in interviews and practical assessments. Each scenario requires creating a stored procedure.

Scenario 1: Employee Search

Problem

A company wants to search employee details by Employee ID without writing the same query repeatedly.

Task

Create a stored procedure that accepts Employee ID and displays employee details.

Concept: IN Parameter

Scenario 2: Department Employees

Problem

The HR department wants to see all employees working in a specific department.

Task

Create a stored procedure that accepts the department name and displays all employees.

Concept: IN Parameter

Scenario 3: Salary Filter

Problem

The finance team wants to identify employees earning more than a specified salary.

Task

Create a stored procedure that accepts a salary amount and displays all employees earning more than that amount.

Concept: IN Parameter

Scenario 4: City-wise Employees

Problem

The company has offices in multiple cities. Management wants to view employees from a particular city.

Task

Create a stored procedure that accepts a city name and displays employees from that city.

Concept: IN Parameter

Scenario 5: Experience-Based Employees

Problem

The recruitment team wants employees who have at least a certain number of years of experience.

Task

Create a stored procedure that accepts years of experience and displays matching employees.

Concept: IN Parameter

Scenario 6: Total Employees

Problem

The CEO wants to know the total number of employees in the company.

Task

Create a stored procedure that returns the total employee count.

Concept: OUT Parameter

Scenario 7: Highest Salary

Problem

The finance department wants to know the highest salary paid in the company.

Task

Create a stored procedure that returns the highest salary.

Concept: OUT Parameter

Scenario 8: Average Salary

Problem

The accounts department wants to calculate the average salary of all employees.

Task

Create a stored procedure that returns the average salary.

Concept: OUT Parameter

Scenario 9: Total Salary Expenditure

Problem

Management wants to know the total monthly salary expenditure.

Task

Create a stored procedure that returns the sum of all employee salaries.

Concept: OUT Parameter

Scenario 10: IT Employee Count

Problem

The IT manager wants to know how many employees are working in the IT department.

Task

Create a stored procedure that returns the number of IT employees.

Concept: OUT Parameter

Scenario 11: Bonus Distribution

Problem

The company announces a ₹5,000 bonus for a specific employee.

Task

Create a stored procedure that accepts Employee ID and Bonus Amount, updates the salary, and returns the updated salary.

Concept: INOUT Parameter

Scenario 12: Salary Increment

Problem

Every year, employees receive an increment.

Task

Create a stored procedure that accepts Employee ID and Increment Amount, updates the salary, and returns the new salary.

Concept: INOUT Parameter

Scenario 13: Tax Deduction

Problem

The payroll department deducts tax from employees' salaries.

Task

Create a stored procedure that accepts Employee ID and Tax Amount, deducts the tax, and returns the updated salary.

Concept: INOUT Parameter

Scenario 14: Promotion**Problem**

An employee gets promoted, resulting in both a salary increase and an increase in experience.

Task

Create a stored procedure that updates the employee's salary and experience and returns the updated salary.

Concept: INOUT Parameter

Scenario 15: Festival Bonus**Problem**

During a festival, employees receive an additional bonus.

Task

Create a stored procedure that adds the bonus to the salary and returns the revised salary.

Concept: INOUT Parameter

Scenario 16: Student Results**Problem**

A college wants to display all students who scored above a specified mark.

Task

Create a stored procedure that accepts the minimum marks and displays qualifying students.

Concept: IN Parameter

Scenario 17: Product Search**Problem**

An e-commerce website allows users to search products by category.

Task

Create a stored procedure that accepts the category name and displays matching products.

Concept: IN Parameter

Scenario 18: Order Summary**Problem**

An online shopping system wants to know the total number of orders placed.

Task

Create a stored procedure that returns the total number of orders.

Concept: OUT Parameter

Scenario 19: Bank Account Deposit**Problem**

A customer deposits money into a bank account.

Task

Create a stored procedure that accepts Account Number and Deposit Amount, updates the balance, and returns the updated balance.

Concept: INOUT Parameter

Scenario 20: Hospital Patient Search**Problem**

A hospital receptionist wants to retrieve patient details using the Patient ID.

Task

Create a stored procedure that accepts Patient ID and displays the patient's information.

Concept: IN Parameter

Scenario 21: Library Management**Problem**

A librarian wants to know how many books are currently available.

Task

Create a stored procedure that returns the total number of available books.

Concept: OUT Parameter

Scenario 22: Hotel Room Booking**Problem**

A hotel receptionist books a room for a customer.

Task

Create a stored procedure that updates the room status from **Available** to **Booked** and returns the updated status.

Concept: INOUT Parameter

Scenario 23: Inventory Management**Problem**

A store sells a product, reducing the available stock.

Task

Create a stored procedure that accepts Product ID and Quantity Sold, updates the stock, and returns the remaining stock.

Concept: INOUT Parameter

Scenario 24: Employee Transfer**Problem**

An employee is transferred to another department.

Task

Create a stored procedure that accepts Employee ID and New Department, updates the department, and displays the updated employee details.

Concept: IN Parameter

Scenario 25: Payroll System

Problem

The payroll department wants to identify employees whose salary falls within a specified range.

Task

Create a stored procedure that accepts a minimum salary and maximum salary and displays all matching employees.

Concept: IN Parameter

Summary

Scenario	Concept
Employee Search	IN
Department	IN
Employees	
Salary Filter	IN
City-wise Employees	IN
Experience Search	IN
Total Employees	OUT
Highest Salary	OUT
Average Salary	OUT
Total Salary	OUT
IT Employee Count	OUT
Bonus Distribution	INOUT
Salary Increment	INOUT
Tax Deduction	INOUT
Promotion	INOUT
Festival Bonus	INOUT
Student Results	IN
Product Search	IN
Order Summary	OUT
Bank Deposit	INOUT
Hospital Search	IN
Library Books Count	OUT
Hotel Booking	INOUT

Inventory Update INOUT
Employee Transfer IN
Payroll Salary Range IN

. Introduction to Triggers

A trigger is a special type of stored program that automatically executes (fires) in response to a specific event — INSERT, UPDATE, or DELETE — occurring on a specified table. Unlike a stored procedure, a trigger is never called directly; the database engine invokes it automatically when the triggering event happens.

General Syntax

```
DELIMITER //

CREATE TRIGGER trigger_name
{BEFORE | AFTER} {INSERT | UPDATE | DELETE}
ON table_name
FOR EACH ROW
BEGIN
    -- trigger logic
    -- use NEW.column_name for inserted/updated values
    -- use OLD.column_name for deleted/previous values
END //

DELIMITER ;
```

Keyword	Meaning
BEFORE	Trigger fires before the triggering event is applied to the table. Often used for validation.
AFTER	Trigger fires after the triggering event has been applied. Often used for auditing/logging.
NEW	Refers to the new row values. Available in INSERT and UPDATE triggers.
OLD	Refers to the row values before the change. Available in UPDATE and DELETE triggers.
FOR EACH ROW	Required in MySQL — the trigger body runs once per affected row.

Database Schema

The examples in this document use the following Employees table and an Employee_Audit table used to log insert, update, and delete activity.

Employees Table:

```
CREATE TABLE Employees (  
    emp_id INT PRIMARY KEY,  
    emp_name VARCHAR(50),  
    department VARCHAR(30),  
    salary DECIMAL(10,2),  
    experience INT,  
    city VARCHAR(30)  
);  
  
INSERT INTO Employees VALUES  
(101,'Rahul','IT',65000,5,'Hyderabad'),  
(102,'Priya','HR',45000,3,'Chennai'),  
(103,'Amit','Finance',70000,6,'Bangalore'),  
(104,'Sneha','IT',80000,8,'Hyderabad'),  
(105,'Kiran','Sales',40000,2,'Pune');
```

Employee_Audit Table:

```
CREATE TABLE Employee_Audit(  
    audit_id INT AUTO_INCREMENT PRIMARY KEY,  
    emp_id INT,  
    action_type VARCHAR(20),  
    old_salary DECIMAL(10,2),  
    new_salary DECIMAL(10,2),  
    action_time DATETIME  
);
```

Trigger Examples

1. BEFORE INSERT (Validation)

Question:

Do not allow employees with salary less than ₹20,000.

Trigger SQL:

```
DELIMITER //  
  
CREATE TRIGGER trg_before_insert_salary  
BEFORE INSERT  
ON Employees  
FOR EACH ROW
```

```
BEGIN
    IF NEW.salary < 20000 THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT='Salary cannot be less than 20000';
    END IF;
END//

DELIMITER ;
```

Test:

```
INSERT INTO Employees
VALUES(106,'Arjun','IT',15000,2,'Delhi');
```

Output:

```
ERROR 1644 (45000):
Salary cannot be less than 20000
```

2. BEFORE INSERT

Question:

Convert employee name into uppercase before inserting.

Trigger SQL:

```
DELIMITER //

CREATE TRIGGER trg_before_insert_uppercase
BEFORE INSERT
ON Employees
FOR EACH ROW
BEGIN
    SET NEW.emp_name=UPPER(NEW.emp_name);
END//

DELIMITER ;
```

Test:

```
INSERT INTO Employees
VALUES(106,'arjun','IT',50000,2,'Delhi');
```

Output:

```
ARJUN
```

3. AFTER INSERT (Auditing)

Question:

Store inserted employee details in the audit table.

Trigger SQL:

```
DELIMITER //
```

```

CREATE TRIGGER trg_after_insert
AFTER INSERT
ON Employees
FOR EACH ROW
BEGIN
    INSERT INTO Employee_Audit
    (emp_id,action_type,new_salary,action_time)
    VALUES
    (NEW.emp_id,'INSERT',NEW.salary,NOW());
END//

DELIMITER ;

```

Test:

```

INSERT INTO Employees
VALUES(107,'Kavya','HR',55000,4,'Chennai');
SELECT * FROM Employee_Audit;

```

4. BEFORE UPDATE (Validation)

Question:

Do not allow salary reduction.

Trigger SQL:

```

DELIMITER //

CREATE TRIGGER trg_before_update_salary
BEFORE UPDATE
ON Employees
FOR EACH ROW
BEGIN
    IF NEW.salary<OLD.salary THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT='Salary cannot be reduced';
    END IF;
END//

DELIMITER ;

```

Test:

```

UPDATE Employees
SET salary=30000
WHERE emp_id=101;

```

Output:

Salary cannot be reduced

5. AFTER UPDATE (Audit Salary Change)

Question:

Store old salary and new salary after updating.

Trigger SQL:

```
DELIMITER //

CREATE TRIGGER trg_after_update
AFTER UPDATE
ON Employees
FOR EACH ROW
BEGIN
    INSERT INTO Employee_Audit
    (emp_id,action_type,old_salary,new_salary,action_time)
    VALUES
    (OLD.emp_id,'UPDATE',OLD.salary,NEW.salary,NOW());
END//

DELIMITER ;
```

Test:

```
UPDATE Employees
SET salary=70000
WHERE emp_id=101;
```

6. BEFORE DELETE (Validation)

Question:

Do not allow deleting IT employees.

Trigger SQL:

```
DELIMITER //

CREATE TRIGGER trg_before_delete
BEFORE DELETE
ON Employees
FOR EACH ROW
BEGIN
    IF OLD.department='IT' THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT='Cannot delete IT employees';
    END IF;
END//

DELIMITER ;
```

Test:

```
DELETE FROM Employees
WHERE emp_id=101;
```

Output:

Cannot delete IT employees

7. AFTER DELETE (Audit)

Question:

Store deleted employee details.

Trigger SQL:

```
DELIMITER //

CREATE TRIGGER trg_after_delete
AFTER DELETE
ON Employees
FOR EACH ROW
BEGIN
    INSERT INTO Employee_Audit
    (emp_id,action_type,old_salary,action_time)
    VALUES
    (OLD.emp_id,'DELETE',OLD.salary,NOW());
END//

DELIMITER ;
```

Test:

```
DELETE FROM Employees
WHERE emp_id=105;
SELECT * FROM Employee_Audit;
```

8. BEFORE UPDATE

Question:

Automatically convert city names into uppercase.

Trigger SQL:

```
DELIMITER //

CREATE TRIGGER trg_city_uppercase
BEFORE UPDATE
ON Employees
FOR EACH ROW
BEGIN
    SET NEW.city=UPPER(NEW.city);
END//

DELIMITER ;
```

9. BEFORE INSERT

Question:

Experience should not be negative.

Trigger SQL:

```
DELIMITER //

CREATE TRIGGER trg_check_experience
BEFORE INSERT
ON Employees
FOR EACH ROW
BEGIN
    IF NEW.experience < 0 THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Experience cannot be negative';
    END IF;
END//

DELIMITER ;
```

10. AFTER INSERT

Question:

Maintain employee count.

Create table:

```
CREATE TABLE Employee_Count(
    total_employees INT
);

INSERT INTO Employee_Count VALUES(5);
```

Trigger:

```
DELIMITER //

CREATE TRIGGER trg_employee_count
AFTER INSERT
ON Employees
FOR EACH ROW
BEGIN
    UPDATE Employee_Count
    SET total_employees = total_employees + 1;
END//

DELIMITER ;
```

11. BEFORE DELETE

Question:

Prevent deletion if employee salary is greater than 70000.

Trigger SQL:

```
DELIMITER //

CREATE TRIGGER trg_high_salary_delete
BEFORE DELETE
ON Employees
FOR EACH ROW
BEGIN
    IF OLD.salary>70000 THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT='High salary employee cannot be deleted';
    END IF;
END//

DELIMITER ;
```

12. AFTER UPDATE

Question:

Store every update in audit.

Trigger SQL:

```
DELIMITER //

CREATE TRIGGER trg_update_log
AFTER UPDATE
ON Employees
FOR EACH ROW
BEGIN
    INSERT INTO Employee_Audit
    (emp_id,action_type,old_salary,new_salary,action_time)
    VALUES
    (NEW.emp_id,'UPDATED',OLD.salary,NEW.salary,NOW());
END//

DELIMITER ;
```

DROP TRIGGER

Drop a specific trigger:

```
DROP TRIGGER trg_after_insert;
```

Drop another trigger:

```
DROP TRIGGER trg_before_update_salary;
```

Check existing triggers:

```
SHOW TRIGGERS;
```

Show trigger definition:

```
SHOW CREATE TRIGGER trg_after_insert;
```

Real-Time Trigger Use Cases

Scenario	Trigger Type
Prevent invalid salary	BEFORE INSERT
Convert names to uppercase	BEFORE INSERT
Validate experience	BEFORE INSERT
Prevent salary reduction	BEFORE UPDATE
Log salary changes	AFTER UPDATE
Prevent deleting important employees	BEFORE DELETE
Log deleted records	AFTER DELETE
Maintain audit history	AFTER INSERT/UPDATE/DELETE
Update employee count	AFTER INSERT
Validate business rules	BEFORE INSERT/UPDATE

Interview Questions

1. Create a trigger to prevent inserting employees with salary less than ₹25,000.
2. Create a trigger to convert employee names to uppercase before insertion.
3. Create a trigger to log every inserted employee into an audit table.
4. Create a trigger to prevent reducing an employee's salary.
5. Create a trigger to log old and new salaries after an update.
6. Create a trigger to prevent deleting employees from the IT department.
7. Create a trigger to store deleted employee details in an audit table.
8. Create a trigger to ensure experience is never negative.

9. Create a trigger to maintain the total employee count after each insertion.
10. Create a trigger to prevent deletion of employees whose salary is greater than ₹70,000.

PART 3 — QUICK REFERENCE

10. Stored Procedure vs Trigger — Comparison

Aspect	Stored Procedure	Trigger
Invocation	Explicitly called using CALL	Fires automatically on INSERT/UPDATE/DELETE
Parameters	Accepts IN, OUT, INOUT parameters	No parameters; uses NEW / OLD
Can be called manually?	Yes	No
Return value	Via OUT/INOUT params or result set	None (cannot return values to caller)
Typical use	Reusable business logic, batch operations	Auditing, validation, enforcing rules automatically
Drop statement	DROP PROCEDURE	DROP TRIGGER

11. Command Cheat-Sheet

Task	Statement
Create a procedure	CREATE PROCEDURE name(params) BEGIN ... END;
Call a procedure	CALL name(args);
Delete a procedure	DROP PROCEDURE [IF EXISTS] name;
View procedure code	SHOW CREATE PROCEDURE name;
List all procedures	SHOW PROCEDURE STATUS WHERE Db = 'db_name';

Create a trigger	CREATE TRIGGER name {BEFORE AFTER} {INSERT UPDATE DELETE} ON table FOR EACH ROW BEGIN ... END;
Delete a trigger	DROP TRIGGER [IF EXISTS] name;
List all triggers	SHOW TRIGGERS;

QUESTION & ANSWER BANK

Question	Answer
Q1. What is a stored procedure?	A precompiled, named set of SQL statements stored in the database that can be executed using the CALL statement. It can accept parameters and contain procedural logic such as variables, conditions, and loops.
Q2. What is the default parameter mode if none is specified in CREATE PROCEDURE?	IN. If no mode is written before a parameter name, MySQL treats it as an IN parameter.
Q3. What is the difference between IN, OUT, and INOUT parameters?	IN passes a value into the procedure only (read-only inside). OUT is set inside the procedure and returned to the caller (initial value ignored). INOUT passes a value in, allows it to be modified, and returns the updated value to the caller.
Q4. Why do we use DELIMITER // before creating a procedure or trigger?	Because the procedure/trigger body itself contains semicolons to separate statements. Changing the delimiter to // (or similar) tells the client to treat the entire block as one statement, ending only at //, so the internal semicolons don't terminate the CREATE statement early.
Q5. How do you call a stored procedure that has an OUT parameter?	Declare a session variable (e.g., @result), pass it in the CALL, then SELECT the variable afterward: CALL proc(@result); SELECT @result;
Q6. How do you permanently remove a stored procedure?	Use DROP PROCEDURE [IF EXISTS] procedure_name;
Q7. What is a trigger in SQL?	A database object that automatically executes a defined block of SQL in response to an INSERT, UPDATE, or DELETE event on a specified table, without being called explicitly.

Q8. What is the difference between a BEFORE trigger and an AFTER trigger?	A BEFORE trigger fires before the change is applied to the table and can validate/modify the incoming row (commonly used for validation). An AFTER trigger fires once the change has been committed and is commonly used for auditing/logging, since it cannot alter the row anymore.
Q9. What do NEW and OLD refer to in a trigger?	NEW refers to the new/incoming row values (available in INSERT and UPDATE triggers). OLD refers to the row's values before the change (available in UPDATE and DELETE triggers). INSERT triggers have no OLD; DELETE triggers have no NEW.
Q10. Can a trigger call a stored procedure or return a value to the caller?	A trigger can call certain stored procedures/functions internally in some databases, but it can never return a value directly to the statement that fired it (unlike a stored procedure called with CALL).
Q11. Why use FOR EACH ROW in a MySQL trigger?	MySQL only supports row-level triggers, so FOR EACH ROW is mandatory — it specifies that the trigger body executes once for every row affected by the INSERT, UPDATE, or DELETE statement.
Q12. How would you reject an invalid INSERT using a trigger?	Create a BEFORE INSERT trigger that checks the NEW row values with an IF condition and raises a custom error using SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'error message'; this aborts the insert.
Q13. Give a real-world example of using a trigger for auditing.	An AFTER UPDATE trigger on an employees table that inserts the emp_id, old_salary, new_salary, and timestamp into a salary_history table every time a salary changes.
Q14. Give a real-world example of using a trigger for validation.	A BEFORE INSERT trigger on a products table that rejects any row where quantity is negative or price is less than or equal to zero, using SIGNAL to raise an error.
Q15. How do you remove a trigger from a database?	Use DROP TRIGGER [IF EXISTS] trigger_name; This removes only the trigger definition, not the table or its data.
Q16. What is the main advantage of stored procedures over sending raw SQL from an application?	Reduced network traffic (single CALL instead of multiple statements), improved performance through precompilation/caching, centralized and reusable business logic, and finer-grained security via EXECUTE privileges instead of direct table access.
Q17. Can a single table have multiple triggers for the same event (e.g., two AFTER INSERT triggers)?	MySQL allows multiple triggers for the same event and timing from version 5.7.2 onward, executed in the order they were created (or as specified with FOLLOWS/PRECEDES); many other databases also support this with similar ordering rules.
Q18. What happens to existing data in a table when its trigger is dropped?	Nothing — DROP TRIGGER only removes the automatic firing logic. All existing rows and data in the table remain completely untouched.

Day-18

NORMALIZATION

Table of Contents

1. Data Redundancy
2. Types of Anomalies
3. What is Normalization?
4. Normalization Forms — Overview
5. First Normal Form (1NF)
6. Second Normal Form (2NF)
7. Third Normal Form (3NF)
8. Boyce-Codd Normal Form (BCNF)
9. Fourth Normal Form (4NF)
10. Fifth Normal Form (5NF)

1. Data Redundancy

- Data redundancy means having the same data stored at multiple places within a database.
- Data redundancy leads to anomalies (errors) in the database.
- It becomes very hard to maintain a database that has redundant data.

Example — Data Redundancy:

NAME	ROLL NO	SECTION
SAI	21	A
SAI	21	A
SHIVA	22	B

Notice that the row for "SAI, 21, A" is repeated — this duplication is an example of data redundancy.

2. Types of Anomalies

Data redundancy causes anomalies. The three types of anomalies are:

- Insertion Anomalies — difficulty inserting data due to missing/dependent information.
- Updation Anomalies — the need to update the same data in multiple places, risking inconsistency if missed.
- Deletion Anomalies — deleting a record unintentionally removes other, unrelated information stored in the same row.

3. What is Normalization?

Normalization is the process of organizing data in a database to reduce data redundancy and improve data integrity.

It involves dividing large tables into smaller ones and defining relationships between them to ensure data consistency.

The main goals of normalization are:

- Eliminate data redundancy (duplicate data).
- Ensure data dependencies are logical.
- Improve data integrity (missing values) and consistency (correctness).

4. Normalization Forms — Overview

Normalization is typically carried out through a series of normal forms (NF), each addressing specific issues:

- First Normal Form (1NF) — Eliminate repeating groups.
- Second Normal Form (2NF) — Eliminate partial dependency.
- Third Normal Form (3NF) — Eliminate transitive dependency.
- Boyce-Codd Normal Form (BCNF) — A stricter version of 3NF.
- Fourth Normal Form (4NF) — Handle multi-valued dependencies.
- Fifth Normal Form (5NF) — Handle join dependencies.

5. First Normal Form (1NF)

- A relation should be in 1NF if it contains only atomic values.
- It states that an attribute of a table cannot hold multiple values.
- 1NF does not allow multi-valued attributes.

Before applying 1NF (multi-valued attribute LANGUAGES_KNOWN):

ROLL NO	NAME	BRANCH	LANGUAGES_KNOWN
101	SAI	CSE	TELUGU, ENGLISH
102	SHIVA	AIML	ENGLISH, HINDI
103	RAJU	DS	ENGLISH, HINDI

After applying 1NF (each value made atomic — one language per row):

ROLL NO	NAME	COURSE	LANGUAGES_KNOWN
101	SAI	CSE	TELUGU
101	SAI	CSE	ENGLISH
102	SHIVA	AIML	ENGLISH
102	SHIVA	AIML	HINDI
103	RAJU	DS	ENGLISH
103	RAJU	DS	HINDI

6. Second Normal Form (2NF)

- The table must already be in 1NF.
- There are no partial dependencies — i.e., no non-prime attribute depends on only part of a composite primary key.
- The primary key should not contain duplicate columns.

Table before applying 2NF (roll no repeats due to multiple languages, a partial dependency on part of the key):

ROLL NO [PK]	NAME	COURSE	LANGUAGES_KNOWN
101	SAI	CSE	TELUGU
101	SAI	CSE	ENGLISH
102	SHIVA	AIML	ENGLISH
102	SHIVA	AIML	HINDI

103	RAJU	DS	ENGLISH
103	RAJU	DS	HINDI

After applying 2NF — split into two related tables:

ROLL NO	NAME	BRANCH
101	SAI	CSE
102	SHIVA	AIML
103	RAJU	DS

ROLL NO	LANGUAGES_KNOWN
101	TELUGU
101	ENGLISH
102	ENGLISH
102	HINDI
103	ENGLISH
103	HINDI

7. Third Normal Form (3NF)

- The table must already be in 2NF.
- There are no transitive dependencies — i.e., no non-prime attribute (N.P.A) depends on another non-prime attribute.

Table before applying 3NF (FEE depends on BRANCH, not directly on ROLL NO — a transitive dependency):

ROLL NO [PK]	NAME	BRANCH [N.P.A]	FEE [N.P.A]
101	SAI	CSE	60000
102	SHIVA	AIML	75000
103	RAMESH	DS	55000
104	SURESH	ECE	30000

After applying 3NF — split into two related tables:

ROLL NO [PK]	NAME [N.P.A]	BRANCH [N.P.A]
101	SAI	CSE
102	SHIVA	AIML
103	SURESH	DS
104	RAMESH	ECE

BRANCH [PK]	FEE [N.P.A]
CSE	60000
AIML	75000
DS	55000
ECE	35000

8. Boyce-Codd Normal Form (BCNF)

- The table must already be in 3NF.
- Every determinant must be a candidate key — i.e., no non-trivial functional dependency should exist where the determinant is not a superkey.
- In other words: if A depends on B, then B should be a super key.

Table before applying BCNF:

ROLL NO [PK]	NAME	BRANCH_ID	BRANCH NAME
101	SAI	121	CSE
102	SIVA	122	AIML
103	RAMESH	123	DS
104	SURESH	121	CSE

After applying BCNF — split into two related tables:

ROLL NO [PK]	NAME
101	SAI

102	SHIVA
103	RAMESH
104	SURESH

BRANCH_ID	BRANCH_NAME
121	CSE
122	AIML
123	DS

9. Fourth Normal Form (4NF)

- The table must already be in BCNF.
- It has no multi-valued dependencies — i.e., attributes are independent of each other and depend only on the primary key.

Table before applying 4NF (COURSE and HOBBY are independent multi-valued facts about the same ROLL NO):

ROLL NO [PK]	COURSE [N.P.A]	HOBBY [N.P.A]
101	CSE	DANCING
102	AIML	SLEEPING
103	DS	SINGING

After applying 4NF — split into two independent tables:

ROLL NO [PK]	COURSE [N.P.A]
101	CSE
102	AIML
103	DS

ROLL NO [PK]	HOBBY [N.P.A]
101	DANCING

102	SLEEPING
103	SINGING

10. Fifth Normal Form (5NF)

- The table must already be in 4NF.
- It does not consist of any join dependencies.
- 5NF is satisfied when tables present in the DBMS are broken down into multiple tables to ensure there is no redundancy when they are re-joined.

Table before applying 5NF:

SUBJECT	FACULTY	YEAR
C	SAI	1
C	SHIVA	1
JAVA	RAJU	2
DBMS	RAMESH	3

After applying 5NF — split into three tables to remove the join dependency:

SUBJECT	FACULTY
C	SAI
C	SHIVA
JAVA	RAJU
DBMS	RAMESH

FACULTY	YEAR
SAI	1
SHIVA	1
RAJU	2
RAMESH	3

SUBJECT	YEAR
C	1
JAVA	2
DBMS	3

ENTITY-RELATIONSHIP (ER) MODEL

Table of Contents

1. The Entity-Relationship (ER) Model
2. Entity
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4. Entity and Attributes — Combined Example
5. Relationships
6. One-to-One (1:1) Relationship
7. One-to-Many (1:M) Relationship
8. Many-to-Many (M:N) Relationship
9. Cardinality
10. Practice Question

1. The Entity-Relationship (ER) Model

The Entity-Relationship (ER) Model is a conceptual framework used to describe data relationships in a structured way.

Components of the ER Model:

- Entities
- Attributes

- Relationships
- Cardinality

2. Entity

An entity represents a real-world object or concept that has a unique existence.

Entities are classified into:

- Strong Entities — exist independently (e.g., Customer, Student, Book).
- Weak Entities — depend on other entities for existence (e.g., Order_Item depends on Order).

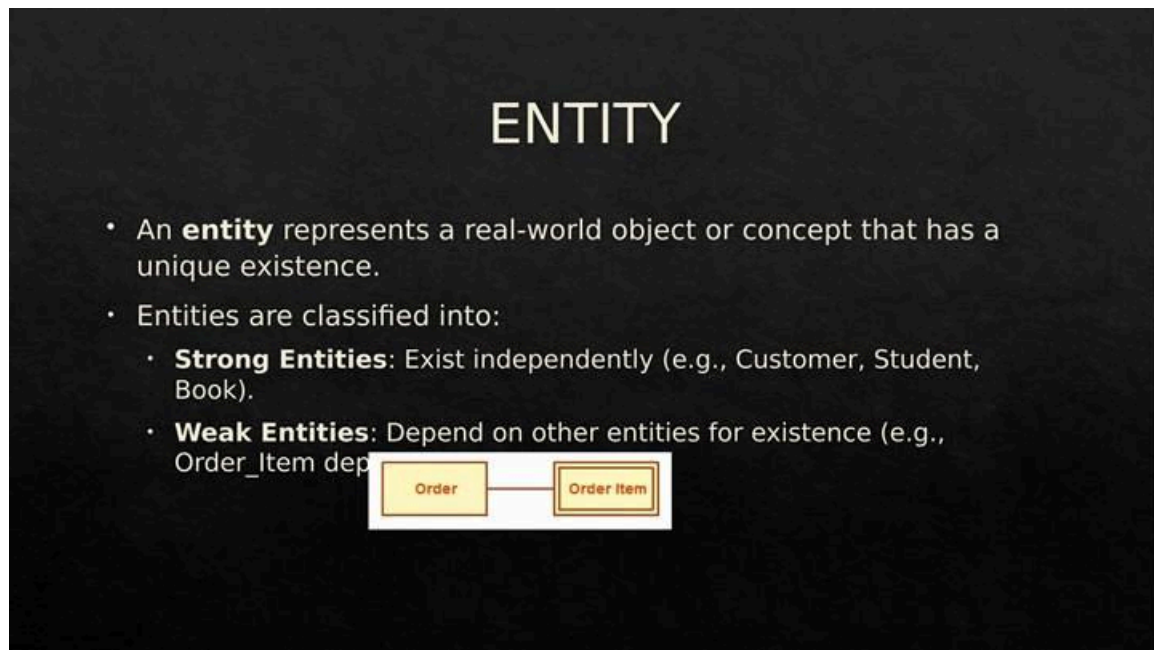


Diagram: Strong Entity vs. Weak Entity (Slide 3)

3. Attributes

Attributes describe properties or characteristics of an entity.

Types of Attributes:

- Simple (Atomic) Attributes — cannot be divided further (e.g., Name, Age).

ATTRIBUTES

- Attributes describe properties or characteristics of an entity.
- **Types of Attributes:**
 - **Simple (Atomic) Attributes:** Cannot be divided further (e.g., Name, Age).



Diagram: Simple (Atomic) Attribute (Slide 4)

- Composite Attribute — can be broken down into sub-attributes (e.g., Address → Street, City, Zip).

ATTRIBUTES

- Composite Attribute: Can be broken down into sub-attributes (e.g., Address → Street, City, Zip).

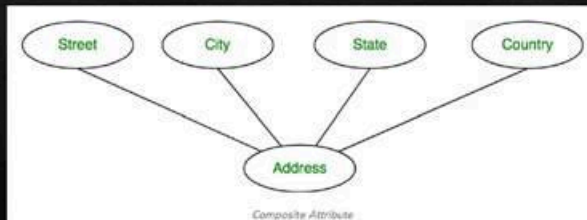


Diagram: Composite Attribute (Slide 5)

- Key Attribute — uniquely identifies an entity (Primary Key).
- Multivalued Attributes — can have multiple values (e.g., Phone Numbers).

- Derived Attributes — calculated from other attributes (e.g., Age from Date of Birth).

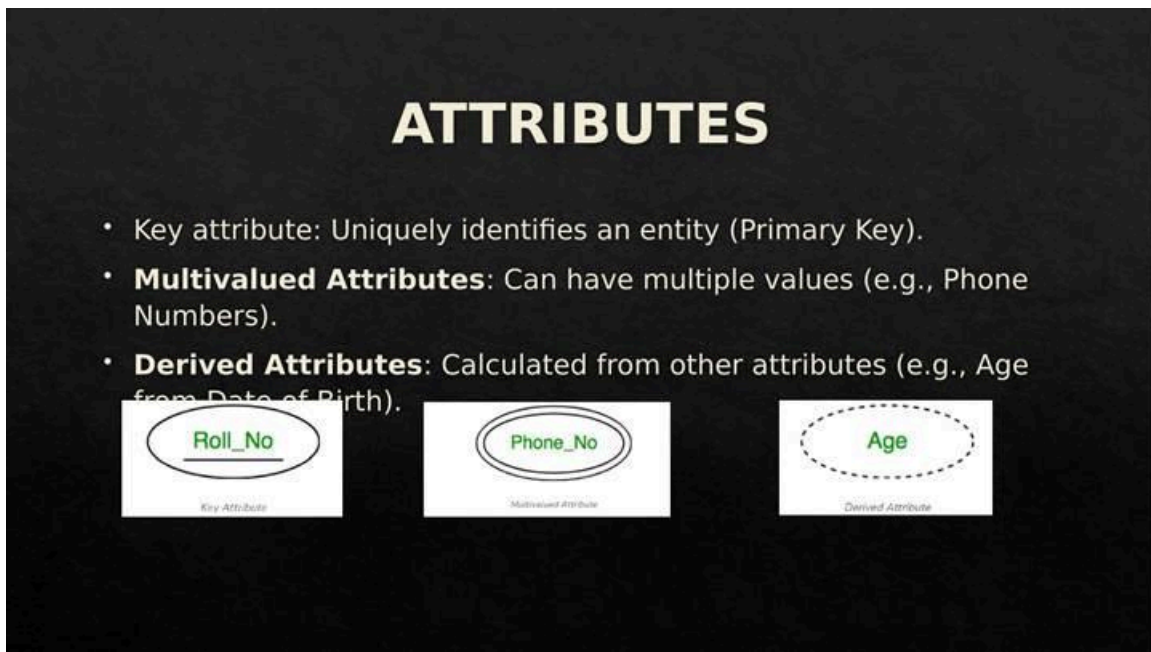


Diagram: Key, Multivalued, and Derived Attributes (Slide 6)

4. Entity and Attributes — Combined Example

The diagram below brings together an entity with its full set of attribute types in a single ER diagram.

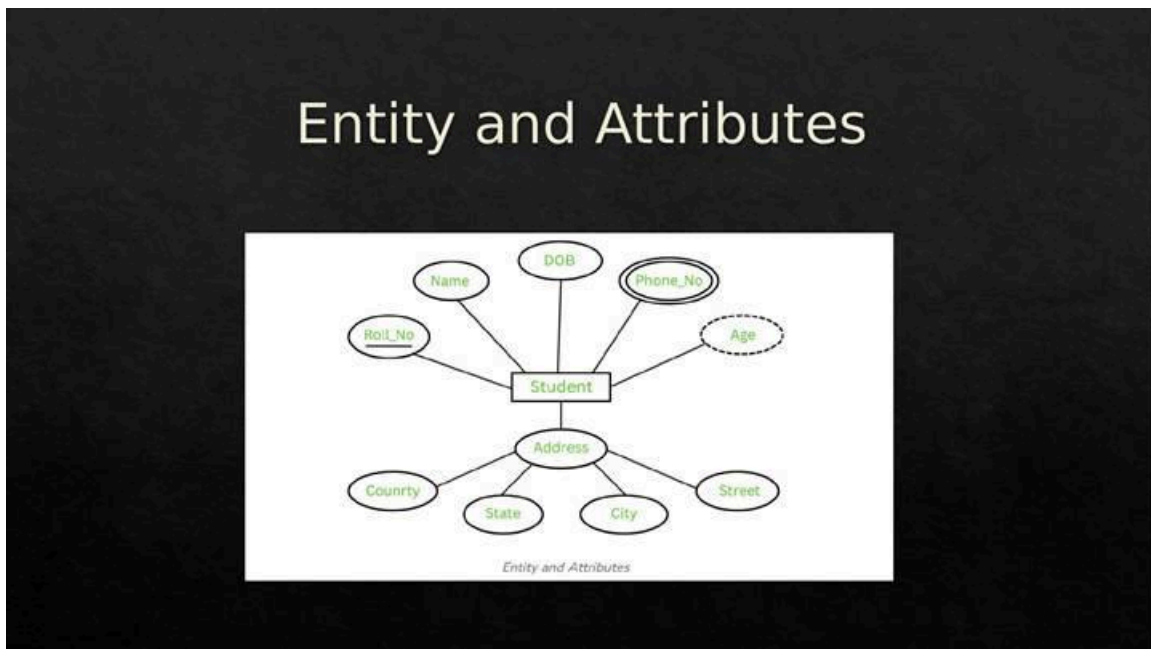


Diagram: Entity with Attributes (Slide 7)

5. Relationships

A relationship represents an association between two or more entities.

Types of Relationships:

- One-to-One (1:1)
- One-to-Many (1:M)
- Many-to-Many (M:N)

6. One-to-One (1:1) Relationship

One-to-One (1:1) → Each entity in A is related to one entity in B.

Example: COUNTRY – PRESIDENT.

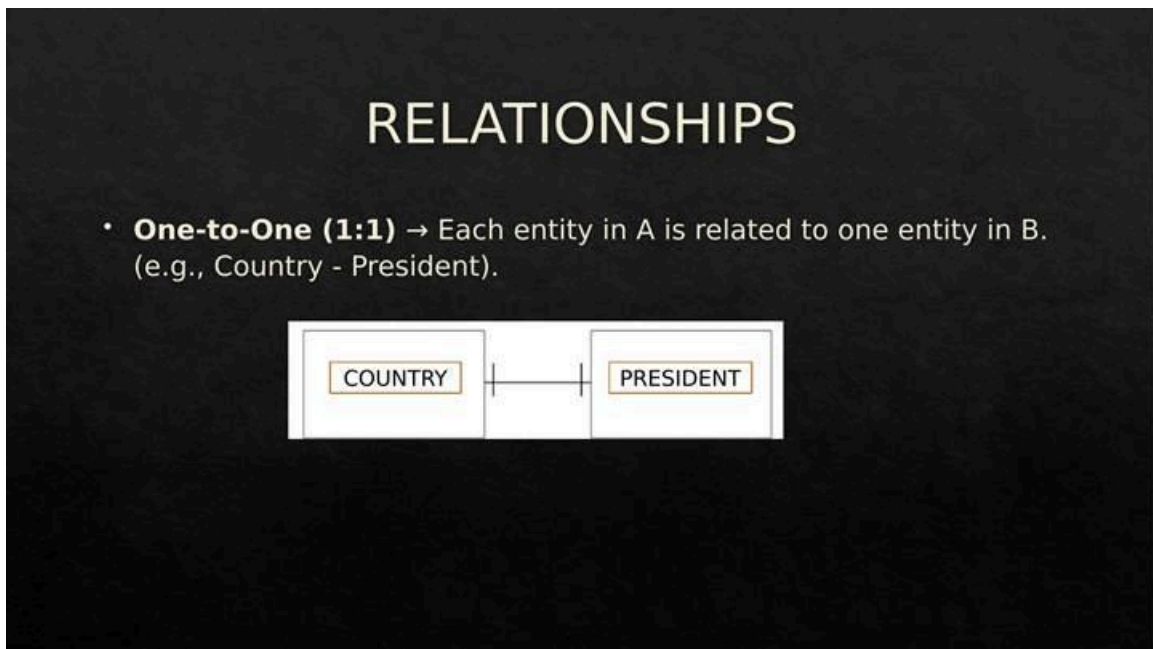


Diagram: One-to-One Relationship — Country and President (Slide 9)

7. One-to-Many (1:M) Relationship

One-to-Many (1:M) → One entity in A can relate to multiple entities in B.

Example: TEACHER – STUDENTS.

RELATIONSHIPS

- **One-to-Many (1:M)** → One entity in A can relate to multiple entities in B. (e.g., Teacher - Students).

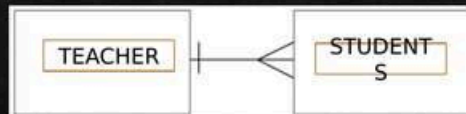


Diagram: One-to-Many Relationship — Teacher and Students (Slide 10)

8. Many-to-Many (M:N) Relationship

Many-to-Many (M:N) → Multiple entities in A relate to multiple entities in B.

Example: STUDENTS – COURSES.

RELATIONSHIPS

- **Many-to-Many (M:N)** → Multiple entities in A relate to multiple entities in B. (e.g., Students - Courses).

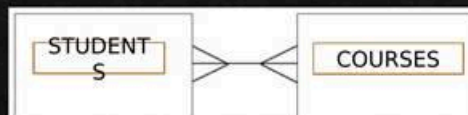


Diagram: Many-to-Many Relationship — Students and Courses (Slide 11)

9. Cardinality

Cardinality defines the number of instances of an entity involved in a relationship.

Types:

- 1:1 (One-to-One)
- 1:M (One-to-Many)
- M:N (Many-to-Many)

10. Practice Question

Generate an ER Diagram based on the scenario below:

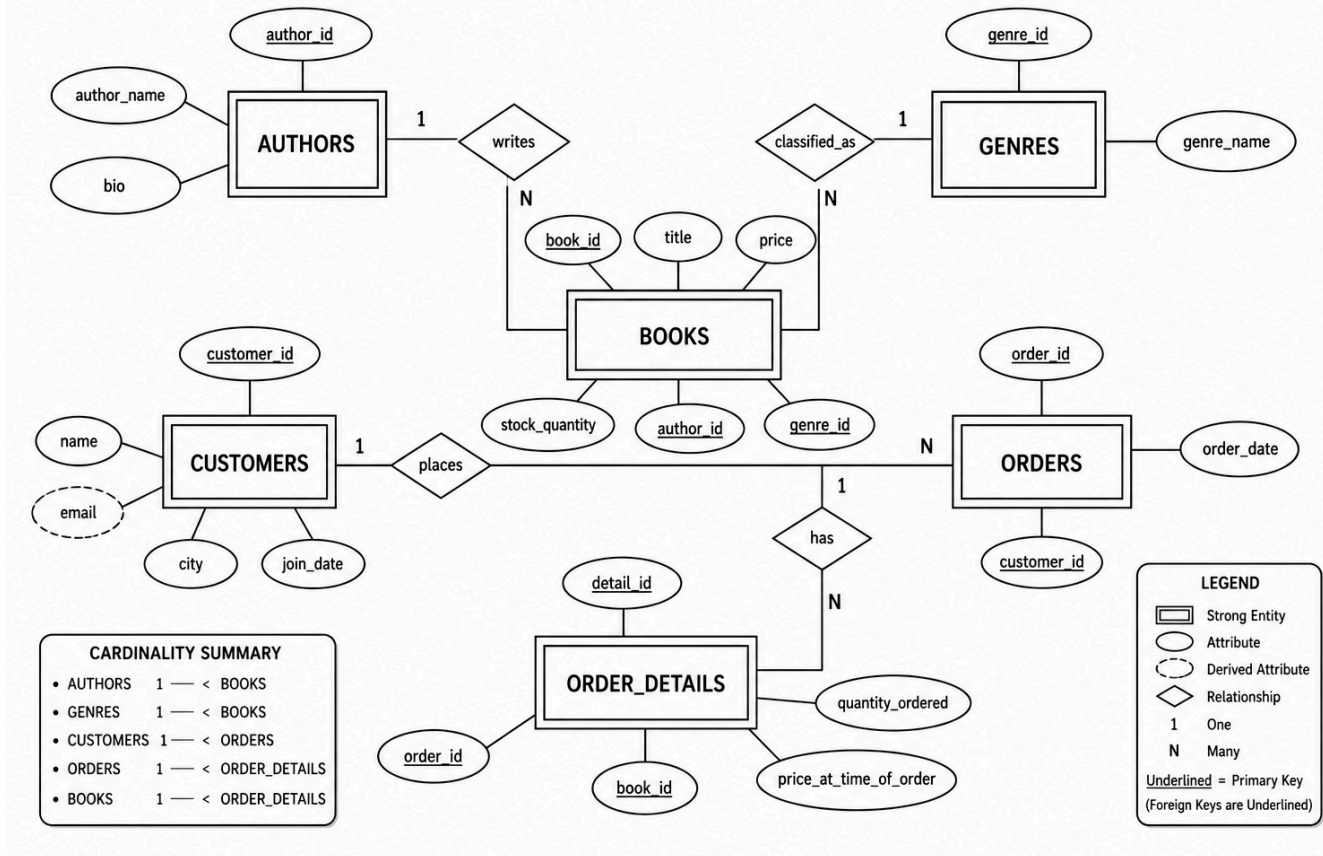
- Patient details will be provided with values: ID [PK], Name, Age, and Visit Date.
- Test Reports will be provided with values: Name [PK], Date, and Result.
- A Patient will visit a Doctor, where Doctor has values: ID [PK], Name, and Specialization.

MySQL Learning Scenario: The "Kitab

Ghar" Bookstore

This document contains a complete MySQL learning scenario based on an Indian online bookstore context. It includes 6 tables, DDL/DML scripts, and 3 levels of practice questions.

BOOKSTORE MANAGEMENT SYSTEM – ER DIAGRAM



1. Database Structure (DDL)

Run these scripts to create your tables.

```
-- 1. Authors
CREATE TABLE Authors (
    author_id INT AUTO_INCREMENT PRIMARY KEY,
    author_name VARCHAR(100) NOT NULL,
    bio TEXT
);
```

-- 2. Genres

CREATE TABLE Genres (

genre_id INT AUTO_INCREMENT PRIMARY KEY,

genre_name VARCHAR(50) NOT NULL

);

-- 3. Customers

CREATE TABLE Customers (

customer_id INT AUTO_INCREMENT PRIMARY KEY,

name VARCHAR(100),

email VARCHAR(100) UNIQUE, city

VARCHAR(50),

join_date DATE

);

-- 4. Books

CREATE TABLE Books (

book_id INT AUTO_INCREMENT PRIMARY

KEY, title VARCHAR(150),

price DECIMAL(10, 2),

stock_quantity INT,

author_id INT, genre_id

INT,

FOREIGN KEY (author_id) REFERENCES Authors(author_id),

FOREIGN KEY (genre_id) REFERENCES Genres(genre_id)

);

-- 5. Orders

CREATE TABLE Orders (

order_id INT AUTO_INCREMENT PRIMARY KEY,

order_date DATE,

customer_id INT,

FOREIGN KEY (customer_id) REFERENCES Customers(customer_id)

);

-- 6. Order_Details

CREATE TABLE Order_Details (

detail_id INT AUTO_INCREMENT PRIMARY KEY,

```
order_id INT, book_id
INT,
quantity_ordered INT,
price_at_time_of_order DECIMAL(10,2),
FOREIGN KEY (order_id) REFERENCES Orders(order_id),
FOREIGN KEY (book_id) REFERENCES Books(book_id)
);
```

2. Sample Data (DML)

Run these scripts to insert data (Indian Context).

```
INSERT INTO Authors (author_name, bio) VALUES
('Chetan Bhagat', 'Indian author known for 5 Point Someone.'),
('Arundhati Roy', 'Winner of the Man Booker Prize.'),
('R.K. Narayan', 'Creator of Malgudi Days.'), ('Amish
Tripathi', 'Known for the Shiva Trilogy.');
```

```
INSERT INTO Genres (genre_name) VALUES ('Fiction'),
('Mythology'), ('History'), ('Romance');
```

```
INSERT INTO Books (title, price, stock_quantity, author_id, genre_id) VALUES
('Five Point Someone', 250.00, 50, 1, 1),
('2 States', 300.00, 40, 1, 4),
('The God of Small Things', 450.00, 20, 2, 1),
('Malgudi Days', 200.00, 100, 3, 1),
('The Immortals of Meluha', 399.00, 60, 4, 2);
```

```
INSERT INTO Customers (name, email, city, join_date) VALUES
('Aarav Sharma', 'aarav@gmail.com', 'Mumbai', '2023-01-10'),
('Diya Patel', 'diya.p@yahoo.com', 'Ahmedabad', '2023-02-15'), ('Vihaan
Reddy', 'vihaan.r@outlook.com', 'Hyderabad', '2023-03-20'), ('Ananya
Gupta', 'ananya.g@gmail.com', 'Delhi', '2023-04-05');
```

```
INSERT INTO Orders (order_date, customer_id) VALUES
('2023-05-01', 1),
('2023-05-03', 2),
('2023-05-05', 1),
('2023-06-01', 3);
```

```
INSERT INTO Order_Details (order_id, book_id, quantity_ordered, price_at_time_of_order)
VALUES
(1, 1, 1, 250.00),
(1, 4, 2, 200.00),
(2, 5, 1, 399.00),
(3, 2, 1, 300.00),
(4, 3, 1, 450.00);
```

3. Practice Questions

Level 1: Beginner

1. List all books that cost less than 300 Rupees.

```
SELECT title, price FROM Books
WHERE price < 300;
```

2. Find the email address of 'Vihaan Reddy'.

```
SELECT email FROM
Customers
WHERE name = 'Vihaan Reddy';
```

3. Display all books in the 'Fiction' genre (ID 1).

```
SELECT title FROM Books
WHERE genre_id = 1;
```

4. List all authors alphabetically.

```
SELECT author_name FROM Authors
ORDER BY author_name ASC;
```

5. Count how many customers are from 'Mumbai'.

```
SELECT COUNT(*)  
FROM Customers  
WHERE city = 'Mumbai';
```

6. Show the most expensive book.

```
SELECT * FROM Books ORDER  
BY price DESC LIMIT 1;
```

Level 2: Intermediate

1. List book titles with Author names.

```
SELECT b.title, a.author_name FROM Books b  
JOIN Authors a ON b.author_id = a.author_id;
```

2. Total amount spent by 'Aarav Sharma'.

```
SELECT c.name, SUM(od.price_at_time_of_order * od.quantity_ordered)  
FROM Customers c  
JOIN Orders o ON c.customer_id = o.customer_id JOIN Order_Details od ON  
o.order_id = od.order_id WHERE c.name = 'Aarav Sharma'  
GROUP BY c.name;
```

3. Genre with highest stock.

```
SELECT g.genre_name, SUM(b.stock_quantity) as Total FROM Genres g  
JOIN Books b ON g.genre_id = b.genre_id GROUP BY g.genre_name  
ORDER BY Total DESC LIMIT 1;
```

4. Orders in May 2023.

```
SELECT * FROM Orders  
WHERE order_date BETWEEN '2023-05-01' AND '2023-05-31';
```


5. Books never ordered.

```
SELECT b.title FROM Books b
LEFT JOIN Order_Details od ON b.book_id = od.book_id WHERE od.detail_id IS NULL;
```

6. Total books sold per author.

```
SELECT a.author_name, SUM(od.quantity_ordered)

FROM Authors a
JOIN Books b ON a.author_id = b.author_id
JOIN Order_Details od ON b.book_id = od.book_id GROUP BY a.author_name;
```

Level 3: Advanced

1. Customer with highest number of orders.

```
SELECT c.name, COUNT(o.order_id) as Cnt FROM Customers c
JOIN Orders o ON c.customer_id = o.customer_id GROUP BY c.name
ORDER BY Cnt DESC LIMIT 1;
```

2. Authors with average book price > 300.

```
SELECT a.author_name, AVG(b.price) FROM Authors a
JOIN Books b ON a.author_id = b.author_id GROUP BY a.author_name
HAVING AVG(b.price) > 300;
```

3. Best selling book.

```
SELECT b.title, SUM(od.quantity_ordered) as Sold FROM Books b
JOIN Order_Details od ON b.book_id = od.book_id GROUP BY b.title
ORDER BY Sold DESC LIMIT 1;
```

4. Customers who bought 'Malgudi Days' AND 'Five Point Someone'.

```

SELECT c.name FROM Customers c WHERE
c.customer_id IN (
    SELECT o.customer_id FROM Orders o
    JOIN Order_Details od ON o.order_id = od.order_id JOIN Books b ON od.book_id =
    b.book_id

    WHERE b.title = 'Malgudi Days'
)
AND c.customer_id IN (
    SELECT o.customer_id FROM Orders o
    JOIN Order_Details od ON o.order_id = od.order_id JOIN Books b ON od.book_id =
    b.book_id
    WHERE b.title = 'Five Point Someone'
);

```

5. Customers with no orders in 6 months.

```

SELECT name FROM Customers WHERE
customer_id NOT IN (
    SELECT customer_id FROM Orders
    WHERE order_date >= DATE_SUB(CURDATE(), INTERVAL 6 MONTH)
);

```

6. Revenue from 'Fiction'.

SQL

```

SELECT SUM(od.quantity_ordered * od.price_at_time_of_order)
FROM Genres g
JOIN Books b ON g.genre_id = b.genre_id
JOIN Order_Details od ON b.book_id = od.book_id WHERE g.genre_name =
'Fiction';

```

Instagram Clone ERD (in the style you want)

Entities and attributes:

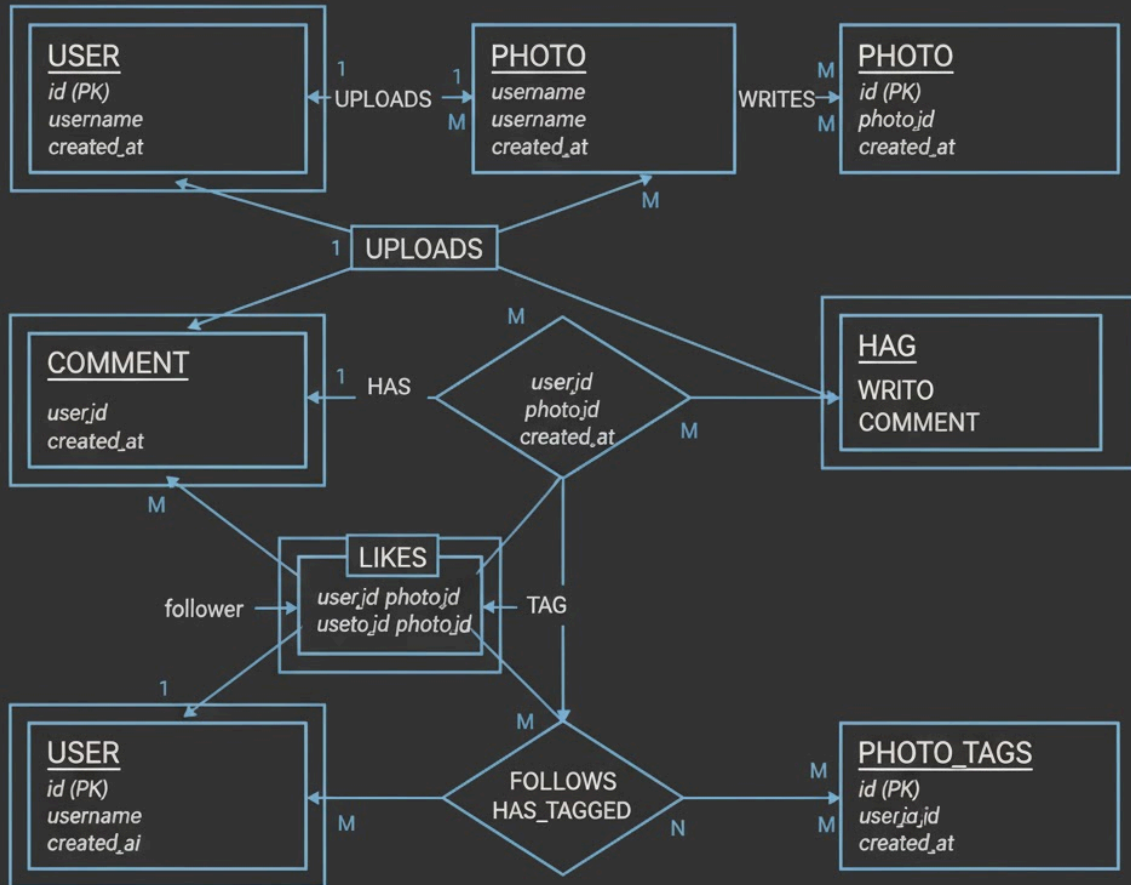
- **User (strong entity)**
Attributes: **id (PK)**, **username**, **created_at**
- **Photo (strong entity)**
Attributes: **id (PK)**, **image_url**, **created_at**
- **Comment (strong entity)**
Attributes: **id (PK)**, **comment_text**, **created_at**
- **Like (weak/composite key entity)**
Attributes: **user_id (PK, FK)**, **photo_id (PK, FK)**, **created_at**
- **Follow (weak/composite key entity, self-relationship)**
Attributes: **follower_id (PK, FK)**, **followee_id (PK, FK)**, **created_at**
- **Tag (strong entity)**
Attributes: **id (PK)**, **tag_name**, **created_at**
- **Photo_Tags (weak/composite key entity)**
Attributes: **photo_id (PK, FK)**, **tag_id (PK, FK)**

◆ **Relationships (with cardinalities like your example):**

- A User uploads Photos → 1 User : M Photos
- A User writes Comments → 1 User : M Comments
- A Photo has Comments → 1 Photo : M Comments
- A User Likes Photos → M:N (resolved via **Likes** weak entity)
- A User Follows another User → M:N (resolved via **Follows** weak entity, self-relationship)
- A Photo has Tags → M:N (resolved via **Photo_Tags**)

Instagram Clone ER Diagram

- Rectangle: Strong Weak Entity
- Double DOBILOND: Identifying Relationship
- Underlined: Primary Key
- Underlined: Simple Attribute
- Relationship
- Underlined: Primary Key



ER diagram — quick summary

- Entities (tables): **users**, **photos**, **comments**, **likes**, **follows**, **tags**, **photo_tags**.
- Primary keys: single or composite as in your DDL.
- Foreign keys: all FKs reference parent PKs (you used **ON DELETE NO ACTION / ON UPDATE NO ACTION**).
- Relationship types:

- users (1) — (N) photos (a user uploads many photos)
- photos (1) — (N) comments (a photo has many comments)
- users (1) — (N) comments (a user authors many comments)
- users (N) — (N) photos via **likes** (many users can like many photos)
- users (N) — (N) users via **follows** (self-referential many-to-many: follower ↔ followee)
- photos (N) — (N) tags via **photo_tags** (a photo can have many tags, tag can label many photos)

Entities & attributes (as in diagram)

1. USERS

- **PK** id : INT UNSIGNED AUTO_INCREMENT
- username : VARCHAR(255) — **UNIQUE**
- created_at : TIMESTAMP DEFAULT NOW()

2. PHOTOS

- **PK** id : INT UNSIGNED AUTO_INCREMENT
- image_url : VARCHAR(255)
- **FK** user_id : INT UNSIGNED → users(id)
- created_at : TIMESTAMP DEFAULT NOW()

3. COMMENTS

- **PK** id : INT UNSIGNED AUTO_INCREMENT

- `comment_text` : VARCHAR(255)
- **FK** `photo_id` : INT UNSIGNED → `photos(id)`
- **FK** `user_id` : INT UNSIGNED → `users(id)`
- `created_at` : TIMESTAMP DEFAULT NOW()

4. **LIKES** (associative/join table implementing many-to-many)

- **PK (composite)** (`user_id`, `photo_id`)
- **FK** `user_id` → `users(id)`
- **FK** `photo_id` → `photos(id)`
- `created_at` : TIMESTAMP DEFAULT NOW()

5. **FOLLOWS** (self-referencing many-to-many between users)

- **PK (composite)** (`follower_id`, `followee_id`)
- **FK** `follower_id` → `users(id)` (role: follower)
- **FK** `followee_id` → `users(id)` (role: followee)
- `created_at` : TIMESTAMP DEFAULT NOW()

6. **TAGS**

- **PK** `id` : INT UNSIGNED AUTO_INCREMENT
- `tag_name` : VARCHAR(255) — **UNIQUE**
- `created_at` : TIMESTAMP DEFAULT NOW()

7. **PHOTO_TAGS** (associative table for photo↔tag many-to-many)

- **PK (composite)** (`photo_id`, `tag_id`)

- **FK** `photo_id` → `photos(id)`
- **FK** `tag_id` → `tags(id)`

Relationships, cardinalities & roles (explicit)

- `users.id` —< `photos.user_id` : **1** : **N** (one user uploads many photos).
- `photos.id` —< `comments.photo_id` : **1** : **N** (one photo has many comments).
- `users.id` —< `comments.user_id` : **1** : **N** (one user authors many comments).
- `users.id` —< `likes.user_id` and `photos.id` —< `likes.photo_id` : **N** : **M** (likes as join table implements many-to-many; composite PK prevents duplicate likes).
- `users.id` —< `follows.follower_id` and `users.id` —< `follows.followee_id` : **N** : **M** self-relationship (follows).
- `photos.id` —< `photo_tags.photo_id` and `tags.id` —< `photo_tags.tag_id` : **N** : **M** (photo-tags).

Instagram clone

```
DROP DATABASE IF EXISTS instagram_clone_db;
```

```
CREATE DATABASE instagram_clone_db;
```

```
USE instagram_clone_db;
```

```
CREATE TABLE users (
  id INT UNSIGNED NOT NULL AUTO_INCREMENT,
  username VARCHAR(255) NOT NULL,
  created_at TIMESTAMP NOT NULL DEFAULT NOW(),
  PRIMARY KEY (id),
  UNIQUE INDEX username_UNIQUE (username ASC) VISIBLE
);
```

```
CREATE TABLE photos (  
  id INT UNSIGNED NOT NULL AUTO_INCREMENT,  
  image_url VARCHAR(255) NOT NULL,  
  user_id INT UNSIGNED NOT NULL,  
  created_at TIMESTAMP NOT NULL DEFAULT NOW(),  
  PRIMARY KEY (id),  
  INDEX user_id_photos_idx (user_id ASC) VISIBLE,  
  CONSTRAINT user_id_photos  
    FOREIGN KEY (user_id)  
    REFERENCES users (id)  
    ON DELETE NO ACTION  
    ON UPDATE NO ACTION  
);
```

```
CREATE TABLE comments (  
  id INT UNSIGNED NOT NULL AUTO_INCREMENT,  
  comment_text VARCHAR(255) NOT NULL,  
  photo_id INT UNSIGNED NOT NULL,  
  user_id INT UNSIGNED NOT NULL,  
  created_at TIMESTAMP NOT NULL DEFAULT NOW(),  
  PRIMARY KEY (id),  
  INDEX photo_id_comments_idx (photo_id ASC) VISIBLE,  
  INDEX user_id_comments_idx (user_id ASC) VISIBLE,  
  CONSTRAINT photo_id_comments  
    FOREIGN KEY (photo_id)  
    REFERENCES photos (id)  
    ON DELETE NO ACTION  
    ON UPDATE NO ACTION,  
  CONSTRAINT user_id_comments  
    FOREIGN KEY (user_id)  
    REFERENCES users (id)  
    ON DELETE NO ACTION  
    ON UPDATE NO ACTION  
);
```

```
CREATE TABLE likes (  
  user_id INT UNSIGNED NOT NULL,  
  photo_id INT UNSIGNED NOT NULL,  
  created_at TIMESTAMP NOT NULL DEFAULT NOW(),  
  PRIMARY KEY (user_id, photo_id),
```



```

INDEX user_id_likes_idx (user_id ASC) VISIBLE,
INDEX photo_id_likes_idx (photo_id ASC) VISIBLE,
CONSTRAINT user_id_likes
    FOREIGN KEY (user_id)
    REFERENCES users (id)
    ON DELETE NO ACTION
    ON UPDATE NO ACTION,
CONSTRAINT photo_id_likes
    FOREIGN KEY (photo_id)
    REFERENCES photos (id)
    ON DELETE NO ACTION
    ON UPDATE NO ACTION
);

CREATE TABLE follows (
    follower_id INT UNSIGNED NOT NULL,
    followee_id INT UNSIGNED NOT NULL,
    created_at TIMESTAMP NOT NULL DEFAULT NOW(),
    PRIMARY KEY (follower_id, followee_id),
    INDEX follower_id_follows_idx (follower_id ASC) VISIBLE,
    INDEX followee_id_follows_idx (followee_id ASC) VISIBLE,
    CONSTRAINT follower_id_follows
        FOREIGN KEY (follower_id)
        REFERENCES users (id)
        ON DELETE NO ACTION
        ON UPDATE NO ACTION,
    CONSTRAINT followee_id_follows
        FOREIGN KEY (followee_id)
        REFERENCES users (id)
        ON DELETE NO ACTION
        ON UPDATE NO ACTION
);

CREATE TABLE tags (
    id INT UNSIGNED NOT NULL AUTO_INCREMENT,
    tag_name VARCHAR(255) NOT NULL,
    created_at TIMESTAMP NOT NULL DEFAULT NOW(),
    PRIMARY KEY (id),
    UNIQUE INDEX tag_name_UNIQUE (tag_name ASC) VISIBLE
);

```

```

CREATE TABLE photo_tags (
  photo_id INT UNSIGNED NOT NULL,
  tag_id INT UNSIGNED NOT NULL,
  PRIMARY KEY (photo_id, tag_id),
  INDEX photo_id_photo_tags_idx (photo_id ASC) VISIBLE,
  INDEX tag_id_photo_tags_idx (tag_id ASC) VISIBLE,
  CONSTRAINT photo_id_photo_tags
    FOREIGN KEY (photo_id)
      REFERENCES photos (id)
    ON DELETE NO ACTION
    ON UPDATE NO ACTION,
  CONSTRAINT tag_id_photo_tags
    FOREIGN KEY (tag_id)
      REFERENCES tags (id)
    ON DELETE NO ACTION
    ON UPDATE NO ACTION
);

```

```

INSERT INTO users (username, created_at)
VALUES ('Kenton_Kirlin', '2017-02-16 18:22:10.846'), ('Andre_Purdy85', '2017-04-02
17:11:21.417'), ('Harley_Lind18', '2017-02-21 11:12:32.574'), ('Arely_Bogan63', '2016-08-13
01:28:43.085'), ('Aniya_Hackett', '2016-12-07 01:04:39.298'), ('Travon.Waters', '2017-04-30
13:26:14.496'), ('Kasandra_Homenick', '2016-12-12 06:50:07.996'), ('Tabitha_Schamberger11',
'2016-08-20 02:19:45.512'), ('Gus93', '2016-06-24 19:36:30.978'), ('Presley_McClure',
'2016-08-07 16:25:48.561'), ('Justina.Gaylord27', '2017-05-04 16:32:15.577'), ('Dereck65',
'2017-01-19 01:34:14.296'), ('Alexandro35', '2017-03-29 17:09:02.344'), ('Jaclyn81', '2017-02-06
23:29:16.394'), ('Billy52', '2016-10-05 14:10:20.453'), ('Annalise.McKenzie16', '2016-08-02
21:32:45.646'), ('Norbert_Carroll35', '2017-02-06 22:05:43.425'), ('Odessa2', '2016-10-21
18:16:56.390'), ('Hailee26', '2017-04-29 18:53:39.650'), ('Delpha.Kihn', '2016-08-31
02:42:30.288'), ('Rocio33', '2017-01-23 11:51:15.467'), ('Kenneth64', '2016-12-27
09:48:17.380'), ('Eveline95', '2017-01-23 23:14:18.569'), ('Maxwell.Halvorson', '2017-04-18
02:32:43.597'), ('Tierra.Trantow', '2016-10-03 12:49:20.774'), ('Josianne.Friesen', '2016-06-07
12:47:00.703'), ('Darwin29', '2017-03-18 03:10:07.047'), ('Dario77', '2016-08-18 07:15:02.823'),
('Jaime53', '2016-09-11 18:51:56.965'), ('Kaley9', '2016-09-23 21:24:20.222'), ('Aiyana_Hoeger',
'2016-09-29 20:28:12.457'), ('Irwin.Larson', '2016-08-26 19:36:22.199'), ('Yvette.Gottlieb91',
'2016-11-14 12:32:01.405'), ('Pearl7', '2016-07-08 21:42:00.982'), ('Lennie_Hartmann40',
'2017-03-30 03:25:21.937'), ('Ollie_Ledner37', '2016-08-04 15:42:20.322'), ('Yazmin_Mills95',
'2016-07-27 00:56:44.310'), ('Jordyn.Jacobson2', '2016-05-14 07:56:25.835'), ('Kelsi26',
'2016-06-08 17:48:08.478'), ('Rafael.Hickle2', '2016-05-19 09:51:25.779'), ('Mckenna17',

```

'2016-07-17 17:25:44.855'), ('Maya.Farrell', '2016-12-11 18:04:45.344'), ('Janet.Armstrong', '2016-10-06 07:57:44.491'), ('Seth46', '2016-07-07 11:40:26.557'), ('David.Osinski47', '2017-02-05 21:23:37.392'), ('Malinda_Streich', '2016-07-09 21:37:07.610'), ('Harrison.Beatty50', '2016-09-02 03:48:38.340'), ('Granville_Kutch', '2016-06-26 03:10:22.202'), ('Morgan.Kassulke', '2016-10-30 12:42:31.387'), ('Gerard79', '2016-08-23 19:47:44.102'), ('Mariano_Koch3', '2017-04-17 14:14:45.662'), ('Zack_Kemmer93', '2017-01-01 05:58:22.276'), ('Linnea59', '2017-02-07 07:49:33.830'), ('Duane60', '2016-12-21 04:43:37.761'), ('Meggie_Doyle', '2017-04-04 12:17:33.931'), ('Peter.Stehr0', '2016-08-22 18:05:42.167'), ('Julien_Schmidt', '2017-02-02 23:12:48.451'), ('Aurelie71', '2016-05-31 06:20:56.909'), ('Cesar93', '2016-10-18 16:42:43.220'), ('Sam52', '2017-03-30 22:03:45.159'), ('Jayson65', '2016-10-14 19:10:52.564'), ('Ressie_Stanton46', '2016-12-20 15:09:08.721'), ('Elenor88', '2016-05-08 01:30:40.677'), ('Florence99', '2016-10-06 23:08:30.626'), ('Adelle96', '2016-10-01 00:37:57.429'), ('Mike.Auer39', '2016-07-01 17:36:14.714'), ('Emilio_Bernier52', '2016-05-06 13:04:29.960'), ('Franco_Keebler64', '2016-11-13 20:09:26.855'), ('Karley_Bosco', '2016-06-24 23:38:52.138'), ('Erick5', '2017-04-05 23:44:47.060'), ('Nia_Haag', '2016-05-14 15:38:50.230'), ('Kathryn80', '2016-10-11 09:01:56.764'), ('Jaylan.Lakin', '2016-06-10 23:58:52.210'), ('Hulda.Macejkovic', '2017-01-25 17:17:27.717'), ('Leslie67', '2016-09-21 05:14:01.207'), ('Janelle.Nikolaus81', '2016-07-21 09:26:09.466'), ('Donald.Fritsch', '2017-01-07 10:05:41.165'), ('Colten.Harris76', '2016-10-10 02:38:52.941'), ('Katarina.Dibbert', '2016-11-03 13:14:10.647'), ('Darby_Herzog', '2016-05-06 00:14:21.191'), ('Esther.Zulauf61', '2017-01-14 17:02:33.511'), ('Aracely.Johnston98', '2016-07-25 18:49:09.996'), ('Bartholome.Bernhard', '2016-11-06 02:31:23.463'), ('Alysa22', '2017-01-01 17:44:42.980'), ('Milford_Gleichner42', '2017-04-30 07:50:51.280'), ('Delfina_VonRueden68', '2017-03-21 12:02:14.358'), ('Rick29', '2017-02-24 11:25:08.160'), ('Clint27', '2016-06-02 21:40:09.555'), ('Jessyca_West', '2016-09-14 23:47:04.780'), ('Esmeralda.Mraz57', '2017-03-03 11:52:27.469'), ('Bethany20', '2016-06-03 23:31:53.322'), ('Frederik_Rice', '2016-07-06 21:56:28.654'), ('Willie_Leuschke', '2017-02-15 01:40:53.310'), ('Damon35', '2016-10-31 14:44:27.239'), ('Nicole71', '2016-05-09 17:30:22.371'), ('Keenan.Schamberger60', '2016-08-28 14:57:28.221'), ('Tomas.Beatty93', '2017-02-11 11:38:55.026'), ('Imani_Nicolas17', '2017-01-31 22:59:34.108'), ('Alek_Watsica', '2016-12-10 07:43:58.083'), ('Javonte83', '2017-03-27 22:06:37.433');

INSERT INTO photos (image_url, user_id)

VALUES ('http://elijah.biz', 1), ('https://shanon.org', 1), ('http://vicky.biz', 1), ('http://oleta.net', 1), ('https://jennings.biz', 1), ('https://quinn.biz', 2), ('https://selina.name', 2), ('http://malvina.org', 2), ('https://branson.biz', 2), ('https://elenor.name', 3), ('https://marcelino.com', 3), ('http://felicity.name', 3), ('https://fred.com', 3), ('https://gerhard.biz', 4), ('https://sherwood.net', 4), ('https://maudie.org', 4), ('http://annamae.name', 6), ('https://mac.org', 6), ('http://miracle.info', 6), ('http://emmet.com', 6), ('https://lisa.com', 6), ('https://brooklyn.name', 8), ('http://madison.net', 8), ('http://annie.name', 8), ('http://darron.info', 8), ('http://saige.com', 9), ('https://reece.net', 9), ('http://vance.org', 9), ('http://ignacio.net', 9), ('http://kenny.com', 10),

('http://remington.name', 10), ('http://kurtis.info', 10), ('https://alisha.com', 11), ('https://henderson.com', 11), ('http://bonnie.info', 11), ('http://kennith.net', 11), ('http://camille.name', 11), ('http://alena.net', 12), ('http://ralph.name', 12), ('https://tyshawn.com', 12), ('https://adella.net', 12), ('https://cielo.info', 13), ('https://easter.net', 13), ('http://golden.org', 13), ('http://kendall.biz', 13), ('https://glenda.info', 13), ('http://dominic.biz', 15), ('http://tressie.info', 15), ('http://estevan.org', 15), ('http://zena.com', 15), ('https://abagail.com', 16), ('https://hershel.com', 16), ('http://collin.com', 16), ('https://clair.com', 16), ('https://deondre.com', 17), ('http://tristin.name', 17), ('http://kirk.org', 17), ('https://modesta.info', 18), ('http://rylan.biz', 19), ('https://noemie.com', 19), ('https://dejon.name', 20), ('https://rigoberto.net', 22), ('http://megane.biz', 23), ('http://emmalee.org', 23), ('http://nathan.net', 23), ('http://lionel.net', 23), ('http://danyka.net', 23), ('https://woodrow.com', 23), ('http://harvey.info', 23), ('http://aiden.org', 23), ('https://tito.name', 23), ('https://julian.net', 23), ('https://mafalda.org', 23), ('http://verner.org', 23), ('https://elmore.org', 26), ('http://kasandra.com', 26), ('https://jarret.info', 26), ('http://einar.net', 26), ('http://terry.info', 26), ('https://holden.com', 27), ('https://jacinto.org', 28), ('https://geoffrey.info', 28), ('http://paxton.com', 28), ('https://trinity.biz', 28), ('http://fabiola.org', 29), ('https://bryce.org', 29), ('http://emery.net', 29), ('https://marian.biz', 29), ('https://kennedi.org', 29), ('https://fanny.net', 29), ('http://lottie.net', 29), ('http://lacy.biz', 29), ('https://jensen.name', 30), ('http://virginia.org', 30), ('https://ariel.net', 31), ('http://roger.info', 32), ('https://carolanne.com', 32), ('https://margarita.info', 32), ('https://kayden.biz', 32), ('https://brook.com', 33), ('https://gust.net', 33), ('http://bridie.name', 33), ('http://barton.name', 33), ('https://karina.biz', 33), ('https://mariam.com', 35), ('https://trycia.com', 35), ('https://everette.biz', 37), ('http://boris.biz', 38), ('http://arthur.name', 38), ('https://cesar.com', 39), ('http://charlie.com', 40), ('https://lina.biz', 42), ('https://darwin.net', 42), ('https://aliyah.biz', 42), ('http://euna.info', 43), ('https://maymie.net', 43), ('http://joanie.name', 43), ('http://whitney.net', 43), ('http://garrison.name', 43), ('https://olga.org', 44), ('https://donavon.org', 44), ('http://moses.biz', 44), ('http://shannon.org', 44), ('http://kendrick.net', 46), ('https://carey.com', 46), ('http://lia.biz', 46), ('https://celestine.name', 46), ('http://laila.info', 47), ('http://buddy.com', 47), ('http://americo.biz', 47), ('http://lurline.info', 47), ('http://kailee.org', 47), ('https://edyth.com', 48), ('https://isaias.biz', 50), ('http://rosetta.net', 50), ('https://marianna.info', 50), ('https://roel.org', 51), ('http://julia.info', 51), ('https://seamus.org', 51), ('https://earline.info', 51), ('http://quinten.info', 51), ('https://dolores.org', 52), ('https://aimee.info', 52), ('https://roderick.name', 52), ('https://jarret.name', 52), ('https://cathrine.com', 52), ('https://adela.com', 55), ('https://demarco.com', 56), ('https://lila.biz', 58), ('http://ruben.net', 58), ('https://alia.name', 58), ('http://rickey.name', 58), ('http://janie.com', 58), ('http://myron.org', 58), ('http://pearl.info', 58), ('http://victoria.biz', 58), ('https://carmella.com', 59), ('http://leo.com', 59), ('http://jedediah.net', 59), ('http://sedrick.net', 59), ('https://clare.org', 59), ('https://malcolm.name', 59), ('https://lesly.com', 59), ('https://roberta.net', 59), ('https://estell.info', 59), ('http://kaleigh.biz', 59), ('http://blanca.org', 60), ('http://delpha.com', 60), ('http://annabelle.org', 61), ('http://ephraim.com', 62), ('https://dameon.name', 62), ('https://julianne.org', 63),

('https://jasmin.biz', 63), ('https://delbert.net', 63), ('https://norval.com', 63), ('https://jessy.name', 64), ('https://nolan.name', 64), ('http://devon.info', 64), ('http://ora.net', 64), ('https://vivien.com', 64), ('https://hilda.info', 65), ('https://dorcas.biz', 65), ('http://hettie.net', 65), ('http://rico.biz', 65), ('http://jody.biz', 65), ('http://sheila.com', 67), ('http://germaine.name', 67), ('https://mariane.org', 67), ('http://reva.com', 69), ('https://llewellyn.info', 70), ('http://ellen.info', 72), ('https://anahi.info', 72), ('https://javonte.org', 72), ('http://florine.org', 72), ('http://marcellus.info', 72), ('http://stan.biz', 73), ('http://jarred.net', 77), ('http://jovanny.net', 77), ('http://linnie.com', 77), ('http://dino.name', 77), ('https://jane.net', 77), ('http://laney.name', 77), ('http://trever.org', 78), ('http://emilie.net', 78), ('http://ted.info', 78), ('http://keshawn.org', 78), ('http://ole.info', 78), ('https://ford.biz', 79), ('https://humberto.org', 82), ('http://penelope.info', 82), ('https://dimitri.info', 84), ('http://cynthia.info', 84), ('http://princess.org', 85), ('https://ruthie.info', 85), ('https://evan.name', 86), ('http://deontae.org', 86), ('https://timmy.info', 86), ('http://susie.info', 86), ('http://hollis.com', 86), ('https://colt.org', 86), ('http://lucio.org', 86), ('http://germaine.org', 86), ('http://howard.net', 86), ('http://helena.name', 87), ('https://celestino.name', 87), ('http://violet.info', 87), ('http://nigel.biz', 87), ('https://noble.net', 88), ('https://emilia.org', 88), ('http://destini.net', 88), ('https://mariano.net', 88), ('https://brittany.org', 88), ('https://madaline.name', 88), ('https://devonte.name', 88), ('https://blanche.net', 88), ('http://cali.net', 88), ('http://mekhi.name', 88), ('http://adela.com', 88), ('https://devan.com', 92), ('https://jarrett.name', 92), ('https://sid.biz', 92), ('https://jadyn.name', 93), ('https://erik.com', 93), ('http://freeda.biz', 94), ('http://ayden.name', 95), ('https://kathleen.biz', 95), ('https://helmer.org', 96), ('https://maggie.info', 96), ('https://cecilia.net', 96), ('http://ayla.org', 97), ('https://elyssa.biz', 97), ('http://jennie.com', 98), ('http://ryleigh.info', 99), ('https://darien.name', 99), ('https://xzavier.org', 99), ('https://kaela.name', 100), ('http://dedrick.info', 100);

INSERT INTO follows (follower_id, followee_id)

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INSERT INTO tags (tag_name)

VALUES ('sunset'), ('photography'), ('sunrise'), ('landscape'), ('food'), ('foodie'), ('delicious'), ('beauty'), ('stunning'), ('dreamy'), ('lol'), ('happy'), ('fun'), ('style'), ('hair'), ('fashion'), ('party'), ('concert'), ('drunk'), ('beach'), ('smile');

INSERT INTO photo_tags (photo_id, tag_id)

VALUES (1, 18), (1, 17), (1, 21), (1, 13), (1, 19), (2, 4), (2, 3), (2, 20), (2, 2), (3, 8), (4, 12), (4, 11), (4, 21), (4, 13), (5, 15), (5, 14), (5, 17), (5, 16), (6, 19), (6, 13), (6, 17), (6, 21), (7, 11), (7, 12), (7, 21), (7, 13), (8, 17), (8, 21), (8, 13), (8, 19), (9, 18), (10, 2), (11, 12), (11, 21), (11, 11), (12, 4), (13, 13), (13, 19), (14, 1), (14, 20), (17, 19), (17, 13), (17, 18), (19, 5), (21, 20), (21, 3), (21, 1), (21, 4), (22, 7), (22, 5), (22, 6), (23, 18), (23, 19), (23, 13), (23, 21), (24, 12), (24, 13), (24, 21), (25, 8), (25, 21), (25, 10), (26, 20), (26, 21), (26, 8), (26, 9), (26, 10), (28, 9), (28, 10), (29, 8), (29, 21), (29, 9), (29, 20), (30, 7), (30, 6), (30, 5), (31, 7), (31, 5), (32, 10), (32, 9), (32, 21), (32, 8), (32, 20), (34, 11), (34, 13), (34, 12), (35, 6), (35, 7), (35, 5), (36, 5), (36, 7), (36, 6), (37, 18), (37, 21), (37, 13), (38, 3), (38, 2), (43, 11), (43, 21), (43, 12), (44, 20), (44, 10), (45, 3), (45, 1), (47, 20), (49, 21), (49, 18), (50, 5), (51, 21), (51, 8), (51, 20), (53, 8), (55, 12), (56, 11), (57, 13), (57, 18), (57, 21), (57, 19), (57, 17), (58, 8), (59, 20), (59, 9), (59, 10), (59, 21), (60, 16), (60, 15), (60, 17), (61, 20), (61, 10), (62, 2), (63, 6), (64, 12), (64, 13), (64, 21), (64, 11), (65, 7), (65, 5), (66, 15), (66, 17), (68, 11), (68, 21), (70, 21), (70, 12), (70, 11), (71, 14), (71, 15), (71, 16), (72, 20), (72, 8), (72, 21), (73, 13), (73, 21), (74, 16), (74, 15), (75, 1), (75, 20), (75, 4), (76, 11), (76, 12), (78, 16), (78, 17), (78, 15), (79, 6), (79, 5), (79, 7), (83, 1), (83, 3), (84, 17), (84, 18), (84, 13), (85, 20), (85, 1), (85, 3), (86, 13), (87, 12), (87, 13), (87, 21), (88, 18), (88, 17), (90, 14), (90, 17), (91, 3), (91, 4), (91, 1), (92, 19), (93, 17), (93, 21), (93, 13), (93, 18),

(94, 19), (94, 18), (94, 13), (94, 17), (94, 21), (95, 18), (95, 19), (95, 17), (95, 21), (95, 13), (96, 15), (96, 17), (97, 7), (97, 5), (97, 6), (98, 16), (98, 15), (98, 14), (98, 17), (100, 20), (101, 14), (101, 16), (102, 14), (102, 16), (102, 15), (103, 5), (107, 11), (107, 12), (107, 21), (108, 11), (109, 9), (109, 8), (109, 10), (110, 2), (111, 17), (111, 15), (111, 14), (111, 16), (112, 21), (112, 9), (112, 20), (116, 7), (117, 21), (118, 20), (118, 4), (118, 1), (120, 21), (120, 19), (120, 17), (121, 21), (121, 13), (121, 11), (121, 12), (122, 15), (123, 17), (123, 14), (123, 15), (124, 15), (124, 16), (124, 17), (125, 11), (125, 12), (126, 2), (126, 3), (126, 4), (127, 21), (131, 5), (132, 19), (132, 17), (132, 13), (132, 18), (132, 21), (133, 15), (133, 17), (133, 16), (133, 14), (134, 21), (134, 11), (134, 12), (134, 13), (135, 17), (136, 8), (136, 9), (136, 21), (136, 20), (136, 10), (137, 11), (137, 13), (137, 21), (137, 12), (138, 9), (138, 20), (138, 10), (139, 5), (140, 11), (140, 12), (141, 17), (142, 21), (142, 20), (145, 18), (145, 13), (145, 19), (145, 17), (145, 21), (147, 5), (148, 11), (149, 20), (149, 4), (149, 1), (149, 2), (152, 16), (152, 17), (152, 15), (152, 14), (153, 15), (153, 16), (153, 17), (153, 14), (154, 21), (155, 10), (155, 21), (155, 20), (155, 9), (155, 8), (156, 6), (156, 5), (157, 13), (157, 11), (157, 12), (157, 21), (158, 15), (159, 5), (160, 5), (161, 11), (162, 15), (162, 14), (162, 17), (163, 21), (163, 13), (163, 19), (166, 13), (166, 17), (168, 6), (168, 5), (169, 12), (171, 10), (171, 21), (171, 9), (171, 8), (171, 20), (172, 15), (172, 16), (172, 14), (173, 16), (173, 15), (173, 17), (174, 5), (175, 17), (175, 18), (177, 15), (177, 17), (178, 16), (181, 5), (181, 7), (182, 8), (182, 10), (182, 20), (182, 21), (183, 5), (184, 13), (184, 21), (184, 12), (185, 20), (185, 8), (185, 21), (185, 9), (185, 10), (188, 13), (188, 21), (188, 11), (189, 10), (189, 8), (189, 20), (190, 17), (190, 18), (191, 5), (191, 6), (194, 1), (197, 13), (197, 18), (197, 21), (197, 19), (199, 11), (199, 13), (201, 3), (201, 4), (201, 2), (201, 1), (201, 20), (202, 21), (202, 18), (202, 19), (202, 17), (203, 17), (203, 19), (203, 18), (205, 18), (205, 21), (205, 17), (206, 14), (207, 13), (207, 21), (208, 4), (208, 20), (209, 20), (209, 3), (209, 2), (210, 2), (210, 1), (211, 18), (211, 13), (212, 8), (212, 21), (212, 20), (212, 9), (214, 14), (214, 15), (214, 17), (214, 16), (216, 3), (216, 1), (218, 8), (218, 20), (219, 17), (219, 19), (219, 13), (219, 21), (219, 18), (220, 2), (220, 3), (220, 20), (220, 4), (221, 12), (221, 13), (221, 21), (222, 7), (225, 7), (226, 12), (226, 13), (226, 21), (226, 11), (227, 2), (227, 3), (227, 4), (227, 1), (227, 20), (228, 20), (230, 10), (230, 21), (230, 9), (230, 8), (230, 20), (231, 3), (231, 4), (231, 1), (232, 8), (232, 10), (232, 20), (232, 21), (232, 9), (233, 3), (233, 1), (233, 2), (233, 20), (235, 10), (237, 10), (237, 20), (238, 1), (238, 4), (238, 3), (238, 2), (238, 20), (239, 17), (239, 21), (239, 18), (240, 20), (240, 4), (242, 1), (242, 20), (242, 4), (242, 2), (243, 1), (244, 7), (244, 5), (244, 6), (247, 16), (247, 14), (248, 10), (248, 9), (249, 7), (249, 5), (250, 14), (250, 16), (250, 15), (251, 11), (253, 7), (254, 4), (254, 2), (254, 20), (254, 3), (257, 13), (257, 19), (257, 18);

Project challenges

=====

/* Finding 5 oldest users */

```
SELECT ID, USERNAME, CREATED_AT
FROM users
ORDER BY CREATED_AT ASC;
```

```
SELECT * FROM users
    ORDER BY created_at
    LIMIT 5;
```

/**

* We need to figure out when to schedule an ad campaign.

* What days of the week do most users register on?

**/

```
SELECT
    DAYNAME(created_at) AS day_of_the_week,
    COUNT(*) AS total_count
FROM users
GROUP BY day_of_the_week
ORDER BY total_count DESC;
```

/**

* We want to target our inactive users with an email campaign.

* Find the users who have never posted a photo?

**/

```
SELECT
    users.id,
    username,
    users.created_at as user_joining_date
FROM users
    LEFT JOIN photos
        ON users.id = photos.user_id
WHERE photos.user_id IS NULL;
```

/**

* We are running a new contest to see who can get the most likes on a single photo.

* Find out who won the contest? 🙌

**/

SELECT

users.id AS user_id,

username,

photos.id AS photo_id,

photos.image_url,

COUNT(*) AS total_likes_count

FROM photos

JOIN likes

ON photos.id = likes.photo_id

JOIN users

ON users.id = photos.user_id

GROUP BY photos.id

ORDER BY total_likes_count DESC

LIMIT 1;

/**

* Our Investors would like to know...

* How many times does the average user post?

**/

SELECT

ROUND(

(SELECT COUNT(*) FROM photos) / (SELECT COUNT(*) FROM users),

2

) AS avg_user_post;

/**

* A Brand wants to know which hashtags to use in a Post.

* What are the Top 5 most commonly used hashtags?

**/

SELECT


```

tags.id AS tag_id,
tags.tag_name,
COUNT(*) as total_tags_count
FROM tags
JOIN photo_tags
ON tags.id = photo_tags.tag_id
GROUP BY tags.id
ORDER BY total_tags_count DESC
LIMIT 5;

```

```

*****

```

```

*****

```

```

/**

```

```

* We have a small problem with bots on our site...

```

```

* Find number of users who have liked every single photo on the site?

```

```

**/

```

```

SELECT
users.id AS user_id,
users.username,
COUNT(*) AS total_user_likes
FROM users
JOIN likes
ON users.id = likes.user_id
GROUP BY users.id
HAVING total_user_likes = (
SELECT COUNT(*) FROM photos
);

```